

What Is Valuable to Know?

A Statistical Design Document for Investment-Worth Modeling
on PitchBook/FactSet Deal Data

hawkes-sim design document — the master reference for model and code decisions

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How to read this document

This document is the master reference for the `hawkes-sim` modeling effort. Every future code change in `hawkes_calibration/` should trace back to a numbered decision, assumption, or protocol stated here. It replaces the earlier working notes (`THEORY.md`, `MATH_NOTES.md`, `STRATEGY.md`) as the single source of truth. The design chapters do not quote empirical results: validation on synthetic data is planned in Chapter 19, and real-data results belong in Chapter 20.

The document is organized around one question — *what is valuable to know?* — posed in Chapter 1 and answered differently by every subsequent part. Chapter 2 fixes, once, what the PitchBook/FactSet data delivers and the six ways observation corrupts it. Chapters 6 to 8, 10 and 11 build the continuous-time machinery: point-process foundations, Poisson and exponential-Hawkes models with covariate-driven excitation and inhibition, the treatment of untrusted and censored deal times, and goodness-of-fit checks. Chapter 12 treats the covariate staleness that PitchBook imposes. Chapters 13 and 14 develop the discrete-time bucket formulation solved with gradient boosting and the ranking view evaluated by recall at K . Chapter 15 handles deals arriving from companies outside the tracked universe. Chapters 16 and 17 close the loop: from model outputs to investment decisions, and from decisions to code.

Three reading conventions. First, *assumptions are contracts*: every model result is stated under numbered assumptions, and the text says which assumptions are believed, which are convenient fictions, and which are known to be false in the data (with the consequence of the violation). Second, *Design decision* boxes record commitments: a choice among alternatives, the rationale, and the trigger that would reverse it. Third, *Code hooks* boxes at the end of each chapter map the mathematics onto modules of `hawkes_calibration/`, existing or planned; these boxes, together with Chapter 17, are the implementation contract.

Proofs are given in the running text when they are short and instructive; longer arguments are deferred to the ends of chapters. Citations are restricted to work we actually rely on — for a theorem, a method we implement, or a warning we heed — and the bibliography is annotated in that spirit throughout the text.

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Part I

The Question and the Data

Chapter 1

The Central Question: What Is Valuable to Know?

1.1 Worth, formally

The business objective is to understand which private-market investments are “worth it.” That phrase hides a decision problem. The first rule of this document is to make the decision problem explicit *before* any model is chosen. An investment thesis about a private company is a bet on a few future observables: whether the company raises again, when, at what valuation, and whether the eventual exit repays the capital deployed today. A model is useful only if it improves one of those bets. So the central question is not “which model is best” but:

What is valuable to know?

Formally, let $\mathcal{D}$ be a space of actions available to the investor (allocate attention to a company, enter a diligence process, price a secondary position, pass), let ω denote the future evolution of the market — the marked point process of deals defined in Chapter 2 — and let $u(d, \omega)$ be the utility of action $d \in \mathcal{D}$ under outcome ω . A predictive object π (a probability, an intensity, a ranking) is *valuable* exactly to the extent that

$$\text{VoI}(\pi) = \mathbb{E}[u(d_\pi^*, \omega)] - \mathbb{E}[u(d_0^*, \omega)], \quad d_\pi^* = \arg \max_{d \in \mathcal{D}} \mathbb{E}[u(d, \omega) \mid \pi], \quad (1.1)$$

is positive and large, where d_0^* is the action taken without the model. This is the standard value-of-information decomposition. Its practical force here is negative: it rules out models whose outputs no action consumes. Every model family in this document is admitted only because a concrete action consumes its output, and each family’s evaluation metric is chosen as a proxy for Equation (1.1) under that action, not as a generic accuracy score.

1.2 The four answers and the targets they induce

The sourcing and monitoring workflows that would consume model output give four answers to “what is valuable to know.” Each induces a different mathematical target on the same underlying data.

T1 — Identity: *which* company gets funded next. The sourcing action is a short list: given finite diligence capacity, which K companies should be looked at now because they are most likely to be the next ones to raise? The natural target is the conditional distribution of the next deal’s identity,

$$\mathbb{P}(\text{next deal in sector } s \text{ goes to firm } i \mid \mathcal{F}_{t-}), \quad (1.2)$$

and the natural metric is ranking quality — recall at K against the next deals actually observed (Chapter 14). Crucially, Equation (1.2) conditions on *a deal occurring*: the timing of deals cancels from the target, which is why identity models can survive data defects that destroy timing models (Section 1.3).

T2 — Timing: *when the deals arrive*. Pricing, portfolio pacing, and capital-deployment planning consume arrival times: how many deals will sector s produce next quarter, and how is firm i ’s next round distributed in time? The target is the conditional intensity $\lambda(t)$ of the deal process (Chapter 6) and functionals of it (expected counts, waiting-time distributions), evaluated by likelihood on held-out windows and by calibration diagnostics (Chapter 11). This is the target the Poisson and Hawkes machinery of Chapters 7 and 8 serves.

T3 — Horizon events: *will firm i raise within Δ* . Many consuming actions are binary and horizon-bound: enter diligence now if the company is likely to raise within the next six months; deprioritize if no event is plausible within two years. The target is the fixed-horizon event probability

$$p_i(t, \Delta) = \mathbb{P}(N_i(t + \Delta) - N_i(t) \geq 1 \mid \mathcal{F}_t), \quad \Delta \in \{6M, 1Y, 2Y\}, \quad (1.3)$$

a discrete-time object that can be attacked directly with supervised learning on a bucketed panel (Chapter 13) — at the price of label-construction subtleties that Chapter 13 treats in detail.

T4 — Marks: *size, valuation, terms*. Finally, worth depends on *what* the deal is, not only that it happens: round size, pre/post valuation, up/down/flat classification. These are mark distributions conditional on an event, $\mathbb{P}(m \in \cdot \mid \text{deal at } t \text{ by } i, \mathcal{F}_{t-})$. This document develops T4 only to the extent needed by the decision layer (Chapter 16); its estimation reuses the regression machinery of Chapter 13 and inherits every covariate caveat of Chapter 12.

These four targets are not competing models of the same quantity. They are different marginals and conditionals of one object — the marked point process — consumed by different actions. The failure mode this document is designed to prevent is answering one target’s question with another target’s model and metric (for instance, selling a horizon classifier’s probability, T3, as a timing model, T2; or judging a ranking model, T1, by likelihood alone).

1.3 One process, two factorizations

The unifying object is the marked point process of deals: ground times t_m (when a deal happens), marks (s_m, i_m, m_m) (which sector, which firm, what terms). Its likelihood factorizes *exactly* as

$$p(\text{times, sectors, firms, terms}) = \underbrace{p(\text{times, sectors})}_{\text{ground process}} \times \underbrace{p(\text{firm} \mid \text{sector, time, history})}_{\text{mark factor}} \times \underbrace{p(\text{terms} \mid \text{firm, time})}_{\text{deal marks}} \quad (1.4)$$

This identity is established as Theorem 6.7; it is the decomposition labeled M1 in the repository’s earlier theory notes, and the whole design rests on it. It is an identity of likelihoods, not an approximation: all approximations live *inside* one factor at a time, and data defects can be assigned to the factor they damage. Three consequences organize the document.

First, *the targets map onto factors*. T2 lives in the ground process; T1 lives in the mark factor; T4 lives in the deal marks; T3 is a horizon functional of ground $\times$ mark. A team can therefore improve one factor without touching the others, and Chapter 17 assigns code modules factor by factor.

Second, *the mark factor is robust where the ground process is fragile*. The mark factor is a conditional-choice likelihood in which the sector-time baseline cancels (Chapter 14); it survives untrusted deal dates, weekly aggregation, and even moderate intensity misspecification. The ground process absorbs precisely those defects. When the data cannot support a trustworthy ground model, the mark factor still delivers T1. This asymmetry, developed in Chapters 10 and 14, is why ranking is the most defensible deliverable on current data quality.

Third, *discrete-time models are the same object at coarser resolution*. The bucketed panel of Chapter 13 is the ground+mark object aggregated to intervals: Theorem 13.4 shows the bucket-probability model is exactly a discrete-time hazard model, and recovers the continuous-time intensity as bucket width shrinks. XGBoost on buckets and Hawkes in continuous time are therefore the same object at different resolutions. The choice between them is decided by the target and by the data defects of Chapter 2.

1.4 From questions to models: the decision map

Table 1.1 is the document in one table. Each row is an answer to “what is valuable to know”; the columns give the mathematical target, the model family this document commits to for that target, the evaluation metric that proxies Equation (1.1), and the chapter that develops it.

Valuable to know	Target	Committed model	Metric	Ch.
Which firm raises next (identity)	$\mathbb{P}(i \mid \text{deal}, s, t, \mathcal{H})$	risk-set conditional logit $\rightarrow$ gradient-boosted ranker	recall@ K , MRR, held-out conditional NLL	14
When/how many deals (timing)	ground intensity $\lambda_s(t)$	covariate NHPP $\rightarrow$ exponential Hawkes with engineered excitation/inhibition	held-out NLL, time-rescaling, PIT	7–11
Will i raise within Δ (horizon)	$p_i(t, \Delta)$, $\Delta \in \{6M, 1Y, 2Y\}$	discrete-hazard panel + XGBoost with staleness features	calibrated probabilities; PR-AUC; lift@ K	13
Deal terms (marks)	$p(m \mid i, t)$	mark regressions on deal covariates	per-mark scores	16

Table 1.1: The decision map: answers to “what is valuable to know,” the targets they induce, and the models and metrics this document commits to.

Within the point-process answer there is also a non-log-linear, likelihood-based option: the intensity comes from a boosted-tree, random-forest, or small-network model on engineered covariates (Chapter 9). It trades interpretability and closed-form theory for flexibility, and it applies when deal times are trusted (uncensored).

Design decision (D1.1 — target before model).

No model is built, modified, or benchmarked in this repository without first naming which target (T1–T4) it serves and which action consumes its output. Pull requests that improve a metric not attached to a target are rejected by policy. Rationale: Equation (1.1); the failure mode it prevents is optimizing likelihood in a factor whose output nothing consumes. Reversal trigger: none anticipated.

Design decision (D1.2 — both stacks, one object).

We maintain *both* the continuous-time stack (NHPP/Hawkes ground process + risk-set mark factor) and the discrete-time stack (bucketed panel + XGBoost), as views of the same marked process at different resolutions, rather than choosing one ideology. Rationale: we expect the covariate-driven baseline to carry most of the predictive mass for timing, with excitation adding a smaller increment, and we expect a gradient-boosted learner on equal features to beat the linear conditional logit on identity. Both expectations are hypotheses to be checked on the synthetic world (Chapter 19). Each stack should dominate somewhere. Reversal trigger: if, after the stale-covariate remedies of Chapter 12, one stack dominates on *every* consuming action for two consecutive evaluation campaigns, the other is demoted to benchmark-only status.

1.5 What “worth it” will mean operationally

The decision layer of Chapter 16 composes the four targets into the quantities the investment workflow consumes: a ranked sourcing list with horizon-qualified event probabilities attached (T1 + T3), sector pacing forecasts with uncertainty bands (T2), and expected-terms context (T4). Until that chapter, “worth it” is used in the narrow sense of Equation (1.1): a model output is worth producing if and only if the action that consumes it gains utility, and the evaluation protocols are designed so that offline metrics are monotone proxies for that gain. The chapters in between make those outputs trustworthy on the data we actually have — the subject of the next chapter.

Code hooks.

The decision map of Table 1.1 is mirrored in code by keeping the factors of Equation (1.4) in separate modules with separate evaluation harnesses: ground-process fitters (hawkes_calibration/sector_stability.py, hawkes_calibration/eventtime/), mark-factor fitters (hawkes_calibration/sector_ranker.py, hawkes_calibration/models/hawkes_ranker.py), and (to be written) the bucketed-panel builder and horizon classifier of Chapter 13 (hawkes_calibration/panel/), the staleness state layer of Chapter 12 (hawkes_calibration/covstate/), and the decision layer of Chapter 16 (hawkes_calibration/decision/). Chapter 17 gives the full mapping with migration steps.

Chapter 2

The Data: PitchBook Events, FactSet Macro, and What Observation Does to Them

2.1 Sources and window

Two data sources feed every model in this document.

PitchBook: the event-driven layer. The deal history of the private market, delivered as three relational tables whose column-level dictionary is given in Appendix B. The *deals* table is the event log: one row per financing event, keyed by `deal_id` and `company_id`, carrying the deal date, sector taxonomy (`primary_industry_sector / group / code`), round descriptors (`vc_round`, `deal_type`, `deal_class`, up/down/flat flags), economics (`deal_size` with a status flag distinguishing Actual/Estimated/Undisclosed, `pre_money_valuation`, `post_valuation`, `raised_to_date`, invested capital and ownership fields), firm financials as reported around the deal (`revenue`, `ebitda`, `gross_profit`, `net_income`, employees, growth since last deal), and geography. The *companies* table is a snapshot per tracked firm: identity, sector, founding year, business status, and last-observed-deal summaries (`last_financing_date`, `_size`, `_valuation`, `_deal_type`). The *investors* table links firms to their investors with first-investment dates. The universe flag `company_universe` marks whether a deal belongs to a firm PitchBook tracks; deals do occur with `company_universe = 'Out of Universe'`, whose firms have *no row* in the companies table — defect (C6) below.

FactSet: the market-clock layer. A small panel of macro series in FactSet Standardized Economics form: policy and effective rates (`FFUNDT`, `FFUND`, daily), inflation (`CPI_RAW` level and `CPI_YOY`, monthly), unemployment (`RUNEMP`, monthly), each published with a lag of days to weeks after its reference period. These series are the exogenous covariates of the ground process: they move on calendar time regardless of deal activity, which makes them the only covariates in the system that are *never stale* in the sense of Chapter 12 — but they are *lagged*, and the lag matters for point-in-time discipline (Section 2.4).

Window. The joined history runs from January 2011 through June 2024: $T \approx 13.5$ years, i.e. roughly 704 weeks of joint deal-and-macro history. All development described in this document uses a temporal split: a training window, a validation window for model selection, and a strictly held-out test window at the end (the split protocol is fixed in Chapter 11). The 2011 start postdates the crisis regime shift of 2008–2010 by design; the June 2024 end is an administrative censoring time in the precise sense of defect (C1).

2.2 The formal observation model

We now fix, once, the probabilistic description of what the data is. Every later chapter states its assumptions relative to this model.

Definition 2.1 (Market process). Work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$. The market is a marked point process

$$\{(t_m, s_m, i_m, m_m)\}_{m \geq 1}, \quad t_1 \leq t_2 \leq \dots,$$

where $t_m \in [0, T]$ is the (true) time of the m -th deal, $s_m \in \mathcal{S}$ its sector, $i_m \in \mathcal{I} \cup \mathcal{I}^\dagger$ the receiving firm — $\mathcal{I}$ the tracked universe, $\mathcal{I}^\dagger$ the untracked remainder — and m_m the deal marks (size, valuation, round type, ...). For each sector s , $N_s(t)$ counts deals in s up to t ; for each tracked firm i , $N_i(t)$ counts firm i 's deals. The filtration $\mathcal{F}_t$ contains the internal history of the marked process and the paths of all exogenous covariates up to t .

Definition 2.2 (Covariate processes). Three covariate processes enter the models:

- (i) *Macro covariates* $x^{\text{mac}}(t) \in \mathbb{R}^p$: exogenous, càdlàg, observed for all t but only through their *published* version $x_{\text{pub}}^{\text{mac}}(t) = x^{\text{mac}}(t - L(t))$, where $L(t)$ is the publication lag process (Section 2.4).
- (ii) *Firm covariates* $x_i(t) \in \mathbb{R}^q$ for $i \in \mathcal{I}$: the true, latent, evolving state of firm i (scale, financials, capitalization). The data does *not* deliver $x_i(t)$; it delivers the last value recorded at firm i 's most recent deal,

$$\tilde{x}_i(t) = x_i(\tau_i(t)), \quad \tau_i(t) = \sup\{t_m < t : i_m = i\}, \quad (2.1)$$

the last-observation-carried-forward (LOCF) covariate, whose observation times are themselves event times of the modeled process. This is defect (C5), the subject of Chapter 12.

- (iii) *Engineered history covariates*: deterministic functionals of the observed history $\mathcal{H}_t$ — chosen by us — time since firm i 's last deal, deal counts in trailing windows, sector activity summaries, staleness ages. These are always available and always point-in-time correct; Chapter 8 uses them to build excitation *and* inhibition into intensity models, and Chapter 13 feeds them to gradient boosting.

Definition 2.3 (Conditional intensity and at-risk indicator). For a counting process N_e (e a sector, firm, or firm-transition index), a *conditional intensity* $\lambda_e(t)$ is a nonnegative predictable process such that

$$M_e(t) = N_e(t) - \int_0^t Y_e(u) \lambda_e(u) du \quad (2.2)$$

is a local martingale, where $Y_e(t) \in \{0, 1\}$ is the at-risk indicator: the unit is under observation and eligible for the event at t^- (Andersen et al., 1993). All likelihoods in this document are instances of the point-process (Jacod) likelihood

$$\ell(\vartheta) = \sum_m \log \lambda_{e_m}(t_m; \vartheta) - \sum_e \int_0^T Y_e(u) \lambda_e(u; \vartheta) du, \quad (2.3)$$

whose martingale estimation theory (unbiased score at the truth, Fisher information as predictable variation, asymptotic normality) is developed in Chapter 6 following Andersen et al. (1993) and Andersen and Gill (1982).

Most of the data's pathology concentrates in the at-risk indicator: nearly every defect below corrupts either Y (who is really at risk, and since when), or the event times t_m (when did it really happen), or the covariates the intensity conditions on.

2.3 The defect ledger

Six defects, (C1)–(C6), exhaust the ways this dataset deviates from the ideal of Definition 2.1. For each: the formal statement, the consequence if ignored, and the chapter that owns the treatment. The first four restate, in final form, defects identified in the repository’s earlier theory notes; (C5) and (C6) are treated as first-class defects here because the data review showed they bind hardest.

(C1) Administrative right censoring at the window end

Every firm alive in June 2024 has its next round unobserved. Formally, censoring is the predictable process $C_i(t) = \mathbf{1}\{t \leq T\}$; under independent censoring the observed process $\int C_i dN_i$ has intensity $C_i(t)\lambda_i(t)$ and the likelihood Equation (2.3) computed on observed data is a genuine partial likelihood (Andersen et al., 1993, III.2). Administrative censoring at fixed T is independent by construction. The censored spell contributes exactly its survival factor $\exp(-\int \lambda)$: no term dropped, no term invented. This is the benign defect — *provided open spells are kept*.

Warning 2.4. Dropping open (censored) spells converts right censoring into right *truncation*: selection on the event occurring in-window, which oversamples short inter-deal gaps and biases every fitted hazard upward. This is not classical length bias (which oversamples *long* spells in prevalent-cohort designs); the correct treatment of sampling schemes and truncation is Kalbfleisch and Prentice (2002). Policy: every dataset builder in the repository must emit censored spells with an explicit censoring flag, and every likelihood must consume them. Chapters 7 and 10 implement this for Poisson and Hawkes calibration respectively; Chapter 13 implements the bucketed analogue (censored buckets are labeled, not silently dropped).

(C2) Left truncation: firms enter observation by raising

A firm is invisible to PitchBook until its first recorded financing: it enters observation at $V_i =$ first-deal date, and enters *because of an event*. Mechanically this is delayed entry (Keiding, 1990): risk sets are $Y_i(t) = \mathbf{1}\{V_i \leq t < U_i\}$ and every integral in Equation (2.3) starts at V_i . Statistically it is a selection: the observed universe is “firms conditioned on having raised at least once,” so unconditional statements about companies are not identified from tracked data alone. Chapter 15 is where this selection is modeled rather than ignored.

(C3) Unlabeled exits: the immortal-firm problem

Failures and quiet wind-downs are mostly unrecorded (`business_status` is coarse and laggy). A dead firm keeps $Y_i(t) = 1$ in any naive risk set. Three facts must be kept distinct. (i) The contamination bias — dead firms held at risk deflate every fitted hazard — arises under *any* unlabeled-exit mechanism, even exit independent of funding propensity: it is a misrecorded Y_i , not a dependence effect. (ii) Because exit also correlates with the latent quality that drives funding, we do not expect the defect to be correctable from observables: even a good exit classifier is unlikely to reconstruct truthful risk sets, and we expect risk-set truthfulness to be one of the largest sensitivities in the whole system. Both expectations are to be tested on the synthetic world (Chapter 19). (iii) If exits *were* labeled, removing firms at exit would be valid for cause-specific funding intensities with no independence assumption at all — the standard competing-risks fact (Putter et al., 2007). The problem is that exits are unobserved, not that they are dependent. Treatments — exposure windows, mover–stayer mixtures (Farewell, 1982), and ranking-layer mitigations — are developed where they bind: Chapters 7, 13 and 14.

(C4) Untrusted deal dates and aggregation

Two distinct corruptions of t_m . First, *date jitter*: PitchBook records announced vs. closed dates that can differ by days, and a nontrivial fraction of dates are back-filled or rounded; the recorded $\hat{t}_m$ is best modeled as t_m observed through a small-window error. Second, *deliberate aggregation*: for modeling at weekly resolution we replace times by bin counts, which is interval censoring by construction. Both are instances of one statement: the event count over an interval is trusted, the position within the interval is not. Chapter 10 builds the corresponding likelihoods: interval-censored NHPP calibration in closed form, and interval-censored Hawkes via the mean-intensity (mean-Poisson-process) approximation of Rizoïu et al. (2022), with the information loss quantified — branching-scale parameters survive aggregation, timescale parameters do not (the κ - θ ridge of Chapter 11; Fisher information in the decay $\rightarrow 0$ as bin width grows past the kernel half-life).

(C5) Stale firm covariates, updated only at own deals

Defect Equation (2.1): firm covariates are observed only at the firm’s own deals, so at any t the available $\tilde{x}_i(t)$ has age $a_i(t) = t - \tau_i(t)$, and the age is largest exactly for the firms that have gone longest without raising — the observation times are *informative*, being event times of the process under study. This is simultaneously a measurement-error problem (attenuation of covariate effects; Carroll et al., 2006; Prentice, 1982) and an internal-covariate problem in the sense of Kalbfleisch and Prentice (2002). Chapter 12 formalizes both faces and commits to a layered remedy: staleness-age features always; an explicit covariate-evolution state layer where it pays; joint-model escalation (Rizopoulos, 2012; Tsiatis and Davidian, 2004) only if a trigger fires. The size of the attenuation is to be measured on the synthetic world (Chapter 19).

(C6) Out-of-universe deals: events with no covariates

Deals occur from firms not in the tracked universe (`company_universe = 'Out of Universe'`, no companies-table row, no covariates, no history). Consequences: sector deal counts are complete but firm-attributable counts are thinned; any firm-level risk set is missing members who *do* compete for deals; and the mark factor must own an outside option or it will overstate every tracked firm’s probability. Chapter 15 develops the treatment: an explicit newcomer/outside component in the mark factor (with the discipline that *newcomer probability* — first appearance in the data — must never be conflated with *entrant probability* — economic entry), a ghost/external component in the ground process, and unseen-mass estimation in the Good–Turing spirit (Good, 1953; Gale and Sampson, 1995) as the model-free check on how much deal mass the tracked universe can see.

2.4 Point-in-time discipline

A model that consumes information not available at prediction time is worthless regardless of its metrics. The repository already enforces the following discipline, and this document makes it policy (implemented in `synthetic/loaders.py`, to be preserved verbatim by every real-data loader):

Protocol 2.5 (Point-in-time construction).

- (a) *Macro covariates are as-of joined on publish date*: the value of $x_{\text{pub}}^{\text{mac}}(t)$ is the last value whose *publication* date (not reference date) precedes t .
- (b) *Firm features are LOCF from past deals only*: $\tilde{x}_i(t)$ uses deals strictly before t ; the staleness age $a_i(t)$ is always carried alongside as a feature, never discarded.

Defect	What is corrupted	If ignored	Owned by
(C1)	spells open at June 2024	upward-biased hazards if open spells dropped	Chapters 7, 10 and 13
(C2)	entry into observation (Y_i before first deal)	selection mistaken for dynamics	Chapter 15
(C3)	Y_i after unrecorded death	deflated hazards, inverted rankings	Chapters 7, 13 and 14
(C4)	event times t_m	spurious timing estimates; broken θ	Chapters 10 and 11
(C5)	firm covariates $x_i(t)$	attenuated, staleness-confounded effects	Chapter 12
(C6)	firm identity i_m , risk-set membership	overstated tracked-firm probabilities	Chapter 15

Table 2.1: The defect ledger. Every modeling chapter states which defects it is robust to and which assumptions it needs about the rest.

- (c) *Risk-set membership is inferred from observables*: founding year and first-deal date determine entry; any exit proxy used to close spells must itself be point-in-time.
- (d) *Deduplication and hygiene run before everything*: duplicate deal rows (same firm within a few days, jittered size) are collapsed by the documented rule; sector mislabels and undisclosed sizes are handled by the imputation rules of Appendix B, never ad hoc.
- (e) *Evaluation splits are temporal*, with model selection strictly inside the training era. No hyperparameter, feature transform, or calibration map may see post-split data.

2.5 The synthetic world

The real data’s defects cannot be switched off, so the repository maintains a synthetic world for validation: generated PitchBook-style tables (deals/companies/investors) and a FactSet-style macro feed with publication lags, passed through an observation layer that reproduces defects (C1)–(C6) with known ground truth (**synthetic/**). Two regimes matter: regime A is generated by the exact two-stage model family we fit (a well-specified test), and regime B is a daily firm-level block-Hawkes world with self-inhibition and latent quality, deliberately *not* in the fitted family (a misspecification stress test). The synthetic world is a planned validation tool, not a source of standing results. The experiments it supports are specified in Chapter 19, under a fixed transfer convention: synthetic outcomes set *priors and vetoes* for real-data work, never substitutes for real-data evaluation.

Design decision (D2.1 — defects are assumptions, stated per model).

Every model chapter opens by declaring, defect by defect against Table 2.1, whether the model (i) is robust to the defect, (ii) corrects it under stated assumptions, or (iii) is vulnerable and accepts the bias with a documented sign. No silent vulnerability is permitted. Rationale: we expect the largest errors to come not from wrong models but from unstated exposure to (C3) and (C5); the synthetic experiments of Chapter 19 are designed to test this.

Code hooks.

Real-data ingestion will mirror `synthetic/loaders.py`: an adapter `hawkes_calibration/data/pitchbook.py` (to be written) emitting the same fitter-ready arrays — weekly sector counts, as-of macro matrix, LOCF firm features *plus staleness ages*, at-risk masks, and the event list — so that every fitter downstream is source-agnostic. The defect ledger of Table 2.1 becomes a validation suite (`hawkes_calibration/data/audits.py`): open-spell retention check, entry-date consistency, duplicate-collapse idempotence, as-of-join look-ahead test, and universe-share reporting (`off_universe_share` is already a first-class output of the synthetic loader and must be one for the real loader).

Part II

The Transparent Regime: When Fundamentals Are Observable

Chapter 3

Defining and Predicting Success from Fundamentals

This chapter opens the transparent-regime track of the decision map (Section 1.4). In the European jurisdictions, statutory filings make firm fundamentals — revenue, EBITDA, payroll, balance-sheet items — observable on a fiscal-year clock. The question “is this firm succeeding?” therefore has a directly measurable answer and needs none of the point-process machinery of the later parts: the target is a scalar success label, and the model that serves it is supervised learning on a firm-year panel. Everything here is still downstream of Chapter 2: the deals table and macro feed supply features, and the point-in-time discipline of Protocol 2.5 governs every join. What changes is that a fundamentals panel joins them — and with it new ways to leak the future.

Against the defect ledger of Table 2.1, this regime trades defects rather than escaping them. (C1) reappears as administrative censoring of recent cohorts’ labels. (C3) reappears as the survivor trap of Section 3.4. (C5) is largely *absent* — the statutory clock refreshes covariates whether or not the firm transacts, which is what makes the regime transparent — but a publication lag of 9–12 months takes its place, and Section 3.6 shows it is dangerous on its own. (C4) and (C6) do not bind: fiscal years are trusted dates, and the filing universe is near-complete for limited-liability companies.

3.1 The fundamentals panel

For each EU firm i and fiscal year y we observe, from registry filings,

$$F_{i,y} = (\text{revenue}_{i,y}, \text{EBITDA}_{i,y}, \text{payroll}_{i,y}, \text{employees}_{i,y}, \text{assets}_{i,y}, \text{equity}_{i,y}, \text{debt}_{i,y}), \quad (3.1)$$

at yearly frequency, with near-complete coverage of limited-liability companies. The filing for fiscal year y is published at a date $\tau_{i,y}$ that lags fiscal year end by 9–12 months in typical jurisdictions (statutory deadlines plus registry processing); $\tau_{i,y}$ is recorded in the registry metadata and is itself firm-informative (late filers are disproportionately distressed). Filings exist in *vintages*: the as-reported figures published at $\tau_{i,y}$, and later restatements that overwrite them in most commercial extracts.

Warning 3.1 (Use as-reported vintages). Restated figures incorporate information (audits, corrections, insolvency administrators’ revisions) that did not exist at $\tau_{i,y}$. Training features on restated fundamentals violates the point-in-time discipline of Protocol 2.5 exactly as consuming the companies-table snapshot columns of Section 2.1 would, and the bias is not neutral: restatements are concentrated among firms that later failed, so restated features leak failure. Features must be built from the as-reported vintage; restated data may be used for

labels only when the label is explicitly defined on final figures, and then the choice must be stated.

3.2 Candidate success metrics

Fix a horizon h (years) and write $E_y = \text{EBITDA}_{i,y}$, $R_y = \text{revenue}_{i,y}$, $L_y = \text{employees}_{i,y}$, suppressing i .

EBITDA growth. The naive candidate $\log(E_{y+h}/E_y)$ is undefined whenever $E_y \leq 0$ or $E_{y+h} \leq 0$ — for private growth companies, roughly a third of the sample. Dropping those firm-years selects on profitability and destroys the target’s meaning. We instead use the *arc (symmetric) growth rate*:

Definition 3.2 (Arc growth). For a fundamental F with values F_y, F_{y+h} not both zero,

$$g_y^{(h)} = \frac{F_{y+h} - F_y}{\frac{1}{2}(|F_{y+h}| + |F_y|)} \in [-2, 2], \quad (3.2)$$

with $g_y^{(h)} = 0$ when $F_{y+h} = F_y = 0$.

Proposition 3.3. *The arc growth rate is defined for all sign patterns, is bounded in $[-2, 2]$, is antisymmetric under swapping $F_y \leftrightarrow F_{y+h}$, and for $F_y, F_{y+h} > 0$ satisfies $g_y^{(h)} = \log(F_{y+h}/F_y) + O(\log^3(F_{y+h}/F_y))$, so it agrees with log-growth to second order near zero growth.*

Proof sketch. Boundedness and antisymmetry are immediate from Equation (3.2). For the expansion, set $u = \log(F_{y+h}/F_y)$; then $g = 2(e^u - 1)/(e^u + 1) = 2 \tanh(u/2) = u - u^3/12 + O(u^5)$. The full argument, including the characterization of when $|g| = 2$ is attained, is in Appendix C. $\square$

Example 3.4 (Arc growth across a sign change). A firm improves from $E_y = -0.4\text{m}$ to $E_{y+2} = +1.2\text{m}$. Log-growth is undefined; arc growth gives $g = (1.2 - (-0.4))/\frac{1}{2}(1.2 + 0.4) = 2.0$, the upper bound: sign-crossing improvements saturate the scale, which is the intended reading (crossing into profitability is a categorical success, and the metric does not pretend to rank magnitudes across the sign change). By contrast a doubling from 0.6m to 1.2m gives $g = 0.6/0.9 = 0.667$ against $\log 2 = 0.693$: in the ordinary regime the two scales agree to within 4%, as Proposition 3.3 predicts.

Boundedness is useful: a firm going from $E_y = 10\text{k}$ to $E_{y+h} = 5\text{m}$ is a success, but not five hundred times the success of a doubling, and a bounded target prevents a few small-denominator observations from dominating a squared-error fit. A sign-preserving alternative when unbounded scale matters is the difference of inverse-hyperbolic-sine transforms, $\text{asinh}(F_{y+h}) - \text{asinh}(F_y)$ with $\text{asinh}(x) = \log(x + \sqrt{x^2 + 1})$, which behaves like the log-difference for $|F| \gg 1$ and remains defined through zero. In either case we winsorize the realized target at its 1st and 99th percentiles, computed *within training folds only*, to control residual tail leverage.

Revenue CAGR. $\text{CAGR}_y^{(h)} = (R_{y+h}/R_y)^{1/h} - 1$, defined since $R_y > 0$ for operating firms (pre-revenue firm-years are flagged and excluded from this metric, itself a selection to be aware of). Robust to margin accounting; blind to profitability.

Margin expansion. $\Delta m_y^{(h)} = E_{y+h}/R_{y+h} - E_y/R_y$, defined where both revenues are positive. Captures operating-leverage improvement; extremely heavy-tailed for small R and sensitive to cost-capitalization choices.

Employment growth. Arc growth Equation (3.2) applied to L_y . Nearly immune to accounting discretion and currency; economically coarser.

Survival. $S_y^{(h)} = \mathbf{1}\{\text{firm } i \text{ is not insolvent or liquidated by fiscal year } y+h\}$, from registry status events. Unambiguous and cheap, but nearly uninformative in the bulk: most firms survive two years, so the indicator separates the left tail only.

Metric	Interpretability	Acct. robustness	Tail behaviour	ML target
EBITDA arc growth	high	moderate	bounded	high
EBITDA log-growth	high	moderate	undefined $E \leq 0$	unusable
Revenue CAGR	high	high	heavy right tail	moderate
Margin expansion	moderate	low	very heavy	low
Employment arc growth	moderate	high	bounded	high
Survival indicator	high	high	degenerate bulk	low (imbalanced)

Table 3.1: Candidate success metrics. “Acct. robustness” is robustness to discretionary accounting choices (depreciation schedules, cost capitalization, revenue recognition); “ML target” summarizes definedness, tail control, and informativeness for ranking.

3.3 The formal estimand and its timing

Definition 3.5 (Success label). Fix a metric g from Section 3.2 and horizon h . The label of firm i at base year y is $Y_{i,y} = g(F_{i,y}, F_{i,y+h})$, the metric evaluated over $(y, y+h]$. The *information date* of the pair is $\tau_{i,y}$, the publication date of the year- y filing; a feature vector $Z_{i,\tau_{i,y}}$ is admissible if it satisfies Protocol 2.5 at $\tau_{i,y}$.

The predictive estimand of the second half of this chapter is $\mathbb{E}[Y_{i,y} \mid Z_{i,\tau_{i,y}}]$; the causal estimand of Chapter 4 replaces conditioning with intervention on a financing exposure, $\mathbb{E}[Y_{i,y} \mid \text{do}(A_{i,y}=a)]$. Both require Y to be a single well-defined scalar attached to a single well-defined date (Hernán and Robins, 2020). Four label-engineering pitfalls recur in practice:

1. **Horizon misalignment.** Features dated at fiscal year end y instead of at $\tau_{i,y}$ grant the model 9–12 months of foresight; conversely a label read at $\tau_{i,y+h}$ but indexed as “ h -year growth” actually spans up to $h+1$ years of calendar time. We index everything by τ and record the realized calendar span.
2. **Fiscal-year heterogeneity.** Fiscal year ends vary across firms and jurisdictions; “year y ” is not a common calendar interval. Cross-sectional comparisons and time-based validation splits must use $\tau_{i,y}$, not y .
3. **Consolidation changes.** A firm switching between solo and consolidated accounts (or acquiring a subsidiary) exhibits spurious jumps in every fundamental. We flag consolidation-scope changes from registry metadata and either exclude the spanning label or difference within a constant scope.
4. **Currency.** Fundamentals are filed in local currency. Growth rates of the form Equation (3.2) are currency-invariant only if both endpoints are in the same currency; we compute growth in local currency and never convert at two different dates’ exchange rates.

3.4 Censoring and the survivor trap

A label $Y_{i,y}$ at horizon h requires the year- $(y+h)$ filing. It is missing for two very different reasons: the cohort is young (the calendar has not yet produced the filing — administrative censoring, the panel analogue of defect (C1)) or the firm exited (insolvency, acquisition, migration out of filing obligations — the analogue of (C3), here at least partially *labeled* by registry status). Simply dropping unlabelled firm-years conditions the training distribution on surviving h more years, which biases every estimate toward the survivor population — precisely the firms for which success is overrepresented, in the same way that conditioning on the risk set must respect who is still observable (Cox, 1972). The risk-set machinery reappears for event data in Chapter 8; here a simple reweighting suffices.

Let $C_{i,y} = \mathbf{1}\{F_{i,y+h} \text{ observed}\}$ and let $\hat{G}(x_{i,y})$ estimate $\mathbb{P}(C_{i,y} = 1 \mid x_{i,y})$ from pre-base-year covariates $x_{i,y}$ (age, size, sector, filing punctuality), e.g. by logistic regression within calendar cohorts. The inverse-probability-of-censoring (IPC) weighted loss

$$\hat{L}(f) = \frac{1}{n} \sum_{(i,y)} \frac{C_{i,y}}{\hat{G}(x_{i,y})} \ell(Y_{i,y}, f(Z_{i,\tau_{i,y}})) \quad (3.3)$$

is unbiased for the full-population risk when censoring is independent of Y given $x_{i,y}$ — an assumption to state, not to assume silently: exit by insolvency is plainly outcome-dependent. Practically we (i) use Equation (3.3) for administrative censoring, which is mechanically independent of Y given the cohort; (ii) treat insolvency exits not as censored but as *observed failures*, assigning $Y = -2$ (the lower bound of arc growth) with a sensitivity band; and (iii) treat acquisition exits, which are plausibly successes, as a competing outcome reported separately rather than folded into Y . Truncating weights at $\hat{G} \geq 0.1$ keeps the variance of Equation (3.3) controlled.

3.5 Stylized facts to verify on any real panel

Before fitting anything, three distributional facts should be confirmed on the panel at hand; each has a design consequence, and their absence usually indicates a data problem rather than an interesting economy.

1. **Heavy-tailed growth.** Firm growth rates are far from Gaussian: the cross-sectional distribution of Equation (3.2) is sharply peaked, with tails well approximated by a Laplace or heavier law. Consequence: squared error concentrates on the tails; prefer pseudo-Huber or quantile losses and report rank metrics alongside RMSE (Section 3.9).
2. **Mean reversion.** Successive growth rates correlate negatively: extreme years partially reverse. Consequence: last year's growth is a strong but *sign-flipping* feature, and naive momentum readings of feature importances mislead; evaluation must be out-of-time, not just out-of-sample.
3. **Size dependence.** Growth dispersion falls with size, roughly $\text{sd}(g) \propto R_y^{-\beta}$ with $\beta \approx 0.15$ (a well-documented deviation from Gibrat's law). Consequence: any loss that ignores this heteroskedasticity overweights small firms; stratify evaluation by size decile and consider variance-stabilizing weights.

Table 3.2 lists illustrative moments of the pattern these facts predict. Verifying them on the synthetic EU panel (Section 2.5) is part of the validation plan of Chapter 19; the same four columns should also be computed on any real panel before the first model is fit.

These facts jointly motivate the loss and cross-validation architecture of the prediction layer below: heavy tails pick the loss, mean reversion picks out-of-time splits, size dependence picks the stratification.

Metric ($h = 2$)	Skewness	Excess kurtosis	Lag-1 autocorr.	Undefined share
EBITDA arc growth	−0.3	9.8	−0.21	0.0%
EBITDA log-growth	—	—	—	34%
Revenue CAGR	+2.1	14.6	−0.12	6%
Margin expansion	−0.8	27.3	−0.35	6%
Employment arc growth	+0.4	6.2	−0.09	0.0%

Table 3.2: Illustrative stylized-fact moments (winsorization off, all firm-years with both endpoints observed), to be verified on the synthetic EU panel (Chapter 19). “Undefined share” is the fraction of firm-years on which the metric does not exist; the log-growth row is blank because its moments are conditional on a selected subsample and hence not comparable.

3.6 Worked example: labelling a cohort

The definitions above compress into a short routine that is easy to get wrong. We run it once, by hand, for a single firm whose EBITDA path crosses zero. Table 3.3 gives the firm’s as-reported figures and registry publication dates; fiscal years end on December 31.

Fiscal year y	FY end	EBITDA E_y (EUR m)	Published $\tau_{i,y}$
2016	2016-12-31	−0.8	2017-10-14
2017	2017-12-31	−0.3	2018-09-30
2018	2018-12-31	+0.5	2019-11-22
2019	2019-12-31	+1.1	2020-10-05
2020	2020-12-31	+1.4	2021-09-18

Table 3.3: Timeline of the worked firm. Publication lags fiscal year end by 9–11 months; the late 2018 filing (almost 11 months) is itself informative (Section 3.1).

The label. Take base year $y = 2017$ and the primary horizon $h = 2$. By Definition 3.5 with arc growth Equation (3.2),

$$Y_{i,2017} = \frac{E_{2019} - E_{2017}}{\frac{1}{2}(|E_{2019}| + |E_{2017}|)} = \frac{1.1 - (-0.3)}{\frac{1}{2}(1.1 + 0.3)} = 2.0, \quad (3.4)$$

the saturated value: the firm crossed into profitability over (2017, 2019], a categorical success in the reading of Example 3.4.

Information-date alignment. The pair is indexed at $\tau_{i,2017}$, i.e. 2018-09-30, the instant the year-2017 figures become public. Admissible features there: the as-reported 2016 and 2017 filings, deal rows with `deal_date` before that date, and macro values published before it. The label needs E_{2019} , public only on 2020-10-05: any training set assembled earlier carries this pair as administratively censored (Section 3.4), and the realized calendar span — information date to label availability — is 24.2 months, recorded alongside the nominal $h = 2$.

Warning 3.6 (The not-yet-published filing). Consider the neighbouring pair, base year 2018. Dating its features at fiscal year end 2018-12-31 — the natural join key in a panel indexed by fiscal year — makes $E_{2018} = +0.5\text{m}$ a feature eleven months before its publication on 2019-11-22. The leaked content is precisely the label’s content: the sign crossing. A backtest learns that “latest EBITDA just turned positive” predicts a saturated label; in production at the end of 2018 the newest public figure is $E_{2017} = -0.3\text{m}$ and the signal does not exist. The fix is mechanical, not statistical: filings join as-of on $\tau_{i,y} \leq t$, never on fiscal year — the exact

analogue of joining macro series on `publish_date` in Protocol 2.5(a) — and the staleness of the newest published filing enters as a feature, so the model sees lateness instead of the future.

The cohort routine is the by-hand computation industrialized: for every firm and every base year whose filing is published, emit the pair at $\tau_{i,y}$; attach the label when the year- $(y+h)$ filing publishes; score insolvency exits at $Y = -2$ and route acquisition exits to the competing-outcome tally (Section 3.4); winsorize realized labels within training folds only. Every row of the resulting table carries $(i, y, \tau_{i,y}, Y, C)$, and the validation splits of Section 3.10 operate on τ , never on y .

3.6.1 Robustness of the label choice

The label is a construction, so before committing we ask how much rides on it. Table 3.4 states the comparison to run on the synthetic panel (Chapter 19), with the pattern we expect: the primary label against three alternatives, on three stability criteria — Spearman rank correlation of firm rankings with the primary label; sensitivity of top-decile membership to moving the winsorization quantiles from 1%/99% to 0.5%/99.5%; and tail behaviour of the realized target.

Label ($h = 2$)	Rank corr.	Winsor. sensitivity	Tail behaviour
EBITDA arc growth (primary)	1.00	low	bounded, mass at ± 2
asinh-difference of EBITDA	0.86	moderate	unbounded, moderate tails
Revenue CAGR	0.55	high	heavy right tail
Employment arc growth	0.48	low	bounded, thin tails

Table 3.4: Expected stability of the label choice, to be verified on the synthetic panel (Chapter 19). “Rank corr.” is the Spearman correlation of firm rankings with the primary label; “winsor. sensitivity” is the extent to which top-decile membership changes when the winsorization quantiles move from 1%/99% to 0.5%/99.5%.

The expected pattern is the same on any real panel: the asinh alternative reorders little (it is a monotone reshaping of the same fundamental), while revenue- and employment-based labels agree with the primary only moderately — they measure different economics, not noisy copies of one number. Rank correlations near 0.5 mean that model comparisons, feature importances, and top-decile readings can flip across definitions. We therefore re-run the headline models of this chapter and Chapter 4 under one pre-declared alternative — employment arc growth, chosen as the most accounting-immune and the most distant — and report both. A conclusion that survives both labels is about firms; one that does not is about the metric. Declaring the single alternative before results are seen keeps this a robustness check rather than metric shopping, in the same spirit as controlling a family of tests instead of reporting the best one (Romano et al., 2019).

The committed primary metric — EBITDA arc growth at $h = 2$, winsorized within folds, insolvency exits scored at -2 — is recorded with its rationale as decision D-EU.1 at the end of the chapter. Three points of the rationale matter for the rest of the chapter. First, $h = 2$ balances signal against censoring: at $h = 1$ the label is dominated by accounting noise and mean reversion (Section 3.5); at $h = 3$ the administratively censored share of recent cohorts grows large and the IPC weights of Equation (3.3) get large. Second, EBITDA rather than revenue keeps the metric sensitive to the margin dimension investors are paid on, while the arc transform removes the sign-definedness problem that rules out log-growth. Third — and decisively for Chapter 4 — a composite index $Y^{\text{comp}} = \sum_j w_j Y_j$ degrades identifiability of the causal estimand: each component has its own confounding structure, so a backdoor set valid for the composite need not exist even when one exists per component; treatment effects of

opposite sign on components cancel uninterpretably; and the weights w_j become untestable modelling choices inside the estimand itself. One outcome, one horizon, one information date: everything downstream inherits its identifiability from this choice.

3.7 The learning problem

With the label fixed, the first modelling layer is plain supervised learning. A row is a firm-year (i, y) . The label is $Y_{i,y} = w_d(g_{i,y}^{(h)})$, the arc-growth of EBITDA between statement years y and $y + h$ with $h = 2$, winsorized at level c exactly as above. The feature vector $X_{i,y} \in \mathbb{R}^p$ is assembled *point-in-time*: it may use only filings published and macro prints released by the information date $t_{\text{info}}(i, y) = \tau_{i,y}$ of the row, respecting Protocol 2.5. We estimate two objects:

$$m(x) = \mathbb{E}[Y \mid X = x], \quad q_\tau(x) = \inf\{a : \mathbb{P}(Y \leq a \mid X = x) \geq \tau\}, \quad (3.5)$$

the conditional mean (point screening) and a small grid of conditional quantiles, $\tau \in \{0.05, 0.10, 0.50, 0.90, 0.95\}$. The quantiles matter: the outcome is heavy-tailed even after winsorization, upside screening acts on $q_{0.9}$ rather than the mean, downside risk control acts on $q_{0.1}$, and the pair $(q_{0.05}, q_{0.95})$ is the input to the conformal intervals of Section 3.11.

Two dependence structures dominate. *Within-firm*: rows (i, y) and (i, y') share persistent firm effects (management quality, niche economics), so errors are serially correlated within i . *Within-era*: rows of a given year y share the macro shock, so errors are cross-sectionally correlated within y . The first dictates firm-grouped resampling; the second dictates era-blocked validation. Neither is optional, and both recur — in bucketed form — in Chapter 13.

3.8 Feature catalogue

Table 3.5 lists the feature families. All are computable from the fundamentals panel $F_{i,y}$, the deals table where coverage exists, and the macro feed; all obey the as-of rule.

Table 3.5: Feature catalogue for the EU prediction layer. All features are as-of $t_{\text{info}}(i, y)$; macro features use publication-lagged vintages per Protocol 2.5(a).

Family	Examples	Construction
Lagged levels	EBITDA, revenue, payroll, assets at $y, y-1, y-2$	$\log(1 + \cdot)$ where positive; raw for signed EBITDA
Growth rates	1–3y arc-growth of revenue, EBITDA, payroll	same arc transform as the target, unwinsorized
Ratios	EBITDA margin, leverage, asset turnover, payroll share	clipped at 1st/99th pctile per sector-year
Peer position	within-sector-year rank and z-score of margin, growth, size	computed per (s, y) cell, rank in $[0, 1]$
Sector-year aggregates	median growth, IQR, firm count	cell-level, merged back
Firm statics	age at y , size bucket, country	age in years, log assets bucket
Macro as-of	policy rate level and 12m change, CPI YoY, unemployment level/change	vintage at t_{info} ; level <i>and</i> change
Deal history	raised-to-date, round count, months since last round	zero-filled with indicator where firm never in deals

Missingness. Fundamentals gaps are informative: small companies file abbreviated accounts, distressed companies file late, and the propensity to be missing depends on the (unobserved) value — textbook MNAR. We therefore do *not* mean-impute: imputation overwrites the

signal carried by absence and biases every downstream split. Instead we rely on the native sparsity handling of gradient boosting, in which each split learns a default direction for missing values from the data (Chen and Guestrin, 2016), and we add explicit indicators (`ebitda_missing_lag1`, `never_in_deals`) for gaps whose economic meaning we can state. Trees can then interact the indicator with size and sector: “missing EBITDA at small size” differs from “missing EBITDA at large size”, and the model should be free to learn that.

3.9 Why boosted trees here — and what they cannot do

The learner is gradient-boosted trees in the XGBoost formulation (Chen and Guestrin, 2016; Friedman, 2001); the second-order boosting step, gain formula, and sparsity-aware default directions are derived once in Chapter 13 (Proposition 13.11) and are not repeated here. The case for boosted trees on *this* problem has four parts: (i) the signal is riddled with interactions (margin $\times$ sector, growth $\times$ rate regime) that we cannot enumerate a priori and trees discover automatically; (ii) the features are mixed continuous/ordinal/categorical with wildly different scales, and trees are invariant to monotone transforms of each feature, so we spend no effort on marginal calibration; (iii) native missingness handling, as above; (iv) strong, well-understood regularization (depth, shrinkage, subsampling, ℓ_2 on leaf weights).

The case against has two entries, and both matter here. First, *no extrapolation*: a tree ensemble is piecewise constant, so its prediction outside the convex hull of the training features is the value at the nearest boundary cell. If every training era has falling policy rates, the fitted rate response is a flat extension — the model cannot form an opinion about a rising-rate regime it has never seen. Second, *unstable attribution*: correlated features split importance unpredictably across refits, so feature rankings are not portable conclusions.

We mitigate rather than solve. Era-aware validation (Section 3.10) at least measures degradation across regimes instead of hiding it. Monotone constraints encode signs we are willing to defend economically — e.g. growth non-increasing in the 12-month rate *change*, downside quantile non-increasing in leverage — which reduces variance in thin regions and prevents sign flips there.

Remark 3.7 (Constraints are priors, not facts). A monotone constraint changes the model class, not the world. Impose one only where the sign survives scrutiny (the rate *level* is ambiguous: it proxies both financing cost and the demand cycle; the *change* is the cleaner candidate). A wrong constraint is worse than none, because it silently converts local noise into global bias. The same discipline governs the bucketed panel of Chapter 13, where an apparently obvious monotonicity in staleness age is deliberately declined.

For the conditional-mean model, heavy tails argue against squared error even after win-sorization. We use the pseudo-Huber objective

$$\ell_\delta(r) = \delta^2 \left(\sqrt{1 + (r/\delta)^2} - 1 \right), \quad (3.6)$$

which is quadratic for $|r| \ll \delta$ and linear for $|r| \gg \delta$, with δ set to roughly one cross-sectional standard deviation of Y . Note the fitted functional then sits between conditional mean and median; we report which loss produced which model and never mix them in evaluation.

3.10 Validation design

The general temporal-evaluation constitution of this document — fixed development split, rolling-origin headlines, no post-split information anywhere — is stated once in Section 11.6 (Protocol 11.15) and applies here unchanged. What is EU-specific is the shape the protocol takes on a firm-year panel with overlapping two-year labels, and why anything weaker fails.

Never random K-fold. Random row-level K-fold fails three ways on this panel. (i) *Firm leakage*: rows of the same firm land on both sides of the split, and the persistent firm effect makes validation loss optimistic — the model is partly memorizing firms. (ii) *Target overlap*: with $h = 2$, the labels of years y and $y + 1$ share a statement year, so adjacent rows carry overlapping information about each other’s targets. (iii) *Era leakage*: the model trains on the future macro regime it will be asked to predict, which no deployed model enjoys. All three inflate measured skill precisely where the deployment risk is largest.

Rolling-origin splits with an embargo. We order eras (label years) $y_1 < \dots < y_E$ by information date and build folds $k = 1, \dots, K$: train on eras up to y_{e_k} , *embargo* the next h eras (their labels overlap the training window), validate on the era after the embargo. Every fold mimics deployment: fit on the past, predict a disjoint future. The same firm may appear in train (early years) and validation (later years) — that is legitimate and realistic; the leakage to prevent is label overlap, handled by the embargo, not firm recurrence across time. The embargo is the one ingredient the general protocol of Section 11.6 does not need to state, because only horizon labels overlap this way.

Hyperparameter protocol. Random search (100–200 draws) over the ranges in Table 3.6; each draw is scored by its *average validation loss across all rolling folds*, with the number of boosting rounds chosen by early stopping on the *most recent* fold only — the one knob that benefits most from recency. The final model refits on all data with the tuned configuration and the early-stopped round count scaled by the ratio of full to training rows. Tuning on an average across eras deliberately trades a little peak skill for regime stability.

Table 3.6: Search ranges. Depth is kept shallow: firm-year panels reward many weak interactions over few deep ones.

Parameter	Range	Note
max depth	3–6	log-uniform integer
learning rate	0.01–0.1	log-uniform
min child weight	5–100	guards thin sector cells
row subsample	0.6–0.9	per boosting round
column subsample	0.5–0.9	decorrelates trees
ℓ_2 leaf penalty	0.5–20	log-uniform
rounds	early stopping	patience 50, most recent fold

3.11 Uncertainty quantification

Three tools, three targets: per-firm heteroskedastic intervals (quantile regression), finite-sample coverage repair (conformal), and model-level uncertainty of scalar summaries (cluster bootstrap). Table 3.7 states when each is the right instrument.

3.11.1 Quantile regression with the pinball loss

For $\tau \in (0, 1)$ the pinball (check) loss is

$$\rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\}) = \tau u_+ + (1 - \tau) u_-, \quad (3.7)$$

with $u_+ = \max(u, 0)$, $u_- = \max(-u, 0)$. Its population minimizer is exactly the quantile we want:

Proposition 3.8 (The pinball minimizer is the τ -quantile). *Let Y have distribution function F and finite first moment, and let $\varphi(a) = \mathbb{E}[\rho_\tau(Y - a)]$. Then any a^* with $F(a^{*-}) \leq \tau \leq F(a^*)$ minimizes φ ; if F is continuous and strictly increasing, $a^* = F^{-1}(\tau)$ uniquely.*

Proof. φ is convex, and for a a continuity point of F , $\varphi'(a) = -\tau\mathbb{P}(Y > a) + (1 - \tau)\mathbb{P}(Y < a) = F(a) - \tau$. A convex function is minimized where its (sub)derivative changes sign, i.e. where $F(a^-) - \tau \leq 0 \leq F(a) - \tau$. The version with all measure-theoretic care (general F , subgradient characterization) is in Appendix C. $\square$

We fit one boosted model per τ with objective Equation (3.7) applied conditionally. Separately fitted quantiles can cross in finite samples; we repair by sorting the predicted grid per row (monotone rearrangement), which never worsens pinball loss.

3.11.2 Conformalized quantile regression

Quantile models are calibrated only insofar as the model class and sample size allow; conformal prediction repairs marginal coverage with finite-sample guarantees (Romano et al., 2019).

Algorithm (split CQR).

1. Split the available rows into a proper training set I_1 and a calibration set I_2 of size n_2 , with the split at the firm level and I_2 chosen as a *block*: the most recent complete era(s) before the deployment date.
2. On I_1 , fit lower and upper quantile models $\hat{q}_{\alpha/2}$, $\hat{q}_{1-\alpha/2}$ as above.
3. For each $j \in I_2$ compute the conformity score $E_j = \max\{\hat{q}_{\alpha/2}(X_j) - Y_j, Y_j - \hat{q}_{1-\alpha/2}(X_j)\}$.
4. Let $Q_{1-\alpha}$ be the $\lceil (n_2 + 1)(1 - \alpha) \rceil$ -th smallest of $\{E_j\}_{j \in I_2}$.
5. Report the interval $C(x) = [\hat{q}_{\alpha/2}(x) - Q_{1-\alpha}, \hat{q}_{1-\alpha/2}(x) + Q_{1-\alpha}]$.

Theorem 3.9 (CQR marginal coverage, Romano et al., 2019). *If the calibration pairs $\{(X_j, Y_j)\}_{j \in I_2}$ and the test pair (X, Y) are exchangeable, then $\mathbb{P}\{Y \in C(X)\} \geq 1 - \alpha$; if the scores are almost surely distinct, also $\mathbb{P}\{Y \in C(X)\} \leq 1 - \alpha + 1/(n_2 + 1)$.*

The proof, via an exchangeable-rank lemma, is in Appendix C.

Warning 3.10 (Temporal data are not exchangeable). Theorem 3.9 assumes the test row is exchangeable with the calibration rows. A macro-era shift between calibration block and deployment breaks this, and no finite-sample guarantee survives. Practice: calibrate on the most recent contiguous block (as above), *treat the nominal level as approximate*, measure empirical coverage per era on every historical fold, and recalibrate at every model vintage. If per-era coverage drifts systematically with the macro state, widen intervals by the worst observed era's deficit rather than trusting the nominal α .

3.11.3 Cluster bootstrap over firms

Quantile and conformal intervals describe *per-row* outcome uncertainty. A different question — how uncertain is a scalar summary of the fitted model, such as an era MAE, a decile-spread statistic, or the skill gap over a benchmark — requires resampling the fitting process itself (Efron and Tibshirani, 1994).

Algorithm (firm-level percentile bootstrap).

1. Let $\mathcal{I}$ be the set of firms. For $b = 1, \dots, B$ (we use $B \in [200, 500]$): draw $|\mathcal{I}|$ firms from $\mathcal{I}$ with replacement and assemble the replicate panel from *all rows* of each drawn firm (duplicates kept).
2. Re-run the frozen pipeline on the replicate: feature construction, the tuned hyperparameter configuration (not re-searched), fit, and compute the scalar summary $\hat{\theta}_b$ on the replicate’s validation eras.
3. Report the percentile interval $[\hat{\theta}_{(\lceil \alpha B/2 \rceil)}, \hat{\theta}_{(\lceil (1-\alpha/2)B \rceil)}]$.

Resampling firms, not rows, is the key choice: within-firm correlation means the effective sample size is closer to the number of firms than the number of rows, and a row-level bootstrap understates variance accordingly.

Table 3.7: Choosing the uncertainty tool.

Tool	Quantifies	Guarantee	Cost
Quantile regression	per-row outcome spread	none per se	1 fit per τ
Split CQR	per-row interval coverage	finite-sample marginal, approximate under drift	1 calibration pass
Firm bootstrap	model-level scalars (MAE, skill gaps)	asymptotic, within the observed design	B refits

3.12 Evaluation protocol and benchmarks

We report MAE and R^2 *by era* and by horizon, never pooled alone: a pooled MAE averages over exactly the regime heterogeneity the model is weakest on. The median model (pinball at $\tau = 0.5$) is evaluated with MAE — matched loss — and the pseudo-Huber mean model with both MAE and RMSE. Quantile calibration is checked per era with reliability diagrams: for each τ , the empirical fraction of outcomes below $\hat{q}_\tau(X)$ plotted against τ ; a well-calibrated model tracks the diagonal, a regime-sensitive one shows era-dependent slope. Subgroup stability: per-cell MAE by sector and size tercile relative to pooled MAE, flagging cells where a benchmark beats the boosted model — such cells are candidates for exclusion from deployment, not for silent averaging. This is the panel-regression face of the reporting discipline of Section 11.7: no number without the era, subgroup, and benchmark that qualify it.

Warning 3.11 (Attributions are associational). SHAP-style attributions decompose the *fitted* $\hat{m}$, i.e. they redistribute predicted differences among correlated inputs of an associational object. A large attribution on `months_since_last_round` does not mean financing drives growth; it means firms that recently raised look different, which is precisely the selection effect Chapter 4 is built to separate from the causal one. Use attributions for debugging (leakage detection, sanity of signs) and never in an investment memo as an effect size.

Every fold carries three benchmarks: *persistence* (predict the firm’s last observed growth), *sector-year mean* (predict the cell average), and *ridge* regression on the identical feature matrix (with the missingness indicators, since ridge has no native sparsity handling). The shipping rule is strict: the boosted model is deployed only where its out-of-era skill gap over ridge exceeds the half-width of the firm-bootstrap interval on that gap. A gap inside the interval is noise, and ridge is cheaper, more stable, and partially extrapolates. Table 3.8 records the situations in which gradient boosting is the wrong tool altogether.

Table 3.8: When gradient boosting is the wrong tool.

Situation	Symptom	Use instead
Tiny sample	$\lesssim 5k$ firm-years	ridge on curated features
Regime extrapolation	deployment macro outside training support	linear/constrained macro term + trees on the rest
Regulatory interpretability	auditable decision rule required	scorecard or shallow monotone model
Skill gap within noise	bootstrap CI covers zero	ridge; revisit features

The model of this chapter answers “which firms will grow?” conditional on everything observable. It is silent on “what happens if a firm raises?” — the fitted dependence on deal-history features mixes selection with effect. Chapter 4 takes up that question with the same data and an explicitly causal contract.

Design decision (D-EU.1 — the transparent-regime target and its gate).

The primary success metric of the transparent-regime track is *EBITDA arc growth at horizon $h = 2$* (Definition 3.2), winsorized at the within-fold 1%/99% quantiles, with insolvency exits scored at -2 , acquisition exits reported as a competing outcome, and administrative censoring handled by the IPC weights of Equation (3.3); no composite index is used, and one pre-declared alternative label (employment arc growth) is re-run for every headline result. The committed learner is gradient-boosted trees (pseudo-Huber mean plus a pinball quantile grid) under the embargoed rolling-origin validation of Section 3.10, and the deployment gate is the ridge benchmark: the boosted model ships only in cells where its out-of-era skill gap over ridge exceeds the firm-bootstrap half-width on that gap. Rationale: signal-vs-censoring balance at $h = 2$, sign-definedness of the arc transform, and single-outcome identifiability for Chapter 4; Table 3.4 for why the alternative label is mandatory. Reversal trigger: if the pre-declared alternative label flips the sign or ordering of a headline conclusion in two consecutive evaluation campaigns, the label choice is reopened and this box is rewritten.

Code hooks.

The cohort routine of Section 3.6 becomes a planned panel builder `hawkes_calibration/eu/panel.py`, emitting rows $(i, y, \tau_{i,y}, Y, C)$ with as-reported vintages only, mirroring the loader contract of `synthetic/loaders.py` (as-of joins, staleness ages carried as features). Label and metric implementations (arc growth, asinh difference, within-fold winsorization) belong in a planned `hawkes_calibration/eu/labels.py` with the stylized-fact audit of Section 3.5 as its test suite; the synthetic EU-style panel behind Tables 3.2 and 3.4 will be generated from the synthetic world in `handbook/code/synthetic/` (`firms.py`, `marks.py`, `observe.py`). The boosted stack — embargoed rolling-origin folds, pinball grid, split CQR, firm-level bootstrap — is shared infrastructure with the bucketed panel of Chapter 13 and should live once, in the planned `hawkes_calibration/panel/` learner package, not twice.

Chapter 4

A Causal Layer: Does Financing Cause Growth?

On the decision map of Section 1.4, this chapter serves the one consuming action that prediction cannot: capital-allocation post-mortems and targeting (“did our financings work?”, “where would a round move the needle?”). Chapter 3 predicts; managers ask a different question: *if* a company raises a round, does growth follow — or do growing companies simply raise? The fitted $\hat{m}(x)$ cannot answer this, because the dependence it learns on deal-history features mixes the effect of financing with the selection of good firms into financing. This chapter builds the minimal causal layer the transparent regime supports. The regime is favourable for one reason: the statutory fundamentals panel of Chapter 3 — joined point-in-time to the deals and macro tables of Chapter 2 — gives a *rich, audited, pre-treatment* covariate history, close to the dossier an investor actually screens on, so adjustment has a realistic chance. Every estimate ships with cluster-bootstrap uncertainty, an E-value, and two falsification tests, and the whole pipeline is to be validated on the synthetic world of Section 2.5, where the true effect is known by construction (Chapter 19). Against the defect ledger of Table 2.1, the exposures are those of Chapter 3 — survivor selection in the outcome (handled by the same censoring policy) — plus one new and dominant one: confounding by latent quality, which is not in the ledger because it is not a data defect at all, and which this chapter treats as its central object.

4.1 Estimands and assumptions

Treatment: $A_{i,y} = \mathbf{1}\{\text{firm } i \text{ closed a PE/VC equity round in year } y\}$, from the deals table. Outcome: $Y_{i,y}$, the $h = 2$ winsorized EBITDA arc-growth of Definition 3.5, measured over $(y, y + h]$ so it is strictly post-treatment. Each unit carries potential outcomes $Y_{i,y}(1)$ and $Y_{i,y}(0)$: growth with and without a round in year y .

Assumption 4.1 (SUTVA). (i) No interference: $Y_{i,y}(a)$ does not depend on other firms’ treatments; (ii) single version of treatment: $A_{i,y} = 1$ denotes one well-defined intervention (Imbens and Rubin, 2015).

Assumption 4.2 (Consistency). $Y_{i,y} = A_{i,y}Y_{i,y}(1) + (1 - A_{i,y})Y_{i,y}(0)$ (Hernán and Robins, 2020).

Both parts of Assumption 4.1 are approximations here: a round can hurt a direct competitor’s growth (interference; we flag narrow niches), and rounds differ in size and investor (versions; we deliberately coarsen to “raised institutional equity” and treat dose–response as an extension, not the estimand).

The estimands, in decreasing ambition of the conditioning:

$$\tau_{\text{ATE}} = \mathbb{E}[Y(1) - Y(0)], \quad \tau_{\text{ATT}} = \mathbb{E}[Y(1) - Y(0) \mid A = 1], \quad \tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]. \quad (4.1)$$

For the portfolio question (“did our financings work?”) the ATT is the object; for targeting (“where would a round move the needle?”) the CATE $\tau(x)$; the ATE is the reporting default because it is the easiest to estimate stably. Identification requires:

Assumption 4.3 (Conditional exchangeability and positivity). There exists an observed pre-treatment set Z with $(Y(0), Y(1)) \perp\!\!\!\perp A \mid Z$ and $0 < e(z) = \mathbb{P}(A = 1 \mid Z = z) < 1$ for all z in the support (Imbens and Rubin, 2015; Hernán and Robins, 2020).

Assumption 4.3 is not a modelling choice; it is a claim about the world that the next section argues from a graph, Section 4.6 stress-tests, and Section 4.7 verifies where the truth is known.

4.2 The causal graph

Five variables. U : latent firm quality (management, product, moat) — unobserved. X : the observed pre-treatment fundamentals history (levels, growth path, margins, leverage, payroll over the three years up to y), plus sector and age. M : the macro state as of year y (rates, CPI YoY, unemployment, as-of vintages). A : the treatment. Y : the outcome. The edges: $U \rightarrow A$ (investors select quality), $U \rightarrow Y$ (quality drives growth), $U \rightarrow X$ (quality leaves tracks in the filings), $X \rightarrow A$ (investors screen on the observable dossier), $X \rightarrow Y$ (fundamentals persist), $M \rightarrow A$ (funding cycles), $M \rightarrow Y$ (macro drives growth), and the causal edge of interest $A \rightarrow Y$. Table 4.1 enumerates every A – Y path.

Table 4.1: Paths between A and Y in the working graph.

Path	Type	Status after conditioning on (X, M)
$A \rightarrow Y$	causal	kept (the estimand)
$A \leftarrow X \rightarrow Y$	backdoor	blocked by X
$A \leftarrow M \rightarrow Y$	backdoor	blocked by M (sector-year cells)
$A \leftarrow U \rightarrow Y$	backdoor	<i>not</i> blockable directly; attenuated via the proxy path $U \rightarrow X$
$A \leftarrow U \rightarrow X \rightarrow Y$	backdoor	blocked by X

The whole argument of the transparent-regime track sits in the fourth row. U is unobserved, so the path $A \leftarrow U \rightarrow Y$ cannot be blocked exactly. But investors do not observe U either: they observe the dossier, which is essentially X — audited statements, growth trajectory, margins, leverage. To the extent that the selection $U \rightarrow A$ flows *through* what is observable about quality, conditioning on a rich X intercepts most of it, and the residual confounding is the part of quality that moves investors without leaving any track in three years of filings. In the transparent regime that residual is plausibly small and, crucially, *boundable* (Section 4.6). In the opaque regime there is no fundamentals panel, X collapses to deal marks and coarse firmographics, and the argument fails — which is why Chapter 5 and everything after it change the estimand rather than pretend the backdoor is closed.

4.3 Identification: the backdoor criterion

Definition 4.4 (Backdoor criterion, Pearl, 1995). A set of observed variables Z satisfies the backdoor criterion relative to (A, Y) if (i) no element of Z is a descendant of A , and (ii) Z

blocks every path between A and Y that contains an arrow into A .

Theorem 4.5 (Adjustment formula, Pearl, 1995). *If Z satisfies the backdoor criterion relative to (A, Y) , then*

$$\mathbb{E}[Y \mid \text{do}(A = a)] = \mathbb{E}_Z[\mathbb{E}[Y \mid A = a, Z]], \quad (4.2)$$

the outer expectation over the marginal law of Z , and consequently $\tau_{\text{ATE}} = \mathbb{E}_Z[\mathbb{E}[Y \mid A = 1, Z] - \mathbb{E}[Y \mid A = 0, Z]]$.

The proof — via the truncated factorization, an interventions-do-not-move-non-descendants lemma, and the invariance half of the do-calculus — is given in Appendix C, together with the three-line potential-outcomes special case under Assumption 4.3 that this chapter actually operates under.

In the working graph, $Z = (X, M)$ satisfies the criterion *up to* the residual U -path discussed above: formally we proceed under Assumption 4.3 with $Z = (X \text{ history over 3 years, sector-year cell})$ — the cells absorb M and sector-level funding cycles — and quantify the violation rather than assume it away.

Warning 4.6 (Bad controls). More conditioning is not more safety. Three failure modes, all present in this dataset. (i) *Mediators*: post-treatment employee counts sit on the path $A \rightarrow \text{hiring} \rightarrow Y$; conditioning on them subtracts part of the effect being estimated. (ii) *Colliders/descendants*: a subsequent valuation mark is caused both by the round (A) and by quality (U); conditioning on it opens the path $A \rightarrow \text{mark} \leftarrow U \rightarrow Y$ and manufactures spurious association. (iii) *M-bias*: even a *pre-treatment* variable can hurt — if some early covariate is a common effect of one latent cause that drives selection and another latent cause that drives growth, then conditioning on it opens a previously closed path between those latents. The rule is graphical, not chronological: the adjustment set is chosen by Definition 4.4, never by predictive power, and nothing dated after the round enters Z .

4.4 Estimators, in increasing robustness

Throughout, nuisance functions are fitted with the boosted-tree machinery of Chapter 3 (same features, same era discipline), written $\mu_a(z) = \mathbb{E}[Y \mid A = a, Z = z]$ and $e(z) = \mathbb{P}(A = 1 \mid Z = z)$.

(1) Outcome regression (g-formula). Fit $\hat{\mu}_1, \hat{\mu}_0$ and average: $\hat{\tau}_{\text{OR}} = n^{-1} \sum_{i,y} \{\hat{\mu}_1(Z_{i,y}) - \hat{\mu}_0(Z_{i,y})\}$. Two flavours: the T-learner (separate models per arm; honest but noisy in the small treated arm) and the S-learner (A as one feature among p ; stable but regularization can shrink the A -splits, biasing $\hat{\tau}$ toward zero). Both inherit every extrapolation weakness of trees precisely in low-overlap regions, where $\hat{\mu}_1$ is evaluated at control-like z and vice versa.

(2) Inverse propensity weighting. Fit $\hat{e}$, then the Hajek form

$$\hat{\tau}_{\text{IPW}} = \frac{\sum_i A_i Y_i w_i^1}{\sum_i A_i w_i^1} - \frac{\sum_i (1 - A_i) Y_i w_i^0}{\sum_i (1 - A_i) w_i^0}, \quad w^1 = \frac{1}{\hat{e}(Z)}, \quad w^0 = \frac{1}{1 - \hat{e}(Z)}.$$

Mandatory diagnostics before any estimate is read: the histogram of $\hat{e}$ by treatment arm (overlap plot); trimming to $\hat{e} \in [0.02, 0.98]$ with the trimmed fraction reported — if it exceeds a few percent, the estimand has silently changed to a trimmed-population effect and we say so; the effective sample size $\text{ESS} = (\sum w)^2 / \sum w^2$ per arm. IPW uses the treatment model only, so it fails independently of (1) — but its variance explodes near the trimming boundary.

(3) AIPW with cross-fitting (the default). The doubly robust score combines both nuisances (Chernozhukov et al., 2018):

$$\psi(O; \mu, e) = \mu_1(Z) - \mu_0(Z) + \frac{A(Y - \mu_1(Z))}{e(Z)} - \frac{(1 - A)(Y - \mu_0(Z))}{1 - e(Z)}, \quad \hat{\tau}_{\text{AIPW}} = \frac{1}{n} \sum \psi_i. \quad (4.3)$$

Proposition 4.7 (Double robustness). *Under Assumptions 4.2 and 4.3, $\mathbb{E}[\psi(O; \mu, e)] = \tau_{\text{ATE}}$ if either $\mu = (\mu_1^*, \mu_0^*)$ is the true outcome function or $e = e^*$ is the true propensity (with the other nuisance arbitrary, subject to positivity).*

Proof. Consider the $a = 1$ half, $\psi_1 = \mu_1(Z) + A(Y - \mu_1(Z))/e(Z)$. If $\mu_1 = \mu_1^*$: $\mathbb{E}[A(Y - \mu_1^*(Z)) | Z] = e^*(Z) \mathbb{E}[Y - \mu_1^*(Z) | A = 1, Z] = 0$, so $\mathbb{E}[\psi_1] = \mathbb{E}[\mu_1^*(Z)] = \mathbb{E}[Y(1)]$ by Assumption 4.3. If $e = e^*$: $\mathbb{E}[\psi_1 | Z] = \mu_1(Z) + \{e^*(Z) \mathbb{E}[Y | A = 1, Z] - e^*(Z)\mu_1(Z)\}/e^*(Z) = \mathbb{E}[Y | A = 1, Z] = \mathbb{E}[Y(1) | Z]$, and the arbitrary μ_1 cancels. The $a = 0$ half is symmetric; subtract. (Integrability details, and the influence-function variance that makes AIPW efficient, are in Appendix C.) $\square$

Remark 4.8 (Rate condition). Beyond consistency, valid $\sqrt{n}$ inference informally requires the product of nuisance errors to vanish fast: $\|\hat{\mu}_a - \mu_a^*\|_2 \cdot \|\hat{e} - e^*\|_2 = o_P(n^{-1/2})$ (Chernozhukov et al., 2018). Each nuisance may converge slowly (as boosted trees do) as long as the product is fast — this, plus the removal of own-observation overfitting bias by cross-fitting, is why (3) is the default and not merely a hedge.

Algorithm (cross-fitted AIPW).

1. Partition *firms* into $K = 5$ folds; a firm's rows never straddle folds (within-firm correlation would otherwise leak nuisance fits into own scores).
2. For each fold k : fit $\hat{\mu}_1^{(-k)}, \hat{\mu}_0^{(-k)}, \hat{e}^{(-k)}$ on the other $K - 1$ folds, with the era-aware tuning of Chapter 3 frozen once, outside the loop.
3. Trim: drop rows of fold k with $\hat{e}^{(-k)} \notin [0.02, 0.98]$; record the trimmed fraction.
4. Compute $\psi_{i,y}$ for the surviving rows of fold k via Equation (4.3) using only out-of-fold nuisances.
5. Pool: $\hat{\tau} = \text{mean of all } \psi_{i,y}$; variance by Section 4.5; report alongside the overlap plot and trimmed fraction.

4.5 Inference under firm clustering

Rows recur within firms across years, and the scores $\psi_{i,y}$ of a firm are correlated through persistent firm effects. An i.i.d. bootstrap (or the naive variance $n^{-2} \sum (\psi_{i,y} - \hat{\tau})^2$) treats n rows as n independent draws and understates the variance — the effective sample size is nearer the number of firms. Two consistent routes; we run both and report the wider.

(a) Cluster-robust influence-function variance. Let $S_i = \sum_y (\psi_{i,y} - \hat{\tau})$ sum a firm's centred scores. Then

$$\widehat{\text{Var}}(\hat{\tau}) = \frac{1}{n^2} \sum_{i \in \mathcal{I}} S_i^2, \quad (4.4)$$

which is the sandwich variance with firms as clusters; normal intervals $\hat{\tau} \pm z_{1-\alpha/2} \widehat{\text{Var}}^{1/2}$.

(b) Cluster bootstrap (Efron and Tibshirani, 1994).

1. For $b = 1, \dots, B$ (200–500): resample firms with replacement; a drawn firm contributes all its rows.
2. Re-run the *entire* cross-fitting algorithm on the replicate — nuisance refits included, tuning still frozen — to obtain $\hat{\tau}_b$.
3. Report the percentile interval of $\{\hat{\tau}_b\}$.

The bootstrap additionally propagates nuisance-estimation noise that Equation (4.4) treats as fixed; disagreement between (a) and (b) beyond $\sim 20\%$ of interval width is itself a diagnostic of nuisance instability.

4.6 Sensitivity and falsification

Assumption 4.3 cannot be verified from the data it licenses. It can be stress-tested. Every reported effect must clear the checklist of Table 4.2; an estimate that fails any row does not ship.

E-values. The E-value (VanderWeele and Ding, 2017) is the minimum strength of association, on the risk ratio scale, that an unmeasured confounder would need with *both* treatment and outcome, above and beyond Z , to explain away the observed association. For an observed risk ratio $RR \geq 1$,

$$E = RR + \sqrt{RR(RR - 1)}. \quad (4.5)$$

Our outcome is continuous, so we standardize: with effect $d = \hat{\tau}/\hat{\sigma}_Y$ (cluster-bootstrap CI carried through), the conventional conversion $RR \approx \exp(0.91 d)$ feeds Equation (4.5), and the E-value of the CI limit closer to the null is reported alongside that of the point estimate. An E-value of 1.8 reads: a latent-quality confounder must nearly double both the odds of raising and of high growth, conditional on the full fundamentals history, to explain the estimate away — a statement a domain expert can assess directly.

Placebo treatment. Define $A_{i,y}^{\text{fut}} = \mathbf{1}\{\text{round in year } y + 3\}$ and estimate its “effect” on $Y_{i,y}$ (which is fully realized before the placebo round) with the identical AIPW pipeline. Under Assumption 4.3 the estimate must be null: a future round cannot cause past growth, so any nonzero adjusted estimate measures exactly the selection-on-trends that Z failed to absorb.

Negative-control outcome. Symmetrically, estimate the “effect” of $A_{i,y}$ on *pre-treatment* growth (years $y - 2$ to y , excluded from Z for this test only). Nonzero means the treated were already on a different trajectory in a way Z does not capture.

4.7 Synthetic validation protocol

The synthetic world of Section 2.5 generates firms with a known latent quality driving both funding intensity and fundamentals (the regime-B generator makes the quality–funding link explicit), so a true τ can be injected by construction and confounding by quality is present by design. The causal layer is accepted only if it passes the following protocol there; the results are to be reported in Chapter 19, under the transfer convention fixed there — synthetic results set priors and vetoes, never substitutes for real-data evaluation.

Table 4.2: Mandatory pre-shipping checklist for any causal estimate.

Check	Pass criterion	On failure
Overlap plot	common support, trimmed fraction < 5%	redefine estimand (ATT, trimmed population)
Placebo (future round)	CI covers 0	enrich Z (longer history, trend features); else do not ship
Negative control (pre-growth)	CI covers 0	same as placebo
E-value	> plausible latent-quality association given rich X	report as bounded, not as an effect
Variance agreement	IF vs bootstrap within 20%	investigate nuisance instability
Synthetic recovery	Table 4.3 ranking reproduced	pipeline bug; fix before touching real data

1. *Generate* an EU-style panel: fundamentals as noisy, lagged functions of latent quality q_i and macro; treatment assignment with probability increasing in q_i and in the observable dossier; outcome with a known additive effect τ^* of treatment.
2. *Estimate* with four pipelines: naive difference in means; g-formula (T-learner); IPW; cross-fitted AIPW — each with firm-cluster bootstrap CIs and E-values.
3. *Verify* the qualitative ranking of Table 4.3: the naive contrast overstates τ^* badly (quality confounding); all adjusted estimators land near τ^* when the generated X is rich; ablating X columns (hiding the dossier, i.e. simulating the opaque regime of Chapter 5) re-opens the gap, with AIPW degrading most gracefully.
4. *Report* empirical CI coverage of τ^* across Monte Carlo replications and check that the naive estimator’s E-value correctly flags fragility while the adjusted estimators’ E-values are informative about the injected residual confounding.

Table 4.3: Expected qualitative behaviour on the synthetic testbed (Section 2.5). “ $\approx$ ” means CI covers τ^* at nominal rate.

Estimator	Rich X (transparent)	Impoverished X (opaque)
Naive difference	$\gg \tau^*$	$\gg \tau^*$
g-formula (XGB)	$\approx \tau^*$	biased toward naive
IPW (trimmed)	$\approx \tau^*$, wider CI	biased, unstable
AIPW cross-fitted	$\approx \tau^*$, correct coverage	residual bias, flagged by placebo and E-value

This protocol makes the chapter testable: the code that must reproduce Table 4.3 is byte-identical to the code run on the real panel, so a pipeline that cannot recover a known effect is caught before it is believed.

4.8 Limits

Two limits are structural. First, *selection on unobservables is not solved* — it is attenuated by a rich X , bounded by E-values, and probed by placebo and negative-control tests, but a confounder invisible to three years of audited filings and strong enough to survive the checklist remains logically possible. The claim is calibrated accordingly: adjusted estimates with reported sensitivity, not oracle effects. Second, *external validity across macro regimes*:

M enters both selection and outcome, all estimates average over the funding cycles in sample, and a CATE-by-era analysis is data-hungry because treated firms are scarce within any single era. An effect estimated across 2015–2021 need not transport to a tight-money regime, for the same no-extrapolation reasons as in Chapter 3. The opaque regime sharpens both limits at once: with no fundamentals panel there is no rich X , and the honest response is to change the estimand and the observation model — the programme of Chapter 5, completed by the point-process machinery of the following part.

Design decision (D-EU.2 — AIPW behind a falsification gate).

The committed causal estimator for the transparent regime is cross-fitted AIPW Equation (4.3) with firm-level folds, propensity trimming to $[0.02, 0.98]$, and both cluster-robust influence-function and cluster-bootstrap intervals (the wider is reported); g-formula and IPW are retained as diagnostics only. No causal estimate ships without clearing every row of Table 4.2 — overlap, placebo, negative control, E-value, variance agreement, and synthetic recovery. Rationale: double robustness (Proposition 4.7) and the rate condition of Remark 4.8 are the only route to $\sqrt{n}$ inference with boosted-tree nuisances, and the checklist converts the unverifiable Assumption 4.3 into auditable failure modes. Reversal trigger: if the placebo or negative-control tests fail persistently even after enriching Z with longer histories and trend features, the transparent-regime causal claim is demoted to the bounded/associational reporting of Chapter 3 and this box is rewritten to say so.

Code hooks.

The pipeline of Section 4.4 is a planned module `hawkes_calibration/eu/causal.py`: firm-level fold assignment, out-of-fold nuisance fitting (reusing the boosted stack of Chapter 3 with tuning frozen), trimming with recorded fractions, AIPW scores, and the two variance routes of Section 4.5. The checklist of Table 4.2 is an artifact contract: the module emits the overlap histogram, placebo and negative-control estimates, E-values, and variance-agreement ratio as machine-readable outputs alongside $\hat{\tau}$, in the `results/` conventions of Protocol 11.15(d). The synthetic protocol of Section 4.7 builds its confounded panel from `synthetic/firms.py` and `synthetic/regime_b.py` (latent quality already drives funding intensity there) and belongs in the test suite, not in a notebook: recovery of τ^* is a regression test that gates changes to the causal module.

Chapter 5

When Fundamentals Are Latent: Proxies and Selection

This chapter is the hinge of the transparent-regime part. Its place on the decision map of Section 1.4 is to explain why the rest of the document exists. The two preceding chapters assumed fundamentals arrive on a statutory clock: every firm, every fiscal year, a filed income statement. In the US-centric PitchBook universe of Chapter 2 that clock does not exist. The vendor delivers instead the firm’s *transaction record*: a sequence of deal events, each carrying marks — a valuation, a round classification, an employee count, occasionally a revenue or EBITDA figure. Between events we learn nothing. This chapter re-derives the transparent-regime pipeline under that observation model. Three things change: the *target* (EBITDA growth must be replaced by proxies built from marks), the *unit of analysis* (the firm-year panel row is replaced by the consecutive-deal pair), and — most consequentially — the *bias structure*: outcomes are observed only when the firm transacts again, so every naive estimate is filtered through survival-to-next-round. Against the defect ledger of Table 2.1, this is where (C5) and (C3) bind hardest: the covariates are the LOCF process of Equation (2.1) by construction, and the outcomes inherit the immortal-firm selection. The repair for that selection is a model of the deal-arrival process itself, which is why the continuous-time machinery of the next part completes this chapter.

5.1 The observation model

Let $S_{i,t} \in \mathbb{R}^d$ denote the latent state of firm i : fundamentals (revenue, margin trajectory, unit economics) plus soft state (team quality, runway). In the transparent regime we observed a noisy annual census of S . Here we observe marks only at the firm’s own deal times $T_{i,1} < T_{i,2} < \dots$:

$$m_{i,k} = (V_{i,k}, \text{round_type}_{i,k}, E_{i,k}, R_{i,k}, B_{i,k}), \quad (5.1)$$

where V is post-money valuation (populated only for priced rounds), E the reported employee count, and R, B the revenue and EBITDA fields — populated for roughly 45% and 25% of deal rows respectively in typical vendor coverage (Chapter 2, Appendix B). The marks are noisy, truncated views of $S_{i,T_{i,k}}$ in the classical errors-in-variables sense (Carroll et al., 2006): valuation confounds fundamentals with the market multiple, employees confound productivity with cash deployment, and the revenue field is populated selectively.

What the modeler works with between deals is exactly the LOCF panel formalized once in Section 2.2: the carried-forward mark vector $\tilde{S}_{i,t}$ of Equation (2.1) together with the staleness clock $a_i(t) = t - \tau_i(t)$. We do not restate that formalization here; Chapter 12 owns it, including the proof that the pair $(\tilde{S}_i, a_i)$ is predictable and therefore a legitimate covariate, and the analysis of what staleness does to fitted effects. Two consequences matter for this chapter. The

staleness clock carries real information: a valuation carried forward 4 years is a different object from one carried forward 4 months, and models must be allowed to discount stale marks. And the deeper problem is *when* the panel refreshes:

Warning 5.1 (Sampling times are informative). The observation times $T_{i,k}$ are not an exogenous measurement schedule; they are outcomes of the firm’s own trajectory. A firm raises when it is doing well (an up round at a rich multiple) or when it must (runway exhaustion, a down round); firms that are neither raise nothing and go dark. Formally, the event indicator $dN_i(t)$ is not independent of $S_{i,t}$ given the past: the missingness mechanism of the LOCF panel is *missing-not-at-random*. Any panel method that treats $\tilde{S}_{i,t}$ as if generated by random inspection times — imputation, mixed models, naive regressions on the LOCF columns — inherits a bias whose sign and size depend on the unmodelled sampling mechanism. The only honest formulations either (i) condition explicitly on the event frame (Section 5.3) and correct for selection (Section 5.4), or (ii) model the sampling process jointly with the state (Rizopoulos, 2012; Tsiatis and Davidian, 2004) — the escalation rung of Chapter 12, built on the intensity machinery of Chapters 7 and 8.

5.2 What replaces EBITDA growth

The transparent-regime success labels of Chapter 3 are defined on filed fundamentals. We now construct their surrogates, all defined on a pair of consecutive deals $(T_{i,k}, T_{i,k+1})$ with gap $\Delta_{i,k} = T_{i,k+1} - T_{i,k}$.

(a) Implied valuation growth. When both endpoints are priced rounds,

$$g_{i,k}^{\text{val}} = \frac{\log V_{i,k+1} - \log V_{i,k}}{\Delta_{i,k}}, \quad (5.2)$$

an annualized log growth rate. Writing valuation as fundamentals times a sector multiple, $\log V_{i,k} = \log F_{i,T_{i,k}} + \log m_{s(i),T_{i,k}} + \varepsilon_{i,k}$, gives the decomposition

$$g_{i,k}^{\text{val}} = \underbrace{g_{i,k}^F}_{\text{fundamentals growth}} + \underbrace{g_{i,k}^m}_{\text{multiple expansion}} + \underbrace{\nu_{i,k}}_{\text{pricing noise}}. \quad (5.3)$$

Only g^F is a success signal; g^m is a market-wide repricing that the firm did not earn. We identify the multiple component by the public-market-comparables logic: regress g^{val} on sector $\times$ time-bucket fixed effects plus macro controls (rate changes, public-index sector returns over the gap); the fitted cell means estimate g^m , and the residual is the firm-level proxy $\hat{g}_{i,k}^F$. The fixed effects must be estimated on all pairs in the cell, not per firm, or they absorb the signal.

(b) Round direction as an ordinal outcome. The field `vc_round_up_down_flat` classifies each round against the previous one as {down, flat, up}. It is coarse but far better populated than paired valuations (it is defined even when one endpoint is unpriced) and robust to level noise in V . We treat it as an ordinal target (proportional-odds or ordinal gradient boosting); its drawback is the vendor’s flat band, which is a definitional artifact rather than an economic category.

(c) Employee growth. $g_{i,k}^{\text{emp}} = (\log E_{i,k+1} - \log E_{i,k})/\Delta_{i,k}$ is a real-activity proxy available for most pairs. Its bias is mechanical: headcount growth immediately after a raise reflects cash deployment, not demonstrated productivity, so it leads fundamentals optimistically by construction.

(d) **Reported revenue growth.** `revenue_growth_since_last_deal` is the most direct surrogate when populated, but with $\sim 45\%$ endpoint coverage the pair-level coverage is roughly 20%, and reporters skew to larger, later-stage firms — the missingness is again not at random.

Proxy	Pair coverage	Noise	Dominant bias
g^{val} Equation (5.2)	moderate (both rounds priced)	high (negotia- tion, multiple)	upward; confounds multi- ple expansion unless decom- posed via Equation (5.3)
<code>vc_round_ up_down_flat</code>	high	low but coarse	flat band definitional; cen- sors magnitude
g^{emp}	high	moderate	upward after raises (cash- fueled hiring)
<code>revenue_growth_ since_last_deal</code>	low ($\sim 20\%$)	low given re- ported	reporters skew large/late- stage; MNAR

Table 5.1: Growth proxies on the consecutive-deal pair frame. All four are additionally subject to the survival-to-next-round selection of Section 5.4, which no choice among them avoids.

In practice we fit all four and read them jointly: agreement across proxies with different bias directions is the closest available substitute for a filed income statement.

5.3 Supervised learning on the event frame

The prediction unit is the ordered pair $(i, k) \rightarrow (i, k + 1)$: one row per consecutive-deal pair, outcome one of the proxies of Section 5.2. Features come in two blocks with different epistemic status.

Firm-history block, measurable at $T_{i,k}$: the full mark history through the k -th deal (valuation path, round sequence, employee path, deal sizes, investor counts), the staleness pattern, firm age, sector, and gap lengths between *past* deals. Nothing dated inside $(T_{i,k}, T_{i,k+1}]$ from the firm’s own record enters — there is nothing there to use except the second deal itself, which is the outcome’s timestamp.

Gap macro block, aggregates of exogenous series over the gap: mean policy rate, change in CPI year-on-year, change in unemployment, and sector deal heat (sector deal counts per week, normalized by trailing average), all built from point-in-time vintages per Protocol 2.5. Because these are external to the single firm, conditioning on them does not leak the firm’s outcome; but it does change the estimand. With the gap block included, the regression is an *attribution* model — growth conditional on the realized macro path and the realized gap length. For deployment as a forecast at $T_{i,k}$, the gap block must be replaced by scenario paths or integrated over a gap-length model; the gap length $\Delta_{i,k}$ is itself informative (short gaps select hot firms), and we return to precisely this point below.

The learner is the Chapter 3 stack unchanged: gradient boosting with blocked cross-validation (folds split by firm *and* time, under the constitution of Section 11.6), early stopping on the temporal holdout, and conformalized quantile intervals for uncertainty; we do not repeat that machinery here. Two adaptations matter. First, conformal calibration sets must be blocked by firm, since pairs from one firm share latent quality and violate exchangeability across naive random splits. Second, once the selection weights of the next section are introduced, calibration must be weight-aware (weighted conformal quantiles), otherwise coverage holds for the selected population only.

5.4 The selection problem

This section is the core of the chapter. A pair (i, k) exists in the training frame only if the firm raised again within the observation window: define

$$D_{i,k} = \mathbf{1}\{T_{i,k+1} \leq \tau_{\text{end}}\}, \quad (5.4)$$

where τ_{end} is the data cutoff. Every proxy of Section 5.2 is observed conditional on $D_{i,k} = 1$: we see the growth of firms *that came back*. The observed distribution is $P(g \mid D=1)$, not $P(g)$, and the gap between them has an exact expression:

$$\mathbb{E}[g \mid D=1] - \mathbb{E}[g] = \frac{\text{Cov}(g, \mathbf{1}\{D=1\})}{\mathbb{P}(D=1)}. \quad (5.5)$$

Both selection channels have positive covariance with *appearing* in the frame but opposite implications for the observed growth mix: winners raise again (pulling observed g up), and cash-constrained firms raise again at down rounds (pulling it down), while the quiet failures — firms that never transact again — contribute nothing at all. The left tail of $P(g)$ is therefore systematically missing: shutdown is the one outcome that generates no second deal. Estimates of mean growth, of feature effects, and of ranking quality are all computed on the surviving subpopulation — and reading the survivors’ serial-raising pattern as firm-level state dependence rather than selection is the classical trap (Heckman and Borjas, 1980).

Bounds. If g is bounded to $[g_{\text{lo}}, g_{\text{hi}}]$ (after winsorization, or by construction for the ordinal outcome), then with $p = \mathbb{P}(D=1)$,

$$\mathbb{E}[g] \in [p \mathbb{E}[g \mid D=1] + (1-p) g_{\text{lo}}, p \mathbb{E}[g \mid D=1] + (1-p) g_{\text{hi}}], \quad (5.6)$$

the worst-case (no-assumption) bounds. With p well below one — deal records concentrate on a minority of serially raising firms (Figure 5.1) — these bounds are wide. They are still worth reporting: they state what the data alone say, and any point estimate inside them relies on an assumption.

Assumption 5.2 (Selection on observed history). $g_{i,k} \perp\!\!\!\perp D_{i,k} \mid \mathcal{H}_{T_{i,k}}$, where $\mathcal{H}_{T_{i,k}}$ collects the firm’s mark history through deal k and the macro state: given everything observed up to the first deal of the pair, whether the firm raises again carries no further information about its growth.

The practical fix: inverse-probability-of-observation weights. Under Assumption 5.2, define the observation probability

$$\pi_{i,k} = \mathbb{P}(D_{i,k}=1 \mid \mathcal{H}_{T_{i,k}}) = 1 - \exp\left(-\int_{T_{i,k}}^{\tau_{\text{end}}} \lambda_i(u) du\right), \quad (5.7)$$

where λ_i is the firm’s next-deal intensity given its history. Weighting each observed pair by $1/\hat{\pi}_{i,k}$ (stabilized and trimmed in the usual way) makes the weighted frame representative of all at-risk firms: $\mathbb{E}[Dg/\pi] = \mathbb{E}[g]$. This is inverse-probability weighting for informative censoring in the sense of Hernán and Robins (2020), with the hazard-model lineage of Cox (1972); validity requires $\pi_{i,k} > 0$ and a well-specified intensity, and residual selection on latent quality beyond $\mathcal{H}$ is not repaired — which is why we report Equation (5.6) alongside. The selection identities Equations (5.5) and (5.6) and the unbiasedness of the weighted mean are proved in Appendix C.

Remark 5.3 (The bridge to the machinery). Equation (5.7) is the structural link between this document’s two halves. The quantity $\lambda_i(u)$ — the conditional intensity of the firm’s deal-arrival process — is exactly what the next part calibrates: inhomogeneous Poisson with macro covariates (Chapter 7), self-exciting and inhibiting extensions (Chapter 8), and the censored variants that untrusted dates force (Chapter 10). The count model is therefore not a separate deliverable that happens to share the data: it *repairs the growth regression*. A team that skips the point-process layer does not merely lose the deal-timing forecast; it loses the only principled weights for its growth model. The combined pipeline — intensity fit, weights, weighted growth regression — is to be validated on the synthetic world of Section 2.5, where $P(g)$ is known, with results reported in Chapter 19.

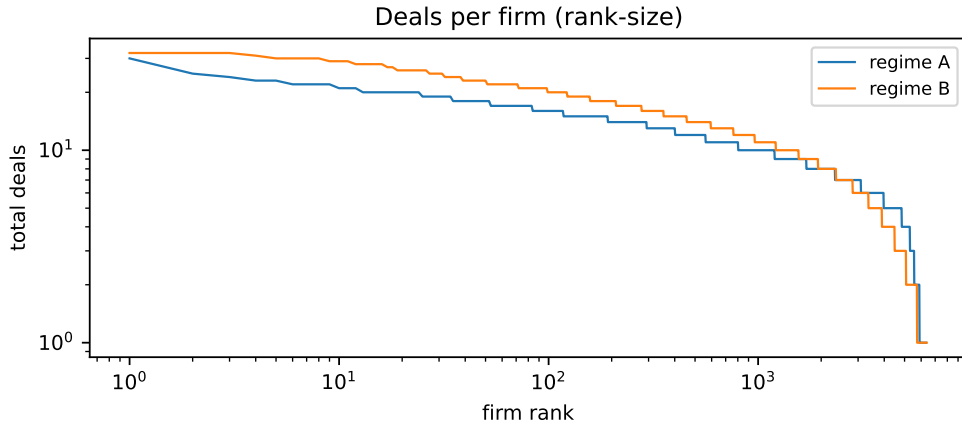


Figure 5.1: Rank-size plot of deals per firm in the two synthetic regimes (illustrative; the generator reproduces the concentration typical of deal records). A small minority of serially raising firms contributes most consecutive-deal pairs, while the modal firm contributes one deal and zero pairs. The pair frame therefore oversamples the firms with the highest historical deal hazard; unweighted estimates describe serial raisers, not the tracked universe. Firm-level cluster bootstrap is mandatory for all standard errors on this frame.

5.5 Macro factors as drivers

With firm-level outcomes this sparse and selected, much of the macro question — do rate changes move private-market activity and pricing? — is better asked at the sector level. The workhorse is a sector-exposure (shift-share style) design. Let $y_{s,w}$ be a sector-week outcome (log deal count, or the cell-mean valuation growth from Equation (5.3)), Δr_w the rate change, and e_s a *predetermined* exposure measure (share of pre-revenue firms in the sector, a duration proxy for how far cash flows sit in the future). We fit

$$y_{s,w} = \alpha_s + \gamma_w + \delta(e_s \cdot \Delta r_w) + u_{s,w}, \quad (5.8)$$

with sector and time fixed effects and two-way clustered standard errors. The time effect γ_w absorbs every common shock — including the *average* effect of the rate change itself — so δ is identified purely from differential sensitivity: do high-exposure sectors move more when rates move?

Three interpretation limits apply. First, the common component of the macro effect is not identified by design; Equation (5.8) estimates relative sensitivities only. Second, latent risk appetite co-moves with rates; if its sector loadings correlate with e_s , the differential estimate

is contaminated too. Third, exposure must be predetermined and not itself an outcome of recent sector performance. A design-based reading in the spirit of Imbens and Rubin (2015) is available only under an explicit exogeneity assumption on the exposure shares, which we state and plan to stress-test on the synthetic testbed (Chapter 19) rather than assume silently. Where valuation marks are the outcome, all of Section 5.4’s caveats apply on top.

The forward pointer of the section: deal *counts* are complete (a deal either appears in the feed or does not), while marks are sparse and selection-laden. The cleanest macro inference therefore lives at the count layer — intensities modulated by macro covariates, the covariate-NHPP baseline that Chapter 7 builds and that every excitation story must beat.

5.6 Decision table and the bridge out

Aspect	Transparent track (Chapters 3 and 4)	Latent adaptation (this chapter)
Target	EBITDA/revenue growth from filings	proxy set: $\hat{g}^F$ from Equation (5.3), ordinal round direction, employee growth, reported revenue growth
Unit	firm $\times$ fiscal year	consecutive-deal pair (i, k)
Feature clock	fiscal-year as-of with publication lags	event-time as-of with staleness clock (Section 2.2, Chapter 12)
Dominant bias	leakage via publication timing	survival-to-next-round selection; MNAR sampling times
Primary fix	point-in-time joins, blocked CV	IP-of-observation weights from the next-deal intensity Equation (5.7); worst-case bounds Equation (5.6)
Uncertainty	conformal intervals, firm-blocked	weight-aware conformal, firm-cluster bootstrap
Macro layer	covariates in the growth model	shift-share differentials Equation (5.8); count-layer inference in Chapter 7

Table 5.2: Method mapping between observation regimes. Every row of the right column degrades gracefully to the left column when fundamentals become observable; the converse direction is the content of this chapter.

Table 5.2 summarizes the chapter and closes the transparent-regime part with an explicit hand-off. The selection problem exposed here is not patched within this chapter’s toolkit: its repair *is* the hazard and point-process machinery that follows — the next-deal intensity of Equation (5.7) is fitted by the Poisson and Hawkes layers of Chapters 7 and 8, and the discrete-time face of the same repair (censored buckets labeled, never dropped; eligibility masks) is the bucketed-panel discipline of Chapter 13. Likewise, the staleness of carried-forward marks is not formalized here — the LOCF observation model belongs to Section 2.2, and its consequences (attenuation, informative observation times, the remedy ladder) are the subject of Chapter 12. Nothing in this chapter’s adaptation is exotic: each entry is a transparent-regime method re-derived under a harsher observation model, with one genuinely new ingredient — the deal-arrival intensity — that the next part constructs.

Design decision (D-EU.3 — no unweighted pair-frame estimate ships).

Proxy outcomes on the consecutive-deal pair frame are always reported as a joint read across the four proxies of Table 5.1, and any population-level statement (mean

growth, feature effect, ranking quality) is computed with inverse-probability-of-observation weights Equation (5.7) derived from a fitted next-deal intensity, flanked by the worst-case bounds Equation (5.6); unweighted estimates are labeled as describing the serial-raiser subpopulation and never leave the diagnostic tier. Rationale: Equation (5.5) is an identity, not a risk — the selected and full-population means differ whenever growth covaries with returning to market, and the frame concentration of Figure 5.1 makes the covariance large. Reversal trigger: if fitted observation probabilities are uniformly near one within a segment (e.g. late-stage, short-horizon pairs), the weights degrade to the identity there and this box’s requirement reduces to reporting that fact.

Code hooks.

The pair-frame builder is a planned module `hawkes_calibration/pairs/`: it consumes the event list and LOCF features emitted by the loaders (`synthetic/loaders.py` now, `hawkes_calibration/data/pitchbook.py` later), enforces the firm-history/gap-macro block separation of Section 5.3, and emits $(i, k, \Delta_{i,k}, g', D_{i,k})$ rows with staleness ages attached. The observation weights Equation (5.7) are computed from the intensity fitters of the next part — `hawkes_calibration/eventtime/` for firm-level gap models, `hawkes_calibration/sector_hazard.py` for the sector layer — so the module depends on Chapters 7 and 8 deliverables by design. The multiple-decomposition regression Equation (5.3) and the shift-share design Equation (5.8) share the sector-week frame of `hawkes_calibration/sector_stability.py`. Validation of the combined weights-plus-regression pipeline against known $P(g)$ will run on `synthetic/` ground truth (regimes A and B), with results reported in Chapter 19.

Part III

Continuous-Time Machinery

Chapter 6

Temporal Point Processes: Foundations

6.1 One machinery, many models

Every probabilistic model in this document is an instance of one object: a counting process characterized by its conditional intensity. The intensity induces the likelihood we fit, its compensator induces the residuals we check, and the same calculus justifies the simulators. This chapter builds that machinery once, so that Chapters 7, 8, 10, 11 and 14 can instantiate Theorem 6.5 with their own λ and move on.

Five results carry the weight. First, the point-process likelihood (Theorem 6.5): one formula, an event term plus an exposure term, that every fitter in `hawkes_calibration/` implements. Second, the ground/mark factorization (Theorem 6.7): the exact identity, promised in Section 1.3, that splits the marked deal process into a timing factor and an identity factor and thereby licenses the repository’s two-layer sector→firm architecture. Third, martingale estimation theory (Section 6.6): why maximum likelihood works for dependent event data, and which standard errors to trust when the model is wrong. Fourth, the time-rescaling theorem (Theorem 6.16): the goodness-of-fit engine of Chapter 11. Fifth, simulation correctness (Proposition 6.18): why the generators planned for the synthetic world of Section 2.5 produce the processes they claim to. The chapter closes, as Chapter 2’s decision D2.1 requires, with an explicit account of which data defects (C1)–(C6) this machinery absorbs and which it cannot.

6.2 Counting processes, histories, and conditional intensity

Definition 6.1 (Counting process). On the filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$ of Definition 2.1, a *counting process* N is an adapted, right-continuous process with $N(0) = 0$, nondecreasing piecewise-constant paths, jumps of size +1, and $N(t) < \infty$ a.s. for every $t \leq T$. A *multivariate counting process* is a finite family $(N_e)_{e \in \mathcal{E}}$ of counting processes, no two of which jump at the same time.

The index e will range over sectors, firms, or firm–transition pairs as each chapter requires; the filtration $\mathcal{F}_t$ contains the internal history $\mathcal{H}_t$ of the marked deal process *and* the observed covariate paths, as fixed in Definition 2.1. Two pieces of structure make the theory work. We name both precisely, because a violation of either shows up first as a software bug.

Predictability. A process H is *predictable* if it is measurable with respect to the σ -algebra on $\Omega \times [0, T]$ generated by left-continuous adapted processes. The operational content: the value $H(t)$ must be computable from information available strictly before t , i.e. from $\mathcal{F}_{t-}$.

Left-continuity in t is the working test — an adapted process with left-continuous paths is predictable, and every intensity model in this document is built from such pieces (LOCF covariates held constant between deals, kernel sums evaluated as left limits, at-risk indicators of the form $\mathbf{1}\{V_i \leq t < U_i\}$ which are left-continuous in t).

Warning 6.2 (Predictability is point-in-time discipline). Predictability of λ is the continuous-time statement of Protocol 2.5. If any input to $\lambda(t)$ carries a timestamp not strictly before t — a covariate as-of joined on reference date instead of publish date, a kernel sum that includes the event at t itself, a risk-set flag closed using information revealed at t — then λ is not predictable, $N - \int Y\lambda$ is not a martingale with respect to the filtration actually used, and everything downstream fails *silently*: the score of Proposition 6.12 acquires a bias, the information identities break, and held-out likelihoods flatter the model. This is why `hawkes_calibration/eventtime/likelihood.py` decays its excitation state to *just before* each event when evaluating $\log \lambda$ at that event.

Compensators. The second piece of structure is the decomposition of a counting process into signal and noise.

Theorem 6.3 (Doob–Meyer decomposition for counting processes; stated, not proved). *Let N be a counting process with $\mathbb{E}N(T) < \infty$. There exists an a.s. unique predictable, nondecreasing process Λ with $\Lambda(0) = 0$, the compensator, such that $M = N - \Lambda$ is a martingale. Under local integrability the same holds with “local martingale” (Andersen et al., 1993, Ch. II).*

Definition 2.3 specializes this to the absolutely continuous case that this document lives in: a *conditional intensity* for N_e is a nonnegative predictable process λ_e such that

$$M_e(t) = N_e(t) - \int_0^t Y_e(u) \lambda_e(u) du \quad (6.1)$$

is a local martingale, with Y_e the at-risk indicator. The heuristic that gives the definition its meaning, and that every proof in this chapter leans on, is the infinitesimal-bin reading of the martingale property:

$$\lambda_e(t) dt = \mathbb{P}(N_e \text{ jumps in } [t, t + dt) \mid \mathcal{F}_{t-}) \quad \text{on } \{Y_e(t) = 1\}. \quad (6.2)$$

The intensity is the instantaneous event rate given *everything known just before now*: internal history, covariate paths, risk-set composition. It is a modeling object (we write λ down) and simultaneously a probabilistic one (Equation (6.1) pins down what writing it down commits us to).

Assumption 6.4 (Standing regularity). Throughout this chapter $(N_e)_{e \in \mathcal{E}}$ is a multivariate counting process such that (i) each N_e admits the intensity $Y_e \lambda_e(\cdot; \vartheta)$ of Equation (6.1), with λ_e predictable, locally bounded, and strictly positive at the jump times of N_e ; (ii) $\mathbb{E} \int_0^T Y_e \lambda_e du < \infty$; (iii) $\lambda_e(t; \cdot)$ is twice continuously differentiable in ϑ on a neighborhood of the truth, with predictable, locally bounded derivative processes, and differentiation under the integral sign is valid; (iv) no two components jump simultaneously. All models fitted in this repository satisfy (i)–(iv) by construction (exponential-family intensities, piecewise-constant covariates, finite windows).

6.3 The point-process likelihood

Theorem 6.5 (Point-process (Jacod) likelihood). *Under Assumption 6.4, let the data on $[0, T]$ be the event times and types (t_m, e_m) , $m = 1, \dots, n$. Then, up to an additive term not*

depending on ϑ (the dominating measure), the log-likelihood is

$$\ell_T(\vartheta) = \sum_{m=1}^n \log \lambda_{e_m}(t_m; \vartheta) - \sum_{e \in \mathcal{E}} \int_0^T Y_e(u) \lambda_e(u; \vartheta) du, \quad (6.3)$$

which is Equation (2.3) of Definition 2.3.

Proof. We give the product-integral derivation over infinitesimal bins; it is the argument to remember, and it becomes rigorous as a product-integral limit (Daley and Vere-Jones, 2008; Andersen et al., 1993). Partition $[0, T]$ into K bins $B_k = ((k-1)\delta, k\delta]$ with $\delta = T/K$, and K large enough that no bin contains two events. By the chain rule of conditional probability, the probability of the observed configuration is the product over bins of the conditional probability of what happened in B_k given $\mathcal{F}_{(k-1)\delta}$. By Equation (6.2) and Assumption 6.4, a bin containing the event (t_m, e_m) contributes $\lambda_{e_m}(t_m) \delta + o(\delta)$; an empty bin contributes $1 - \sum_e Y_e \lambda_e \delta + o(\delta)$; two or more events in one bin have probability $o(\delta)$. Hence

$$\log \mathbb{P}(\text{data}) - n \log \delta = \sum_{m=1}^n \log \lambda_{e_m}(t_m) + \sum_{k: B_k \text{ empty}} \log \left(1 - \sum_e Y_e \lambda_e \delta \right) + o(1).$$

Since $\log(1-x) = -x + O(x^2)$, the middle sum is a Riemann sum of $-\sum_e \int_0^T Y_e \lambda_e du$ and converges to it as $\delta \rightarrow 0$. The diverging term $n \log \delta$ is the same for every ϑ : it cancels in every likelihood ratio and belongs to the dominating measure (concretely, Equation (6.3) exponentiated and multiplied by e^T is the density of the observed process with respect to a unit-rate Poisson process). This yields Equation (6.3). $\square$

The two terms have distinct jobs. The *event term* $\sum_m \log \lambda_{e_m}(t_m)$ rewards the model for placing intensity where events actually occurred; it is evaluated exactly at the jumps, on the log scale, so assigning near-zero intensity to a realized event is very costly. The *compensator term* $\sum_e \int Y_e \lambda_e$ charges the model for exposure: every unit of intensity claimed over every firm-week at risk is paid for linearly, whether or not an event arrives. A model can only win the event term by spending in the compensator term, and the optimal spending pattern is the true intensity — the content of the score identity in Section 6.6. Two consequences of the shape of Equation (6.3) recur constantly. An event-free spell $(a, b]$ contributes exactly the survival factor $\exp(-\int_a^b Y \lambda)$; censored spells therefore carry information and must never be dropped (Warning 2.4, Proposition 6.20). And the event term is the only place event *times* enter; if the times are corrupted, the compensator term survives while the event term degrades — the starting point for the interval-censored treatment of Chapter 10.

6.4 The ground/mark factorization

The market process of Definition 2.1 is a *marked* point process: ground times t_m with marks m_m (in our data the mark is the triple sector–firm–terms). The likelihood of a marked process factorizes exactly into a timing part and a mark part. This identity is the formal license for the two-layer architecture, so we set it up carefully and prove it.

Definition 6.6 (Marked intensity kernel). Let marks live in a measurable space $(\mathcal{M}, \nu)$ with ν a σ -finite reference measure (counting measure for discrete marks such as sector and firm identity, Lebesgue for continuous terms). A *marked point process* with intensity kernel $\lambda(t, m)$ is one for which, for every measurable $A \subseteq \mathcal{M}$, the counting process $N(t, A) = \#\{m : t_m \leq t, m_m \in A\}$ has conditional intensity $\int_A \lambda(t, m) \nu(dm)$. The *ground process* is $N_g(t) = N(t, \mathcal{M})$ with *ground intensity* $\lambda_g(t) = \int_{\mathcal{M}} \lambda(t, m) \nu(dm)$, and the *mark density* is

$$p(m \mid t, \mathcal{H}_{t-}) = \frac{\lambda(t, m)}{\lambda_g(t)} \quad \text{whenever } \lambda_g(t) > 0, \quad (6.4)$$

a probability density in m with respect to ν , predictable in t .

Theorem 6.7 (Ground/mark factorization). *Under Assumption 6.4 applied to the family $\{N(\cdot, A)\}$, the log-likelihood of the marked data $\{(t_m, m_m)\}_{m=1}^n$ on $[0, T]$ satisfies, exactly and for every parameter value and every realization,*

$$\ell_T = \underbrace{\sum_{m=1}^n \log \lambda_g(t_m) - \int_0^T \lambda_g(u) du}_{\text{ground factor: timing}} + \underbrace{\sum_{m=1}^n \log p(m_m | t_m, \mathcal{H}_{t_m^-})}_{\text{mark factor: identity}}. \quad (6.5)$$

Proof. Take first the case of finitely many mark values, which covers the sector–firm part of our data. The marked process is then the multivariate counting process $(N(\cdot, \{m\}))_{m \in \mathcal{M}}$ with intensities $\lambda(t, m)$, and Theorem 6.5 gives

$$\ell_T = \sum_{m=1}^n \log \lambda(t_m, m_m) - \sum_{m \in \mathcal{M}} \int_0^T \lambda(u, m) du.$$

Substitute $\lambda(t, m) = \lambda_g(t) p(m | t, \mathcal{H}_{t-})$ from Equation (6.4). The event term splits into $\sum_m \log \lambda_g(t_m) + \sum_m \log p(m_m | t_m, \mathcal{H}_{t_m^-})$. The compensator term collapses:

$$\sum_m \int_0^T \lambda_g(u) p(m | u, \mathcal{H}_{u-}) du = \int_0^T \lambda_g(u) \underbrace{\sum_m p(m | u, \mathcal{H}_{u-})}_{=1} du = \int_0^T \lambda_g(u) du,$$

because the mark density integrates to one at every u . For general mark spaces, replace the sum over m by $\int \cdot \nu(dm)$; the computation is identical and the underlying likelihood formula is Jacod’s formula for marked processes (Daley and Vere-Jones, 2008). This proves Equation (6.5). $\square$

The crux is worth stating in words: *the mark density leaves no trace in the compensator*, because it integrates to one under the exposure integral. The mark factor therefore contributes only at event times — it is a product of genuine conditional probabilities “given a deal at t_m , which mark” — and involves intensities only through the ratio Equation (6.4), in which any common timing factor cancels. That is the mathematical root of the robustness asymmetry announced in Section 1.3: distortions that rescale or warp λ_g (untrusted dates, aggregation, baseline misspecification) damage the ground factor and cancel from the mark factor. Chapters 10 and 14 quantify this.

Remark 6.8 (Identity, not approximation). Equation (6.5) assumes no independence, no Poisson structure, and no correct specification; it is algebra performed on the likelihood of Theorem 6.5, valid realization by realization. Every approximation made in this document lives *inside* one factor: the weekly MBPP approximation of Chapter 10 approximates the ground factor; the linear-index softmax of Chapter 14 approximates the mark factor; neither touches the other factor, and a data defect can be assigned to the factor whose term it corrupts. This bookkeeping — which factor is damaged — is how the defect ledger of Table 2.1 becomes actionable.

Corollary 6.9 (The two-layer sector→firm architecture). *Specialize the mark to $m = (s, i, m')$: sector, receiving firm, deal terms. Applying Theorem 6.7 and factoring the mark density by the chain rule, $p(s, i, m' | t) = p(s | t) p(i | s, t) p(m' | s, i, t)$, the likelihood splits into the three factors of Equation (1.4). Equivalently, defining sector processes N_s with intensities $\lambda_s(t) = \lambda_g(t) p(s | t, \mathcal{H}_{t-})$, every tracked firm’s intensity factorizes as*

$$\lambda_i(t) = \lambda_{s(i)}(t) \mathbb{P}(i | s(i), t, \mathcal{H}_{t-}), \quad i \in \mathcal{I}, \quad (6.6)$$

and the log-likelihood is the sum of a sector-timing part (in λ_s only) and a firm-identity part (in the conditional choice probabilities only).

Proof. Apply the theorem twice: once with mark (s, i, m') , once viewing i as a mark of the sector- s ground process N_s ; the chain-rule factorization of the joint mark density is the elementary identity for conditional densities, and Equation (6.6) is Equation (6.4) read at the firm level. $\square$

Design decision (D3.1 — the factorization is the architecture).

The repository’s two-layer design — a sector-level ground model (`hawkes_calibration/sector_stability.py`, `hawkes_calibration/eventtime/`) and a firm-level risk-set choice model (`hawkes_calibration/sector_ranker.py`) — is Corollary 6.9 implemented literally, with the two factors parameterized by variation-independent parameter blocks so that maximizing the sum Equation (6.5) decouples into two exact, separate maximizations. Rationale: the factorization is an identity, the factors’ metrics align with distinct targets (T2 vs. T1 of Section 1.2), and the factors have opposite robustness profiles under the defect ledger. Reversal trigger: evidence of material parameter sharing across factors — e.g. a jointly estimated timing-identity model beating the separate fits on *both* held-out ground NLL and identity recall@ K in two consecutive evaluation campaigns — would move us to joint estimation; the factorized *likelihood* would remain, only the separability of its maximization would be given up.

6.5 Superposition and the identity of the next event

Two small results connect the multivariate and marked views, and both are used constantly — in the ranking chapter as the modeling target, and in the simulation section as the attribution step.

Proposition 6.10 (Superposition). *Under Assumption 6.4, the pooled process $\bar{N} = \sum_e N_e$ is a counting process with conditional intensity $\bar{\lambda}(t) = \sum_e Y_e(t) \lambda_e(t)$.*

Proof. $\bar{N}$ is right-continuous, nondecreasing, and has unit jumps because no two components jump together; and $\bar{N} - \int \bar{\lambda} = \sum_e M_e$ is a finite sum of local martingales, hence a local martingale. Since $\bar{\lambda}$ is a sum of predictable processes it is predictable, so it is the conditional intensity of $\bar{N}$. $\square$

Proposition 6.11 (Which type fires). *Under Assumption 6.4, conditionally on $\mathcal{F}_{t-}$ and on the event that the pooled process jumps at t ,*

$$\mathbb{P}(\text{the jump is of type } e \mid \mathcal{F}_{t-}, \Delta \bar{N}(t) = 1) = \frac{Y_e(t) \lambda_e(t)}{\sum_{e'} Y_{e'}(t) \lambda_{e'}(t)}. \quad (6.7)$$

Proof sketch. By Equation (6.2), $\mathbb{P}(\text{type-}e \text{ jump in } [t, t + dt) \mid \mathcal{F}_{t-}) = Y_e \lambda_e(t) dt$ and $\mathbb{P}(\text{some jump}) = \bar{\lambda}(t) dt$; the ratio gives Equation (6.7) as $dt \rightarrow 0$. Structurally, the statement is Theorem 6.7 read backwards: view the type e as the mark of the pooled process; then the right-hand side of Equation (6.7) is exactly the mark density Equation (6.4). The full measure-theoretic proof, together with its operational consequences — the risk-set conditional logit, its partial-likelihood justification (Cox, 1975; McFadden, 1974), and the evaluation protocol built on it — is developed in Chapter 14, which restates Equation (6.7) as the definition of the ranking target. $\square$

Equation (6.7) is the entire T1 target of Section 1.2 in one line: ranking firms by $\lambda_i(t)/\sum_j \lambda_j(t)$ is ranking by conditional next-event probability, and by Equation (6.6) the sector timing factor cancels from the ratio within a sector. It is also the attribution rule of the thinning simulator below: accept a proposed event, then label it type e with probability $\lambda_e/\bar{\lambda}$.

6.6 Martingale estimation theory

Why does maximizing Equation (6.3) work, given that events are dependent, non-stationary, and censored? The answer is not the classical i.i.d. theory but the martingale structure of Equation (6.1); this section develops exactly the pieces later chapters invoke.

Differentiating Equation (6.3) in ϑ (legitimate under Assumption 6.4(iii)) and writing $dN_e - Y_e \lambda_e du = dM_e$ gives the *score process*

$$U(\vartheta, t) = \sum_e \int_0^t \partial_{\vartheta} \log \lambda_e(u; \vartheta) [dN_e(u) - Y_e(u) \lambda_e(u; \vartheta) du] = \sum_e \int_0^t \partial_{\vartheta} \log \lambda_e dM_e(\vartheta), \quad (6.8)$$

where the second equality holds when ϑ is the true value: the score is a stochastic integral of a predictable process against the compensated noise.

Proposition 6.12 (Unbiased score at the truth). *Let ϑ_0 be the true parameter. Under Assumption 6.4, $U(\vartheta_0, \cdot)$ is a mean-zero local square-integrable martingale; under the integrability of Assumption 6.4(ii)–(iii) it is a martingale, and in particular $\mathbb{E}U(\vartheta_0, T) = 0$.*

Proof. At ϑ_0 , each M_e in Equation (6.8) is a local martingale by Equation (6.1), and the integrand $H_e = \partial_{\vartheta} \log \lambda_e$ is predictable and locally bounded by assumption. A stochastic integral of a predictable locally bounded process against a local martingale is a local martingale (Andersen et al., 1993, Ch. II), and $U(\vartheta_0, 0) = 0$. The moment condition upgrades the local martingale to a martingale, whence the expectation is zero at every t , in particular at T . What breaks when the assumptions fail is instructive: if λ peeks at time- t information (Warning 6.2), H_e is not predictable and the integral is no longer a martingale — the score acquires a bias of the same sign as the leak; if the model is misspecified, $M_e(\vartheta)$ is not a martingale for any ϑ in the family, and the score is instead centered at the pseudo-true parameter ϑ_* — the Kullback–Leibler projection of the truth onto the family — which is what the sandwich variance below is built for. $\square$

Proposition 6.13 (Information identities). *Define the observed information $J(\vartheta) = -\partial_{\vartheta}^2 \ell_T(\vartheta)$ and the predictable variation of the score at the truth,*

$$\langle U(\vartheta_0, \cdot) \rangle(T) = \sum_e \int_0^T (\partial_{\vartheta} \log \lambda_e)(\partial_{\vartheta} \log \lambda_e)^{\top} Y_e \lambda_e du. \quad (6.9)$$

Then $J(\vartheta_0) - \langle U(\vartheta_0, \cdot) \rangle(T) = -\sum_e \int_0^T \partial_{\vartheta}^2 \log \lambda_e dM_e$ is a mean-zero martingale term, and consequently (the second Bartlett identity) $\mathbb{E}J(\vartheta_0) = \mathbb{E}\langle U \rangle(T) =: \mathcal{I}(\vartheta_0)$, the expected Fisher information.

Proof. Differentiate Equation (6.3) twice: $-J = \sum_e [\int \partial^2 \log \lambda_e dN_e - \int Y_e \partial^2 \log \lambda_e du]$. Substitute $\partial^2 \log \lambda_e = \lambda_e \{ \partial^2 \log \lambda_e + (\partial \log \lambda_e)(\partial \log \lambda_e)^{\top} \}$ in the second integral and collect terms: $J = -\sum_e \int \partial^2 \log \lambda_e dM_e + \langle U \rangle(T)$. The first term is a stochastic integral of a predictable process against a martingale, mean zero at ϑ_0 by the argument of Proposition 6.12; take expectations. That the compensator of the score’s quadratic variation is Equation (6.9) is the standard predictable-variation computation for counting-process integrals (Andersen et al., 1993, Ch. II). $\square$

Sandwich variance. The MLE $\hat{\vartheta}$ solves $U(\hat{\vartheta}, T) = 0$; a Taylor expansion around the (pseudo-)truth gives $\hat{\vartheta} - \vartheta_* \approx J(\vartheta_*)^{-1} U(\vartheta_*, T)$, hence the asymptotic variance

$$\text{Var}(\hat{\vartheta}) \approx A^{-1} B A^{-1}, \quad A = \mathbb{E} J(\vartheta_*), \quad B = \text{Var} U(\vartheta_*, T). \quad (6.10)$$

When the model is correctly specified, Proposition 6.13 gives $B = A = \mathcal{I}(\vartheta_0)$ and Equation (6.10) collapses to the familiar $\mathcal{I}^{-1}$. When it is not — and we treat misspecification as the operating condition; regime B of the synthetic world (Section 2.5) is misspecified by construction — the equality fails and only the sandwich is valid. In practice B is estimated by the optional-variation analogue of Equation (6.9) (outer products of per-event score contributions), clustered at the sector level because latent frailty is expected to correlate firms within a sector; the frailty experiments planned in Chapter 19 will measure how much this matters.

Theorem 6.14 (Rebolledo’s martingale CLT; statement only). *Let $U^{(n)}$ be a sequence of mean-zero local square-integrable martingales. If (i) $\langle U^{(n)} \rangle(t) \rightarrow V(t)$ in probability for each t , with V deterministic and continuous, and (ii) for each $\varepsilon > 0$ the predictable variation contributed by jumps of magnitude exceeding ε vanishes in probability, then $U^{(n)}$ converges weakly to a continuous Gaussian martingale with variance function V (Andersen et al., 1993; Andersen and Gill, 1982).*

Rebolledo’s theorem is the replacement for the classical CLT in all counting-process asymptotics: applied to the normalized score with $V =$ the limit of Equation (6.9), and combined with the Taylor expansion above, it yields asymptotic normality of $\hat{\vartheta}$ with the variance Equation (6.10). Condition (ii) holds here because individual events contribute vanishing shares of the total information as the window or the cross-section grows.

Theorem 6.15 (Consistency and asymptotic normality of the point-process MLE; statement only). *For stationary ergodic point processes with intensity families satisfying identifiability, smoothness, and moment conditions, $\hat{\vartheta}_T \rightarrow \vartheta_0$ a.s. and $\sqrt{T}(\hat{\vartheta}_T - \vartheta_0) \Rightarrow \mathcal{N}(0, \mathcal{I}_1(\vartheta_0)^{-1})$ as $T \rightarrow \infty$, with $\mathcal{I}_1$ the per-unit-time information (Ogata, 1978). The counting-process regime with many units observed on a fixed window — our firm panel — is covered by the same martingale program (Andersen and Gill, 1982; Andersen et al., 1993).*

Both asymptotic regimes are present in this repository and should not be conflated: sector-level timing models see $S = 12$ fixed components over a long window (Ogata’s $T \rightarrow \infty$ regime), while the mark factor sees thousands of firms per risk set on a fixed window (the Andersen–Gill $n \rightarrow \infty$ regime). Standard errors quoted in later chapters state which regime they invoke.

Design decision (D3.2 — martingale-grade inference outputs).

Every likelihood fitter in `hawkes_calibration/` must expose (i) the analytic score Equation (6.8), (ii) an observed-information or numerical Hessian, and (iii) sandwich standard errors Equation (6.10) with sector-level clustering as the default reported uncertainty; model-based $\mathcal{I}^{-1}$ errors may be shown only alongside. Rationale: misspecification is the operating condition (regime B), under which only the sandwich is valid for the pseudo-true parameter; and exposing the score is what makes the gradient checks and the GOF machinery of Chapter 11 cheap. Reversal trigger: for a model family that the diagnostics of Chapter 11 certify as well specified on real data across two campaigns, model-based errors may be promoted to primary for that family only.

6.7 The time-rescaling theorem

If the fitted intensity is right, rescaling time by the fitted compensator turns the data into a unit-rate Poisson process. This is the goodness-of-fit engine of Chapter 11; we state and prove it here.

Theorem 6.16 (Time rescaling). *Let N have conditional intensity λ under Assumption 6.4, with compensator $\Lambda(t) = \int_0^t Y \lambda \, du$ continuous and strictly increasing on the observation window, and let $t_1 < t_2 < \dots < t_n$ be the event times on $[0, T]$. Then the rescaled times $\tau_m = \Lambda(t_m)$ form a unit-rate Poisson process on $[0, \Lambda(T)]$; equivalently, the rescaled gaps*

$$\Delta_m = \Lambda(t_m) - \Lambda(t_{m-1}) = \int_{t_{m-1}}^{t_m} Y(u) \lambda(u) \, du, \quad t_0 = 0, \quad (6.11)$$

are i.i.d. $\text{Exp}(1)$, and $\Lambda(T) - \Lambda(t_n)$ is an independent right-censored $\text{Exp}(1)$ variable (Brown et al., 2002; Ogata, 1988).

Proof. We give the hazard argument in the case where, between events, the intensity path is determined by the history at the last event together with the observed covariate paths — true of every model fitted in this document, whose $\lambda(u)$ for $u \in (t_{m-1}, t_m]$ is an explicit functional of $\mathcal{H}_{t_{m-1}}$, the no-further-event continuation, and covariates that $\mathcal{F}$ carries. Condition on $\mathcal{F}_{t_{m-1}}$ and let $g(u)$ denote that determined intensity path. By Equation (6.2), the waiting time to the next event has hazard g : $\mathbb{P}(t_m \in [u, u + du) \mid t_m \geq u, \mathcal{F}_{t_{m-1}}) = g(u) \, du$, hence the conditional survivor function $\mathbb{P}(t_m > u \mid \mathcal{F}_{t_{m-1}}) = \exp(-\int_{t_{m-1}}^u g)$. Now fix $x \geq 0$ and let $u_x = \inf\{u : \int_{t_{m-1}}^u g = x\}$, finite and uniquely attained because Λ is continuous and strictly increasing. The events $\{\Delta_m > x\}$ and $\{t_m > u_x\}$ coincide, so

$$\mathbb{P}(\Delta_m > x \mid \mathcal{F}_{t_{m-1}}) = \exp\left(-\int_{t_{m-1}}^{u_x} g(u) \, du\right) = e^{-x}.$$

The conditional law of Δ_m given the entire past is $\text{Exp}(1)$, free of the past; iterating over m makes $(\Delta_1, \dots, \Delta_n)$ i.i.d. $\text{Exp}(1)$, which is the gap characterization of the unit-rate Poisson process. The same computation with x at the censoring boundary gives the final censored gap. In full generality — intensity paths influenced by additional external randomness between events — the result is the random-time-change theorem: $M(\Lambda^{-1}(\tau))$ is a martingale by optional stopping, so the time-changed counting process has deterministic continuous compensator τ , and Watanabe’s characterization identifies it as unit-rate Poisson (Daley and Vere-Jones, 2008). $\square$

The practical reading: *a correct model makes its own residuals exponential*, whatever the dependence structure of the original data. The resulting protocol — rescale with the fitted compensator, then test the gaps against $\text{Exp}(1)$ by Kolmogorov–Smirnov and QQ plots, as in Ogata’s residual analysis (Ogata, 1988) — is fixed as this repository’s primary event-time diagnostic in Chapter 11, including the treatment of the censored last gap. Two caveats travel with it. The test consumes exact event times, so under defect (C4) — weekly aggregation — it degrades by construction; the interval-censored regime uses bucket-level Pearson and dispersion residuals instead (Chapter 11, `hawkes_calibration/mbpp/gof.py`). And rescaling with an intensity fitted on the *same* data is optimistic; the split protocol of Chapter 11 rescales held-out windows with frozen parameters.

6.8 Simulation: inversion and thinning

Simulation matters here because the synthetic world of Section 2.5, the parameter-recovery studies planned in Chapter 19, and the posterior predictive checks of Chapter 11 all rest on generators whose output provably has the intended intensity. Two algorithms cover our needs.

Corollary 6.17 (Inverse-compensator simulation). *Let $E_1, E_2, \dots$ be i.i.d. $\text{Exp}(1)$. Setting, recursively,*

$$t_m = \inf \left\{ u > t_{m-1} : \int_{t_{m-1}}^u \lambda(v) dv \geq E_m \right\},$$

with λ evaluated along the history generated so far, produces a point process with conditional intensity λ .

Proof. This is Theorem 6.16 read backwards: the construction forces the rescaled gaps to be the i.i.d. $\text{Exp}(1)$ inputs, and the hazard computation in that proof, run in reverse, shows the waiting time generated has exactly the conditional hazard λ . $\square$

Inversion is exact and fast whenever the conditional compensator inverts in closed form — the covariate-driven Poisson models of Chapter 7 with piecewise-constant covariates are the leading case. When the compensator does not invert cheaply, we thin.

Proposition 6.18 (Correctness of Ogata’s modified thinning). *Suppose that on each inter-refresh interval (between consecutive accepted events, covariate breakpoints, or bound refreshes) a constant $\bar{\lambda}$ is available with $\lambda(u) \leq \bar{\lambda}$ for all u in the interval, where λ is evaluated along the accepted history and its no-new-event continuation. Generate proposals on the interval from a homogeneous Poisson process of rate $\bar{\lambda}$, and accept a proposal at τ independently with probability $\lambda(\tau)/\bar{\lambda}$, where $\lambda(\tau)$ is computed from the accepted history up to τ^- . Then the accepted points form a point process with conditional intensity λ (Lewis and Shedler, 1979; Ogata, 1981).*

Proof. Work on the enlarged filtration generated by the accepted history, the proposal process, and the acceptance variables. Given the accepted history $\mathcal{F}_t^{\text{acc}}$, the probability of an accepted point in $[t, t + dt)$ is the probability of a proposal there, $\bar{\lambda} dt + o(dt)$, times the acceptance probability $\lambda(t)/\bar{\lambda}$ — which is $\mathcal{F}_t^{\text{acc}}$ -measurable (predictable), because $\lambda(t)$ depends only on accepted points before t and on covariates; two proposals in the bin have probability $o(dt)$. The product is $\lambda(t) dt + o(dt)$, so the accepted counting process minus $\int \lambda$ is a martingale on the enlarged filtration, and since λ is adapted to the accepted history, it is the conditional intensity of the accepted process (Equation (6.1)). The bound being valid on the whole inter-refresh interval is what makes the acceptance probability well defined (≤ 1) throughout; the multivariate version accepts with $\bar{\lambda}^{-1} \sum_e \lambda_e(\tau)$ and then labels the accepted point type e with probability $\lambda_e(\tau) / \sum_{e'} \lambda_{e'}(\tau)$, which is correct by Proposition 6.11. $\square$

Warning 6.19 (The bound is part of the algorithm’s correctness). Thinning is exact only while $\bar{\lambda}$ truly dominates. For exponential Hawkes with nonnegative kernels ($h(t) = \kappa \theta e^{-\theta t}$), the intensity strictly decreases between events, so the value just after the last event is a valid bound — unless the baseline jumps upward inside the interval, as it does whenever a piecewise-constant covariate steps. A generator that fails to refresh the bound at covariate breakpoints silently under-simulates events after every upward step, biasing any simulation-based conclusion toward weaker covariate effects. `hawkes_calibration/eventtime/simulate.py` refreshes at accepted events and at covariate breakpoints for exactly this reason, and verifies its recursive intensity against a naive recomputation.

For the exponential kernel there is also an exact, thinning-free simulator that exploits the Markov property of the intensity itself (cluster and shot-noise structure), due to Dassios and Zhao (2013); it and the branching (cluster) representation are developed where they are used, in Chapter 8.

6.9 What the machinery absorbs, and what it cannot

Decision D2.1 of Chapter 2 requires every model chapter to declare its position against the defect ledger of Table 2.1. This chapter contributes the declaration for the machinery itself: which defects the counting-process framework absorbs *by construction*, and which no amount of likelihood theory can repair.

Proposition 6.20 (Independent censoring preserves the likelihood; (C1)). *Let C_e be $\{0, 1\}$ -valued predictable censoring processes, and let $\mathcal{G}_t \supseteq \mathcal{F}_t$ be the filtration enlarged with the censoring information. Assume independent censoring: each N_e has the same intensity $Y_e \lambda_e$ with respect to $\{\mathcal{G}_t\}$ as with respect to $\{\mathcal{F}_t\}$. Then the observed processes $N_e^c(t) = \int_0^t C_e dN_e$ have $\mathcal{G}$ -intensity $C_e Y_e \lambda_e$, and Equation (6.3) computed on the observed events with exposure $C_e Y_e$ is a genuine (partial) likelihood: its score retains the martingale property of Proposition 6.12, and all of Section 6.6 applies unchanged (Andersen et al., 1993, III.2).*

Proof sketch. Since C_e is predictable, its value at t is known given $\mathcal{G}_{t-}$, so $\mathbb{E}[dN_e^c | \mathcal{G}_{t-}] = C_e(t) \mathbb{E}[dN_e | \mathcal{G}_{t-}] = C_e Y_e \lambda_e dt$, the middle equality being the independent-censoring hypothesis. Hence $N_e^c - \int C_e Y_e \lambda_e$ is a $\mathcal{G}$ -martingale, and Theorem 6.5 applied to (N_e^c) yields Equation (6.3) with Y_e replaced by $C_e Y_e$; at observed events $C_e = 1$, so the event term is untouched. The factor of the joint likelihood governing the censoring mechanism itself contains no ϑ and is dropped — that is the sense in which the result is a *partial* likelihood. Administrative censoring at fixed T has $C_e(t) = \mathbf{1}\{t \leq T\}$ deterministic, hence predictable, and trivially preserves the intensity: it is independent by construction. What breaks without predictability is the first equality; what breaks without independence is the second — if censoring anticipates the process (firms leave observation *because* their funding prospects changed), $\mathbb{E}[dN_e | \mathcal{G}_{t-}] \neq Y_e \lambda_e dt$ and the observed-data likelihood is biased. That failure mode is exactly defect (C3). $\square$

Operationally, Proposition 6.20 is the license already exercised in Warning 2.4: a spell open at June 2024 contributes its survival factor $\exp(-\int \lambda)$ through the compensator term, no term is dropped and none invented, *provided the open spell is kept in the data*. The machinery is robust to (C1).

(C2) Left truncation. Delayed entry is absorbed by the at-risk indicator: setting $Y_i(t) = \mathbf{1}\{V_i \leq t < U_i\}$ makes every compensator integral in Equation (6.3) start at the entry time V_i , and the likelihood is valid *conditionally on the pre-entry history* (Keiding, 1990; Kalbfleisch and Prentice, 2002). The mechanical fix is complete; the statistical selection is not fixed at all: because entry occurs *by raising*, every estimand is conditional on “at least one deal before T ,” and unconditional statements about companies require the open-universe treatment of Chapter 15. The machinery *corrects (C2) under the stated conditional interpretation*, and silently changes the estimand if that interpretation is forgotten.

(C3) Unlabeled exits. Here the machinery is *vulnerable, with a known sign*. A dead firm with $Y_i \equiv 1$ contributes exposure and no events — indistinguishable, inside the likelihood, from a live quiet firm — so no score, residual, or information identity can detect the corruption: Equation (6.3) is being evaluated with the wrong Y , and every fitted intensity is biased *downward* while ranking risk sets are contaminated with phantom competitors. We expect the damage to be large; the synthetic study of Chapter 19 is designed to measure it. Treatments are exposure windows and mover–stayer structure in Chapters 7 and 13 and ranking-layer mitigations in Chapter 14; none is a consequence of the machinery, all are assumptions imported to compensate for it.

(C4) Untrusted times and aggregation. The event term of Equation (6.3) — and only the event term — consumes exact times, and Theorem 6.16 and Proposition 6.18 consume them too. Date jitter perturbs $\log \lambda(t_m)$ where λ moves fastest, i.e. it attacks timescale parameters first; full aggregation deletes the event term and requires the interval-censored likelihoods of Chapter 10 (Rizoiu et al., 2022). Our working hypothesis, to be tested in Chapter 19, is that κ survives aggregation while θ does not. The machinery is vulnerable to (C4); Chapter 10 is the correction.

(C5) and (C6). Stale covariates leave the machinery *valid but reinterpreted*: LOCF covariates as constructed under Protocol 2.5 are predictable, so Equation (6.3) is a genuine likelihood for the model that conditions on the *observed* covariates — but its coefficients are attenuated relative to the true-covariate model, which is Chapter 12’s subject. Out-of-universe deals leave sector-level ground counts complete but thin the firm-level mark factor: without an explicit outside option, the normalization in Equation (6.7) overstates every tracked firm’s probability — Chapter 15 owns the fix.

In summary: the machinery of this chapter is robust to (C1), corrects (C2) under a conditional reading, and is structurally blind to (C3), (C4) on the event term, (C5) in interpretation, and (C6) in the mark normalization. That blindness is the agenda of Chapters 7, 10, 12 and 15.

Code hooks.

Theorem 6.5 is implemented in `hawkes_calibration/eventtime/likelihood.py`: `log_likelihood` assembles the event term by recursive exponential-kernel sums decayed to each event’s left limit (Warning 6.2) plus a closed-form compensator term; `log_likelihood_and_grad` returns the analytic score Equation (6.8), satisfying decision D3.2. Proposition 6.18 and Warning 6.19 are implemented in `hawkes_calibration/eventtime/simulate.py` (`simulate_multivariate_hawkes`): Ogata thinning with bound refresh at accepted events and covariate breakpoints, with a naive-recomputation cross-check of the recursive intensity. Theorem 6.16 lives in `hawkes_calibration/mbpp/gof.py` (`time_rescaling_residuals`, `ks_test_exp1`, `qq_exp1`), with bucket-level Pearson/dispersion residuals as the (C4)-regime substitute. Proposition 6.11 is the modeling target of `hawkes_calibration/sector_ranker.py` (Chapter 14). Proposition 6.20 and the (C2) risk-set rule become validation checks in the planned `hawkes_calibration/data/audits.py` of Chapter 2: open-spell retention and entry-date consistency are preconditions for the likelihood to be the one this chapter proved things about.

Chapter 7

Poisson Models: The Baselines That Must Be Beaten

7.1 Why the ladder starts here

The timing target T2 of Chapter 1 asks for the ground intensity $\lambda_s(t)$: how many deals, in which sector, next week, with calibrated uncertainty. This chapter builds the two simplest answers — the homogeneous Poisson process, which asserts that nothing is valuable to know beyond the level of activity, and the inhomogeneous Poisson process (NHPP) with covariates, which asserts that the *market clock* is valuable but the market’s own history is not. Their role in the design is adversarial. Every richer timing model in this document — the history-modulated GLM previewed in Section 7.7, the exponential Hawkes machinery of Chapter 8, the interval-censored variants of Chapter 10 — is admitted into production only by beating a well-covariates NHPP on held-out likelihood under the protocols of Chapter 11. Our working hypothesis is that covariates carry most of the held-out gain over a seasonal baseline, and that everything added on top of them — lags, excitation, cross-sector terms — contributes far less; the decomposition benchmarks planned in Chapter 19 will test this. A Hawkes fit whose covariate baseline is impoverished will absorb the macro cycle into its kernel and report spurious contagion; the only defense is a baseline that already knows the macro cycle, which is what this chapter calibrates.

The second reason to start here is that this is the one rung whose likelihood is *exactly right* under the observation scheme of Chapter 2. At sector resolution, right censoring costs nothing (Section 7.5 proves this), interval censoring by weekly aggregation costs nothing (Theorem 7.10), and sectors neither enter late nor die, so (C2) and (C3) do not bind. Calibration discipline is cheap at this rung. Every later deviation from it must be priced against a baseline whose own accounting is exact.

7.2 The two models and their assumptions

Throughout, units e are sectors $s \in \mathcal{S}$ (the primary case) or, for the event-time path, tracked firms $i \in \mathcal{I}$; N_e is the counting process of Definition 2.3 with at-risk indicator Y_e , and all likelihoods specialize the Jacod likelihood Equation (2.3) whose martingale estimation theory is developed in Chapter 6. The homogeneous model sets $\lambda_e(t) \equiv \mu_e$: conditionally on nothing, events arrive at a constant rate, the MLE is $\hat{\mu}_e = N_e(T) / \int_0^T Y_e$, and the model’s only message is the level. It is the null model for timing; its calendar-dummy refinement (a rate per quarter or per season) is the “seasonal baseline” against which covariate gains are measured. The inhomogeneous model is the first calibrated one, and we now fix its assumptions as numbered contracts.

Assumption 7.1 (Conditional Poisson structure). Given the covariate paths, the intensity of N_e is a deterministic function of time: $\lambda_e(t; \vartheta) = \lambda_e(t)$ measurable with respect to the covariate history alone, with no dependence on the internal history $\mathcal{H}_{t-}$ of the deal process. Equivalently, conditionally on covariates, N_e is an inhomogeneous Poisson process: counts over disjoint sets are independent, and the count over a set A is Poisson with mean $\int_A Y_e(u) \lambda_e(u) du$ (Kingman, 1993; Daley and Vere-Jones, 2003).

Assumption 7.2 (Log-linear intensity). $\lambda_s(t) = \exp(\beta^\top x_s(t))$, where $x_s(t) \in \mathbb{R}^d$ stacks a sector intercept (equivalently, sector dummies), the published as-of macro values of Protocol 2.5(a), and deterministic seasonality and calendar terms; per-sector slopes are interactions of the macro block with the sector dummies. The exponential link is the canonical one; Section 7.2.1 defends the choice against its alternatives.

Assumption 7.3 (Piecewise-constant weekly covariates). $x_s(\cdot)$ is constant on the cells of a partition of $[0, T]$ into weekly bins: macro as-of series are step functions by construction (they jump at publication dates, which we align to the weekly grid), calendar terms are constant within a week, and sector dummies are constant always. Consequently $\lambda_s(\cdot)$ is a step function.

Assumption 7.4 (External covariates). The covariates of Assumption 7.2 are *external* in the sense of Kalbfleisch and Prentice (2002): their evolution does not depend on the deal stream. Macro series, calendar terms, and sector dummies qualify. Lagged deal counts, exponentially filtered event streams, and any functional of $\mathcal{H}_{t-}$ do *not*; adding them changes the model class (Section 7.7) and voids Assumption 7.1.

Assumption 7.5 (Spanning). Writing c for the sector-week cells of Assumption 7.3, $w_c = \int_{\text{cell}} Y(u) du$ for the exposure of cell c and x_c for its covariate vector, the family $\{x_c : w_c > 0\}$ spans $\mathbb{R}^d$; equivalently $\sum_c w_c x_c x_c^\top \succ 0$.

Assumption 7.1 is a known fiction — Section 7.6 measures its failure and upgrades the noise model — while Assumptions 7.3 and 7.4 are believed exactly true for the design of Assumption 7.2, and Assumption 7.5 is checkable from the design matrix before any fit.

7.2.1 Why the exponential link

Three properties recommend $\exp$. *Positivity for free*: any $\beta \in \mathbb{R}^d$ yields a valid intensity, so estimation is unconstrained smooth optimization — no boundary, no barrier, no projected gradient. *Concavity*: the log-likelihood is globally concave in β (Theorem 7.6), so the fitted object is a reproducible global optimum rather than a local one an optimizer happened to find. *Multiplicative interpretation*: a unit move in covariate j multiplies the rate by e^{β_j} uniformly across sectors and weeks, so macro effects read as rate ratios — the natural language for “tightening moved weekly late-stage deal flow down 12%” — and sector dummies give sector-proportional baselines. The log link is also *aggregation-coherent* under Assumption 7.3: binning to weeks leaves the functional form intact with the bin width absorbed into the intercept (Theorem 7.10), which is precisely what makes the weekly GLM an exact NHPP calibration rather than a discretization of one.

The alternatives are worth stating because the repository uses one of them. The *identity link* $\lambda = \beta^\top x$ makes effects additive — deals per week per unit of covariate — which is the natural scale for excitation terms with a branching interpretation, and it tolerates within-bin covariate variation exactly (the integral of a linear function is linear in the averaged covariate, so no Jensen gap ever opens). Its price is positivity: the likelihood is concave only on the polyhedron $\{\beta : \beta^\top x_c > 0 \forall c\}$, the MLE can sit on the boundary where standard asymptotics fail (Self and Liang, 1987), and every fit needs constraints. The *softplus link* $\lambda = \log(1 + e^\eta)$ keeps positivity free and grows only linearly for large η , which reduces the exponential extrapolation risk of

the log link outside the training covariate support; it remains log-concave, but the coefficients lose their rate-ratio reading and the fit leaves standard GLM software. The stabilized sector layer of Chapter 8 is a deliberate hybrid: exponential link for the covariate baseline (this chapter's object, with all three advantages), identity link for the added excitation term (where the branching reading and the stability constraint live).

Design decision (D4.1 — exponential link for the covariate baseline).

The covariate-driven baseline intensity of every ground-process model in this document uses the log-linear form of Assumption 7.2. Rationale: unconstrained concave estimation (Theorem 7.6), exact weekly-GLM equivalence (Theorem 7.10), multiplicative macro interpretation, and sign freedom for later history terms (the exp-link route is one of the three legal ways to express inhibition; Chapter 8). Reversal trigger: if held-out evaluation shows exponential extrapolation failures — predictions pinned at the support-envelope guard of `hawkes_calibration/sector_stability.py` on more than isolated weeks — the baseline link is switched to softplus and this chapter's concavity and equivalence results are re-derived for it (both survive in weakened form; the GLM packaging does not).

7.3 Likelihood, score, and the geometry of the optimum

Under Assumptions 7.1 and 7.2, the Jacod likelihood Equation (2.3) specializes, for the pooled sector process on $[0, T]$, to

$$\ell(\beta) = \sum_m \beta^\top x_{s_m}(t_m) - \sum_{s \in \mathcal{S}} \int_0^T Y_s(u) e^{\beta^\top x_s(u)} du, \quad (7.1)$$

the sum running over observed deals m with sector s_m and time t_m . Under Assumption 7.3 the integral collapses over cells: with y_c the event count, w_c the exposure, and x_c the covariate of cell c ,

$$\ell(\beta) = \sum_c (y_c \beta^\top x_c - w_c e^{\beta^\top x_c}). \quad (7.2)$$

Differentiating Equation (7.2) (or Equation (7.1); the cell form is the general one with the integral discretized exactly):

$$\nabla \ell(\beta) = \sum_c (y_c - w_c e^{\beta^\top x_c}) x_c, \quad (7.3)$$

$$\nabla^2 \ell(\beta) = - \sum_c w_c e^{\beta^\top x_c} x_c x_c^\top. \quad (7.4)$$

The score Equation (7.3) is the familiar “observed minus expected” weighted by covariates; evaluated at the truth it is the terminal value of the martingale $\int x (dN - Y \lambda dt)$, which is the engine of the asymptotics below. The Hessian Equation (7.4) does not involve the counts y_c at all — a canonical-link property: observed information equals conditional expected information, so Newton's method, Fisher scoring, and iteratively reweighted least squares are the same algorithm here.

Theorem 7.6 (Strict log-concavity under the exponential link). *Under Assumptions 7.1 to 7.3, the map $\beta \mapsto \ell(\beta)$ of Equation (7.2) is concave on all of $\mathbb{R}^d$: $\nabla^2 \ell(\beta) \preceq 0$ for every β . If in addition Assumption 7.5 holds, it is strictly concave: $\nabla^2 \ell(\beta) \prec 0$ for every β .*

Proof. For any $v \in \mathbb{R}^d$,

$$v^\top \nabla^2 \ell(\beta) v = - \sum_c w_c e^{\beta^\top x_c} (v^\top x_c)^2 \leq 0,$$

since every summand is a product of nonnegative factors: concavity, with no condition. Equality holds iff $v^\top x_c = 0$ for every cell with $w_c > 0$ (the weights $w_c e^{\beta^\top x_c}$ are strictly positive on exactly those cells). Under Assumption 7.5 no $v \neq 0$ annihilates all exposed x_c , so the quadratic form is strictly negative for all $v \neq 0$ and all β . When Assumption 7.5 fails — say two macro series are exactly collinear over the training window — the likelihood is flat along the offending direction and β is identified only up to it; the fitted *intensity* is still unique, which is why ridge regularization plus inference on well-identified linear combinations, rather than coefficient surgery, is the prescribed response to the near-failure that smooth, correlated macro series actually produce. $\square$

Corollary 7.7 (Global optimum and solver guarantees). *Under the assumptions of Theorem 7.6 with Assumption 7.5, the MLE, when it exists, is the unique global maximizer of ℓ , every stationary point is that maximizer, and any ascent method with standard line-search or trust-region safeguards — Newton, Fisher scoring, L-BFGS — converges to it from any starting point. Two fits of the same data that report different optima are evidence of a software defect, not of multimodality.*

Proof. Strict concavity on $\mathbb{R}^d$ makes the superlevel sets convex and any stationary point the unique global maximum; convergence of safeguarded ascent methods on smooth concave objectives with nonsingular Hessian is standard. The content worth stating is the last sentence: at this rung, optimizer nondeterminism is a bug detector. This guarantee is exactly what is forfeited — first partially (block concavity), then wholly — as excitation parameters with free decays enter in Chapter 8. $\square$

Asymptotics. At the true β_0 the score is a martingale integral with predictable variation $\mathcal{I}(\beta_0) = \sum_c w_c e^{\beta_0^\top x_c} x_c x_c^\top$ — the Fisher information, equal to minus the Hessian Equation (7.4). Rebolledo’s martingale CLT then gives $\hat{\beta} \approx \mathcal{N}(\beta_0, \mathcal{I}^{-1})$ as information accumulates, through a longer window or more sectors; this is the counting-process estimation theory of Andersen et al. (1993) and Andersen and Gill (1982), and the stationary-point-process version of Ogata (1978), specialized to the concave case. What matters practically is that information is *pooled*: a d -dimensional β needs adequate total information $\sum_c w_c e^{\beta^\top x_c} x_c x_c^\top$, not adequate counts in every sector-week cell — sparse cells are a separation risk (Proposition 7.8), not an asymptotics failure per se.

7.3.1 Existence: separation and sparse cells

Concavity guarantees uniqueness, not existence. The Poisson analogue of logistic separation is the one genuine pathology of this rung.

Proposition 7.8 (Nonexistence of the MLE under separation). *Under the assumptions of Theorem 7.6 with Assumption 7.5, the MLE fails to exist iff there is a direction $v \neq 0$ with*

- (i) $v^\top x_c \leq 0$ for every cell with $w_c > 0$, and
- (ii) $v^\top x_c = 0$ for every cell with $y_c > 0$.

Along any such v the likelihood increases strictly and monotonically to a finite supremum that is not attained; the fitted intensity on the cells with $v^\top x_c < 0$ degenerates to zero.

Proof. Suppose such v exists. By Assumption 7.5, $v^\top x_c \neq 0$ for at least one exposed cell, which by (ii) has $y_c = 0$ and by (i) has $v^\top x_c < 0$. Along $\beta + \tau v$, the event term of Equation (7.2) is constant in τ by (ii), while each compensator term $-w_c e^{\beta^\top x_c + \tau v^\top x_c}$ is nondecreasing in τ by (i) and strictly increasing on the cell just exhibited. Hence ℓ increases strictly to the finite limit obtained by deleting the vanishing compensator terms, and no finite β attains it. Conversely, if no such v exists, then along every ray $\beta + \tau v$ either some exposed cell has $v^\top x_c > 0$, in which case the compensator term drives $\ell \rightarrow -\infty$ exponentially (dominating the linearly growing event term), or all exposed cells have $v^\top x_c \leq 0$ with strict inequality at some event cell, in which case the event term drives $\ell \rightarrow -\infty$ linearly. So ℓ is coercive on $\mathbb{R}^d$ and, being continuous, attains its maximum. $\square$

Warning 7.9 (Sparse cells produce infinite coefficients). The separating directions of Proposition 7.8 are not exotic: any indicator covariate whose active cells contain *zero events* — a small sector crossed with a rare calendar dummy, an interaction active only in a few quiet weeks — yields $v = \text{minus that coordinate}$, and the fitted coefficient runs to $-\infty$ (the MLE of a zero rate). At weekly resolution with twelve sectors and rich calendar interactions this *will* occur. Remedies, in the order the repository applies them: the ridge penalty already standard in the fitters (a strictly concave perturbation restores existence unconditionally; the penalty must be reported next to the fit); the Firth adjustment, maximizing $\ell(\beta) + \frac{1}{2} \log \det \mathcal{I}(\beta)$, which removes the leading-order bias of the MLE and keeps estimates finite (Firth, 1993), with the rare-events analysis of King and Zeng (2001) as the companion warning that unpenalized rates estimated from a handful of events are biased as well as fragile; and design hygiene — collapse or drop cells/dummies that the training window cannot inform. Related but distinct is *extrapolation* risk: a log-linear intensity evaluated outside the training covariate support is an exponential extrapolation even when the MLE exists; the support-envelope winsorization in `hawkes_calibration/sector_stability.py` is the implemented guard, and train/test covariate ranges are reported next to every held-out NLL (Chapter 11).

7.4 The binned-GLM equivalence theorem

The repository fits weekly sector counts with a Poisson GLM. The following theorem says this *is* NHPP maximum likelihood — exactly, not approximately — and identifies precisely what binning discards (nothing, under this chapter’s assumptions).

Theorem 7.10 (Piecewise-constant NHPP = Poisson GLM with log-exposure offset). *Let Assumptions 7.1 to 7.3 hold, so that λ is constant, equal to $\lambda_c = e^{\beta^\top x_c}$, on each cell c of the weekly partition, and let $w_c = \int_{\text{cell}} Y(u) du$ be the (possibly partial) exposure of cell c . Then:*

- (i) (Distribution.) *Conditionally on covariates and exposures, the cell counts are independent with $y_c \sim \text{Poisson}(w_c e^{\beta^\top x_c})$.*
- (ii) (Likelihood identity.) *The point-process log-likelihood Equation (7.1) and the log-likelihood of the Poisson GLM with log link, design vectors x_c , and offset $\log w_c$ differ by a constant free of β :*

$$\ell_{\text{pp}}(\beta) = \sum_c \left(y_c \log(w_c e^{\beta^\top x_c}) - w_c e^{\beta^\top x_c} - \log y_c! \right) + \text{const.} \quad (7.5)$$

- (iii) (Sufficiency.) *The vector of cell counts $(y_c)_c$ is sufficient for β : given the counts, the event times within each cell are distributed as i.i.d. uniform draws on the at-risk subset of the cell, a law free of β .*

Consequently the NHPP MLE, score, observed information, and all likelihood-based inference coincide with those of the binned Poisson GLM; fitting the weekly sector count GLM is an NHPP calibration, not a discretization of one.

Proof. (i) is the independent-increments property of Assumption 7.1 applied to the disjoint at-risk subsets of the cells: the count over the at-risk subset A_c of cell c is Poisson with mean $\int_{A_c} \lambda_c du = w_c \lambda_c$ (Kingman, 1993).

(ii) Because x is constant on cells, the event term of Equation (7.1) groups as $\sum_m \beta^\top x(t_m) = \sum_c y_c \beta^\top x_c$, and the compensator integral collapses to $\sum_c w_c e^{\beta^\top x_c}$, giving Equation (7.2). Expanding the GLM log-likelihood on the right of Equation (7.5):

$$\sum_c \left(y_c \log w_c + y_c \beta^\top x_c - w_c e^{\beta^\top x_c} - \log y_c! \right) = \ell_{\text{pp}}(\beta) + \sum_c (y_c \log w_c - \log y_c!),$$

and the bracketed sum is a statistic of the data only.

(iii) Conditionally on $y_c = n$, the n points of a Poisson process with *constant* rate on A_c are distributed as the order statistics of n i.i.d. uniforms on A_c , for every value of the rate (Kingman, 1993); the conditional law is therefore free of β , and sufficiency follows from the factorization $L(\beta; \text{times}) = L(\beta; \text{counts}) \times p(\text{times} \mid \text{counts})$. The final claims follow by differentiating an identity that holds up to an additive constant. $\square$

Corollary 7.11 (The repository’s weekly GLM is an NHPP fit). *With equal-width weekly bins and fully exposed sectors ($w_c \equiv w$), the offset $\log w$ is absorbed into the sector intercepts, and the weekly sector Poisson GLM — `fit_sector_count_model` of `hawkes_calibration/sector_stability.py` with `n_lags=0`, and its vectorized synthetic-world twin `fit_sector_glm_fast(n_lags=0)` — computes exactly the NHPP MLE of Assumptions 7.2 and 7.3. Partial exposure (a unit entering or leaving observation mid-week) is handled exactly by the offset, with no change to the estimator’s form.*

Two remarks bound the theorem’s reach. First, parts (i) and (iii) never used the log link: the reduction of a piecewise-constant point process to independent bin counts holds for *any* positive rate parameterization; the log link is what makes the reduced problem a canonical-link GLM with the concavity of Theorem 7.6. Second, exactness requires Assumption 7.3. If covariates vary within a bin and the fit uses a bin-representative value $\bar{x}_c$, a Jensen gap opens — the true cell mean $\int e^{\beta^\top x(u)} du$ exceeds $w_c e^{\beta^\top \bar{x}_c}$ whenever x genuinely varies — and the GLM becomes an approximation. At weekly bins with monthly macro series and calendar terms the gap is identically zero; the exactness is a property of our design, not a general fact about binning.

7.5 Right censoring: the complete treatment at this rung

What must Poisson calibration do about right censoring? The answer, proved here in both of the bookkeeping styles used by the codebase, is: *nothing beyond keeping the censored exposure*. There is no correction term, no weighting, no imputation; the survival factor that a censored spell must contribute is already computed by the compensator integral, provided open spells are kept and exposure is integrated to the end of observation.

Proposition 7.12 (Intensity view: censoring is already in the likelihood). *Let unit e be observed on $[V_e, C_e]$ with $C_e = \min(T, U_e)$, where T is the administrative window end and U_e a recorded exit time, and let censoring be independent in the sense of Chapter 6 (the intensity of N_e with respect to the filtration enlarged by censoring information is unchanged; Andersen et al., 1993, III.2). Administrative censoring at the deterministic T is independent by construction; removal at a recorded exit is valid for the cause-specific funding intensity with*

no independence assumption at all (Chapter 2, defect (C3)(iii); [Putter et al., 2007](#)). Then the observed-data log-likelihood is exactly Equation (7.1) with $Y_e(u) = \mathbf{1}\{V_e \leq u \leq C_e\}$; in particular the event-free portion $(t_{\text{last}}, C_e]$ of an open spell contributes

$$- \int_{t_{\text{last}}}^{C_e} \lambda_e(u) du$$

to ℓ — the log of its survival factor — and nothing else. In the binned form of Theorem 7.10, the entire treatment is the exposure $w_c = \int_{\text{cell}} \mathbf{1}\{V_e \leq u \leq C_e\} du$: keeping open spells and integrating exposure to $\min(T, U_e)$ is the complete and exact treatment of (C1).

Proof. Under independent censoring the observed counting process $\int \mathbf{1}\{u \leq C_e\} dN_e(u)$ has intensity $Y_e(t)\lambda_e(t)$ with respect to the enlarged filtration (Chapter 6; [Andersen et al., 1993](#), III.2), so the Jacod likelihood of the observed data is Equation (2.3) with this intensity — which is Equation (7.1) restricted to $[V_e, C_e]$. Splitting the compensator integral at the observed event times isolates the term over $(t_{\text{last}}, C_e]$, an interval containing no event: its entire contribution is $-\int \lambda_e$, with no $\log \lambda$ term to drop or invent. The binned statement follows from Theorem 7.10, whose exposure w_c integrates precisely this at-risk indicator. The proposition fails exactly when censoring is *informative* — observation ends for reasons correlated with the intensity given covariates. That is not (C1) but (C3), the unlabeled-exit defect: an unrecorded death is a censoring event we never see and therefore cannot put into C_e ; Section 7.8 states the resulting bias. $\square$

Proposition 7.13 (Waiting-time view: the same computation). *Under Assumption 7.1, for a spell starting at a (an entry or an observed event) the waiting time to the next event has survival function*

$$S(c) = \mathbb{P}(N(a, c] = 0) = \exp\left(-\int_a^c \lambda(u) du\right), \quad c \geq a, \quad (7.6)$$

and density $f(c) = \lambda(c)S(c)$. The likelihood assembled spell-by-spell — density factors for completed spells, survival factors $S(C)$ for spells censored at C — equals the Jacod likelihood of Proposition 7.12 exactly:

$$\prod_{m=1}^n \lambda(t_m) e^{-\int_{t_{m-1}}^{t_m} \lambda(u) du} \times e^{-\int_{t_n}^C \lambda(u) du} = \left(\prod_{m=1}^n \lambda(t_m) \right) e^{-\int_{t_0}^C \lambda(u) du}, \quad t_0 = V.$$

Proof. Equation (7.6) is the Poisson count law of Assumption 7.1 evaluated at zero; differentiating $1 - S(c)$ gives $f(c) = \lambda(c)S(c)$. Taking logs of the displayed product, the survival exponents telescope over the partition $t_0 < t_1 < \dots < t_n < C$ into the single compensator integral $-\int_{t_0}^C \lambda$, and the event factors contribute $\sum_m \log \lambda(t_m)$: this is Equation (7.1) for the unit. The two views are therefore not two treatments to be reconciled but one number computed with two bookkeepings. Practical corollary: a spell-table implementation (one row per spell with a censoring flag, survival factor for flagged rows) and an exposure-table implementation (one row per unit-week cell with exposure w_c) must return identical log-likelihoods on the same data — a cheap and mandatory cross-implementation test. In the homogeneous case Equation (7.6) is the exponential waiting time $e^{-\mu(c-a)}$, and the renewal view is classical survival analysis of exponential spells; for recurrent events the counting-process formulation is the one that generalizes ([Cook and Lawless, 2007](#)). $\square$

Corollary 7.14 (Interval censoring is free at this rung). *Suppose deal dates are trusted only at resolution δ (the week): the bin of each event is correct, its position within the bin is not. Under the assumptions of Theorem 7.10 with bin width δ , the binned Poisson likelihood Equation (7.5) is the exact likelihood of the trusted data — not an approximation — because by part (iii)*

the within-bin positions carry no information about β : the corrupted coordinates are exactly the ancillary ones. Boundary cells truncated by entry or censoring retain exactness through their partial exposure w_c . None of this survives history dependence: for a Hawkes process the intensity after t depends on the within-bin positions of earlier events, binning destroys likelihood information, and the interval-censored machinery of Chapter 10 (Rizoiu et al., 2022) becomes necessary. The contrast is the precise sense in which weekly aggregation is free at rung 1 and costly afterwards — and it is why (C4) appears as “robust” in this chapter’s defect table and as a first-class problem in Chapters 8 and 10.

Warning 7.15 (Dropping open spells is right truncation, not censoring). The exactness of Propositions 7.12 and 7.13 holds only if censored spells are *kept*. Discarding them — fitting only spells that end in an observed event — conditions on the event landing in-window, which oversamples short gaps and biases every fitted rate upward; the observation scheme becomes right truncation, a different and harder problem (Kalbfleisch and Prentice, 2002). This restates Warning 2.4 at the point of use. For the weekly sector GLM the trap is nearly impossible to fall into (every sector-week cell exists whether or not it contains events — the zero counts *are* the censored exposure), which is itself an argument for the cell representation; for spell-table firm-level code the trap is one careless `dropna` away, and the open-spell retention audit of `hawkes_calibration/data/audits.py` exists to catch it.

7.6 Overdispersion: from Poisson to negative binomial

Assumption 7.1 forces $\text{Var}(y_c) = \mathbb{E}(y_c)$ cell by cell. We expect funding counts to violate this: latent heterogeneity (unmodeled sector-week quality, vendor coverage drift) and genuine history dependence both inflate variance beyond the mean. The synthetic studies of Chapter 19 will measure the dispersion this produces under controlled misspecification. Two upgrades present themselves, one inferential and one distributional, and the design needs both for different purposes.

The inferential upgrade is *quasi-Poisson*: keep the mean model and the estimating equation Equation (7.3), posit $\text{Var}(y_c) = \phi \mu_c$, estimate $\hat{\phi}$ by the Pearson statistic $\hat{\phi} = \frac{1}{n-d} \sum_c (y_c - \hat{\mu}_c)^2 / \hat{\mu}_c$, and scale standard errors by $\hat{\phi}^{1/2}$ (or use the sandwich variance, which is robust to variance misspecification beyond the proportional form). Because the score Equation (7.3) is unbiased whenever the *mean* is correct, $\hat{\beta}$ keeps its consistency under overdispersion; only its reported precision was wrong. Quasi-Poisson repairs the standard errors at zero modeling cost, but it supplies no predictive distribution — and the evaluation protocols of Chapter 11 score predictive distributions (PIT, proper scoring rules), not point forecasts. Hence the distributional upgrade.

Proposition 7.16 (Gamma frailty makes a negative binomial). *Let $y \mid Z \sim \text{Poisson}(Z\mu)$ with frailty $Z \sim \text{Gamma}(\alpha, \alpha)$ (shape and rate α , so $\mathbb{E}Z = 1$, $\text{Var} Z = 1/\alpha$). Then y is negative binomial:*

$$\mathbb{P}(y = k) = \frac{\Gamma(k + \alpha)}{\Gamma(\alpha) k!} \left(\frac{\alpha}{\alpha + \mu} \right)^\alpha \left(\frac{\mu}{\alpha + \mu} \right)^k, \quad k = 0, 1, 2, \dots, \quad (7.7)$$

with $\mathbb{E}y = \mu$ and $\text{Var} y = \mu(1 + \mu/\alpha)$. As $\alpha \rightarrow \infty$ the law converges to $\text{Poisson}(\mu)$. Moreover, with $\mu_c = w_c e^{\beta^\top x_c}$ and α fixed, the negative binomial log-likelihood remains concave in β .

Proof. Integrating the Poisson pmf against the Gamma density,

$$\mathbb{P}(y = k) = \int_0^\infty \frac{e^{-z\mu} (z\mu)^k}{k!} \frac{\alpha^\alpha z^{\alpha-1} e^{-\alpha z}}{\Gamma(\alpha)} dz = \frac{\mu^k \alpha^\alpha}{k! \Gamma(\alpha)} \int_0^\infty z^{k+\alpha-1} e^{-(\mu+\alpha)z} dz = \frac{\mu^k \alpha^\alpha}{k! \Gamma(\alpha)} \frac{\Gamma(k + \alpha)}{(\mu + \alpha)^{k+\alpha}},$$

which rearranges to Equation (7.7). The moments follow from the tower rule: $\mathbb{E}y = \mathbb{E}[Z\mu] = \mu$ and $\text{Var} y = \mathbb{E}[Z\mu] + \text{Var}(Z\mu) = \mu + \mu^2/\alpha$. For $\alpha \rightarrow \infty$, $Z \rightarrow 1$ in probability and the

mixture collapses to the Poisson. For concavity, write $\eta_c = \log w_c + \beta^\top x_c$; the β -dependent part of the log-likelihood per cell is $y_c \eta_c - (y_c + \alpha) \log(\alpha + e^{\eta_c})$. The first term is linear, and $\log(\alpha + e^\eta) = \text{logsumexp}(\log \alpha, \eta)$ is convex in η , hence $-(y_c + \alpha) \log(\alpha + e^{\eta_c})$ is concave in the linear function η_c of β ; sums of concave functions are concave. $\square$

The mixture derivation is not a convenience trick; it is the model's meaning. Z is a multiplicative *frailty* — the latent common factor of Vaupel et al. (1979), developed for multivariate event data by Hougaard (2000) — attached here to each sector-week cell to absorb heterogeneity the covariates miss. That reading also sets the limits of the upgrade. Independent frailties per cell fix the *marginal* variance but assert no dynamics; if the missing structure is autocorrelated (a persistent latent factor, or genuine event-to-event excitation), Pearson residuals will still be serially correlated after the NB upgrade, and that residual autocorrelation — not overdispersion itself — is the escalation signal toward Chapter 8. The distinction matters because overdispersion is produced by frailty *and* by contagion alike; a dispersion statistic can demand escalation but can never certify excitation. If clustered variance is read as contagion, branching estimates inflate; the frailty experiments of Chapter 19 are designed to measure that inflation. The frailty-vs-contagion identification problem is treated properly in Chapter 8.

Design decision (D4.2 — negative binomial is the default count noise).

The default observation model for sector-week counts is the negative binomial Equation (7.7) with mean $w_c e^{\beta^\top x_c}$ and a fitted dispersion α (per sector, unless pooling is justified by the diagnostics), replacing the pure Poisson wherever a predictive count distribution is consumed — held-out NLL, PIT calibration, decision outputs. Rationale: overdispersion is expected from heterogeneity and history dependence (to be measured in Chapter 19); an exact frailty interpretation (Proposition 7.16) that absorbs heterogeneity without asserting dynamics; preserved mean structure and β interpretation; preserved concavity in β at fixed α . Quasi-Poisson/sandwich standard errors remain the reporting standard for coefficient inference in plain-Poisson fits. Dispersion diagnostics — $\hat{\phi}$, fitted $1/\alpha$, and randomized PIT — are computed with every count fit under the protocols of Chapter 11. Reversal triggers: if real-data fits drive $1/\alpha \rightarrow 0$ after the covariate baseline (equidispersion), the upgrade is dropped as needless; if NB residuals stay serially correlated, the deficiency is dynamic and the remedy is history terms (Section 7.7, Chapter 8), not a fatter marginal tail.

7.7 History as covariates: the self-modulated GLM, previewed

One step remains inside the GLM family, and the repository's sector layer already uses it. Add the recent past to the design: lagged own- and cross-sector counts (or exponentially filtered versions of them) enter the linear predictor,

$$\log \Lambda_{s,t} = a_s + \beta^\top x_t + \sum_{r \in \mathcal{S}} \sum_{\ell=1}^L b_{sr\ell} y_{r,t-\ell}, \quad (7.8)$$

which is a *log-linear Poisson autoregression* — the count time series family of Fokianos et al. (2009), log-linear variant Fokianos and Tjøstheim (2011). Everything mechanical from this chapter survives: the parameters sit in a linear predictor, so the likelihood is concave (Theorem 7.6 verbatim), the solver guarantees of Corollary 7.7 hold, and the exp link grants *sign freedom* — negative b coefficients express inhibition, one of the three legal routes to it

catalogued in Chapter 8. This is rung 3 of the ladder. Fixed lags (or fixed-decay filters) play the role of a fixed kernel: they capture the predictive content of excitation without estimating the timescale, the one parameter that weekly data cannot identify (the κ - θ ridge; Chapter 11).

Two prices are paid, and both are conceptual rather than computational. First, lagged counts are *internal* covariates (Assumption 7.4 is voided): the model is a legitimate conditional intensity and one-step-ahead prediction is clean, but multi-step forecasts and any scenario analysis must simulate the system jointly — plugging today’s history into a formula for next quarter is not a probability of anything (Kalbfleisch and Prentice, 2002). Second, a stability condition that must be stated explicitly:

Warning 7.17 (Stability of log-link feedback is an operating-point condition). For the *additive* excitation model of Chapter 8, with nonnegative lag matrices summing to G , subcriticality is the coefficient condition $\rho(G) < 1$, enforced in `hawkes_calibration/sector_stability.py` through the row-sum bound $\rho_{\max} = 0.95$. That certificate does *not* transfer to Equation (7.8). Under the log link one past event multiplies next week’s rate by e^b , so the marginal effect of an event on the conditional mean is $b\Lambda$ — coefficient *times operating level*. A feedback matrix that is harmless at 0.3 events per sector-week can be explosive at 3; no inequality on the b ’s alone, spectral or otherwise, certifies stability. The applicable theory is the contraction/ergodicity analysis of Fokianos and Tjøstheim (2011), whose conditions constrain the feedback of *bounded or slowly varying* transforms of the past; raw counts in the exponent sit outside the Lipschitz structure that the theory requires. We expect simulated log-link feedback models to produce metastable excursions even when the nominal spectral radius is subcritical; the stress tests of Chapter 19 probe exactly this. Design rules that follow: history enters exponents only through bounded transforms (EWMA levels, $\log(1 + \cdot)$ counts); any simulation or multi-step use of a fitted self-modulated GLM must verify stability *at the operating point*; and the branching reading of excitation is reserved for the additive parameterization. The full treatment — both parameterizations, their stability certificates, and what each can and cannot claim about contagion — is Chapter 8.

7.8 Defect declaration

Per Design decision D2.1 (Chapter 2), the Poisson models of this chapter declare their position against the defect ledger of Table 2.1. Table 7.1 summarizes; the signs are the chapter’s own results.

The asymmetry in Table 7.1 between sector and firm rows is the quantitative reason the ground process is calibrated at sector resolution first: at that resolution the Poisson rung is exactly right under (C1), (C2), (C4) and immune to (C3), (C6), so its held-out likelihood is a trustworthy floor. Firm-level Poisson calibration through the event-time path inherits the (C3) deflation and (C5) attenuation with the stated signs, and every later firm-level model must beat it *while carrying the same declared exposures*.

Code hooks.

Existing. `hawkes_calibration/sector_stability.py` (`fit_sector_count_model`): with `n_lags=0` the additive excitation vanishes and the fit is exactly the weekly NHPP GLM of Theorem 7.10 and Corollary 7.11; its ridge `l2_beta` and baseline support-envelope winsorization implement the remedies of Warning 7.9; `sector_rate_at` is the one-step predictor. `synthetic/fast_fit.py::fit_sector_glm_fast(n_lags=0)` is the vectorized synthetic-world twin for the decomposition benchmarks planned in Chapter 19. `hawkes_calibration/eventtime/estimate.py`

Defect	Status at this rung	Mechanism and sign	Treatment
(C1)	corrected <i>exactly</i>	censored exposure contributes its survival factor through the compensator; no bias (Proposition 7.12)	this chapter
(C2)	corrected mechanically	risk sets start at V_i ; exposure w_c integrates from entry. Residual selection (“firms that have raised”) is an interpretation limit, not a likelihood error	Chapter 15
(C3)	<i>vulnerable</i> (firm level)	unrecorded exits inflate exposure w_c with dead time; fitted hazards <i>deflated</i> (downward sign), worst for long-quiet firms. Sector level immune: sectors do not die	exposure windows, Chapter 13
(C4)	robust at weekly resolution	binned likelihood is exact; within-bin positions are ancillary (Corollary 7.14); day-level jitter within weeks is immaterial	Chapter 10
(C5)	<i>vulnerable</i>	stale firm covariates attenuate their fitted effects toward zero; macro covariates unaffected (published as-of, never stale). Bites the event-time firm path, not the sector GLM	Chapter 12
(C6)	sector immune / firm vulnerable	sector counts are complete (out-of-universe deals still carry sectors); firm-attributable intensity misses competitors and thins counts	Chapter 15

Table 7.1: Defect declaration for the Poisson rung (D2.1 compliance). “Immune”/“robust” claims hold under this chapter’s stated assumptions; signed vulnerabilities are accepted and carried to the chapters that own the remedies.

(`fit_multivariate_with_covariates`) is the day-resolution NHPP path when excitation is disabled, with `hawkes_calibration/eventtime/covariates.py` (`PiecewiseConstantCovariate.integrate_exp_gamma`) computing the compensator $\int e^{\beta^\top x}$ in closed form cell by cell — the continuous-time twin of the GLM offset. *To be written.* `hawkes_calibration/sector_negbin.py`: the D4.2 negative-binomial layer — NB likelihood Equation (7.7) with log-mean $\log w_c + \beta^\top x_c$, per-sector α , quasi-Poisson $\hat{\phi}$ and PIT diagnostics emitted in the Chapter 11 harness format. `hawkes_calibration/data/audits.py` (exposure integrity, per Chapter 2): open-spell retention (Warning 7.15), entry-date consistency of w_c integration limits, spell-table vs. exposure-table log-likelihood parity (Proposition 7.13), and train/test covariate-support overlap reporting (Warning 7.9).

Chapter 8

Exponential Hawkes: Excitation, Inhibition, and Covariates

8.1 Model and assumptions

The Poisson models of Chapter 7 treat deal flow as exogenously modulated: nothing that happens in the market changes the future of the market. If their residual diagnostics show clustering beyond what covariates explain — a hypothesis the studies of Chapter 19 will probe under controlled conditions — the Hawkes process (Hawkes, 1971) is the minimal extension in which events beget events. This chapter develops the exponential-kernel Hawkes layer end to end: the branching representation that organizes everything (Section 8.2), the likelihood and the recursion that makes it computable (Section 8.3), the EM algorithm on the latent genealogy (Section 8.4), simulation (Section 8.5), and then the central section: what to do about *inhibition*, which a nonnegative-kernel Hawkes process provably cannot express, and which we expect in this market — a firm that has just raised is temporarily *less* likely to raise again (Section 8.6). The chapter closes with the frailty-versus-contagion identifiability problem (Section 8.7) and the defect declaration required by D2.1 (Section 8.8).

Assumption 8.1 (Exponential Hawkes ground process). On the filtered space of Definition 2.1, the univariate model for a single stream is

$$\lambda(t) = \mu(t) + \sum_{t_m < t} h(t - t_m), \quad h(u) = \kappa \theta e^{-\theta u}, \quad \kappa \geq 0, \theta > 0, \quad (8.1)$$

and the multivariate, sector-indexed model is, for $s \in \mathcal{S}$,

$$\lambda_s(t) = \mu_s(t) + \sum_{r \in \mathcal{S}} \sum_{\substack{t_m < t \\ s_m = r}} \kappa_{sr} \theta_{sr} e^{-\theta_{sr}(t - t_m)}, \quad \mu_s(t) = \exp(\beta_s^\top x_s(t)), \quad (8.2)$$

where the covariate process $x_s(t)$ (macro series, sector activity summaries, seasonal terms) is predictable and locally bounded, constructed under the point-in-time protocol Protocol 2.5. The *branching matrix* is

$$G_{sr} = \int_0^\infty \kappa_{sr} \theta_{sr} e^{-\theta_{sr} u} du = \kappa_{sr}, \quad (8.3)$$

and we impose *subcriticality*: $\rho(G) < 1$.

Two virtues of the (κ, θ) parameterization justify committing to it over the equally common $\alpha e^{-\beta t}$ form. First, $\kappa = \int_0^\infty h$ is a *unitless branching mass*: by Section 8.2 it is the expected number of direct offspring per event, invariant to the time unit, while θ is a pure timescale with half-life $\log 2 / \theta$. Second, the two *decouple in aggregation analysis*: binning event times

at width Δ (defect (C4)) destroys information about θ once $\Delta\theta$ is large, but the mass κ remains estimable from count balance — the κ – θ ridge analyzed in Chapter 11. Our working hypothesis, to be tested in Chapter 19, is that κ survives aggregation while θ does not. A parameterization in which the kernel height $\alpha = \kappa\theta$ is a free parameter entangles mass and timescale in exactly the direction the data cannot resolve; keeping (κ, θ) explicit keeps the ill-conditioned direction visible. (The event-time fitters currently store $\alpha = \kappa\theta$ and $\beta = \theta$ internally; the mapping is recorded in the code hooks.)

Subcriticality is the standing stability constraint. For constant baselines it is exactly the condition for a stationary version to exist (Hawkes, 1971; Brémaud and Massoulié, 1996); for the covariate-driven baseline of Equation (8.2) the process is defined conditionally on the covariate paths and is non-explosive on finite windows regardless (Remark 8.3), but $\rho(G) < 1$ still governs cluster sizes, forecast amplification, and the sanity of any simulation. Because $G \geq 0$, Perron–Frobenius gives $\rho(G) \leq \max_s \sum_r G_{sr}$, so a row-sum bound is a convex sufficient condition — this is precisely the constraint the repository’s discrete-face fitter enforces with $\rho_{\max} = 0.95$ (`hawkes_calibration/sector_stability.py`), and the continuous-time layer inherits the same certificate. Multivariate Hawkes models with spectral-radius stability conditions are standard in financial applications (Embrechts et al., 2011; Bacry et al., 2015).

8.2 The cluster representation

Everything tractable about the linear Hawkes process follows from one theorem: the process is a Poisson cluster process (Hawkes and Oakes, 1974).

Theorem 8.2 (Hawkes–Oakes cluster representation). *Let $\mu_s(t) \geq 0$ be locally integrable baselines and $h_{sr} \geq 0$ kernels with $G_{sr} = \int h_{sr} < \infty$. Construct a marked point process as follows: immigrants of type s arrive as an inhomogeneous Poisson process with rate $\mu_s(t)$, independently across types; recursively, each event of type r at time t_m (immigrant or offspring) generates direct offspring of type s on (t_m, ∞) as an inhomogeneous Poisson process with rate $h_{sr}(t - t_m)$, independently of everything else; the process is the superposition of all generations. Then the superposition is a Hawkes process with intensity Equation (8.2). Conversely, every Hawkes process with these parameters is equal in law to this cluster process.*

Proof of the construction direction. Let $\{\mathcal{G}_t\}$ be the filtration generated by the labeled construction (all events with their genealogy). Fix t and consider arrivals of type s in $(t, t + dt]$. They are of two kinds. New immigrants form a Poisson process with deterministic rate μ_s , independent of $\mathcal{G}_t$, hence contribute $\mathcal{G}_t$ -intensity $\mu_s(t)$. For each event $t_m < t$ of type r already born, its type- s offspring process is inhomogeneous Poisson with rate $h_{sr}(\cdot - t_m)$; by independent increments of the Poisson process, its future beyond t is conditionally independent of $\mathcal{G}_t$ with intensity $h_{sr}(t - t_m)$. Events born after t contribute nothing, since the kernels are supported on positive ages. The streams are independent by construction, and the intensity of a superposition of conditionally independent streams is the sum of the intensities (additivity of compensators), so the type- s superposed counting process has $\mathcal{G}_t$ -intensity $\mu_s(t) + \sum_r \sum_{t_m < t, s_m=r} h_{sr}(t - t_m)$. Finally, this expression is a functional of the unlabeled event times alone, hence measurable with respect to the internal filtration $\mathcal{F}_t \subseteq \mathcal{G}_t$; by the innovation (projection) theorem the $\mathcal{F}_t$ -intensity is the optional projection of the $\mathcal{G}_t$ -intensity, which here equals itself (Daley and Vere-Jones, 2003, Ch. 7). So the unlabeled superposition has the Hawkes intensity Equation (8.2). The converse (existence and uniqueness in law of the cluster decomposition for a given Hawkes law) is Hawkes and Oakes (1974); we use only the construction direction, which is what simulation and EM consume. $\square$

Remark 8.3 (Non-explosion does not need subcriticality). On a finite window $[0, T]$ the mean intensity $\xi_s(t) = \mathbb{E}[\lambda_s(t)]$ satisfies the Volterra equation $\xi = \mu + h * \xi$ componentwise; since

the exponential kernels are bounded by $\kappa_{sr}\theta_{sr}$, Grönwall's inequality gives $\xi_s(t) \leq \bar{\mu} e^{ct} < \infty$ for constants depending only on the parameters. The process is therefore well defined and a.s. finite on any finite window for *any* κ ; what $\rho(G) < 1$ buys is stationarity, finite cluster sizes in expectation, and bounded long-run amplification — the subject of the corollaries below. This distinction matters later: for the *nonlinear* models of Section 8.6 even finite-window good behavior is no longer automatic.

Corollary 8.4 (Subcriticality and cluster size). *In the cluster construction, the offspring cascade of a single type- r event is a multitype Galton–Watson process with mean-offspring matrix G . The expected total cluster size (all generations, all types, including the ancestor) is finite iff $\rho(G) < 1$, in which case the expected number of type- s events in the cluster of a type- r ancestor is $[(I - G)^{-1}]_{sr}$. In the univariate case the expected total cluster size is $1/(1 - \kappa)$.*

Proof. The number of direct type- s offspring of a type- r event is Poisson with mean $\int h_{sr} = G_{sr}$, independently across events, so generation sizes form a multitype branching process: the expected type-composition vector of generation n descending from one type- r ancestor is $G^n e_r$ (induction on n , using linearity of expectation and independence of offspring processes). The expected cluster composition is $\sum_{n \geq 0} G^n e_r$. For a nonnegative matrix the Neumann series $\sum_n G^n$ converges iff $\rho(G) < 1$ (Gelfand's formula), with limit $(I - G)^{-1}$; if $\rho(G) \geq 1$ the series diverges because $\mathbf{1}^\top G^n e_r \not\rightarrow 0$ along the Perron direction. Univariate: $\sum_n \kappa^n = 1/(1 - \kappa)$. $\square$

Corollary 8.5 (Stationary rate and endogenous share). *Let the baselines be constant, $\mu_s(t) \equiv \bar{\mu}_s$, and $\rho(G) < 1$. The stationary mean rate vector is*

$$\bar{\lambda} = (I - G)^{-1} \bar{\mu}, \quad \text{univariate: } \bar{\lambda} = \frac{\bar{\mu}}{1 - \kappa}, \quad (8.4)$$

and the stationary fraction of events that are offspring rather than immigrants (the endogenous share) is $1 - \mathbf{1}^\top \bar{\mu} / \mathbf{1}^\top \bar{\lambda}$; in the univariate case it equals κ .

Proof. Taking expectations in Equation (8.2) under stationarity, $\bar{\lambda}_s = \bar{\mu}_s + \sum_r G_{sr} \bar{\lambda}_r$ because $\mathbb{E} \sum_{t_m < t, s_m = r} h_{sr}(t - t_m) = \int_0^\infty h_{sr}(u) \bar{\lambda}_r du$ by Campbell's formula; solving gives Equation (8.4). Immigrants arrive at rate $\bar{\mu}_s$ by construction, so the immigrant share of events is $\mathbf{1}^\top \bar{\mu} / \mathbf{1}^\top \bar{\lambda}$ and the endogenous share is its complement; univariate, $1 - \bar{\mu} / \bar{\lambda} = 1 - (1 - \kappa) = \kappa$. Equivalently: each immigrant's cluster has expected size $1/(1 - \kappa)$ (Corollary 8.4), of which exactly one event is the immigrant. $\square$

Remark 8.6 (A common accounting error). It is tempting to reason “rate $\bar{\mu}$ of immigrants, each with κ children, hence total rate $\bar{\mu}(1 + \kappa)$ and endogenous share $\kappa/(1 + \kappa)$.” This counts only the first generation: children beget children, the geometric series compounds to $1/(1 - \kappa)$, and the correct endogenous share is κ , not $\kappa/(1 + \kappa)$. At $\kappa = 0.5$ the difference is 50% versus 33% of deal flow attributed to contagion — a large error. All endogeneity reporting in this repository uses Corollary 8.5, and (better) the posterior parentage diagnostic of Section 8.4, subject to the upper-bound discipline of Section 8.7.

8.3 Likelihood and the Ogata recursion

The Jacod likelihood Equation (2.3) specializes to Equation (8.1) as

$$\ell(\beta, \kappa, \theta) = \sum_{m=1}^n \log(\mu(t_m) + \kappa \theta A(m)) - \int_0^T \mu(u) du - \kappa \sum_{m=1}^n (1 - e^{-\theta(T-t_m)}), \quad (8.5)$$

where

$$A(m) := \sum_{l < m} e^{-\theta(t_m - t_l)} \quad (8.6)$$

is the decayed event count seen by event m . The compensator term follows by exact integration: $\int_{t_m}^T \kappa \theta e^{-\theta(u-t_m)} du = \kappa(1 - e^{-\theta(T-t_m)})$, summed over events; the baseline integral is available in closed form for the piecewise-constant covariates used in the repository (`hawkes_calibration/eventtime/covariates.py`). Computed naively, the $A(m)$ cost $O(n^2)$; the exponential kernel — alone among common kernels — makes them recursive (Ozaki, 1979).

Proposition 8.7 (Ogata recursion). *With $A(1) = 0$, the quantities Equation (8.6) satisfy*

$$A(m) = e^{-\theta(t_m - t_{m-1})} (1 + A(m-1)), \quad m \geq 2, \quad (8.7)$$

so Equation (8.5) is computable in $O(n)$ after sorting.

Proof. By induction on m . Base case $m = 2$: $A(2) = e^{-\theta(t_2 - t_1)} = e^{-\theta(t_2 - t_1)}(1 + A(1))$ since $A(1) = 0$. Inductive step: assume $A(m-1)$ equals Equation (8.6) at index $m-1$. Splitting off the $l = m-1$ term and factoring the common decay over $[t_{m-1}, t_m]$,

$$A(m) = \sum_{l \leq m-1} e^{-\theta(t_m - t_l)} = e^{-\theta(t_m - t_{m-1})} \left(e^0 + \sum_{l < m-1} e^{-\theta(t_{m-1} - t_l)} \right) = e^{-\theta(t_m - t_{m-1})} (1 + A(m-1)),$$

which is Equation (8.7). The recursion works because the exponential kernel satisfies $h(u+v) = h(u)e^{-\theta v}$: the whole past compresses into one number that decays deterministically between events. This is the Markov-embedding property; it fails for power-law kernels, which admit no finite recursion, and it is why this document fixes the exponential family as the parametric default. $\square$

In the multivariate model the same recursion runs per source–target pair: $A_{sr}(m) = e^{-\theta_{sr}(t_m - t_{m-1})} (\mathbf{1}\{s_{m-1} = r\} + A_{sr}(m-1))$ on the merged event stream, giving $O(n|\mathcal{S}|^2)$ per likelihood pass; `hawkes_calibration/eventtime/likelihood.py` implements exactly this, in a block-cumsum form that is numerically stable over long horizons, together with analytic gradients. Asymptotic normality of the MLE in the stationary ergodic case is Ogata (1978).

Positivity transforms. The fitters optimize unconstrained by reparameterizing $\kappa = e^a$ (in code, $\alpha = e^a$ with $\alpha = \kappa\theta$), so gradient methods never propose negative branching and standard errors return via the delta method (`hawkes_calibration/eventtime/estimate.py`). Two consequences deserve note. First, the transform is a *modeling* statement, not a numerical trick: it hard-codes the nonnegative-kernel assumption whose consequences Section 8.6 confronts — a very negative a is near-zero excitation, never inhibition. Second, the boundary $\kappa = 0$ is pushed to $a \rightarrow -\infty$, so “no excitation” is a flat direction of the transformed likelihood, and testing $H_0: \kappa = 0$ is a boundary problem whose LRT null is the mixture $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ (Self and Liang, 1987); the testing protocol lives in Chapter 11. When the number of sectors makes the $|\mathcal{S}|^2$ branching entries data-hungry, the ℓ_1 -penalized fitter with exact zeros (`hawkes_calibration/eventtime/lasso.py`) implements the penalized multivariate excitation recovery of Hansen et al. (2015).

Warning 8.8 (The likelihood is not concave in (κ, θ)). For *fixed* θ and a linearly parameterized baseline, Equation (8.5) is concave: each event term is the log of an affine function of the parameters and the compensator is linear. Jointly in (κ, θ) it is not — the event terms involve $\kappa\theta e^{-\theta\Delta}$, and the likelihood surface develops the κ – θ ridge analyzed in Chapter 11: data observed at resolution coarser than the kernel half-life constrain roughly the product of mass and effective within-window decay, not each factor. With the exp-link baseline of Equation (8.2) even the fixed- θ problem loses global concavity (the event terms $\log(e^{\beta^\top x} + c)$ are convex in β), retaining concavity only in the excitation block given the baseline. Practice therefore fixes a grid of decays, maximizes at each grid point with multistart, and reports the *profile* likelihood over θ — flagging ridge flatness rather than hiding it. The EM algorithm of the next section is the complementary tool: it restores block-concavity through augmentation and yields the diagnostics no direct MLE provides.

8.4 EM on the latent genealogy

Theorem 8.2 says every event has a latent *parent*: either the baseline (immigrant) or an earlier event. Making the genealogy explicit turns Hawkes estimation into a missing-data problem with an exact E-step and closed-form M-steps — the cascades-of-Poisson-processes formulation of Simma and Jordan (2010), identical in structure to the ETAS EM of Veen and Schoenberg (2008). We derive it in the univariate model; the multivariate bookkeeping is indicated at the end.

Let $u_m \in \{0, 1, \dots, m-1\}$ denote the parent of event m ($0 =$ immigrant, $l \geq 1 =$ event t_l).

Lemma 8.9 (Complete-data likelihood). *Given the genealogy $u = (u_1, \dots, u_n)$, the complete-data log-likelihood on $[0, T]$ is*

$$\ell_c(\beta, \kappa, \theta; u) = \sum_{m: u_m=0} \log \mu(t_m) - \int_0^T \mu + \sum_{m: u_m=l \geq 1} [\log \kappa + \log \theta - \theta(t_m - t_l)] - \kappa \sum_{l=1}^n (1 - e^{-\theta(T-t_l)}). \quad (8.8)$$

Proof. Under the cluster construction the labeled data decompose into independent inhomogeneous Poisson processes: the immigrant process with rate μ on $[0, T]$, contributing $\sum_{u_m=0} \log \mu(t_m) - \int_0^T \mu$; and, for each event l , its offspring process with rate $h(\cdot - t_l)$ observed on $(t_l, T]$, contributing $\sum_{m: u_m=l} \log h(t_m - t_l) - \int_0^{T-t_l} h(v) dv$. Summing, and evaluating $\int_0^{T-t_l} \kappa \theta e^{-\theta v} dv = \kappa(1 - e^{-\theta(T-t_l)})$, gives Equation (8.8). Note the compensator does not depend on u : parents determine only which factor each event's log-intensity lands in. $\square$

The observed-data likelihood Equation (8.5) is recovered by summing e^{ℓ_c} over genealogies — the log-of-sum that destroys concavity. Augmentation splits it back apart.

Proposition 8.10 (E-step: exact posterior parentage). *Under parameters (β, κ, θ) , the posterior over genealogies given the event times factorizes across events, $\mathbb{P}(u \mid t_{1:n}) = \prod_m p_{m, u_m}$, with*

$$p_{m0} = \frac{\mu(t_m)}{\mu(t_m) + \kappa \theta A(m)}, \quad p_{ml} = \frac{\kappa \theta e^{-\theta(t_m - t_l)}}{\mu(t_m) + \kappa \theta A(m)}, \quad 1 \leq l < m, \quad (8.9)$$

with $A(m)$ as in Equation (8.6) — so the E-step reuses the Ogata recursion and costs $O(n)$ beyond the pairwise weights.

Proof. By Lemma 8.9, $e^{\ell_c(u)} = C \prod_m w_{m, u_m}$ where $w_{m0} = \mu(t_m)$, $w_{ml} = \kappa \theta e^{-\theta(t_m - t_l)}$, and C collects the compensator factors, which do not involve u . Hence $\mathbb{P}(u \mid t_{1:n}) \propto \prod_m w_{m, u_m}$, which factorizes with per-event normalizers $\mu(t_m) + \kappa \theta A(m)$. Probabilistically this is the thinning identity: given $\mathcal{F}_{t-}$, the event at t_m belongs to one of the superposed streams with probability proportional to that stream's intensity. $\square$

Proposition 8.11 (M-step). *Let $P_0 = \sum_m p_{m0}$ and $P_1 = n - P_0 = \sum_m \sum_{l \geq 1} p_{ml}$ be the expected immigrant and offspring counts, and $D = \sum_{m, l \geq 1} p_{ml}(t_m - t_l)$ the expected total parent-child gap. Maximizing $Q(\cdot) = \mathbb{E}_{u \sim p}[\ell_c]$ gives, in the interior approximation that replaces the truncated exposures $1 - e^{-\theta(T-t_l)}$ by 1:*

$$\hat{\mu} = \frac{P_0}{T} \quad (\text{constant baseline}), \quad \hat{\kappa} = \frac{P_1}{n}, \quad \hat{\theta} = \frac{P_1}{D}, \quad (8.10)$$

i.e. *branching* = expected children per event, *decay* = reciprocal of the p -weighted mean parent-child gap. With exact exposures the κ -update becomes $\hat{\kappa}(\theta) = P_1 / \sum_l (1 - e^{-\theta(T-t_l)})$ and $\hat{\theta}$ solves the scalar equation $P_1/\theta - D = \hat{\kappa}(\theta) \sum_l (T - t_l) e^{-\theta(T-t_l)}$.

Proof. Q inherits the separable form of Equation (8.8) with indicators replaced by posteriors. The κ -part is $P_1 \log \kappa - \kappa \sum_l (1 - e^{-\theta(T-t_l)})$, concave with the stated root; the θ -part is $P_1 \log \theta - \theta D - \kappa \sum_l (1 - e^{-\theta(T-t_l)})$, whose stationarity condition is the displayed scalar equation; setting the exposure terms to 1 (events far from the window edge) decouples them into Equation (8.10). For the exp-link baseline, the β -part of Q is $\sum_m p_{m0} \beta^\top x(t_m) - \int_0^T e^{\beta^\top x}$ — a weighted Poisson GLM, globally concave. The augmentation thus restores concavity to the very block that broke it in Warning 8.8. $\square$

Theorem 8.12 (Monotone ascent). *Let $\vartheta^{(k+1)}$ maximize $Q(\cdot \mid \vartheta^{(k)})$. Then $\ell(\vartheta^{(k+1)}) \geq \ell(\vartheta^{(k)})$, with equality only if $\vartheta^{(k)}$ already maximizes $Q(\cdot \mid \vartheta^{(k)})$.*

Proof. Write $\pi_\vartheta(u) = \mathbb{P}(u \mid t_{1:n}; \vartheta)$ for the posterior of Proposition 8.10. For any ϑ' , $\ell(\vartheta') = \mathbb{E}_{\pi_\vartheta}[\ell_c(\vartheta'; u)] - \mathbb{E}_{\pi_\vartheta}[\log \pi_{\vartheta'}(u)]$, since $\ell_c(\vartheta'; u) = \ell(\vartheta') + \log \pi_{\vartheta'}(u)$ holds for every u (Bayes) and the expectation of a constant is itself. Subtracting the same identity at ϑ ,

$$\ell(\vartheta') - \ell(\vartheta) = [Q(\vartheta' \mid \vartheta) - Q(\vartheta \mid \vartheta)] + \text{KL}(\pi_\vartheta \parallel \pi_{\vartheta'}) \geq Q(\vartheta' \mid \vartheta) - Q(\vartheta \mid \vartheta) \geq 0$$

for $\vartheta' = \vartheta^{(k+1)}$, using nonnegativity of Kullback–Leibler divergence (Jensen) and the definition of the M-step. Equality forces both terms to vanish. $\square$

Three practical points. *Boundary corrections.* The exact-exposure updates of Proposition 8.11 are not optional refinements: near the right window edge an event’s offspring are partially unobserved, and the interior approximation deflates κ while the orphaned early-window events (parents before time 0) inflate μ . The right edge is corrected *exactly* inside the M-step — the offspring process of an observed parent really is Poisson observed on $(t_l, T]$ — while the left edge is only mitigated, by discarding a burn-in of several kernel half-lives from the immigrant accounting (Veen and Schoenberg, 2008). *The E-step is the endogeneity diagnostic.* The honest answer to “how much of deal flow is contagion” is not the plug-in $\hat{\kappa}$ of Corollary 8.5 but the posterior expected offspring share $\frac{1}{n} \sum_m (1 - p_{m0})$, together with the distribution of attributed parent–child gaps: genuine contagion concentrates attributions at the kernel timescale, while a latent common factor inflates $\hat{\kappa}$ yet produces attributions with no timescale structure (Section 8.7). *Constraints are not automatic.* EM keeps every $\kappa \geq 0$ by construction, but nothing in the parent accounting bounds $\rho(\hat{G})$ below one — a rare source sector attributed many children can push a column past criticality — so the subcriticality certificate must be checked and enforced after each fit, exactly as the discrete-face fitter does. In the multivariate model the updates read $\hat{G}_{sr} = \sum_{m:s_m=s} \sum_{l:s_l=r} p_{ml} / \sum_{l:s_l=r} (1 - e^{-\theta(T-t_l)})$, expected type- s children per unit of type- r exposure. If profile likelihoods over the θ -grid show the exponential shape itself misfits (time-rescaling diagnostics of Chapter 11), the escalation is the histogram-kernel nonparametric EM of Lewis and Mohler (2011) — same E-step, kernel M-step replaced by a binned estimator.

Protocol 8.13 (Estimation ladder for the continuous-time sector layer). (a) Fix a grid of decays θ spanning well-separated half-lives from weeks to quarters; do not free θ jointly with κ except on the final reported grid point. (b) At each grid point, fit by EM (Propositions 8.10 and 8.11) with right-edge exposure corrections and a burn-in of at least four half-lives, multistarted; cross-check with the direct MLE of Section 8.3. (c) Report the profile likelihood over θ with its flatness (the ridge diagnostic of Chapter 11), never a single $(\hat{\kappa}, \hat{\theta})$ point alone. (d) Report $\rho(\hat{G})$, the posterior endogenous share, and the parent–child gap histogram; enforce $\rho(\hat{G}) \leq \rho_{\max} = 0.95$ as in the discrete face. (e) All branching estimates carry the upper-bound language of Protocol 8.20.

8.5 Simulation

Simulation is not an afterthought: multi-step forecasts require it (history covariates are *internal*, so scenario propagation must draw events and update features jointly (Kalbfleisch and Prentice, 2002)), and the goodness-of-fit machinery of Chapter 11 consumes simulated envelopes. Three routes, all exact in law.

Lemma 8.14 (Thinning bound for exponential kernels). *On any interval containing no events and no baseline increase, $t \mapsto \lambda_s(t)$ of Equation (8.2) is nonincreasing. Hence $\lambda^* = \sum_s \lambda_s(\tau^+)$ evaluated just after the most recent event or baseline breakpoint τ dominates the total intensity until the next event or breakpoint.*

Proof. The excitation part is a nonnegative combination of $e^{-\theta_{sr}(t-t_m)}$, each strictly decreasing in t ; the baseline is constant on the interval by hypothesis. A sum of nonincreasing functions is nonincreasing. $\square$

Ogata thinning (Lewis and Shedler, 1979; Ogata, 1981) then specializes cleanly: from the current time propose a candidate gap $\sim \text{Exp}(\lambda^*)$, accept with probability $\sum_s \lambda_s(t_{\text{cand}})/\lambda^*$, attribute the accepted event to sector s with probability proportional to λ_s , refresh the bound at every event *and* at every covariate breakpoint (where the piecewise-constant baseline may jump up). The Markov embedding keeps the per-event cost at $O(|\mathcal{S}|^2)$ state updates; `hawkes_calibration/eventtime/simulate.py` implements exactly this, including the breakpoint refresh. *Exact inversion* (Dassios and Zhao, 2013) removes rejection entirely for the exponential kernel: $(t, \lambda(t))$ is a piecewise-deterministic Markov process, and the next inter-arrival time is the minimum of a baseline-driven $\text{Exp}(\bar{\mu})$ draw and an excitation-driven time whose survival function $\exp(-\frac{Y}{\theta}(1 - e^{-\theta u}))$, Y the current excess intensity, inverts in closed form (defective with probability $e^{-Y/\theta}$: the cluster may die). This is the preferred engine for the long forecast horizons where thinning’s bound refreshes dominate cost. *Cluster simulation* (Theorem 8.2: draw immigrants, then recursive Poisson offspring) is the third route; it is the one that exposes the genealogy, which makes it the reference implementation for validating EM attributions and edge corrections on synthetic data. All three must agree in law; the test suite compares them distributionally.

8.6 Inhibition and excitation via engineered covariates

We expect the market to have a signed dynamic: sector-level deal flow *self-excites* (capital attention, momentum), while a firm’s own recent raise *inhibits* (a funded firm leaves the market for months). Regime B of the synthetic world builds in exactly this structure, with a 30-day inhibition half-life against a 7-day contagion half-life (Section 2.5). The linear Hawkes family of Assumption 8.1 cannot represent the inhibitory half, and the proof is one paragraph.

Proposition 8.15 (Monotonicity: nonnegative kernels cannot inhibit). *Let $\lambda(t; \mathcal{H}) = \mu(t) + \sum_{t_m \in \mathcal{H}, t_m < t} h(t - t_m)$ with $h \geq 0$. Then for every history $\mathcal{H}$, every additional event time $t_0 < t$, and every t : $\lambda(t; \mathcal{H} \cup \{t_0\}) \geq \lambda(t; \mathcal{H})$, and $\lambda(t; \mathcal{H}) \geq \mu(t)$. Consequently no parameter choice makes the intensity after an added event lower than before it, nor ever lower than the baseline: a post-deal decrease in a firm’s own intensity is inexpressible in any linear Hawkes model with nonnegative kernels, univariate or multivariate, for any (κ, θ) .*

Proof. The intensity is additive in history: $\lambda(t; \mathcal{H} \cup \{t_0\}) = \lambda(t; \mathcal{H}) + h(t - t_0) \geq \lambda(t; \mathcal{H})$ since $h \geq 0$; dropping the whole excitation sum gives $\lambda \geq \mu$. The infimum over parameters of the added term is 0 (attained as $\kappa \rightarrow 0$), so the family’s expressive range for the effect of an own event is $[0, \infty)$: “no effect” is its best imitation of inhibition. The positivity transform $\kappa = e^a$

of Section 8.3 makes the restriction syntactically invisible — the optimizer happily drives a very negative — which is why this proposition must be stated rather than discovered in production. $\square$

Three legal routes remain. Each is developed with its model, validity argument, estimation story, and stability discussion; the decision box at the end commits.

Route (a): log-link self-modulated intensity

Move history inside the link:

$$\lambda(t) = \exp(\beta^\top x(t) + \gamma^\top r(t)), \quad (8.11)$$

where $r(t)$ is a vector of *engineered history covariates*: recency indicators and ages (a cooldown flag $\mathbf{1}\{t - \tau_i(t) \leq W\}$ with $\tau_i(t)$ the last own-deal time, the gap transform $\log(1 + t - \tau_i(t))$, and decayed event counts

$$H_\theta(t) = \sum_{t_m < t} e^{-\theta(t-t_m)}, \quad dH_\theta = -\theta H_\theta dt + dN(t), \quad (8.12)$$

at a small grid of *fixed* (or profiled) decays, per source sector in the multivariate case. Negative components of γ are inhibition, positive are excitation: the sign restriction of Proposition 8.15 evaporates because positivity is carried by the exponential, not by the kernel. This is the point-process GLM of Truccolo et al. (2005), and the discrete-time weekly face of it is the repository’s current sector model.

Proposition 8.16 (Validity and concavity of the log-link model). *Let each component of r be a left-continuous functional of $\mathcal{H}_t^-$ as above. Then (i) λ in Equation (8.11) is predictable, strictly positive, and a.s. finite on the observation window, so the likelihood Equation (2.3) is well defined with no positivity constraints on (β, γ) ; (ii) $(H_\theta, t - \tau_i(t))$ is a finite-dimensional Markov state, so Equation (8.11) depends on history only through a computable summary; (iii) for fixed decays and windows, the log-likelihood is globally concave in (β, γ) .*

Proof. (i) Each feature is adapted and left-continuous (built from events strictly before t), hence predictable; $\exp$ makes λ positive; $H_\theta(t) \leq N(t^-) < \infty$ a.s. keeps it finite on the window. (ii) is the ODE in Equation (8.12) plus the trivial age dynamics. (iii) The likelihood is $\sum_m (\beta^\top x + \gamma^\top r)(t_m) - \int_0^T \exp(\beta^\top x + \gamma^\top r)(u) du$: the event sum is linear in the parameters, and the compensator is an integral of $\exp(\text{affine})$, convex in the parameters for every u , so its negative is concave; a sum of concave functions is concave. Estimation is a GLM: the entire fragility of Hawkes fitting lived in the decay, and fixing it returns the problem to convex territory. Note what (i) does *not* claim: global-in-time stability of the process. That is Warning 8.18. $\square$

The log-link model contains the linear Hawkes model to first order.

Proposition 8.17 (First-order nesting of linear Hawkes). *Fix θ and constant baseline μ , and write $x(t) = \kappa\theta H_\theta(t)/\mu \geq 0$ for the excitation load. The linear Hawkes intensity $\lambda_{\text{lin}} = \mu(1 + x)$ and the log-link intensity $\lambda_{\text{log}} = \mu e^x$ (i.e. Equation (8.11) with feature H_θ and coefficient $\gamma = \kappa\theta/\mu$) satisfy*

$$0 \leq \log \lambda_{\text{log}}(t) - \log \lambda_{\text{lin}}(t) = x(t) - \log(1 + x(t)) \leq \frac{1}{2} x(t)^2. \quad (8.13)$$

Thus a decayed-count feature with $\gamma > 0$ reproduces linear-Hawkes excitation to first order in the load, with a second-order overshoot; and $\gamma < 0$ delivers the mirrored soft inhibition that Proposition 8.15 forbids the linear family. The GLM route nests soft versions of both.

Proof. $\log \lambda_{\text{lin}} = \log \mu + \log(1 + x)$ and $\log \lambda_{\text{log}} = \log \mu + x$; the elementary bounds $x - \frac{x^2}{2} \leq \log(1 + x) \leq x$ for $x \geq 0$ give Equation (8.13). If, as we expect, covariates dominate and the excitation load is small, the surrogate error is second order in an already small quantity; Chapter 19 will measure the load. The overshoot has a sign, however: the log-link *amplifies* large excursions that the linear model would meet additively, and that is the source of the instability below. $\square$

Warning 8.18 ($\rho(G) < 1$ does not stabilize a log-link count model (Mechanism B)). For the linear model, subcriticality is a level-free certificate: it depends only on kernel masses. For log-link feedback it has no such force. In the weekly discrete face $\log m_{s,t} = a_s + \sum_r B_{sr} y_{r,t-1}$ (counts y in the exponent), the deterministic skeleton $m \mapsto \exp(a + Bm)$ has Jacobian $\text{diag}(m^*)B$ at a fixed point m^* : local stability requires $\rho(\text{diag}(m^*)B) < 1$, an *operating-point* condition that scales with the level m^* . A stochastic excursion raises the operating point, the local certificate can silently fail, and the process runs away — explosion is an *absorbing rare event*, invisible to in-sample diagnostics. We expect simulation to show exactly this failure mode: long metastable excursions in configurations whose linear-style spectral radius is nominally far subcritical. The stress tests of Chapter 19 are designed to exhibit it. The theory agrees from both sides: geometric ergodicity for log-linear Poisson autoregressions holds only under contraction conditions on the feedback coefficients (Fokianos and Tjøstheim, 2011), and the continuous-time stability theory for nonlinear Hawkes requires a Lipschitz link (Brémaud and Massoulié, 1996) — $\exp$ is not Lipschitz. Design rules that follow: history enters the exponent only through *bounded or saturating* transforms (EWMA levels, $\log(1 + \cdot)$ counts, clipped features), never raw counts; every simulation carries event caps; and the operating-point diagnostic $\rho(\text{diag}(\hat{m}_t)\hat{B})$ is monitored along simulated paths, not certified once at fit time.

Route (b): rectified additive intensity

Keep the kernel signed and restore positivity by rectification:

$$\lambda(t) = \left[\mu(t) + \sum_{t_m < t} h^+(t - t_m) - \sum_{t_m < t} h^-(t - t_m) \right]_+, \quad h^\pm \geq 0. \quad (8.14)$$

Validity is immediate ($[\cdot]_+$ of a predictable process is predictable and nonnegative), and the mathematics is genuinely available: viewing $\lambda = \phi(\mu + \int h_{\text{signed}} dN)$ with $\phi(x) = x_+$ Lipschitz(1), existence, uniqueness, and stability of a stationary version hold when the total variation of the kernels is subcritical (Brémaud and Massoulié, 1996), and the renewal structure of signed-exponential-kernel processes is worked out in Costa et al. (2020). Maximum likelihood is feasible for the exponential kernel because the times at which the rectifier binds (intensity hits zero, then restarts) are computable in closed form, giving an exact compensator (Bonnet et al., 2021). The costs are structural, not numerical. The likelihood is nonsmooth in the parameters (kinks wherever a zero-crossing time moves), estimation loses the GLM concavity of route (a), and — decisively for us — *expectation and rectification do not commute*: $\mathbb{E}[X]_+ \neq [\mathbb{E}X]_+$ whenever the rectifier binds with positive probability. The mean-intensity Volterra equation $\xi = \mu + h * \xi$, whose closed-form exponential-kernel solution is the engine of the interval-censored MBPP machinery of Chapter 10, therefore has no analogue under Equation (8.14). Since defect (C4) forces much of our fitting through interval-censored likelihoods, route (b) would cut off the machinery that Chapter 10 needs. It is held in reserve as the escalation for a genuinely signed continuous-time ground kernel.

Route (c): move inhibition to the mark factor

The factorization of Theorem 6.7 offers the cleanest exit: keep the *ground* process linear Hawkes over sectors — positive momentum only, which is the economically natural sign at the sector level — and let firm-level inhibition live in the mark factor, where the ranker of Chapter 14 already consumes a cooldown covariate with a negative coefficient. The firm-level intensity is

$$\lambda_i(t) = \lambda_{s(i)}(t) p(i | s(i), t, \mathcal{H}_{t-}), \quad p(i | s, t) = \frac{e^{q_i(t)}}{\sum_{j \in \mathcal{R}_{s,t}} e^{q_j(t)}}, \quad q_i(t) = w^\top z_i(t) + \eta c_i(t), \quad (8.15)$$

with $c_i(t) = \mathbf{1}\{t - \tau_i(t) \leq W\}$ the cooldown flag and $\eta \leq 0$.

Proposition 8.19 (Cooldown in the mark factor is firm-level inhibition). *Under Equation (8.15) with $\eta < 0$: (i) λ_i is predictable and positive, and the joint likelihood factorizes exactly as ground $\times$ mark, so both factors are fitted by their own valid likelihoods; (ii) holding other firms' features fixed, the switch $c_i : 0 \rightarrow 1$ multiplies $p(i | s, t)$ by a factor in $(e^\eta, 1)$ — a strict decrease — hence λ_i strictly decreases while λ_s is untouched; (iii) the ground layer retains nonnegative kernels, so the cluster representation, the $\rho(G)$ certificate, and the MBPP aggregation theory of Chapter 10 all survive verbatim.*

Proof. (i) is Theorem 6.7 plus predictability of c_i (built from events strictly before t). For (ii), write $p_i(\eta) = e^{q+\eta}/(D+e^{q+\eta})$ with $q = w^\top z_i$ and $D = \sum_{j \neq i} e^{q_j} > 0$: the map $x \mapsto x/(D+x)$ is strictly increasing, so $\eta < 0$ strictly decreases p_i ; the ratio $p_i(\eta)/p_i(0) = e^\eta(D+e^q)/(D+e^{q+\eta})$ lies strictly between e^η and 1. (iii) is by construction: no signed object ever touches the ground intensity. What (c) *cannot* express is a firm's own deal suppressing the *sector aggregate*: the mark factor only redistributes a fixed sector flow across firms. If aggregate post-deal suppression is real, it belongs to route (a) at the sector level, or ultimately to route (b). $\square$

Route (c) is the current production architecture (`hawkes_calibration/sector_stability.py` + `hawkes_calibration/sector_ranker.py`). We expect the self-inhibition *ordering* recovered through the ranker cooldown to stay robust even when the parametric shape of the cooldown is wrong; the misspecification experiments of Chapter 19 test this.

Design decision (D5.1 — signed dynamics without signed kernels).

We commit to routes (a) and (c), with division of labor: (c) *owns firm-level inhibition* — the linear sector Hawkes ground layer plus the ranker cooldown covariate, as implemented today — because it preserves the cluster representation, the subcriticality certificate, and the MBPP interval-censored machinery that defect (C4) makes indispensable; (a) *owns sector-level covariate modulation and timing-feature models* — fixed-decay log-link intensities with bounded history features, as in `hawkes_calibration/models/event_block_hawkes.py` — because it keeps concavity and sign freedom at the price of the operating-point stability discipline of Warning 8.18. Route (b), the rectified additive model (Bonnet et al., 2021; Costa et al., 2020), is the recorded escalation if a genuinely signed *continuous-time ground kernel* is ever required. Reversal triggers: escalate to (b) if the time-rescaling and PIT diagnostics of Chapter 11 show persistent post-burst *overprediction of sector aggregates* that route-(a) saturating features cannot absorb (aggregate suppression, which (c) cannot express by Proposition 8.19), or if the decision layer of Chapter 16 demands calibrated signed amplification in continuous time. Restrict (a) — or shift its mass to (c) — if the Mechanism-B operating-point monitor fires in production simulation.

8.7 Frailty versus contagion

A fitted $\hat{\kappa} > 0$ is not proof of contagion. Clustered, overdispersed event flow has two structural explanations: events beget events (Hawkes), or a latent common factor modulates everyone’s rate (frailty; the doubly-stochastic alternative). Both produce overdispersion, autocorrelated counts, and cross-sector comovement, and from a single realization of moderate length they are close to observationally equivalent — the point-process face of the state-dependence-versus-heterogeneity problem of Heckman and Borjas (1980). The mechanism of confounding is concrete: if a positively autocorrelated factor $\xi(t)$ multiplies the true baseline and the model omits it, the score of an excitation loading at zero has expectation

$$\mathbb{E}_0[\partial_\gamma \ell|_{\gamma=0}] \propto \int_0^T \text{Cov}(H_\theta(t), \xi(t)) dt > 0, \quad (8.16)$$

because the decayed count H_θ is built from past events that themselves loaded on ξ : the fitted excitation absorbs the common factor. The frailty world of Chapter 19 is designed to measure the size of this inflation under a persistent latent factor.

The doctrine that follows is layered. *Covariates first* (Duffie et al., 2007): every observable common driver belongs in $\mu_s(t)$; comovement explained by covariates was never contagion, and our ladder runs baseline-first by construction (Chapter 7 precedes this chapter). *Latent-factor escalation* (Duffie et al., 2009): with many sectors, frailty and contagion have different fingerprints — a common factor loads simultaneously across sectors, contagion propagates along the event stream at the kernel’s lag structure — so a latent-factor model (EM or particle methods over ξ) is genuinely estimable; it is an escalation beyond this chapter that we have deliberately not built. *Both coexist* (Azizpour et al., 2018): the operational conclusion for corporate default data, after conditioning on covariates, was contagion *and* frailty — so the honest posture is not to pick one but to bound. The posterior parentage diagnostic of Section 8.4 is the sharpest single-sample probe we have: attributed parent–child gaps concentrated at the kernel timescale are consistent with contagion; attributions spread evenly in time indict a common factor.

Protocol 8.20 (Branching estimates are upper bounds on contagion). Every reported $\hat{\kappa}$, $\hat{G}$, or endogenous share is accompanied by: (i) the covariate set of the baseline it was fitted against; (ii) the sentence “upper bound on contagion; unobserved common factors inflate this estimate,” citing the frailty calibration of Chapter 19 as the measured severity once that study is run; (iii) the parent–child gap histogram of Section 8.4. No standalone $\hat{\kappa}$ appears in any deliverable.

8.8 Defect declaration (D2.1)

Against the ledger of Table 2.1, for the models of this chapter (linear sector Hawkes; log-link history models; the mark-factor cooldown belongs to Chapter 14).

(C1) *Administrative censoring: robust (exact)*. The likelihood Equation (8.5) integrates the compensator to T ; the open window contributes its survival factor $\exp(-\int \lambda)$ exactly. Unlike the Poisson case, that factor is *path-dependent* — it depends on the realized history through the excitation terms — but for exponential kernels it is available in closed form given the history, so nothing is approximated. Within EM, the right-edge exposure correction of Proposition 8.11 is the same fact seen from the cluster side: an observed parent’s offspring process really is Poisson observed on $(t_l, T]$. The subtleties of history-dependent survival under *interval* censoring are deferred to Chapter 10.

(C2) *Left truncation: corrects under assumptions*. All compensator integrals start at entry times, and risk-set indicators Y multiply intensities as in Equation (2.3). The Hawkes-specific residue is the *left edge*: in-window events whose parents predate the window are structurally

misattributed to immigration, biasing $\hat{\mu}$ up and $\hat{G}$ down. Burn-in of several half-lives mitigates; with 90-day-scale kernel mass and a 13.5-year window the residual bias is small, but short per-sector subwindow fits are not licensed (Protocol 8.13).

(C3) *Unlabeled exits: robust at the sector layer, vulnerable at the firm layer.* Sector counts are complete regardless of zombie firms, so the ground models of this chapter are immune. Any firm-level intensity (the route-(a) block model, the ranker) keeps dead firms at risk: fitted firm hazards are deflated and cooldown effects attenuated, with the documented sign (downward on firm-level intensities). We expect the risk-set sensitivity to be among the largest in the system; Chapter 19 will measure it, and treatments live in Chapter 14.

(C4) *Untrusted dates and aggregation: VULNERABLE for θ , with signed statement.* Date jitter and binning destroy the information that identifies the timescale: Fisher information in θ collapses once bin width exceeds the kernel half-life, while κ remains estimable from count balance. The bias is not a small perturbation of $\hat{\theta}$ but a loss of identification — the κ - θ ridge — and timing claims from weekly data are vetoed outright. This defect is the reason Chapter 10 exists: the interval-censored MBPP likelihood is the tractable replacement for the exact likelihood this chapter derived, and Chapter 11 quantifies the bucket-width rule (safe at ≤ 1 half-life, destroyed at ≥ 4).

(C5) *Stale firm covariates: pointer.* The sector baselines of this chapter consume macro and sector-activity covariates, which are never stale (as-of joined); the engineered history covariates of route (a) are point-in-time correct by construction. Firm-level covariates entering z_i or block-model baselines inherit the attenuation and informative-observation pathologies of Chapter 12.

(C6) *Out-of-universe deals: robust at the ground layer, pointer at the mark layer.* Sector event streams include out-of-universe deals, so both the excitation sources and the counts this chapter models are complete. The mark factor must carry an outside option or overstate every tracked firm's share; that is Chapter 15's treatment.

Code hooks.

Existing. `hawkes_calibration/eventtime/likelihood.py`: exponential-kernel Jacod likelihood with the per-pair Ogata recursions of Proposition 8.7 in numerically stable block-cumsum form, plus analytic gradients. `hawkes_calibration/eventtime/estimate.py`: fixed-decay MLE with the $\alpha = e^a$ positivity transform and delta-method standard errors; note the code parameterizes $h(u) = \alpha e^{-\beta u}$, so $\kappa = \alpha/\beta$, $\theta = \beta$ — migrating to the (κ, θ) convention of this document is a pending hygiene item. `hawkes_calibration/eventtime/simulate.py`: Ogata thinning specialized via Lemma 8.14, with baseline-breakpoint bound refresh. `hawkes_calibration/eventtime/covariates.py`: piecewise-constant exp-link baselines with closed-form compensators. `hawkes_calibration/eventtime/lasso.py`: FISTA-based ℓ_1 recovery of sparse excitation matrices (Hansen et al., 2015). `hawkes_calibration/sector_stability.py`: the current discrete face (weekly log-link count model, nonnegative lag excitation, row-sum constraint $\rho_{\max} = 0.95$, `aggregate_excitation` and `excitation_spectral_radius` diagnostics). `hawkes_calibration/models/event_block_hawkes.py`: route (a) in production form — exp-link event-time intensity with signed recency term ($\rho \geq 0$ on the inhibitory feature) and sector-block excitation, concave by Proposition 8.16. *To be written.* A continuous-time sector Hawkes module (`hawkes_calibration/eventtime/sector_hawkes.py`) implementing Protocol 8.13: EM fitter with posterior-parentage output (Proposition 8.10), right-edge exposure corrections, profile-decay grids, and the $\rho(\hat{G})$ enforcement; the exact Dassios-Zhao simulator (Dassios and Zhao, 2013) alongside thinning; and the

Mechanism-B operating-point monitor of Warning [8.18](#) as a first-class simulation diagnostic.

Chapter 9

Nonparametric Intensities: Boosted Trees, Forests, and Small Networks

9.1 Scope and placement

This chapter keeps the estimation principle of Chapters 7 and 8 and relaxes the functional form. The intensity is still fitted by maximum likelihood — the objective is still the point-process log-likelihood of Theorem 6.5 — but the intensity is no longer log-linear in the features. The model is

$$\lambda_i(t) = g(f(w_i(t))), \quad (9.1)$$

where f is a flexible regression function — a gradient-boosted tree ensemble, a random forest, or a small neural network — and g is a fixed positivity link, either $g(z) = e^z$ or the softplus $g(z) = \log(1 + e^z)$ discussed for the parametric case in Section 7.2.1. The feature vector $w_i(t)$ has two blocks:

- (i) the point-in-time covariates $x_i(t)$ of Protocol 2.5 (macro series, sector dummies, calendar terms, firm attributes); and
- (ii) the *engineered history features* of Section 8.6: the time since the firm’s last deal, decayed event counts $H_\theta(t)$ at several fixed half-lives (Equation (8.12)), and recent-deal counts in the firm’s sector.

Nothing in block (ii) is new. These are the same excitation and inhibition covariates that route (a) of Chapter 8 feeds into a log-linear intensity. What changes is the map from features to log-intensity: f may bend, saturate, and interact them freely.

Assumption 9.1 (Trusted, uncensored deal times). Event times are correct at the resolution of the analysis bins, and no event is interval-censored beyond that resolution. Both the bin assignment of each event and the engineered history features of Section 8.6 consume exact times; neither survives date corruption.

Warning 9.2 (This regime has no defense against untrusted dates). The parametric chapters carry an escape route: when dates are unreliable, the interval-censored likelihoods of Chapter 10 replace the exact likelihood. That route depends on closed-form compensators, which Equation (9.1) does not have. If Assumption 9.1 fails — vendor-imputed dates, month-rounded dates, bulk backfills — do not fit this chapter’s models. Fall back to Chapter 10, or coarsen the bins to the trusted resolution and drop every feature built at a finer one.

These models drop the shape assumptions of the parametric chapters and give up most of what those assumptions buy. Their role is the *challenger* (Section 9.8).

9.2 The reduction to Poisson regression

Machine-learning software minimizes losses over rows; it does not maximize point-process likelihoods. One reduction closes the gap: on a fine enough grid, the point-process likelihood *is* a Poisson regression likelihood over (unit, bin) rows with exposure offsets. This is the point-process GLM framing of [Truccolo et al. \(2005\)](#); Theorem 7.10 is its log-linear special case, and the version we need allows an arbitrary f .

Partition $(0, T]$ into bins $B_k = (a_{k-1}, a_k]$, $k = 1, \dots, K$, with widths at most δ . For each unit i and bin k define the count, the exposure, and the bin feature

$$y_{ik} = \#\{m : t_m \in B_k, i_m = i\}, \quad e_{ik} = \int_{B_k} Y_i(u) du, \quad w_{ik} = w_i(a_{k-1}^+). \quad (9.2)$$

Assumption 9.3 (Piecewise-constant features). $w_i(\cdot)$ is constant on every bin B_k : covariates are LOCF series aligned to the grid, and history features are frozen at each bin's left endpoint. Under Equation (9.1) the intensity is then a step function with values $\lambda_{ik} = g(f(w_{ik}))$.

Proposition 9.4 (Binned Poisson reduction). *Under Assumption 9.3:*

(i) *The point-process log-likelihood of Theorem 6.5 equals*

$$\ell(f) = \sum_{i,k} \left(y_{ik} \log \lambda_{ik} - e_{ik} \lambda_{ik} \right), \quad (9.3)$$

exactly, and Equation (9.3) differs from the log-likelihood of independent counts $y_{ik} \sim \text{Poisson}(e_{ik}\lambda_{ik})$ by an additive constant free of f .

(ii) *Consequently, any learner that minimizes the Poisson deviance over the rows Equation (9.2), with $\log e_{ik}$ as an offset (or exposure weight, in the equivalent rate parameterization), performs maximum likelihood for the point process within the class of intensities measurable with respect to the binned features.*

(iii) *If the true feature path updates continuously between grid points (history features decay as in Equation (8.12)) and is frozen at bin left endpoints, then on every bin containing at most one event the log-likelihood error of the frozen model relative to the continuous-feature model is $O(\delta)$, uniformly over intensity paths bounded away from 0 and ∞ .*

Proof. (i) The intensity is constant on each bin, so the compensator integral splits into $\sum_{i,k} \lambda_{ik} \int_{B_k} Y_i = \sum_{i,k} \lambda_{ik} e_{ik}$. Each event of unit i in bin k contributes $\log \lambda_i(t_m) = \log \lambda_{ik}$ whatever its position in the bin, so the event term groups into $\sum_{i,k} y_{ik} \log \lambda_{ik}$. The Poisson count log-likelihood is $\sum_{i,k} [y_{ik} \log(e_{ik}\lambda_{ik}) - e_{ik}\lambda_{ik} - \log y_{ik}!]$, which equals Equation (9.3) plus $\sum_{i,k} [y_{ik} \log e_{ik} - \log y_{ik}!]$, a statistic of the data alone. (ii) The exposure-weighted Poisson deviance is, up to sign and a data-only constant, the Poisson count log-likelihood of (i). (iii) Between events the features move only by decay, so on B_k the continuous intensity satisfies $\sup_{u \in B_k} |\lambda_i(u) - \lambda_{ik}| = O(\delta)$ whenever $g \circ f$ is Lipschitz on the (bounded) range of the features; the compensator error on the bin is therefore at most $e_{ik} \cdot O(\delta) = O(\delta)$. For the event term, with at most one event in the bin, the feature seen by that event at t_m^- involves no other within-bin event, so the frozen value differs from the true value only by within-bin decay drift, again $O(\delta)$; since the intensity is bounded away from zero, $\log$ is Lipschitz on its range and the log-intensity error is $O(\delta)$. A second event in the same bin would break this: the first event's jump in the history features would be missed. The bin width is therefore chosen below the shortest engineered half-life, which also makes two-event bins rare. $\square$

Remark 9.5 (One table, three model families). Proposition 9.4 is the whole interface. Build the (unit, bin) table once and every model in this chapter trains on it, alongside the log-linear GLM of Chapter 7 as the reference. All report the same held-out point-process NLL, so the comparisons of Section 9.8 are exact.

9.3 Boosted trees

Gradient boosting fits f as a sum of regression trees, each tree fitted to the gradient of the loss at the current fit (Friedman, 2001). XGBoost implements the Poisson case directly (Chen and Guestrin, 2016): objective `count:poisson`, log link, and `base_margin` set to $\log e_{ik}$ so that the exposure enters as a fixed offset rather than a feature.

The mechanics are transparent. Write $u_r = f(w_r)$ for the raw score of row $r = (i, k)$, $o_r = \log e_r$ for the offset, and $m_r = \exp(u_r + o_r) = e_r e^{u_r}$ for the predicted mean count. The per-row negative log-likelihood, gradient, and Hessian in the score are

$$\ell_r(u_r) = m_r - y_r (u_r + o_r) + \log y_r!, \quad \frac{\partial \ell_r}{\partial u_r} = m_r - y_r, \quad \frac{\partial^2 \ell_r}{\partial u_r^2} = m_r. \quad (9.4)$$

The gradient is “predicted minus observed” and the Hessian is the predicted mean — the same canonical-link structure as the GLM score Equation (7.3), evaluated tree by tree. Two controls keep the ensemble economically sensible and temporally honest.

Monotone constraints. Trees will fit a wiggle in any direction the training noise suggests. Where economics fixes a direction, we impose it: `monotone_constraints` restricts every split so that f is monotone in the flagged feature. Candidates: intensity nonincreasing in the policy rate for late-stage deal flow; nondecreasing in the sector’s recent-deal count. Features whose true shape we expect to be non-monotone — time since the firm’s last deal, which should show inhibition then recovery — stay unconstrained; discovering that shape is one reason to run this model. Constraints double as regularization along directions where the sign is already known.

Early stopping on a temporal window. The number of boosting rounds is the main capacity knob. It is chosen by early stopping on a validation window strictly later in time than the training rows, matching Section 11.6. Random row splits are forbidden: rows from the same week share macro covariates and sector history features, so a random split leaks the validation period into training and overstates capacity.

Protocol 9.6 (Boosted-tree intensity fit). (a) Build the (unit, bin) table of Equation (9.2) with bin width at most the shortest engineered half-life. (b) Fit XGBoost with objective `count:poisson`, `base_margin` = $\log e_{ik}$, monotone constraints from the signed-feature registry, and early stopping on the last temporal block of the training window. (c) Report held-out point-process NLL via Equation (9.3), PIT calibration, and the time-rescaling diagnostics of Section 9.6. (d) Report uncertainty by the firm-cluster bootstrap of Section 9.6.

Survival-flavored variants exist: accelerated-failure-time objectives in XGBoost handle censored durations directly (Barnwal et al., 2022), and gradient-boosted Cox models are standard (Pölsterl, 2020). We do not use them: the recurrent-event, time-varying-covariate structure of deal flow maps more cleanly onto Proposition 9.4, which keeps the likelihood identical to the rest of the document.

9.4 Random forests

A random forest fits many deep trees on bootstrap resamples and averages them. Two routes adapt forests to the (unit, bin) table. The clean route is Poisson-deviance splitting: grow each tree with the Poisson deviance as split criterion, weight rows by exposure e_{ik} , and predict the rate y/e within each leaf. The pragmatic route, for software without a Poisson criterion, is exposure-weighted least squares on empirical log rates $\log\{(y_{ik} + c)/e_{ik}\}$ with a small stabilizing

constant c ; that is a quasi-likelihood approximation, not maximum likelihood, and must be flagged as such in any NLL comparison.

An honest account of what averaging does. A forest prediction is an average of training-set leaf values, so the fit never leaves the range of what it saw: forests do not extrapolate. When a macro covariate exits its training range, the response goes flat. That is a defect (a genuine trend continues and the forest will not follow it) and a defense (the log-linear GLM extrapolates *exponentially* in the same situation; Warning 7.9), and flat is often the safer error. The forest is therefore the stability-oriented challenger, the boosted ensemble the capacity-oriented one, and disagreement between them outside the training support is itself a diagnostic.

Out-of-bag evaluation scores each row with the trees that did not see it, an almost-free deviance estimate. It is a screening tool only: OOB rows are interleaved in time with training rows, so OOB deviance inherits the temporal leakage that Section 11.6 exists to prevent. Cross-family model selection still runs on the temporal splits; OOB tunes forest internals (depth, feature subsampling) before that comparison.

9.5 Small neural networks

Neural networks enter twice, once cheaply and once exactly.

Route (i): the binned Poisson route. A small multilayer perceptron replaces the tree ensemble in Proposition 9.4: input w_{ik} , a few narrow hidden layers, and a softplus output head

$$\lambda_{ik} = \log(1 + e^{z_{ik}}), \quad z_{ik} = \text{network}(w_{ik}), \quad (9.5)$$

trained by minimizing the same exposure-weighted Poisson deviance. The softplus head keeps the intensity positive without the exponential amplification that Warning 8.18 warns about. Everything from the tree discussion transfers: temporal early stopping, the shared table, the same held-out NLL. The network adds smooth interpolation and costs optimization noise.

Route (ii): the exact continuous-time likelihood. Nothing forces binning. The exact objective is

$$\ell(f) = \sum_m \log g(f(w_{i_m}(t_m^-))) - \sum_i \int_0^T Y_i(u) g(f(w_i(u))) du, \quad (9.6)$$

and it is computable whenever the integral is. Between events the engineered history features decay exponentially (Equation (8.12)) and the covariates are piecewise constant, so $w_i(u)$ follows a known closed-form curve on each inter-event interval. For log-linear f the integral then has closed pieces — the parametric machinery of Chapter 8 — but for a generic network $g(f(\cdot))$ along a known curve has no closed integral, and each inter-event piece is evaluated by numerical quadrature (a few points per piece suffice when the piece is shorter than the fastest half-life). Automatic differentiation passes through the quadrature, so training is standard.

Route (ii) is the door to neural temporal point processes, which replace the engineered features themselves with a learned history embedding: recurrent embeddings with an exponential-form intensity (Du et al., 2016), continuous-time recurrent intensities (Mei and Eisner, 2017), networks for the compensator whose derivative gives the intensity exactly (Omi et al., 2019), and intensity-free models of the inter-event distribution (Shchur et al., 2020). These are the fully flexible end of the axis this chapter walks. We do not start there, for three plain reasons. Data volume: a typical firm contributes a handful of deals, and learned embeddings need long streams or aggressive pooling that reintroduces, in opaque form, the heterogeneity questions of Section 8.7. Interpretability: the engineered features are the economic quantities we want

shape estimates for — cooldown, momentum, sector spillover — and an embedding answers none of them directly. Calibration risk: flexible intensity models fail by confident misallocation of intensity, and diagnosing that inside an embedding is much harder than inspecting a feature’s partial response. Neural TPPs are the recorded escalation if the challenger wins and its wins trace to history representation rather than covariate nonlinearity.

9.6 What the regime keeps and what it loses

Table 9.1 states the trade explicitly.

Property	Status	Consequence or replacement
Likelihood objective	kept	Proposition 9.4: the same point-process NLL as Chapters 7 and 8, so held-out comparisons across regimes are exact
Temporal evaluation	kept	the protocol of Section 11.6 applies unchanged; early stopping and tuning obey it
Time-rescaling diagnostics	kept	Theorem 6.16 holds for <i>any</i> fitted intensity; see below
Simulation	kept	thinning (Lewis and Shedler, 1979) with piecewise-constant fitted rates; bound refresh at bin edges
Branching interpretation	lost	no κ , no endogenous share, no parent attribution (Theorem 8.2 does not apply)
Closed-form compensators	lost	compensators are bin sums or quadrature; the interval-censored route of Chapter 10 is unavailable
Stability theory	lost	no spectral-radius certificate; only bounded features and simulation monitoring (Warning 8.18 discipline)
Sandwich standard errors	lost	no low-dimensional score; use the firm-cluster bootstrap (Efron and Tibshirani, 1994)
Frailty-vs-contagion reasoning	weakened	see Warning 9.7 and Section 8.7

Table 9.1: What the nonparametric-intensity regime keeps from the parametric chapters and what it gives up.

Two rows deserve expansion. *Time rescaling survives everything.* The time-rescaling theorem (Section 6.7) makes no parametric assumption: if the fitted intensity is right, the rescaled gaps are unit exponential, whether the intensity came from a GLM, a Hawkes kernel, or a thousand trees (Brown et al., 2002; Ogata, 1988). The fitted compensator here is a running sum of $\lambda_{ik}e_{ik}$ over bins, evaluated with frozen parameters on held-out windows as in Chapter 11; this diagnostic holds the flexible models to the same standard as the parametric ones. *Uncertainty comes from the bootstrap.* The sandwich variance of Chapter 6 needs a score and a Hessian in a fixed finite-dimensional parameter, which a tree ensemble does not have. The replacement is the cluster bootstrap (Efron and Tibshirani, 1994): resample *firms* with replacement, keeping each firm’s whole event history and exposure intact so within-firm dependence is never broken; refit; read uncertainty off the resampled predictions. Resampling rows would pretend the (unit, bin) cells are independent, which they are not.

9.7 Hawkes-flavored behavior without a Hawkes model

In this regime, excitation and inhibition enter through one door only: the engineered history features. A positive learned response to the decayed own-sector count looks like self-excitation; a depressed response at small time-since-last-deal looks like inhibition. The flexible f can shape these responses — thresholds, saturation, interactions with the macro state — which the log-linear form cannot. That freedom is the point of the chapter, and it is also where the causal reading breaks.

Warning 9.7 (Learned excitation is not contagion). A latent firm-quality or sector-attractiveness factor that the covariates miss makes past deal counts a proxy for the missing factor. A flexible f finds and uses that proxy more effectively than a linear model: it can carve the feature space into regions where the proxy works best and imitate self-excitation with no contagion present. This is the confound of Section 8.7 with fewer restraints. The parametric chapter could at least bound the damage — branching estimates are upper bounds, parent-attribution timescales provide a probe — and none of that structure exists here: no branching ratio, no genealogy, no timescale decomposition. Any “deals cause deals” claim from this regime is therefore strictly weaker than the corresponding claim from the parametric model. The reporting rule: nonparametric fits make *predictive* statements about history features, never mechanism statements; if the mechanism question matters, the parametric layer answers it with the diagnostics of Chapter 8.

9.8 Evaluation and the challenger role

Everything runs inside the harness of Section 11.1: the same temporal splits, the same held-out point-process NLL (identical by Proposition 9.4), the same calibration reports — PIT histograms for counts and proper scoring rules for predictive distributions (Gneiting and Raftery, 2007). Two comparisons are designated: the log-linear GLM of Chapter 7 *on the same feature table*, so any gap is attributable to functional form rather than information; and the parametric Hawkes layer of Chapter 8, so the value of genuine event-time structure is measured against flexible use of engineered summaries.

The logic of the comparison is asymmetric. The nonparametric model is not a candidate for the production intensity; it is the *challenger*, and its job is to expose structure the log-linear form misses. If it beats the log-linear GLM materially on held-out NLL and calibration, the conclusion is not “deploy the trees”; it is that the log-linear intensity is missing an interaction or a nonlinearity, and the follow-up is to find it: partial-dependence and interaction diagnostics on the fitted ensemble point at candidate terms, which are added to the GLM explicitly and re-tested. If the challenger cannot beat the GLM, the log-linear form is adequate at the current data size and the parametric stack keeps its advantages.

Design decision (D-NP.1 — the nonparametric intensity is the designated challenger).

The models of this chapter are fitted in every evaluation campaign, on the shared (unit, bin) table, under the harness of Section 11.1, next to the log-linear GLM and the parametric Hawkes layer. Their role is challenger, not candidate: a material win over the log-linear intensity on held-out NLL and calibration is treated as evidence of missing structure, triggering a search (interaction and nonlinearity extraction from the ensemble, then explicit GLM terms and a re-run), not a deployment. Rationale: the challenger keeps the likelihood and the evaluation protocol (Proposition 9.4), so its wins isolate functional form; the interpretability, stability certificates, and inference that production requires

live in the parametric stack (Table 9.1). Reversal trigger: if the challenger’s material win persists across two consecutive campaigns *after* the extracted terms are added to the GLM, the deficit is structural, and promotion of a constrained tree or network intensity to production candidate is put on the table, carrying this chapter’s bootstrap-inference and monitoring obligations with it.

9.9 Defect declaration

Per decision D2.1 (Chapter 2), against the ledger of Table 2.1.

(C1) *Right censoring: corrected exactly.* The exposure e_{ik} integrates the at-risk indicator to the end of observation, and Proposition 9.4 carries the survival factor through the bin sums, as in Chapter 7. Open spells must be kept; the zero-count rows are the censored exposure.

(C2) *Left truncation: corrected mechanically.* Exposure starts at entry. The estimand remains conditional on entry into observation; Chapter 15 owns the unconditional question.

(C3) *Unlabeled exits: vulnerable, downward.* Dead firms kept at risk contribute exposure and no events, deflating fitted intensities. The flexible f adds a twist: it can learn that long-quiet firms have low rates and partially *mask* the defect in-sample without fixing it — the phantom exposure is still wrong. Treatments stay where they were: exposure windows and mover–stayer structure (Chapters 7 and 13), ranking-layer mitigations (Chapter 14).

(C4) *Untrusted times: vulnerable by assumption.* This regime consumes exact times twice, in the bin assignment and in the engineered history features, and has no interval-censored fallback (Warning 9.2). Assumption 9.1 is a precondition, not a robustness claim; when it fails, Chapter 10 is the route.

(C5) *Stale covariates: vulnerable, with a tree-specific hazard.* Stale features attenuate learned responses as in Chapter 12. Trees add a hazard the GLM lacks: staleness *patterns* (reporting gaps, vendor refresh cycles) are learnable signals, and a tree will exploit them if they correlate with outcomes. The guards are the point-in-time protocol Protocol 2.5 and explicit staleness-age features, so the dependence is visible rather than implicit.

(C6) *Out-of-universe deals: sector features complete, firm intensities thinned.* Sector-level history features are built from the complete deal stream and remain valid. Firm-level intensities miss unseen competitors exactly as in the parametric chapters; Chapter 15 owns the correction.

Code hooks.

To be written. `hawkes_calibration/nonparam.py`: (1) the (unit, bin) table builder of Equation (9.2), shared with the GLM fitters so that Proposition 9.4(ii) holds by construction; (2) the XGBoost wrapper of Protocol 9.6 — objective `count:poisson`, `base_margin` = $\log e_{ik}$, a `monotone_constraints` map read from the signed-feature registry, temporal early stopping; (3) a forest wrapper with Poisson-deviance splitting, exposure weights, and OOB reporting (screening only); (4) a small-network trainer with the softplus head of Equation (9.5) and exposure-weighted Poisson loss; (5) a time-rescaling adapter that consumes any `predict_rate(unit, bin)` function and emits the diagnostics of Chapter 11; (6) the firm-cluster bootstrap driver. The `xgboost` objective configuration for (2) is a checked-in parameter file, so challenger runs are reproducible across campaigns.

Chapter 10

Untrusted and Censored Deal Times

10.1 What the dates are, and what we choose to trust

Chapter 2 recorded, as defect (C4), that the event times t_m of Definition 2.1 are not delivered to us. What PitchBook delivers is a recorded date $\hat{t}_m$ produced by an editorial process: announced versus closed dates that differ by days, back-filled records whose dates were reconstructed after the fact (often rounded to a month or quarter boundary), and duplicate rows collapsed by the hygiene rules of Protocol 2.5. Every continuous-time likelihood in Chapters 7 and 8 consumes t_m as if it were exact; this chapter decides what to do when it is not. The nonparametric intensity models of Chapter 9 are valid only on trusted, uncensored dates and have no interval-censored counterpart here; when dates cannot be trusted, that regime must fall back to the binned models of this chapter. Our thesis, defended throughout: *the correct response to untrusted positions is to choose the observation resolution honestly, and to use likelihoods that consume only what we trust — counts of events on intervals, never positions within them.* The count is an editorial fact we can audit; the timestamp is an editorial choice we cannot.

We formalize the corruption with three assumptions, kept deliberately weak.

Assumption 10.1 (Bounded date jitter). Each deal m has a true time t_m and a recorded time $\hat{t}_m = t_m + \varepsilon_m$ with $|\varepsilon_m| \leq \delta_m \leq \delta$ for a known trust radius δ . Nothing else is assumed about the errors ε_m : they may be dependent on each other, on the process, and on the marks; they need not be centered. The bound δ is an output of the date audit (Section 10.8), not a modeling choice.

Assumption 10.2 (Recording conventions). The recorded date decomposes as $\hat{t}_m = t_m + b(r_m) + \varepsilon_m$, where $r_m \in \{\text{closed, announced, backfilled}\}$ is an observable record type, $b(\cdot)$ is a systematic convention offset with $b(\text{closed}) = 0$, and backfilled records may additionally exhibit *heaping*: rounding to calendar boundaries, visible as excess mass on month and quarter starts. After auditing to a single convention (subtracting an estimated b , flagging heaped records), the residual error satisfies Assumption 10.1 with a type-specific δ_m .

Assumption 10.3 (Count completeness). After the deduplication of Protocol 2.5, every in-window deal of the tracked process appears exactly once in the record. Corruption lives in the *position* $\hat{t}_m$, never in the *multiplicity*. (Thinning of the record — events missing entirely — is a different defect: (C6) for out-of-universe attribution, (C3) for unobserved exits. This chapter is orthogonal to both; see Section 10.8.)

The deliberate weakness of Assumption 10.1 — adversarial, possibly dependent errors — is what makes the induced modeling choice sharp. If the ε_m were i.i.d. with a known density we could deconvolve; we refuse to assume that, because announced-versus-closed offsets are systematic, not noise. Under adversarial bounded jitter the only event-level statements that

survive are set-membership statements, and the following proposition is the complete list of what we keep.

Proposition 10.4 (What bounded jitter leaves invariant). *Let $0 = o_0 < o_1 < \dots < o_J = T$ be a partition into bins $I_j = (o_{j-1}, o_j]$ of width $\Delta = o_j - o_{j-1}$, let $C_j = \#\{m : t_m \in I_j\}$ be the true bin counts and $\hat{C}_j = \#\{m : \hat{t}_m \in I_j\}$ the observed ones. Under Assumptions 10.1 and 10.3:*

- (i) *every event observed in bin j has its true time in the dilated bin $(o_{j-1} - \delta, o_j + \delta]$;*
- (ii) *$|\hat{C}_j - C_j| \leq \#\{m : \min_k |t_m - o_k| \leq \delta\}$ restricted to the two boundaries of I_j : only events within δ of a bin boundary can be miscounted, and only into an adjacent bin;*
- (iii) *if the process has locally bounded mean rate $\sup_t \mathbb{E}[\lambda(t)] \leq \bar{\lambda}_{\text{sup}}$, then $\mathbb{E}|\hat{C}_j - C_j| \leq 4\delta \bar{\lambda}_{\text{sup}}$, so the expected miscounted fraction per bin is of order $4\delta/\Delta$;*
- (iv) *counts over unions of consecutive bins are more robust still: transfers across interior boundaries cancel, leaving only the two outer boundaries.*

Proof. (i) is immediate from $|\hat{t}_m - t_m| \leq \delta$. (ii): an event with $t_m \in I_j$ and $\hat{t}_m \notin I_j$ requires $\hat{t}_m$ on the far side of o_{j-1} or o_j , hence $|t_m - o_k| \leq \delta$ for a boundary of I_j ; conversely for events entering I_j . (iii): by (ii) and the mean-measure bound $\mathbb{E} \#\{m : t_m \in A\} = \int_A \mathbb{E}[\lambda] \leq |A| \bar{\lambda}_{\text{sup}}$, the expected number of events within δ of a fixed boundary is at most $2\delta \bar{\lambda}_{\text{sup}}$; each bin has two boundaries. Comparing with $\mathbb{E}C_j \asymp \Delta \bar{\lambda}$ gives the fraction. (iv): an event transferred from I_j to I_{j+1} leaves the count of $I_j \cup I_{j+1}$ unchanged. $\square$

The induced choice is now forced. Any inference that reads event *positions* at scales comparable to δ is reading editorial noise; any inference that reads *counts* on bins of width $\Delta \gg \delta$ is reading the process, up to a boundary-transfer error of controlled order $4\delta/\Delta$ that vanishes as the resolution coarsens. We therefore commit to modeling at resolution no finer than the trust radius — in practice $\Delta \geq$ several multiples of δ — at which point *the data is, by construction, interval-censored count data*, and the rest of this chapter builds the likelihoods that consume exactly that and nothing more. One reassurance: the mark factor of Equation (1.4) conditions on a deal occurring and compares candidates within a risk set, so it consumes dates only through risk-set membership and coarse ordering; jitter at the scale of days is invisible to it (Chapter 14 quantifies this). It is the ground process — the timing factor — that absorbs (C4), which is why this chapter is about intensities.

10.2 Right censoring with history-dependent intensities

Before interval censoring, we settle right censoring — defect (C1) — for the Hawkes case, because a doubt naturally arises there that does not arise for Poisson models. For an NHPP the censored-window contribution $\exp(-\int \lambda)$ involves a deterministic function, and the argument of Chapter 7 is elementary. For a Hawkes process the intensity is random and feeds on the history: one might worry that the survival factor for the window $(t_{\text{last}}, T]$ secretly depends on events we have not seen, or that the ABGK independent-censoring argument breaks. It does not, and the reason — predictability — is worth stating precisely, because it is the same property that will *fail* to rescue us under interval censoring.

Assumption 10.5 (Independent, noninformative censoring). Let N be a counting process with $\mathcal{F}_t$ -intensity $Y(t)\lambda(t; \vartheta)$ as in Definition 2.3. There is an enlarged filtration $\mathcal{G}_t \supseteq \mathcal{F}_t$ carrying the censoring information such that (i) the $\mathcal{G}_t$ -intensity of N is still $Y(t)\lambda(t; \vartheta)$; (ii) the

censoring indicator $C(t) \in \{0, 1\}$ (e.g. $C(t) = \mathbf{1}\{t \leq \tau\}$ for a censoring time τ) is $\mathcal{G}_t$ -predictable; (iii) the censoring mechanism's factor in the full likelihood does not involve ϑ (Andersen et al., 1993, III.2). Administrative censoring at the fixed date T (June 2024) satisfies all three with $\mathcal{G} = \mathcal{F}$ and $C(t) = \mathbf{1}\{t \leq T\}$.

Theorem 10.6 (Censored-window validity for history-dependent intensities). *Under Assumption 10.5, with λ any $\mathcal{G}$ -predictable intensity — in particular the Hawkes intensity $\lambda(t) = \mu(t) + \sum_{t_m < t} h(t - t_m)$, which depends on the realized history:*

- (i) *the observed process $N^o(t) = \int_0^t C(u) dN(u)$ has $\mathcal{G}_t$ -intensity $C(t) Y(t) \lambda(t; \vartheta)$;*
- (ii) *the likelihood of the observed data factorizes as the Jacod likelihood Equation (2.3) evaluated with intensity $CY\lambda$, times a factor free of ϑ ; the first factor is therefore a genuine partial likelihood, and a censored spell contributes exactly its survival factor $\exp(-\int_{\text{spell}} \lambda)$ — no term dropped, no term invented;*
- (iii) *the score of this partial likelihood is a $\mathcal{G}$ -martingale evaluated at T , so the estimation theory of Chapter 6 (unbiased score, information identities, asymptotic normality) applies unchanged.*

Proof. (i) $M = N - \int Y\lambda du$ is a local $\mathcal{G}$ -martingale by Assumption 10.5(i). The process C is bounded and $\mathcal{G}$ -predictable by (ii), so the stochastic integral $\int_0^t C dM = N^o(t) - \int_0^t CY\lambda du$ is again a local $\mathcal{G}$ -martingale; since $\int CY\lambda du$ is predictable and nondecreasing, it is the compensator of N^o , i.e. $CY\lambda$ is the intensity of the observed process. Observe what the proof used: boundedness and predictability of C , predictability of λ — nowhere that λ is deterministic. History dependence is immaterial because predictability is preserved under enlargement by Assumption 10.5(i); this is the ABGK III.2 argument verbatim. (ii) is the standard likelihood factorization for counting processes with the intensity from (i) (Andersen et al., 1993, II.7, III.2): the full likelihood is the product integral of the observed-jump and survival factors for N^o times the censoring factor, and (iii) removes the latter from the maximization. (iii) follows because the score is $\int_0^T \partial_\vartheta \log \lambda (dN^o - CY\lambda du)$, a stochastic integral of a predictable process against the martingale of (i). $\square$

What, then, is genuinely different from the Poisson case? Only this: the survival factor is now *random* — but it is a functional of the *observed* data alone. The next proposition makes the spec's “depends on the unobserved post-censoring events only through — nothing” precise.

Proposition 10.7 (Locality of the compensator). *Let $\tau \leq T$ be the censoring time of Assumption 10.5 and let λ be the Hawkes intensity with causal kernel (h supported on $(0, \infty)$). For every $t \leq \tau$, $\lambda(t)$ is measurable with respect to $\mathcal{H}_{t-}$, the history of events strictly before t ; consequently the integrated compensator $\Lambda(\tau) = \int_0^\tau \lambda(u) du$ is measurable with respect to the observed history $\mathcal{H}_\tau$, and the survival factor $e^{-\Lambda(\tau)}$ is a deterministic functional of the data we possess. Events in (τ, ∞) — observed or not — enter neither the event terms nor the compensator of Theorem 10.6(ii).*

Proof. $\lambda(t) = \mu(t) + \sum_{t_m < t} h(t - t_m)$ involves only event times $t_m < t \leq \tau$, all of which are uncensored under Assumption 10.3; measurability of the integral follows from joint measurability of the predictable path $u \mapsto \lambda(u)$ on $(0, \tau]$ and Tonelli. The claim is exactly the statement that the compensator up to a stopping time is determined by the pre-stopping-time path — the defining property of predictability, which the Hawkes feedback respects because the kernel is causal. $\square$

Warning 10.8 (Completeness of the pre-censoring record is now load-bearing). For a Poisson model, an event missing from the record costs exactly its own two terms (one jump term, one

sliver of compensator); the rest of the likelihood is untouched. For a Hawkes model, a missing past event corrupts the intensity *path* at every later time: the reconstructed $\lambda(t)$ omits a kernel contribution $h(t - t_m)$, so every subsequent event term and the entire survival factor are evaluated at the wrong intensity. The bias has no universal sign — the fitted baseline μ and branching scale κ reappportion the missing excitation — which is precisely why Assumption 10.3 (counts complete, positions untrusted) is the boundary of this chapter’s mandate: it protects the history that the intensity feeds on. Where completeness itself fails, we are in (C3)/(C6) territory and no date discipline repairs it.

10.3 Interval censoring for Poisson models: exactness

Coarsening event times to bin counts is interval censoring by construction. For the Poisson rung of the ladder this costs *no approximation at all* — only resolution. We restate the binned-likelihood theorem of Chapter 7 in the form needed here, and prove it by a route chosen deliberately: the same configuration integral, evaluated for a history-dependent intensity in the next section, is exactly where tractability fails.

Proposition 10.9 (Binned NHPP likelihood is exact). *Let N be an NHPP with deterministic, locally integrable intensity $\lambda(t; \vartheta)$ on $(0, T]$, and let $I_1, \dots, I_J$ partition $(0, T]$ with $\Lambda_j(\vartheta) = \int_{I_j} \lambda(u; \vartheta) du$. Then the bin counts $C_1, \dots, C_J$ are independent with $C_j \sim \text{Poisson}(\Lambda_j)$, and the log-likelihood of the interval-censored data,*

$$\ell_{\text{IC}}(\vartheta) = \sum_{j=1}^J \left(C_j \log \Lambda_j(\vartheta) - \Lambda_j(\vartheta) - \log C_j! \right), \quad (10.1)$$

is the exact likelihood of what was observed — not an approximation to the event-time likelihood but the correct likelihood of the coarsened data. Moreover, if $\lambda(\cdot; \vartheta)$ is constant on each bin (the weekly piecewise-constant models of Chapter 7), the counts are sufficient for ϑ and coarsening loses nothing; in general the loss is confined to the within-bin shape of λ .

Proof. Start from the density of the ordered event times under the full model: for n events at $0 < t_1 < \dots < t_n \leq T$, $f(t_{1:n}) = \prod_{m=1}^n \lambda(t_m) e^{-\int_0^T \lambda(u) du}$. The probability of the count vector $C_{1:J}$ (with $n = \sum_j C_j$) is the integral of f over the configuration region $\mathcal{A}(C_{1:J}) = \{0 < t_1 < \dots < t_n \leq T : \#\{m : t_m \in I_j\} = C_j \forall j\}$. Because λ is deterministic, the integrand is a product of factors each depending on one coordinate only, so the ordered integral factorizes across bins, and within bin j the ordered integral of $\prod \lambda$ over the C_j -dimensional ordered simplex equals $\Lambda_j^{C_j} / C_j!$. Hence $\mathbb{P}(C_{1:J}) = \prod_j (\Lambda_j^{C_j} / C_j!) e^{-\Lambda_j}$, which is Equation (10.1) and simultaneously the independence and Poisson-marginal claims. Sufficiency under piecewise-constant λ is the factorization criterion applied to Equation (10.1): the likelihood depends on the data only through $C_{1:J}$. $\square$

Corollary 10.10 (For NHPP calibration, untrusted dates cost only resolution). *Under Assumptions 10.1 and 10.3, fitting any NHPP of Chapter 7 on bin counts at resolution $\Delta \gg \delta$ uses an exact likelihood (Proposition 10.9) applied to data whose corruption is bounded by Proposition 10.4(iii). No approximation enters anywhere; the entire price of (C4) is the inability to resolve λ within bins — a price we were going to pay anyway by modeling weekly.*

This is the sense in which the weekly sector GLM of the current two-layer architecture (`hawkes_calibration/sector_stability.py`) is already fully (C4)-compliant: it never consumed positions in the first place. The classical ancestor of likelihoods under grouped and censored observation is [Turnbull \(1976\)](#); what is distinctive in our setting is not grouping per se but the interaction of grouping with *dependence*, which is the next section’s subject.

10.4 Interval censoring for Hawkes: where exactness fails

Run the proof of Proposition 10.9 again with the Hawkes intensity. The density of the ordered times is still the Jacod product, so the exact interval-censored likelihood is still a configuration integral:

$$L_{\text{IC}}(\vartheta) = \int_{\mathcal{A}(C_{1:j})} \prod_{m=1}^n \lambda(t_m \mid \mathcal{H}_{t_m^-}; \vartheta) \exp\left(-\int_0^T \lambda(u \mid \mathcal{H}_{u^-}; \vartheta) du\right) dt_1 \cdots dt_n, \quad (10.2)$$

with $\lambda(t \mid \mathcal{H}_{t^-}) = \mu(t) + \sum_{t_m < t} h(t - t_m)$ and $n = \sum_j C_j$. But now the factorization step fails, for the precise reason the model was chosen: the intensity at t_m depends on the *positions* of all earlier events, including those in earlier bins and those in the same bin. The integrand does not separate across bins, the within-bin integrals are not simplex moments of a fixed function, and Equation (10.2) is an n -dimensional integral — at the scale of this dataset, dimension $\sim 5 \times 10^4$ — over a combinatorially-constrained region, with no product structure to exploit. Data augmentation (imputing positions by MCMC or EM over Equation (10.2)) is possible in principle and is how the Bayesian literature treats small problems (Rasmussen, 2013); at our scale, and for a likelihood that must be refit across sectors, splits, and validation campaigns, it is not a workable primary route. The likelihood requires event positions inside bins *because intensity feeds back on history*; interval censoring withholds exactly that. Something must give, and the honest thing to give up is the randomness of the intensity — replacing it by its mean — rather than the honesty of the resolution.

10.5 The mean-intensity route

This section is the chapter’s centerpiece: the *mean behavior Poisson process* (MBPP) of Rizoiu et al. (2022), implemented in `hawkes_calibration/mbpp/`. The idea is to fit, instead of the intractable interval-censored Hawkes likelihood, the exact interval-censored likelihood of the NHPP whose deterministic intensity is the *expected* Hawkes intensity. The parameters (μ, κ, θ) are shared, the first moments are matched exactly, and the price — discarded dependence and overdispersion — is stated and bounded.

10.5.1 The mean intensity and its Volterra equation

Theorem 10.11 (Renewal equation for the mean intensity). *Let N be a Hawkes process on $[0, T]$ with intensity $\lambda(t) = \mu(t) + \int_{(0,t)} h(t-u) dN(u)$, where μ is deterministic, locally bounded, and $h \geq 0$ with $\int_0^\infty h = \kappa < 1$. Then the mean intensity $\bar{\lambda}(t) := \mathbb{E}[\lambda(t)]$ is finite and satisfies the linear Volterra equation of the second kind*

$$\bar{\lambda}(t) = \mu(t) + \int_0^t h(t-u) \bar{\lambda}(u) du. \quad (10.3)$$

Proof. Step 1 (finiteness). By the cluster representation of Chapter 8 (Hawkes and Oakes, 1974), N is the superposition of immigrant events arriving with rate μ and, per event, an independent Galton–Watson cascade with offspring mean $\kappa < 1$; the expected total progeny per immigrant is $1/(1-\kappa)$, so $\mathbb{E}N(t) \leq \int_0^t \mu(u) du / (1-\kappa) < \infty$, and $\bar{\lambda}(t) \leq \mu(t) + \kappa \theta \mathbb{E}N(t) < \infty$ for the exponential kernel (and similarly for any bounded kernel).

Step 2 (martingale property of the compensator). For any nonnegative predictable integrand H , the defining property of the intensity gives the compensator formula $\mathbb{E} \int_0^t H(u) dN(u) = \mathbb{E} \int_0^t H(u) \lambda(u) du$, both sides possibly infinite (Andersen et al., 1993, II.4); this is the statement $\mathbb{E}[dN(u)] = \lambda(u) du$ made rigorous. Fix t and apply it to the deterministic (hence predictable)

integrand $H(u) = h(t - u)\mathbf{1}\{u < t\}$:

$$\mathbb{E} \int_{(0,t)} h(t - u) dN(u) = \mathbb{E} \int_0^t h(t - u) \lambda(u) du = \int_0^t h(t - u) \bar{\lambda}(u) du,$$

the last step by Tonelli (nonnegative integrand) and Step 1. Taking expectations through $\lambda(t) = \mu(t) + \int_{(0,t)} h(t - u) dN(u)$ yields Equation (10.3). $\square$

Note what was used: only the compensator formula — the same martingale structure that powers every likelihood in this document — plus subcriticality for finiteness. No stationarity, no exponential form, no constant baseline.

Theorem 10.12 (Existence, uniqueness, Neumann series). *Under the hypotheses of Theorem 10.11, Equation (10.3) has exactly one locally bounded solution on $[0, T]$, namely*

$$\bar{\lambda} = \mu + R * \mu, \quad R := \sum_{n \geq 1} h^{*n}, \quad \int_0^\infty R = \frac{\kappa}{1 - \kappa}, \quad (10.4)$$

where h^{*n} is the n -fold convolution power. Picard iteration $\bar{\lambda}_{k+1} = \mu + h * \bar{\lambda}_k$, $\bar{\lambda}_0 = \mu$, converges to it geometrically: $\sup_{[0,T]} |\bar{\lambda}_k - \bar{\lambda}| \leq \kappa^{k+1} (1 - \kappa)^{-1} \sup_{[0,T]} \mu$.

Proof sketch. The map $\Phi(f) = \mu + h * f$ on $(L^\infty[0, T], \|\cdot\|_\infty)$ satisfies $\|\Phi(f) - \Phi(g)\|_\infty \leq \|h\|_{L^1} \|f - g\|_\infty = \kappa \|f - g\|_\infty$, a contraction since $\kappa < 1$; Banach's fixed point theorem gives existence, uniqueness among bounded solutions, and the geometric rate. Unrolling the iteration gives the Neumann series Equation (10.4), whose mass identity follows from $\|h^{*n}\|_{L^1} = \kappa^n$ and geometric summation. The general theory of Volterra equations — including the resolvent formalism, L^1 well-posedness, and what survives when $\kappa \geq 1$ locally — is [Gripenberg et al. \(1990\)](#). When $\kappa \uparrow 1$ the geometric rate degrades as $(1 - \kappa)^{-1}$ and at criticality the series diverges: the mean-intensity route inherits exactly the stability boundary of the process itself. $\square$

The resolvent R has a probabilistic reading that connects back to Chapter 8: h^{*n} is the arrival density of n -th generation offspring, so Equation (10.4) says the mean intensity is the immigrant rate plus the expected arrival rate of descendants of all generations — the cluster representation, averaged.

10.5.2 Exponential kernel: closed form

Proposition 10.13 (Closed form for the exponential kernel). *Let $h(t) = \kappa\theta e^{-\theta t}$ and $\mu(t) \equiv \mu$ constant, with $\kappa \in (0, 1)$, $\theta > 0$. Then*

$$\bar{\lambda}(t) = \mu + \frac{\kappa\theta\mu}{(1 - \kappa)\theta} (1 - e^{-(1 - \kappa)\theta t}) = \mu + \frac{\kappa\mu}{1 - \kappa} (1 - e^{-(1 - \kappa)\theta t}), \quad (10.5)$$

so $\bar{\lambda}$ relaxes from $\bar{\lambda}(0) = \mu$ to the stationary level $\bar{\lambda}(\infty) = \mu/(1 - \kappa)$ at the effective relaxation rate $r = (1 - \kappa)\theta$. Equivalently, the resolvent is itself exponential, $R(t) = \kappa\theta e^{-(1 - \kappa)\theta t}$, and for time-varying $\mu(\cdot)$ the solution is the single convolution $\bar{\lambda} = \mu + R * \mu$; the integrated version $\bar{\Lambda}(t) = \int_0^t \bar{\lambda}$ obeys the same equation, $\bar{\Lambda} = \int_0^t \mu + h * \bar{\Lambda}$.

Proof. Write $\bar{\lambda}(t) = \mu + g(t)$ with $g(t) = \int_0^t \kappa\theta e^{-\theta(t-u)} \bar{\lambda}(u) du$. The kernel's exponential form makes g differentiable with

$$g'(t) = \kappa\theta \bar{\lambda}(t) - \theta g(t) = \kappa\theta(\mu + g(t)) - \theta g(t) = \kappa\theta\mu - (1 - \kappa)\theta g(t),$$

a linear ODE with $g(0) = 0$, solved by $g(t) = \frac{\kappa\theta\mu}{(1-\kappa)\theta}(1 - e^{-(1-\kappa)\theta t})$; adding μ gives Equation (10.5). *Sanity checks.* At $t = 0$ the excitation has not started: $\bar{\lambda}(0) = \mu$. As $t \rightarrow \infty$, $\bar{\lambda} \rightarrow \mu + \kappa\mu/(1-\kappa) = \mu/(1-\kappa)$ — the immigrant rate times the mean cluster size, agreeing with the cluster representation and with the stationary Hawkes mean of Chapter 8; and substituting the constant $\mu/(1-\kappa)$ into Equation (10.3) verifies it is the fixed point. For the resolvent claim, either solve $R = h + h * R$ by the same ODE argument or sum the Erlang series $h^{*n}(t) = \kappa^n \theta^n t^{n-1} e^{-\theta t} / (n-1)!$ directly; then $\bar{\lambda} = \mu + R * \mu$ is Equation (10.4). The equation for $\bar{\Lambda}$ follows by integrating Equation (10.3) and Fubini. $\square$

The repository implements exactly this structure (`hawkes_calibration/mbpp/core.py`): the impulse response $R(t) = \kappa\theta e^{(\kappa-1)\theta t}$ is Eq. (14) of Rizoïu et al. (2022); their Theorems 1–2 and Corollary 3 give $\bar{\lambda} = s + R * s$ and $\bar{\Lambda} = S + R * S$ in closed form for every exogenous input $s(\cdot)$ in `hawkes_calibration/mbpp/exogenous.py` (constant, rectangle, piecewise-constant, sinusoidal, impulse trains, and covariate-driven baselines), with a numerical finite-convolution solver for kernels without closed forms (power-law). The relaxation rate $r = (1-\kappa)\theta$ of Proposition 10.13 will reappear twice: as the decay rate of cross-bin count correlation, and as the parameter whose product with the bin width governs the κ - θ ridge.

10.5.3 The MBPP working likelihood, and precisely what it discards

Definition 10.14 (Mean behavior Poisson process and IC likelihood). The MBPP associated with a Hawkes model (μ, h) is the NHPP with deterministic intensity $\bar{\lambda}(\cdot)$ solving Equation (10.3). Its interval-censored log-likelihood on the partition of Proposition 10.4 is the exact Poisson form Equation (10.1) with means $\bar{\Lambda}_j(\vartheta) = \int_{I_j} \bar{\lambda}(u; \vartheta) du$:

$$\ell_{\text{MBPP}}(\vartheta) = \sum_{j=1}^J \left(C_j \log \bar{\Lambda}_j(\vartheta) - \bar{\Lambda}_j(\vartheta) \right) + \text{const}, \quad (10.6)$$

maximized over $\vartheta = (\mu, \kappa, \theta, \dots)$ (`hawkes_calibration/mbpp/interval_censored.py`, `fit_mbpp_ic`).

We state exactly what Equation (10.6) gets right and wrong when the data is generated by the Hawkes process itself.

Exact: the first moments. By Theorem 10.11 and the compensator formula, $\mathbb{E}C_j = \mathbb{E} \int_{I_j} \lambda = \bar{\Lambda}_j$ exactly, for every j , transient or stationary. Consequently the expected score of Equation (10.6) at the true parameters is $\sum_j (\mathbb{E}C_j / \bar{\Lambda}_j - 1) \partial_{\vartheta} \bar{\Lambda}_j = 0$: the MBPP maximum-likelihood estimator is an M-estimator with an unbiased estimating equation — a moment-matching estimator dressed as a likelihood — and is consistent under the usual regularity and ergodicity conditions for correctly specified means. Rizoïu et al. (2022, Prop. 8) give the population version: the expected IC log-likelihood is (up to constants) the generalized Kullback–Leibler divergence between the true interval means and the model’s interval compensators, minimized exactly at parameters reproducing the true means — identifiability is preserved *in the KL sense*, to the extent that the interval means themselves separate the parameters (they pin μ and κ through levels and transients; how much of θ they retain is Section 10.5.4).

Discarded: the second moments. Two falsehoods are asserted by Equation (10.6). First, that C_j is Poisson — but the Hawkes random compensator $\int_{I_j} \lambda$ has been replaced by its expectation, and the law of total variance charges for it: $\text{Var}(C_j) = \mathbb{E}[(\text{martingale variation})] + \text{Var}(\int_{I_j} \lambda) + \text{cross terms} > \mathbb{E}C_j$. Second, that $C_1, \dots, C_J$ are independent — but excitation propagates across bin boundaries, and for the exponential kernel the count autocovariance decays geometrically with lag at exactly the AR root $e^{-r\Delta}$, $r = (1-\kappa)\theta$. The next proposition

quantifies the first falsehood; the second is smallest precisely when bins are wide relative to the relaxation time $1/r$.

Proposition 10.15 (Hawkes bin counts are overdispersed). *For a stationary Hawkes process with $h = \kappa\theta e^{-\theta t}$, $\kappa \in (0, 1)$, constant μ , let $F(\Delta) = \text{Var}(C_j)/\mathbb{E}(C_j)$ be the Fano factor of a bin of width Δ . Then*

$$F(\Delta) \rightarrow 1 \quad (\Delta \rightarrow 0), \quad F(\Delta) \rightarrow \frac{1}{(1-\kappa)^2} \quad (\Delta \rightarrow \infty), \quad (10.7)$$

and $F(\Delta) > 1$ for all $\Delta > 0$: true bin counts are strictly overdispersed relative to the Poisson law asserted by Equation (10.6), by a factor approaching $(1-\kappa)^{-2}$ for wide bins.

Proof sketch. Wide bins. By the cluster representation (Hawkes and Oakes, 1974), N is a superposition of i.i.d. clusters attached to Poisson(μ) immigrants; the total cluster size S is the total progeny of a Galton–Watson process with Poisson(κ) offspring, for which $\mathbb{E}S = (1-\kappa)^{-1}$ and $\text{Var } S = \kappa(1-\kappa)^{-3}$. Cluster durations have finite mean (exponential delays), so as $\Delta \rightarrow \infty$ the count over a bin is, up to boundary terms of $O(1)$ against a mean of $O(\Delta)$, a compound Poisson sum $\sum_{i \leq K} S_i$ with $K \sim \text{Poisson}(\mu\Delta)$; hence $F \rightarrow \mathbb{E}S^2/\mathbb{E}S = (\text{Var } S + (\mathbb{E}S)^2)/\mathbb{E}S = \kappa(1-\kappa)^{-2} + (1-\kappa)^{-1} = (1-\kappa)^{-2}$. *Narrow bins.* $\mathbb{E}[C_j(C_j - 1)] = \iint_{I_j^2} \rho_2 = O(\Delta^2)$ for the (bounded) second-order product density ρ_2 of the stationary process, while $\mathbb{E}C_j = \Theta(\Delta)$, so $F = 1 + O(\Delta) \rightarrow 1$. Strict inequality for all Δ follows from positivity of the covariance density of a self-exciting process with nonnegative kernel (Hawkes, 1971; Daley and Vere-Jones, 2003). What breaks if $\kappa \geq 1$: $\mathbb{E}S = \infty$ and the Fano factor diverges — another face of the stability boundary. $\square$

Warning 10.16 (MBPP standard errors are optimistic). The observed-information standard errors of Equation (10.6) (as reported by `fit_mbpp_ic`) price the counts as Poisson and independent. By Proposition 10.15 the true variance is up to $(1-\kappa)^{-2}$ larger per bin — at $\kappa = 0.35$, for example, this derived ceiling is about 2.4, so naïve standard errors can be about $1.5\times$ too small — and positive cross-bin correlation compounds the understatement. Remedies, in committed order: (i) sandwich (robust M-estimation) covariance, $\mathcal{I}^{-1}V\mathcal{I}^{-1}$ with V the empirical score variance — valid because the estimating equation is unbiased regardless of the working variance (van der Vaart, 1998, Ch. 5); (ii) a negative-binomial working likelihood with variance $\bar{\Lambda}_j + \bar{\Lambda}_j^2/\nu$ when the dispersion diagnostic of Chapter 11 (Pearson statistic, `hawkes_calibration/mbpp/gof.py`) exceeds the predicted Fano band; (iii) the Bayesian route of `hawkes_calibration/mbpp/bayes.py`, whose posterior exhibits the κ - θ ridge as a correlation instead of hiding it in a scalar. Never report the naïve SEs alone.

When is the mean-intensity replacement good? Three regimes, all of which hold for our intended use. (i) *Aggregation over many low-feedback units*: sector-level counts superpose many firms' sparse streams; superpositions of many sparse, weakly dependent point processes are close to Poisson (Daley and Vere-Jones, 2003, Ch. 9), so the Poisson working law is close to true even before the mean replacement. (ii) *Subcritical, moderate κ* : the entire second-moment error is bounded by the Fano ceiling $(1-\kappa)^{-2}$, which is $O(1)$ and known; the bound stays modest as long as the fitted κ sits well below one, and the fitted values will be reported in Chapter 20. (iii) *Bin width at or beyond the kernel half-life*: the neglected cross-bin dependence has AR root $e^{-r\Delta}$, small once $\Delta \gtrsim 1/r$; and within-bin position information — the thing MBPP cannot use — was already destroyed by (C4). The regime where MBPP is *bad* is the mirror image: a single high- κ stream at fine resolution, where one should be fitting the event-time likelihood of Chapter 8 instead — on verified dates.

10.5.4 What survives aggregation: κ yes, θ no

The stationary interval mean is $\bar{\Lambda}_j = \mu\Delta/(1 - \kappa)$: it contains κ and *no trace of θ* . All θ -information in counts therefore lives in temporal structure — transients of Equation (10.5) and cross-bin autocovariances — and both enter only through the relaxation root $e^{-r\Delta}$. A Fisher-information computation makes this quantitative: the information for θ carried by counts decays like $\Delta^2 e^{-2(1-\kappa)\theta\Delta}$ — exponentially in the bin width — so binning does not merely blur the timescale; past a few half-lives it erases it. This yields a working rule, stated here as a hypothesis and scheduled for validation on synthetic data in Chapter 19: *bucket width $\leq$ one kernel half-life should be safe for θ ; $\geq$ four half-lives should destroy timing; κ and covariate effects, read from first moments, should survive any width*. The expected failure signature at coarse widths is a θ biased toward the ridge with a deceptively small spread — the mark of an unidentified direction — and the planned recovery experiments must reproduce it before the rule is trusted. The Bayesian implementation (`hawkes_calibration/mbpp/bayes.py`) is the honest reporting vehicle on the ridge: a flat-prior posterior on (κ, θ) stays wide along θ and tight along κ , and a θ prior from a verified-date subset tightens it legitimately.

Warning 10.17 (Never claim timing precision finer than the honest resolution). A θ fitted from weekly counts is a ridge coordinate, not a measurement. Design consequence, binding on every downstream consumer (Chapter 16): no model output may assert within-week timing structure — expected response times, decay half-lives, “momentum windows” — unless θ was estimated from a verified-date subset at event-time resolution, and the honest resolution of the input data is reported next to any timing claim. The failure mode is tempting because the fit *converges*: the optimizer returns a θ with a small naïve SE while sliding along the ridge.

10.6 Single untrusted dates as one-bin censoring

The machinery above treats wholesale aggregation. The same logic handles retail corruption: a *single* deal whose date is trusted only to $\pm\delta_m$ is an event observed in the one-off interval $I_m = (\hat{t}_m - \delta_m, \hat{t}_m + \delta_m]$, i.e. a private one-bin censoring. Mixed records — verified dates for some deals, windows for others — then call for a mixed likelihood.

Proposition 10.18 (Exact mixed likelihood, Poisson case). *Let N be an NHPP with deterministic intensity λ , and suppose the record consists of (a) verified events at exact times $\{t_m\}_{m \in \mathcal{V}}$ and (b) untrusted events known only to lie in windows $\{I_m\}_{m \in \mathcal{U}}$, one event per window, where the windows are pairwise disjoint and contain no verified events (overlapping windows are first merged into a single interval carrying its total count, reducing to Proposition 10.9 locally), and (c) the assertion, by Assumption 10.3, that no further events occurred in $(0, T]$. The exact log-likelihood of this record is*

$$\ell_{\text{mix}}(\vartheta) = \sum_{m \in \mathcal{V}} \log \lambda(t_m; \vartheta) + \sum_{m \in \mathcal{U}} \log \Lambda(I_m; \vartheta) - \int_0^T \lambda(u; \vartheta) du, \quad \Lambda(I) = \int_I \lambda. \quad (10.8)$$

Proof. The restrictions of an NHPP to disjoint sets are independent. Partition $(0, T]$ into the windows $\{I_m\}_{m \in \mathcal{U}}$ and their complement B . On B the observation is exact: points at $\{t_m\}_{m \in \mathcal{V}}$ and none elsewhere, with density $\prod_{\mathcal{V}} \lambda(t_m) e^{-\Lambda(B)}$. On each I_m the observation is the event $\{N(I_m) = 1\}$, with probability $\Lambda(I_m) e^{-\Lambda(I_m)}$. Multiplying and collecting exponents (the Λ 's add to $\int_0^T \lambda$ over the partition) gives Equation (10.8). Every factor is exact; no within-window position was used. $\square$

So for Poisson calibration, mixed trust costs nothing but the windows' internal resolution: verified deals keep their full event terms, untrusted deals degrade gracefully to interval terms, and the survival factor is common. For Hawkes the by-now-familiar obstacle returns — the

intensity after an untrusted event depends on where in I_m it actually happened — and the exact mixed likelihood is again a configuration integral over $\prod_{m \in \mathcal{U}} I_m$. The committed practical recipe, flagged as an approximation, mirrors the MBPP construction locally: keep exact terms where dates are verified, and push untrusted events through their *expected* kernel contribution. Concretely, define the smeared intensity

$$\tilde{\lambda}(t; \vartheta) = \mu(t) + \sum_{m \in \mathcal{V}: t_m < t} h(t - t_m) + \sum_{m \in \mathcal{U}} \frac{1}{|I_m|} \int_{I_m} h(t - u) \mathbf{1}\{u < t\} du, \quad (10.9)$$

(the untrusted event’s excitation averaged uniformly over its window, with only the past part of the window contributing), and maximize Equation (10.8) with $\tilde{\lambda}$ in place of λ . This is exact as $\kappa \rightarrow 0$ (Poisson limit), errs only in the untrusted events’ second-order feedback terms, and must be validated by the protocol of Section 10.7 before use — the same discipline as the MBPP itself. When untrusted events are dense in some sector-period, the recipe degenerates gracefully: merge that period’s windows into bins and the untrusted sum in Equation (10.8) becomes the MBPP likelihood Equation (10.6) restricted to that period.

10.7 The validation loop

Every approximation admitted in this chapter — the mean-intensity replacement, the mixed-likelihood smearing, the choice of resolution — is validated by the same closed loop, which the repository already implements for the MBPP (`hawkes_calibration/mbpp/ic_simulate.py`) and which is hereby made the standing acceptance test.

Protocol 10.19 (Simulate exactly, censor, refit, compare).

- (a) *Simulate exactly* from the continuous-time Hawkes model at the fitted (or stress) parameters: the cluster/branching construction of [Hawkes and Oakes \(1974\)](#) (`simulate_separable_hawkes`), or exact simulation by [Dassios and Zhao \(2013\)](#), or Ogata thinning ([Ogata, 1981](#); [Lewis and Shedler, 1979](#)) — never from the MBPP itself, which would assume the conclusion.
- (b) *Censor as production does*: collapse event times to the production partition with `interval_censor`; optionally inject adversarial jitter at the audited δ before binning, to certify Proposition 10.4 empirically.
- (c) *Refit with the production likelihood* (`fit_mbpp_ic` and variants; the mixed recipe of Section 10.6 where applicable).
- (d) *Compare*: (1) parameter recovery — κ must be recovered within Monte Carlo bands at every candidate resolution; θ only at resolutions the ridge rule permits, and its failure at coarse bins should reproduce the ridge signature predicted in Section 10.5.4; (2) SE honesty — the ratio of Monte Carlo spread to reported SE is the empirical dispersion factor, to be compared against the Fano prediction of Proposition 10.15 and repaired per Warning 10.16; (3) fit diagnostics — Pearson dispersion and PIT-style count calibration ([Czado et al., 2009](#)) via `hawkes_calibration/mbpp/gof.py`, with the event-time time-rescaling check ([Brown et al., 2002](#)) reserved for verified-date subsets (Chapter 11).
- (e) *Gate*: a resolution/likelihood pair is admitted for real-data use only after passing (d) in the synthetic world at matched sample sizes — the transfer convention of Section 2.5. Campaign results will be reported in Chapter 19.

10.8 Defect declaration and commitments

Per Table 2.1 and decision D2.1, this chapter *owns* (C4) and relates to the rest of the ledger as follows. (C1) *corrected here*: Theorem 10.6 and Proposition 10.7 establish that right censoring costs nothing for history-dependent intensities when open spells are kept, closing the Hawkes case left open by Warning 2.4. (C2) *respected mechanically*: every integral and every bin in this chapter starts at the entry time V_i where firm-level objects are involved; the selection content of (C2) is owned by Chapter 15. (C3) *vulnerable, out of scope*: nothing here repairs contaminated risk sets — sector-level count likelihoods are largely insulated (sectors do not die), but any firm-level use of these tools inherits (C3) with the deflation bias documented in Chapter 2, owned by Chapters 7, 13 and 14. (C5) *orthogonal*: covariates enter the MBPP through the exogenous input $s(\cdot)$ (`hawkes_calibration/mbpp/exogenous.py`); their staleness is Chapter 12’s problem and its remedies compose with everything here. (C6) *robust at sector level*: sector deal counts include out-of-universe deals, so sector-level MBPP consumes complete counts; firm-attributable applications inherit (C6) as thinning, in tension with Assumption 10.3, and are owned by Chapter 15.

Design decision (D6.1 — the honest-resolution rule).

The observation resolution for continuous-time ground-process modeling on PitchBook 2011–2024 is *weekly*. Daily resolution is permitted only for subsets whose closed dates the date audit certifies with $\delta \leq 1$ day; back-filled and announced-only records never qualify. Rationale: Proposition 10.4 (boundary-transfer error $\sim 4\delta/\Delta$), Assumption 10.2 (announced-vs-closed spreads of days, heaping in backfills), and the ridge rule of Section 10.5.4: for kernel half-lives near one week, weekly bins should keep κ identifiable and leave θ marginal, a hypothesis Chapter 19 will test. Reversal trigger: a data vintage whose audit certifies $\delta \leq 1$ day for a sector’s majority of deals moves that sector to daily; conversely, evidence of heaping beyond month boundaries (quarter-level backfill) forces that segment to monthly.

Design decision (D6.2 — MBPP is the committed aggregation likelihood).

Any sector-level continuous-time Hawkes model fitted on binned counts uses the MBPP interval-censored likelihood Equation (10.6) (`fit_mbpp_ic` and variants), with standard errors repaired per Warning 10.16 (sandwich by default; negative-binomial or Bayesian escalation on a failed dispersion check), and with every fit gated by Protocol 10.19. Rationale: exact first moments (Theorem 10.11), KL-sense identifiability (Rizoiu et al., 2022), closed forms for the exponential kernel (Proposition 10.13), and a bounded, quantified price (Proposition 10.15) — against an exact alternative Equation (10.2) that is computationally out of reach. Reversal trigger: if dispersion diagnostics persistently exceed the Fano ceiling $(1 - \hat{\kappa})^{-2}$ (pointing at latent-frailty variance the MBPP cannot absorb), the sector-level count model reverts to the discrete autoregressive count family of Chapter 13 (Fokianos et al., 2009), which models the conditional variance directly.

Design decision (D6.3 — event-time likelihoods reserved for verified dates).

The exact event-time likelihood of Chapter 8 (and the time-rescaling diagnostics that need it) may be fitted only on date-audit-verified subsets; θ estimated anywhere else is prior-constrained from such a subset or fixed (decay banks). Mixed records use Equation (10.8): exactly for Poisson (Proposition 10.18), with the flagged smeared-intensity recipe Equation (10.9) for Hawkes. Rationale: Warning 10.17 — timing claims must trace to data that contains timing. Reversal trigger: none for the principle; the boundary of “verified” moves with the date audit.

Code hooks.

The mathematics of this chapter maps onto `hawkes_calibration/mbpp/` as follows. `core.py`: the MBPP object — kernels, the resolvent/impulse response $R(t) = \kappa\theta e^{(\kappa-1)\theta t}$ of Proposition 10.13, closed-form and numeric solutions of Equation (10.3), interval compensators $\bar{\Lambda}_j$, and the Proposition-6/7 compensator bounds used for forecasting. `interval_censored.py`: the IC likelihood Equation (10.6) (`ic_ll`), the SSE/Bregman alternative, and the fitting family `fit_mbpp_ic` / `_multi` / `_covariates` / `_sumexp` / `_excitation`, plus `forecast_counts`. `exogenous.py`: the exogenous inputs $s(\cdot)$ — piecewise-constant, covariate-driven, impulse, LHPP — through which Chapter 12’s covariates and Chapter 7’s baselines enter. `ic_simulate.py`: the branching simulator and `interval_censor`, powering Protocol 10.19. `gof.py`: Pearson/dispersion diagnostics for counts and time-rescaling residuals for verified-date subsets; `bayes.py`: the posterior treatment of the κ – θ ridge. *To be written*: `hawkes_calibration/data/date_audit.py` — the audit behind Assumptions 10.1 and 10.2: classify each deal date as closed/announced/backfilled, estimate convention offsets $b(r)$, detect month/quarter heaping, emit per-event trust radii δ_m and the verified-date flags consumed by D6.1/D6.3 — and the sandwich-covariance wrapper for `fit_mbpp_ic` required by Warning 10.16.

Chapter 11

Goodness of Fit, Model Comparison, and Calibration Discipline

11.1 The evaluation protocol

Every preceding chapter builds an estimator; this chapter fixes the rules by which all of them are judged. Its deliverables are not models but rules: a definition of “calibrated” for each of the four targets of Chapter 1, the test batteries that operationalize those definitions, the comparison ladder that decides which model family is promoted, and the reporting discipline that keeps every number attached to the caveats it needs. We call this the *evaluation protocol* and mean it to be hard to amend: metrics chosen after seeing results are not metrics. We expect the difference between a defensible claim and an embarrassing one to lie rarely in the model and mostly in whether the evaluation respected the risk set, the temporal split, and the identifiability limits of the parameter being quoted; the synthetic campaigns of Chapter 19 are designed to test exactly that.

The word “calibrated” means something different for each target, and conflating the meanings is the failure mode Section 1.2 warns against.

Definition 11.1 (Calibration, per target). Fix an evaluation era and the point-in-time discipline of Protocol 2.5.

- (i) *T2 (timing)*. An intensity or count model is calibrated if it survives the time-rescaling battery of Protocol 11.3 at the finest resolution its event times honestly support, *and* the count-calibration battery of Protocol 11.8 (PIT uniformity, dispersion near 1, reliability per sector-week) at weekly resolution. Both are required: rescaling sees timing structure that counts hide, and PIT sees marginal-distribution failures that rescaling averages out.
- (ii) *T3 (horizon events)*. A horizon classifier is calibrated if its predicted probabilities match conditional frequencies: $\mathbb{P}(\text{event within } \Delta \mid \hat{p} = p) = p$ across the support of $\hat{p}$, assessed by reliability diagram, calibration intercept and slope. Discrimination (PR-AUC, lift at K) is necessary but never sufficient; recalibration maps (Platt, 1999; Zadrozny and Elkan, 2002; Niculescu-Mizil and Caruana, 2005) are legal but are model components, fitted on the validation era only.
- (iii) *T1 (identity)*. A ranker is judged by ranking metrics (recall at K , MRR, computed on the declared risk set) *and* by held-out conditional likelihood (nats per deal of the mark factor). Each alone is gameable: recall at K is blind to the probabilities attached to the ranks it gets right, and conditional NLL can improve while the tail ordering that sourcing consumes inverts. Both must be reported; neither may be traded silently for the other.

- (iv) T_4 (*marks*). Mark models are scored by proper scoring rules per mark (Section 11.3), with the same temporal split.

Design decision (D7.1 — the evaluation protocol).

The definitions in Definition 11.1, the protocols Protocols 11.3, 11.8, 11.14 and 11.15, and the ladder of Section 11.4 are the only admissible evaluation for models in this repository. A new metric enters the protocol only together with a new consuming action, per decision D1.1 of Chapter 1; a metric may never be added because a model happens to do well on it. Rationale: post hoc metric selection is the cheapest way to fool ourselves; with enough scores to choose from, every family looks good on *some* score. Reversal trigger: a change in the decision layer (Chapter 16) that adds or retires a consuming action.

11.2 Continuous-time goodness of fit

11.2.1 Time rescaling in practice

The instrument for event-time fit is the time-rescaling theorem, proved in Chapter 6 from Watanabe’s characterization and restated here in the form the diagnostics use (Brown et al., 2002; Ogata, 1988; Daley and Vere-Jones, 2003).

Theorem 11.2 (Time rescaling, evaluation form). *Let N be a point process on $[0, T]$ with event times $t_1 < t_2 < \dots < t_n$, adapted to $\{\mathcal{F}_t\}$, admitting a strictly positive conditional intensity $\lambda(t)$ whose compensator $\Lambda(t) = \int_0^t \lambda(u) du$ is a.s. finite and continuous. Then the rescaled times $\tau_m = \Lambda(t_m)$ form a unit-rate Poisson process on $[0, \Lambda(T)]$; equivalently, with $\tau_0 = 0$, the gaps*

$$\xi_m = \tau_m - \tau_{m-1} \sim \text{Exp}(1) \quad i.i.d., \quad u_m = 1 - e^{-\xi_m} \sim U(0, 1) \quad i.i.d. \quad (11.1)$$

Proof. Since $\lambda > 0$, Λ is continuous and strictly increasing, so $s \mapsto \Lambda^{-1}(s)$ is a well-defined family of $\{\mathcal{F}_t\}$ -stopping times. Set $\tilde{N}(s) = N(\Lambda^{-1}(s))$ and $\tilde{\mathcal{F}}_s = \mathcal{F}_{\Lambda^{-1}(s)}$. Because $N(t) - \Lambda(t)$ is a local martingale (Chapter 2, Equation (2.2)), optional stopping at $\Lambda^{-1}(s)$ shows $\tilde{N}(s) - s$ is a local $\{\tilde{\mathcal{F}}_s\}$ -martingale: $\tilde{N}$ is a counting process with deterministic continuous compensator s , hence a unit-rate Poisson process by Watanabe’s characterization (Chapter 6). The gap and uniform statements are the standard exponential-spacings and probability-integral facts. The proof fails exactly where its hypotheses do: if λ vanishes on at-risk gaps, rescale over at-risk time only; and if the t_m are jittered (defect (C4)), the theorem still holds for the *true* times but the computed τ_m use corrupted ones — the test then interrogates model and jitter jointly (Section 11.7). $\square$

In practice Λ is the *fitted* compensator $\hat{\Lambda}$, and three details separate a usable diagnostic from a misleading one: the open final interval, the null distribution under estimated parameters, and localization.

Protocol 11.3 (Time-rescaling battery).

- (a) *Residuals.* Compute ξ_m from Equation (11.1) with the fitted compensator, integrating over at-risk time only, per sector and pooled (`hawkes_calibration/mbpp/gof.py`, `time_rescaling_residuals`).

- (b) *The last interval is censored.* The final gap $\hat{\Lambda}(T) - \tau_n$ is a right-censored Exp(1) draw by memorylessness. It is either excluded from the KS sample or included as a censored observation in a likelihood-based check; it is *never* treated as a complete gap. Treating it as complete injects one draw that is stochastically too small per unit — across many sectors this manufactures a spurious excess of short gaps, i.e. phantom clustering.
- (c) *KS and QQ.* Test the complete gaps against Exp(1) (KS, `ks_test_exp1`) and plot QQ coordinates (`qq_exp1`) with envelope bands. Report the statistic per sector and pooled.
- (d) *Localization.* A rejection is the beginning, not the end: identify *which* sectors, which calendar windows, and which covariate strata carry the misfit, using the stratified residuals of Equation (11.2) — this is Ogata’s residual-analysis program (Ogata, 1988).
- (e) *Defect exposure.* Under (C4) this battery is run only at the resolution the dates honestly support (daily-resolved deals in the event-time track; never sub-week on jittered dates); under (C3) the fitted $\hat{\Lambda}$ absorbs dead exposure and the test inherits the optimism of Warning 11.16.

Warning 11.4 (Estimated parameters make standard bands optimistic for the model). The KS null distribution assumes a fixed hypothesized law. When the compensator is fitted on the same data, $\hat{\Lambda}$ adapts to the sample, the KS statistic is stochastically smaller than under a fixed truth, and standard p -values are conservative: a model can *pass* while wrong. The honest null is simulation-based: simulate from the fitted model, refit, recompute the statistic, and compare the observed value to this refit-in-loop distribution. The envelope machinery of Protocol 11.5 provides the harness; held-out-era residuals (fit on train, rescale on test) are the cheap partial fix and are the repository default.

11.2.2 Martingale residuals over strata

Time rescaling compresses the whole fit into one distributional statement; martingale residuals decompose it. For any predictable stratum g — a set of (unit, time) pairs selected by sector, calendar window, or covariate values known at t^- — define

$$R(g) = \sum_m \mathbf{1}\{(e_m, t_m) \in g\} - \sum_e \int_0^T \mathbf{1}\{(e, u) \in g\} Y_e(u) \hat{\lambda}_e(u) du, \quad Z(g) = \frac{R(g)}{\sqrt{\hat{\Lambda}(g)}}, \quad (11.2)$$

where $\hat{\Lambda}(g)$ is the integral term. Under a correctly specified model $R(g)$ is the terminal value of a mean-zero martingale with predictable variation $\Lambda(g)$, so $Z(g)$ is approximately standard normal when $\hat{\Lambda}(g)$ is large (Andersen et al., 1993). The design use is *systematic lack-of-fit localization*: compute $Z(g)$ over a fixed grid of strata — sector $\times$ calendar quarter, staleness-age bands of $a_i(t)$, macro-covariate terciles — and read the sign pattern. A block of positive Z across all sectors in one era is a missing calendar covariate; positive Z concentrated at high staleness age is the (C5) signature that Chapter 12 predicts; one wild cell is usually data hygiene, not model failure. Isolated $|Z| > 2$ values among dozens of strata are expected; structure is what indicts the model.

11.2.3 Simulation-based envelope tests

Likelihood-based diagnostics ask whether the data look probable under the model; envelope tests ask whether the model’s own sample paths look like the data. They are the only GOF instrument that sees *functionals we did not fit*.

Protocol 11.5 (Envelope battery).

- (a) Simulate B (default 999) replicates of the evaluation era from the fitted model using the simulators of Chapter 8: Ogata thinning (Ogata, 1981), the cluster representation (Hawkes and Oakes, 1974), or exact exponential-kernel simulation (Dassios and Zhao, 2013); pass each replicate through the same observation layer (aggregation, at-risk masks) as the real data.
- (b) Compute, per replicate and for the data, a fixed pre-registered statistic vector: weekly-count autocorrelations at lags 1–8, the dispersion index of Proposition 11.7, the coefficient of variation of inter-event gaps, the maximum weekly count, and the sector-level count cross-correlations.
- (c) Reject a statistic at (pointwise) level $\alpha \approx 2/(B+1)$ if the observed value falls outside the central simulated band; report all statistics together and treat the multiplicity honestly — one excursion among a dozen statistics at $B = 999$ is weak evidence.
- (d) *Defect exposure*. Simulating through the observation layer is mandatory: envelopes from the latent (uncorrupted) model test the wrong null. (C1)/(C4) are reproduced by construction; (C3) cannot be reproduced (we do not know who is dead) and the envelope inherits the same contaminated risk set as the fit.

The double-use caveat of Warning 11.4 applies verbatim: parameters estimated from the data pull the envelope toward the data, so envelope non-rejection is weak; envelope *rejection* is strong and localizes what the model cannot reproduce (typically: the real data’s overdispersion or its cross-sector comovement).

11.3 Count-level goodness of fit

At weekly resolution the model emits a predictive distribution $F_{s,w}$ for each sector-week count $Y_{s,w}$, and the calibration question becomes: are the observed counts distributed as the model says? The continuous-data answer — the probability integral transform $F(Y) \sim U(0,1)$ — fails for discrete data: $F(Y)$ is a discrete random variable, stochastically larger than uniform, and naive PIT histograms read miscalibration where there is none, most severely in sparse cells where $\mathbb{P}(Y = 0)$ is large. Two standard repairs exist (Smith, 1985; Brockwell, 2007; Czado et al., 2009).

Proposition 11.6 (Randomized PIT). *Let Y have cumulative distribution F on $\mathbb{N}_0$ with $p_y = F(y) - F(y-1)$, and let $V \sim U(0,1)$ independent of Y . Then*

$$U = F(Y-1) + V p_Y \tag{11.3}$$

satisfies $U \sim U(0,1)$.

Proof. Fix $u \in (0,1)$ and let y^* be the unique value with $F(y^*-1) < u \leq F(y^*)$. Conditioning on $Y = y$: if $F(y) \leq u$ (i.e. $y < y^*$) then $U \leq u$ with probability 1; if $y > y^*$ then $F(y-1) \geq u$ and $U \leq u$ has probability 0; if $y = y^*$ then $\mathbb{P}(U \leq u \mid Y = y^*) = (u - F(y^*-1))/p_{y^*}$. Summing,

$$\mathbb{P}(U \leq u) = \sum_{y < y^*} p_y + p_{y^*} \cdot \frac{u - F(y^*-1)}{p_{y^*}} = F(y^*-1) + u - F(y^*-1) = u. \quad \square$$

Under a correctly calibrated sequence of predictive distributions the $U_{s,w}$ are i.i.d. uniform and the whole continuous toolkit (KS, histograms, drift plots) applies. The price is randomization: two analysts get two dashboards. The non-randomized alternative of Czado et al. (2009)

replaces each draw by its conditional expectation: with $a = F_{s,w}(y_{s,w} - 1)$, $b = F_{s,w}(y_{s,w})$, define the conditional PIT $c(u; a, b) = \min\{\max\{(u - a)/(b - a), 0\}, 1\}$ and the aggregate $\bar{F}(u) = \frac{1}{n} \sum_{s,w} c(u; a_{s,w}, b_{s,w})$; calibration implies $\mathbb{E} \bar{F}(u) = u$, and the histogram of increments of $\bar{F}$ is the deterministic analogue of the PIT histogram. Repository rule: the non-randomized version drives dashboards and regressions (reproducible), the randomized version drives formal tests (exact uniformity).

Proposition 11.7 (Dispersion index and its sampling distribution). *Let $Y_i \sim \text{Poisson}(\mu_i)$ independently, $i = 1, \dots, n$, and*

$$\hat{\phi} = \frac{1}{n - p} \sum_{i=1}^n \frac{(Y_i - \hat{\mu}_i)^2}{\hat{\mu}_i}, \quad (11.4)$$

with p fitted parameters. With known means ($p = 0$, $\hat{\mu}_i = \mu_i$): $\mathbb{E} \hat{\phi} = 1$ and $\text{Var}(\hat{\phi}) = n^{-2} \sum_i (2 + 1/\mu_i)$, so that if $n^{-1} \sum_i 1/\mu_i \rightarrow \bar{m} < \infty$,

$$\sqrt{n}(\hat{\phi} - 1) \xrightarrow{d} \mathcal{N}(0, 2 + \bar{m}).$$

Estimated means reduce the degrees of freedom, whence the divisor $n - p$.

Proof. For $Y \sim \text{Poisson}(\mu)$ the central moments are $m_2 = \mu$, $m_4 = \mu + 3\mu^2$. Hence $\mathbb{E}[(Y - \mu)^2/\mu] = 1$ and $\text{Var}[(Y - \mu)^2/\mu] = (m_4 - m_2^2)/\mu^2 = (\mu + 2\mu^2)/\mu^2 = 2 + 1/\mu$. Independence gives the variance sum; the Lindeberg CLT (bounded $1/\mu_i$ suffices) gives normality. The proof shows what practitioners forget: the null variance is *not* $2/n$ — the $1/\mu_i$ term dominates in sparse cells, and sector-weeks with $\mu_i \lesssim 1$ (common in this data) widen the honest acceptance band. Judging $\hat{\phi}$ against $1 \pm 2\sqrt{2/(n - p)}$ in a sparse panel manufactures false overdispersion alarms. $\square$

Overdispersion ($\hat{\phi} > 1$) is the escalation trigger of the model ladder — it says history or heterogeneity is missing — but it is *never* evidence for excitation specifically, because frailty and contagion produce it identically (Chapter 8, Section 11.5).

Protocol 11.8 (Count-calibration battery). For every count model, on the evaluation era: (a) non-randomized PIT diagram pooled and per sector; (b) randomized-PIT KS test, seeds logged; (c) dispersion Equation (11.4) with the honest band of Proposition 11.7, reported *alongside every count NLL*; (d) reliability per sector-week: bin cells by predicted mean decile, compare observed to predicted bin means with Poisson error bars; (e) PIT drift over calendar time, plotted — drift is the cheapest detector of vendor-coverage change and of (C5) staleness confounding. Defect exposure: complete sector counts make this battery robust to (C6) at sector level; (C3) does not bite sector counts (sectors do not die); (C4) is absorbed by construction — weekly counts are the honest resolution.

Scoring rules for counts. A scoring rule $S(F, y)$ assigns a loss to forecasting F when y materializes; it is *proper* if $\mathbb{E}_{Y \sim G} S(G, Y) \leq \mathbb{E}_{Y \sim G} S(F, Y)$ for all F , and *strictly proper* if equality forces $F = G$ (Gneiting and Raftery, 2007). Propriety is not decoration: an improper score rewards hedged or sharpened forecasts and corrupts every comparison built on it. We use two:

$$\log S(F, y) = -\log f(y), \quad \text{RPS}(F, y) = \sum_{k=0}^{\infty} (F(k) - \mathbf{1}\{y \leq k\})^2. \quad (11.5)$$

Proposition 11.9 (Strict propriety of the log score). *On the class of probability mass functions with common support, $\log S$ is strictly proper.*

Proof. Let g be the true pmf and f any competitor with the same support. Then

$$\mathbb{E}_g[-\log f(Y)] - \mathbb{E}_g[-\log g(Y)] = \sum_y g(y) \log \frac{g(y)}{f(y)} = \text{KL}(g \| f) \geq 0,$$

with equality iff $f = g$, by Jensen's inequality applied to the strictly concave logarithm ($\sum g \log(f/g) \leq \log \sum f = 0$, strict unless f/g is constant). Hence the expected score is uniquely minimized at the truth. If supports differ the score can be infinite — which is a feature: a model that assigns zero mass to an observed count is infinitely wrong, and the repository treats $f(y) = 0$ on observed data as a bug, not an outlier. $\square$

The ranked probability score is proper on distributions with finite mean (Gneiting and Raftery, 2007; Czado et al., 2009) but not local: it rewards mass *near* the outcome. The two scores are complementary — $\log S$ is the likelihood and dominates the ladder (Section 11.4); RPS is distance-sensitive and robust to a single mis-tailed cell — and both are reported, $\log S$ as primary, RPS as the sanity companion.

11.4 Model comparison

11.4.1 The held-out NLL ladder

The primary comparison metric of the repository is held-out negative log likelihood, in two units: *nats per event* for event-time models ($-\ell/n$ over test-era events) and *nats per sector-week cell* for count models ($-\frac{1}{|\text{cells}|} \sum_{s,w} \log \hat{f}_{s,w}(y_{s,w})$). Every model is reported as a difference against three named baselines fitted on the same train era: *random* (homogeneous Poisson, one global rate), *historical-mean* (per-sector constant rate), and *seasonal NHPP* (the covariate/seasonal Poisson model of Chapter 7). A model that has not been run against all three has not been evaluated. The challenger families include the parametric intensity models of Chapters 7 and 8, the count models of Chapter 13, and the nonparametric intensity models of Chapter 9 — the last admitted only on trusted-date subsets, per Chapter 10.

Warning 11.10 (Units are not exchangeable). Nats per event and nats per cell are different currencies: they condition on different information and normalize by different denominators. No number in one unit may be compared to a number in the other. Comparing an event-time model to a count model requires evaluating both at a common, honest resolution — in practice, aggregating the event-time model's predictive distribution to weekly cells and paying the aggregation cost explicitly (Chapter 10).

Design decision (D7.2 — NLL ladder with the covariates-dominate hypothesis).

Held-out NLL against the three named baselines is the primary promotion metric for T2 models, with Protocol 11.15 governing the split. The working hypothesis is that *covariates dominate and excitation is decimal points*; the synthetic campaigns of Chapter 19 will test it, and Chapter 20 will report the real-data ladder. Until it is overturned, any claim that excitation matters must clear the seasonal NHPP *and* the covariate NHPP under rolling-origin evaluation, not merely improve on a weak baseline. Reversal trigger: excitation contributing ≥ 0.5 nats per cell over the covariate NHPP on real data in two consecutive campaigns retires the hypothesis.

AIC and BIC have a narrow legal scope here: quick screening *within* a family, on the train era, when models differ by a few regular interior parameters. Three limits keep them off

headlines. First, they are in-sample devices, and our risk is out-of-era: a temporal regime shift is invisible to them. Second, their penalty calculus assumes regular asymptotics, which fail exactly at the comparison we care most about — excitation present versus absent — because $\kappa = 0$ is a boundary point, where even the effective parameter count is wrong (below). Third, under misspecification neither estimates what its derivation promises. Held-out NLL is the arbiter; information criteria are a screen.

11.4.2 Testing for excitation at the boundary

The natural test of “is there any excitation?” is the likelihood ratio between the covariate NHPP ($\kappa = 0$) and the exponential-Hawkes extension ($\kappa \geq 0$). But $\kappa = 0$ lies on the boundary of the parameter space — negative branching is not in the model — so the classical χ_1^2 asymptotics fail (Self and Liang, 1987).

Theorem 11.11 (Chi-bar-square limit for one bounded parameter). *Let the model have parameter (κ, ψ) with $\kappa \geq 0$ on the boundary at the null, ψ interior nuisance parameters, and the decay θ fixed (a decay bank, Chapter 8). Assume the LAN regularity conditions of Self and Liang (1987): consistency, a nonsingular Fisher information $\mathcal{I}$ at the true value $(0, \psi_0)$, and differentiability in quadratic mean in a neighborhood intersected with the cone $\{\kappa \geq 0\}$. Then the likelihood-ratio statistic $\lambda_n = 2(\sup_{\kappa \geq 0, \psi} \ell - \sup_{\psi} \ell|_{\kappa=0})$ satisfies*

$$\lambda_n \xrightarrow{d} \frac{1}{2} \chi_0^2 + \frac{1}{2} \chi_1^2, \quad \text{i.e.} \quad \mathbb{P}(\lambda_n \leq x) \rightarrow \frac{1}{2} + \frac{1}{2} \mathbb{P}(\chi_1^2 \leq x), \quad (11.6)$$

where χ_0^2 is the point mass at zero.

Proof. Profile out ψ : under LAN, uniformly over $\kappa = O(n^{-1/2})$,

$$\ell(\kappa) - \ell(0) = \kappa \tilde{S}_n - \frac{1}{2} \kappa^2 \tilde{\mathcal{I}}_n + o_p(1),$$

where $\tilde{S}_n$ is the efficient score for κ (the score residualized on the nuisance scores) and $\tilde{\mathcal{I}}_n$ the efficient information, with $Z_n := \tilde{S}_n / \tilde{\mathcal{I}}_n^{1/2} \xrightarrow{d} \mathcal{N}(0, 1)$ at the null. Maximizing the quadratic over the half-line $\kappa \geq 0$ gives $\hat{\kappa} = \max(\tilde{S}_n, 0) / \tilde{\mathcal{I}}_n$ and

$$\lambda_n = (\max(Z_n, 0))^2 + o_p(1).$$

For $Z \sim \mathcal{N}(0, 1)$: $\mathbb{P}(\max(Z, 0)^2 \leq x) = \mathbb{P}(Z \leq 0) + \mathbb{P}(0 < Z \leq \sqrt{x}) = \frac{1}{2} + \frac{1}{2} \mathbb{P}(\chi_1^2 \leq x)$. The mixture is exactly the geometry: with probability $\frac{1}{2}$ the unconstrained optimum wants negative excitation, the constrained optimum sits at the boundary, and the LR is zero. When more than one parameter is bounded (e.g. a nonnegative excitation matrix), the mixture weights become the probabilities that the projected Gaussian lands on each face of the cone (Self and Liang, 1987), and closed forms disappear — simulate instead. $\square$

Practical rule. The p -value for the boundary LRT is $p = \frac{1}{2} \mathbb{P}(\chi_1^2 \geq \lambda_{\text{obs}})$ — halve the naive p -value; the 5% critical value is 2.71, not 3.84. A team using χ_1^2 tables is running a conservative test and will under-detect excitation; the direction of this error is at least safe, which is why the mistake survives unnoticed.

Warning 11.12 (When the decay is free, the mixture is wrong). Under $H_0 : \kappa = 0$ the decay θ is unidentified — every θ yields the same null model — so if θ is estimated, the LR is effectively a supremum over the decay bank and is stochastically larger than Equation (11.6); the 50:50 rule then *over*-detects excitation. Two legal responses: (i) test per fixed decay-bank member (each obeys Theorem 11.11) with a multiplicity correction across the bank; (ii) parametric bootstrap — simulate from the fitted null NHPP, refit both models, and compare λ_{obs} to the simulated null distribution, reusing the envelope harness of Protocol 11.5. Separately: Wald standard errors from the inverse Hessian are invalid for any parameter whose estimate sits at a boundary (fitted zeros in nonnegative excitation matrices accumulate there); report profile-likelihood intervals or held-out NLL instead.

11.5 Identifiability diagnostics

A model comparison is only as meaningful as the parameters it compares are identified. Three diagnostics are standing obligations for the Hawkes layer.

The κ - θ ridge. Define the profile likelihood $\ell_p(\kappa) = \max_{\theta, \psi} \ell(\kappa, \theta, \psi)$ and its two-dimensional analogue over (κ, θ) . On event-time data the profile is curved in both directions at sufficient volume; on aggregated counts it is not, and the reason is elementary.

Proposition 11.13 (Stationary counts pin κ and hide θ). *For a stationary exponential Hawkes process with baseline μ and kernel $h(\tau) = \kappa\theta e^{-\theta\tau}$, $\kappa < 1$, the mean bucket count over width Δ is $\Xi = \mu\Delta/(1-\kappa)$. Hence*

$$\frac{\partial \Xi}{\partial \kappa} = \frac{\mu \Delta}{(1-\kappa)^2} > 0, \quad \frac{\partial \Xi}{\partial \theta} = 0 :$$

first moments of counts identify κ and carry zero information about θ . All θ -information in counts lives in the temporal covariance structure, whose sensitivity to θ collapses at rate $\Delta^2 e^{-2\theta(1-\kappa)\Delta}$ as the bucket widens (Chapter 10).

Proof. The stationary mean intensity solves the first-moment equation $\bar{\xi} = \mu + \bar{\xi} \int_0^\infty h(\tau) d\tau = \mu + \kappa \bar{\xi}$ (Chapter 8), so $\bar{\xi} = \mu/(1-\kappa)$, and $\Xi = \bar{\xi}\Delta$ does not involve θ . Differentiation is immediate. The covariance statement is Chapter 10’s Fisher-information computation, restated here because it is the operative rule: the exponential factor beats any polynomial, so no amount of clever estimation recovers a timescale from buckets much wider than the kernel’s half-life. $\square$

The predicted consequence, to be checked in the recovery experiments of Chapter 19: at coarse bucket widths, θ should collapse onto the ridge — biased, with a deceptively tight spread, the signature of an unidentified direction — while κ stays stable at every width. The working rule, restated from Chapter 10: bucket width at most one kernel half-life is expected to be safe for timing; four or more half-lives is expected to destroy it; in the destroyed regime, fix θ to a decay bank and report κ only. Operationally, every Hawkes fit ships its profile slices: if ℓ_p varies by less than 0.5 nats across an order of magnitude in θ , the fit *declares* θ unidentified and the reporting layer refuses to print a decay estimate.

Frailty versus contagion: the upper-bound rule. Overdispersion and clustering are produced identically by contagion (past deals raise future intensity) and by frailty (persistent latent heterogeneity) (Duffie et al., 2009; Azizpour et al., 2018); a fitted Hawkes model channels both into $\hat{\kappa}$. A planned synthetic campaign (Chapter 19) will quantify the masquerade: fit the Hawkes model to worlds driven by latent frailty with no contagion and measure how much spurious excitation it reports. Since additional structure (shared covariates, factor models) can only remove apparent excitation, every fitted branching ratio in this repository is reported as an *upper bound* on contagion — the sentence “ $\hat{\kappa} = 0.3$ ” is banned; the sentence “excitation plus unmodeled common factors account for at most $\hat{\kappa} = 0.3$ ” is the template (Chapter 8 develops the decomposition attempts and their limits).

Parameter recovery as a shipping gate.

Protocol 11.14 (Parameter-recovery rehearsal). No new model family ships until it passes, on synthetic data, all of:

- (a) *Gradient check.* Analytic gradients of the likelihood agree with finite differences at relative error 10^{-6} – 10^{-8} at random interior points — the repository’s standing discipline; a family whose gradients do not check does not proceed to fitting.
- (b) *Recovery.* Simulate ≥ 20 datasets from the family at parameter values matched to fitted real-data scale, pass them through the synthetic observation layer with defects (C1)–(C6) active (Section 2.5), fit with the production code path, and verify: parameters claimed interpretable are recovered with bias small against their sampling spread; parameters known unidentified under the observation scheme (e.g. θ under coarse aggregation, Proposition 11.13) are *declared* non-recoverable, and the declaration is part of the family’s documentation.
- (c) *Coverage.* Whatever interval procedure the family advertises (profile, bootstrap, posterior) achieves approximately nominal coverage in the rehearsal.
- (d) *Archival.* The rehearsal is an `experiments/` script with logged seed and config, its results filed under the conventions of Protocol 11.15(d).

A family failing (b) for some parameter may still ship *for prediction*, but that parameter is barred from interpretation until a passing rehearsal exists. Defect exposure: the gate is exactly as honest as the observation layer’s reproduction of (C1)–(C6); (C3)’s unlabeled exits are simulated, which is the one place we can rehearse a defect we cannot observe.

11.6 The temporal evaluation protocol

All of the above presumes an evaluation era. Choosing it is not bookkeeping: with 13.5 years of data spanning a zero-rate boom (2021), a rate-shock bust (2022–2023), and a partial normalization (2024), the split determines which regime the headline numbers describe.

Protocol 11.15 (Temporal evaluation).

- (a) *Development split (fixed).* One split serves all model development: train on data through 2019-12-31; validate (model selection, hyperparameters, calibration maps) on 2020-01-01–2021-12-31; test on 2022-01-01–2024-06-30, touched only when a family is frozen. No result computed on the test era before freezing may influence any modeling choice.
- (b) *Rolling-origin evaluation (headline).* Frozen families are reported by rolling origin: expanding-window refits at quarterly steps across the test era, each evaluated on the following quarter; the headline number is the mean across origins with its range. Rolling origins convert one number into a small distribution and are the only defense against a verdict that is an artifact of a single split point.
- (c) *No post-split information, anywhere.* Every transform — imputation, scaling, feature construction, probability recalibration — obeys Protocol 2.5: fitted on train (calibration maps on validation), applied frozen downstream. Covariate support is checked: test-era covariate ranges are reported next to every NLL, because a log-linear intensity extrapolated outside the train support is exponential extrapolation.
- (d) *Reproducibility.* Every evaluation run logs its seed, config hash, split boundaries, and code version; randomized-PIT draws and bootstrap resamples are seeded; results are machine-readable files under `results/`, one per experiment, following the `experiments/` conventions.

- (e) *Defect exposure*. (C1) is respected by construction (the test era ends at the administrative censoring time and open spells carry their survival factors); (C2)/(C3) exposure is inherited from the risk sets used, and every reported metric names its risk-set rule (Section 11.7).

Design decision (D7.3 — default temporal split).

The default development split is train ≤ 2019 , validation 2020–2021, test 2022–June 2024, as in Protocol 11.15(a). Rationale: the validation era contains a violent regime change (COVID shock, then the 2021 boom), so model selection is stress-tested against nonstationarity rather than sheltered from it; the test era contains the bust and normalization, which is the regime a deployed model faces next; and train ends before any of it, so no family is selected on the regime it will be scored on. The known risk is that 2021–2023 spans a structural break, so fixed-split verdicts may be regime artifacts — which is precisely why headlines come from rolling origins, not from the fixed split. Reversal trigger: if rolling-origin family orderings disagree with fixed-split validation orderings in two consecutive campaigns, the fixed split is replaced by a multi-split (nested rolling) development scheme and this box is rewritten.

11.7 Reporting discipline and defect exposure

Design decision (D7.4 — no number without its caveat).

The following pairings are mandatory in every report, dashboard, README table, and paper draft produced from this repository:

- (i) no hazard or intensity estimate without its *risk-set rule* (who was at risk, from when, closed by what);
- (ii) no branching ratio without its *upper-bound caveat* (Section 11.5);
- (iii) no probability without its *calibration plot* (reliability diagram or PIT, per Definition 11.1);
- (iv) no ranking metric without its *risk-set definition*: the risk set, not the model, can be the largest term in a ranking metric, and the synthetic experiments of Chapter 19 will quantify the observed-versus-oracle gap;
- (v) no count NLL without its *dispersion* printed alongside (Protocol 11.8).

Rationale: each naked number can mislead, and its caveat can reverse the conclusion. Reversal trigger: none anticipated; amendments add pairings, they do not remove them.

Defect declaration (per decision D2.1). The “models” of this chapter are the tests themselves, and they are corrupted by the same six defects of Table 2.1 they are meant to police. (C1) *robust*: administrative censoring is handled exactly — the open final rescaled interval is treated as censored (Protocol 11.3b) and censored cells enter count likelihoods

with their survival factors. (C2) *robust at sector level, conditional at firm level*: all firm-level GOF statements are statements about the entered (post-first-deal) universe; the protocols do not correct the selection, they inherit Chapter 15’s scope. (C3) *vulnerable, optimistic* — see Warning 11.16. (C4) *corrects by resolution*: the batteries are run at the honest resolution only — weekly PIT always; event-time rescaling only on the daily-resolved track — so date jitter is absorbed rather than tested against; the cost is that sub-weekly misfit is invisible by design. (C5) *vulnerable, detectable*: stale covariates push lack of fit into the staleness-age strata of Equation (11.2) and into calendar PIT drift; the diagnostics detect the damage but cannot attribute it without Chapter 12’s machinery. (C6) *robust for the ground process* (sector counts are complete), *thinned for the mark factor* (ranker metrics run on tracked risk sets with the outside option of Chapter 15).

Warning 11.16 (GOF cannot detect dead exposure). Under (C3), dead firms held at risk deflate every fitted firm-level hazard — and the *same* contaminated risk set is used at evaluation time, so the deflated model is scored against the corruption that produced it: held-out NLL, time-rescaling, and reliability can all pass while the hazard is wrong for every living firm. Goodness of fit is a consistency check between model and observed data, not between model and truth. This is why risk-set truthfulness has its own diagnostics (Chapters 7 and 14), why an exit classifier alone — however discriminative — cannot certify the repair, and why rule (i) of decision D7.4 exists: the risk-set rule *is* part of the estimate.

Code hooks.

Existing: `hawkes_calibration/mbpp/gof.py` implements `time_rescaling_residuals`, `ks_test_exp1` (numpy-only KS), `qq_exp1`, `poisson_pearson_residuals`, and `dispersion` — the cores of Protocols 11.3 and 11.8. To be written: `hawkes_calibration/eval/`, the evaluation harness that makes the protocol executable: `hawkes_calibration/eval/ladders.py` (held-out NLL versus the three named baselines, nats per event and per cell, decision D7.2); `hawkes_calibration/eval/pit.py` (randomized and non-randomized PIT, reliability binning, PIT-drift series); `hawkes_calibration/eval/rescaling.py` (wraps `mbpp.gof` with censored-final-gap handling, stratified martingale residuals Equation (11.2), bootstrap bands per Warning 11.4); `hawkes_calibration/eval/envelope.py` (simulation envelopes over `hawkes_calibration/eventtime/simulate.py` and `hawkes_calibration/mbpp/ic_simulate.py`, doubling as the bootstrap null for Warning 11.12); `hawkes_calibration/eval/boundary.py` (chi-bar LRT, Theorem 11.11); `hawkes_calibration/eval/rolling.py` (the rolling-origin driver of Protocol 11.15). Conventions: every campaign is an `experiments/expNN_name.py` script emitting seeded, config-stamped JSON to `results/`; the recovery gate of Protocol 11.14 reuses `synthetic/` world generation; profile-likelihood slices and the gradient check attach to the fitters in `hawkes_calibration/sector_stability.py`, `hawkes_calibration/eventtime/estimate.py`, and `hawkes_calibration/mbpp/`.

Part IV

The Covariate Problem

Chapter 12

The Stale-Covariate Problem

12.1 Why staleness owns a chapter

Defect (C5) of Chapter 2 is easy to state and easy to underestimate. The firm covariates that every model in this document consumes — financials, headcount, capitalization — are not observed as processes. PitchBook records them when, and only when, the firm itself transacts: each deal row carries the covariates as reported around that deal, and between deals the pipeline carries the last recorded value forward (Equation (2.1), implemented in `synthetic/loaders.py` and to be mirrored by every real-data loader per Protocol 2.5). Two facts make this more than a measurement-error nuisance. First, the error is not noise but *drift*: the gap between the last recorded value and the current truth grows with the time since the firm’s last deal. Second, the observation times are event times *of the process being modeled*. The firms that have gone longest without a deal have the stalest covariates, and they are exactly the low-intensity firms. Staleness is informative: the age of a firm’s covariates is itself evidence about the firm.

The stakes justify the chapter’s rank. Design decision D2.1 of Chapter 2 predicts that the largest modeling errors will come not from wrong model families but from unstated exposure to (C3) and (C5). The covariates-dominate hypothesis of Chapter 11 places most of the predictive mass in the covariate channel, and a planned covariate-noise campaign (Chapter 19) will measure how staleness erodes fitted structure before it erodes prediction. A defect that silently corrupts the dominant channel, differentially across the risk set, deserves its own chapter.

The chapter proceeds in the document’s standard order: formalize (Section 12.2), prove the damage in a tractable case (Section 12.3), then commit to a remedy ladder — R1, staleness-aware observed-data modeling, committed and always on (Section 12.4); R2, supervised drift modeling with regression calibration (Section 12.5); R3, an explicit covariate state space (Section 12.6); R4, joint modeling, adopted only if a defined trigger fires (Section 12.7) — and close with the bucket-model face of the same mathematics, the planned synthetic validation, and the defect declaration (Sections 12.8 and 12.9).

12.2 Formalization

Throughout, i ranges over tracked firms, N_i is firm i ’s deal counting process, $\mathcal{F}_t$ is the full filtration of Definition 2.1 *enlarged with the latent covariate paths*, and $\mathcal{G}_t \subset \mathcal{F}_t$ is the *observed* filtration: the internal history of the marked point process, the published macro paths, and the covariate values recorded at past deals — everything a production pipeline can know at t , nothing it cannot.

Assumption 12.1 (Latent covariate dynamics). Each tracked firm carries a latent càdlàg covariate process $x_i(t) \in \mathbb{R}^q$, partitioned into *mechanical* components — deterministic functionals

of the firm's own observed event history and marks (raised-to-date, deal count, time-stamped last-round descriptors) — and *drifting* components (revenue, EBITDA, headcount, latent valuation). As a working model, the drifting components follow a mean-reverting Ornstein–Uhlenbeck diffusion (Uhlenbeck and Ornstein, 1930) with sector- and macro-driven mean,

$$dx_i(t) = -\Theta(x_i(t) - m_i(t))dt + \Sigma_W^{1/2}dW_i(t), \quad m_i(t) = m_{s(i)} + \Gamma^\top x^{\text{mac}}(t), \quad (12.1)$$

with $\Theta = \text{diag}(\vartheta_1, \dots, \vartheta_q)$, $\vartheta_k > 0$ (the mean-reversion rates ϑ are unrelated to the excitation decay θ of Chapter 8), Σ_W diagonal, and W_i idiosyncratic across firms given sector and macro. Where stationarity is invoked, coordinate k has stationary variance $\sigma_{X,k}^2 = \Sigma_{W,kk}/(2\vartheta_k)$. Only the qualitative content — mean reversion toward an observable anchor, conditional variance growing in elapsed time — is load-bearing; the Gaussian linear form is what makes the propositions below exact.

Read concretely, Equation (12.1) says: a firm's revenue and headcount track an anchor set by its sector's trajectory and the macro state — rate cuts and sector booms lift the anchor, contractions lower it — while ϑ_k sets the memory of coordinate k , with covariate half-life $\log 2/\vartheta_k$. Nothing in the data pins these half-lives a priori — headcount is plausibly slow, revenue faster, reported EBITDA noisier still — and estimating them is exactly what the drift dataset of Section 12.5 is for. The OU form is chosen because it is the tractable member of its qualitative class: it is the Gaussian Markov process with mean reversion, it closes the filtering recursions of Section 12.6 in closed form, and it makes every distortion formula in Section 12.3 exact rather than first-order. When the working model fails — heavy-tailed revenue jumps, regime-switching growth — the exact constants below degrade to first-order approximations, but the monotone conclusions (attenuation increasing in age, variance increasing in age, survival tilt downward) survive, because they use only the qualitative content.

Assumption 12.2 (LOCF observation scheme). The data delivers, for each firm and each t past its entry V_i ,

$$\tilde{x}_i(t) = x_i(\tau_i(t)), \quad \tau_i(t) = \sup\{t_m < t : i_m = i\}, \quad a_i(t) = t - \tau_i(t), \quad (12.2)$$

with the strict inequality of Protocol 2.5(b): the value used at t comes from deals *strictly before* t , so that $\tilde{x}_i$ and the staleness age a_i are left-continuous in t . At its own deal times the firm's drifting covariates are observed (as reported in the deal row); Section 12.6 adds reporting noise. For mechanical components LOCF is *exact* — they are step functions that only move at own deals — so (C5) binds on the drifting components alone. This split is the basis of the column classification in Appendix B and of decision D8.3 below.

Proposition 12.3 (The observation scheme is internal and informative). *Under Assumptions 12.1 and 12.2:*

- (i) *Each observation time of firm i 's covariates is a jump time of N_i , hence a stopping time of the modeled filtration $\{\mathcal{F}_t\}$; the processes $\tau_i(t)$, $a_i(t)$, and $\tilde{x}_i(t)$ are left-continuous and adapted to $\{\mathcal{G}_t\}$, hence $\mathcal{G}$ -predictable — they are legitimate intensity covariates.*
- (ii) *The staleness age is a deterministic functional of the event history: $\{a_i(t) > u\} = \{N_i([t-u, t]) = 0\}$; it is the backward recurrence time of the modeled process.*
- (iii) *The scheme is informative: whenever the true intensity λ_i depends on x_i , the conditional law of $x_i(t)$ given $(\tilde{x}_i(t), a_i(t))$ differs from the law obtained from the free dynamics Equation (12.1) alone, because $\mathbb{P}(a_i(t) > u \mid \mathcal{F}_t^X \vee \mathcal{F}_\tau) = \exp(-\int_{t-u}^t Y_i(v) \lambda_i(v) dv)$ depends on the latent covariate path over the gap.*

Proof. (i) Jump times of an adapted counting process are stopping times; between jumps τ_i is frozen and a_i grows deterministically, and the strict-inequality convention in Equation (12.2) makes all three processes left-continuous adapted, hence predictable. (ii) is Equation (12.2) read backwards. (iii) The displayed survival probability is the standard conditional-inhomogeneous-Poisson formula given the intensity path; if λ_i is x_i -measurable and nonconstant in it, the event $\{a_i(t) > u\}$ has different probability under different covariate paths, so by Bayes the conditional law of the path given the gap is tilted (Lemma 12.8 computes the tilt). $\square$

In the Kalbfleisch–Prentice taxonomy of time-dependent covariates (Kalbfleisch and Prentice, 2002, Ch. 6), the macro series are *external*: they evolve regardless of the deal process, and survival-type probability statements conditioning on their future paths are legitimate. The pair $(\tilde{x}_i, a_i)$ is *internal*: it is a functional of the firm’s own event history and marks. The consequence, inherited verbatim from the internal-covariate discussion of Chapter 8, is that intensities conditioned on $(\tilde{x}_i, a_i)$ retain their interpretation, while plug-in survival functionals (“probability of no deal in two years at today’s covariates”) do not — the covariate would not stay at today’s value. One-step prediction is clean; multi-step statements must simulate the system jointly, covariate observation process included.

How much information does the age itself carry? We treat this as an empirical question for Chapter 19. The sign is not obvious: in a cooldown world, recent activity can be anti-signal at short range, so a recency-only ranker need not beat even a popularity baseline. The committed choice models carry an explicit gap covariate because the gap is informative, but its fitted functional form is expected to depend on the generating process, so it must be validated rather than assumed. The age is neither ignorable nor monotone in effect: it interacts with everything else, which is why Section 12.4 insists on interactions and why Section 12.8 argues trees are the natural estimator of that surface. The lesson to carry forward: a is simultaneously a nuisance index (how corrupted is $\tilde{x}$) and a signal (how long has the market declined to fund this firm), and a correct treatment must let the data express both roles at once.

12.3 What staleness does to a log-linear intensity

Fix one firm, drop the subscript, and let the true intensity be log-linear in the current latent covariates,

$$\lambda(t) = \exp(\beta^\top x(t)), \quad x(t) = \tilde{x}(t) + \varepsilon(t), \quad (12.3)$$

where $\varepsilon(t) = x(t) - x(\tau)$ is the drift accumulated since the last observation. (A baseline and external covariates can be carried along untouched; they are omitted for legibility.) We fit with $\tilde{x}$ in place of x . Everything in this section is a statement about what that substitution does.

12.3.1 Attenuation in the Gaussian-drift case

Two benchmark drift laws bracket the mechanism. The first switches off mean reversion and the survival information; it isolates the pure variance effect.

Proposition 12.4 (Independent Gaussian drift: multiplicative distortion). *Suppose that, given $(\tilde{x}, a) = (\tilde{x}, a)$ and under the free drift law (survival information ignored), $\varepsilon(t) \sim \mathcal{N}(0, \Sigma(a))$ independent of $\tilde{x}$ — the $\Theta \rightarrow 0$ (Brownian) limit of Equation (12.1), where $\Sigma(a) = \Sigma_W a$. Then the intensity implied for the observed data is*

$$\tilde{\lambda}(t) = \mathbb{E}[\exp(\beta^\top x(t)) \mid \tilde{x}, a] = \exp\left(\beta^\top \tilde{x} + \frac{1}{2} \beta^\top \Sigma(a) \beta\right) : \quad (12.4)$$

a staleness-dependent multiplicative distortion $\exp\{\frac{1}{2} \beta^\top \Sigma(a) \beta\} \geq 1$, increasing in a , with the coefficient on $\tilde{x}$ unbiased at fixed a .

Proof. $\mathbb{E}[\exp(\beta^\top \varepsilon) \mid \tilde{x}, a] = \exp(\frac{1}{2}\beta^\top \Sigma(a)\beta)$ is the Gaussian moment generating function; independence lets it factor out of $\exp(\beta^\top \tilde{x})$. $\square$

Remark 12.5. The structure “truth = observed + independent error” is Berkson-type, and Equation (12.4) shows its signature: no slope attenuation *conditional on a* , but an omitted a -dependent term. Convexity of $\exp$ makes uncertain firms look *more* intense on average — the variance term rewards staleness — which is one half of the ranking distortion of Proposition 12.9. The classical-error benchmark, $\tilde{x} = x + U$ with $U \perp x$, gives the textbook attenuation instead: $\mathbb{E}[x \mid \tilde{x}] = \mu + \Sigma_X(\Sigma_X + \Sigma_U)^{-1}(\tilde{x} - \mu)$, so the induced log-intensity is linear in $\tilde{x}$ with slope $\Sigma_X(\Sigma_X + \Sigma_U)^{-1}\beta$ — in the scalar case $\beta \sigma_X^2/(\sigma_X^2 + \sigma_U^2)$ (Prentice, 1982; Carroll et al., 2006). The planned covariate-noise campaign realizes exactly this benchmark (Section 12.8).

The second benchmark restores mean reversion; it produces genuine, staleness-graded attenuation and, as a corollary, the functional form of remedy R1.

Proposition 12.6 (Stationary OU drift: staleness-graded attenuation). *Let the drifting covariates follow Equation (12.1) with constant mean m , diagonal Θ , stationary initial condition, and survival information ignored. Then, given $(\tilde{x}, a)$,*

$$x(t) \mid \tilde{x}, a \sim \mathcal{N}(m + e^{-\Theta a}(\tilde{x} - m), V(a)), \quad V(a)_{kk} = \sigma_{X,k}^2(1 - e^{-2\vartheta_k a}), \quad (12.5)$$

and the observed-data intensity is

$$\tilde{\lambda}(t) = \exp\left(\underbrace{(e^{-\Theta a}\beta)^\top \tilde{x}}_{\text{attenuated signal}} + \underbrace{\beta^\top (I - e^{-\Theta a})m + \frac{1}{2}\beta^\top V(a)\beta}_{= g_0(a), \text{ staleness offset}}\right). \quad (12.6)$$

At fixed staleness the coefficient on $\tilde{x}_k$ is $e^{-\vartheta_k a}\beta_k$: an attenuation factor $e^{-\vartheta_k a} \in (0, 1)$, decreasing in the age, equal to the regression-calibration factor $\text{Cov}(x_k(t), \tilde{x}_k)/\text{Var}(\tilde{x}_k)$.

Proof. Variation of constants on Equation (12.1) gives $x(t) = m + e^{-\Theta a}(x(\tau) - m) + \int_\tau^t e^{-\Theta(t-u)}\Sigma_W^{1/2}dW(u)$; the stochastic integral is Gaussian, independent of $x(\tau)$, with the stated variance (compute $\int_0^a e^{-2\vartheta_k u}\Sigma_{W,kk}du$ and use $\sigma_{X,k}^2 = \Sigma_{W,kk}/2\vartheta_k$), which proves Equation (12.5). Then Equation (12.6) is the Gaussian mgf as in Proposition 12.4. For the last claim, stationarity makes $(x(\tau), x(t))$ jointly Gaussian with $\text{Var}(\tilde{x}_k) = \sigma_{X,k}^2$ and $\text{Cov}(x_k(t), \tilde{x}_k) = \sigma_{X,k}^2 e^{-\vartheta_k a}$, so the ratio is $e^{-\vartheta_k a}$; equivalently, time reversibility of the stationary OU lets one write $\tilde{x}_k = m_k + e^{-\vartheta_k a}(x_k(t) - m_k) + U_k$ with $U_k \perp x_k(t)$ — the classical shrink-plus-noise surrogate structure of Prentice (1982), with $e^{-\vartheta_k a}$ playing the role of $\sigma_X^2/(\sigma_X^2 + \sigma_U^2)$. $\square$

Corollary 12.7 (Pooled fit: risk-set-weighted attenuation plus an informativeness term). *Take the scalar case with centered covariate ($m = 0$), and fit the misspecified model $\lambda = \exp(c + b\tilde{x})$ across the risk set while the truth is Equation (12.6). In the small-signal (first-order) regime, the quasi-likelihood fit is the exposure-weighted least-squares projection of the true log-intensity onto $(1, \tilde{x})$, so the estimand is*

$$b = \frac{\text{Cov}_w(\tilde{x}, e^{-\vartheta a}\tilde{x})}{\text{Var}_w(\tilde{x})}\beta + \frac{\text{Cov}_w(\tilde{x}, g_0(a))}{\text{Var}_w(\tilde{x})}, \quad (12.7)$$

with weights w proportional to exposure. If $a \perp \tilde{x}$ across the risk set, $b = \mathbb{E}_w[e^{-\vartheta a}]\beta$: pure attenuation by the average factor. Informativeness breaks the independence with a definite sign: if the intensity is increasing in $\beta^\top x$ with $\beta > 0$, then a is stochastically decreasing in $\tilde{x}$ (better last-seen firms raise sooner), so for the increasing offset g_0 the second term is ≤ 0 — the omitted staleness structure biases the fitted slope further toward zero, beyond the average attenuation.

Proof. Linearizing the log-likelihood Equation (2.3) in the small-signal regime reduces the score equations for (c, b) to the normal equations of exposure-weighted least squares of the true log-intensity $e^{-\vartheta a} \beta \tilde{x} + g_0(a)$ on $(1, \tilde{x})$; Equation (12.7) is the resulting projection coefficient. Under independence the first ratio is $\mathbb{E}_w[e^{-\vartheta a}]$ and the second covariance vanishes. For the sign claim: a stochastically decreasing in $\tilde{x}$ means $\mathbb{E}[g_0(a) \mid \tilde{x}]$ is nonincreasing in $\tilde{x}$ for any increasing g_0 , hence $\text{Cov}_w(\tilde{x}, g_0(a)) \leq 0$. $\square$

This corollary is the precise version of the chapter's slogan: staleness does not merely add noise, it attenuates the fitted covariate effects by a factor tied to the age distribution of the risk set, and the informativeness of the ages adds a signed omitted-variable term on top. Both effects disappear from a model that conditions on the age — which is remedy R1.

12.3.2 The informativeness twist: distortion is differential across the risk set

Propositions 12.4 and 12.6 deliberately ignored one piece of information: the firm has *not* raised during $(\tau, t]$. By Proposition 12.3(iii) that silence is evidence. The following lemma pins its direction; to stay tractable it summarizes the intensity over the gap through the endpoint signal, which is adequate for the slowly-varying covariates at issue.

Lemma 12.8 (Survival tilt). *Let $S = \beta^\top x(t)$ have conditional density $p(s)$ given $(\tilde{x}, a)$ under the free drift law, and suppose the integrated intensity over the gap is $\int_\tau^t \lambda(u) du = a \Lambda(S)$ with Λ strictly increasing (e.g. covariates approximately constant over the gap, $\Lambda(s) = e^s$). Then the conditional density given additionally $\{\text{no event in } (\tau, t]\}$ is*

$$p_{\text{surv}}(s) \propto p(s) e^{-a\Lambda(s)},$$

which is dominated by p in the likelihood-ratio order; consequently, for every increasing h with finite means, $\mathbb{E}[h(S) \mid \tilde{x}, a, \text{no event}] \leq \mathbb{E}[h(S) \mid \tilde{x}, a]$, strictly if $a > 0$, $\beta \neq 0$ and S is nondegenerate.

Proof. Bayes gives the displayed density; the ratio $p_{\text{surv}}/p \propto e^{-a\Lambda(s)}$ is strictly decreasing in s , which is likelihood-ratio dominance, and likelihood-ratio dominance implies the expectation inequality for increasing h (Chebyshev's correlation inequality applied to the decreasing weight $e^{-a\Lambda(s)}$ and increasing h). When the monotone-summary assumption fails — strongly oscillating paths — the pointwise ordering can fail locally, but the mechanism survives: the tilt $\exp(-\int \lambda)$ always downweights high-intensity paths. $\square$

Proposition 12.9 (Sign of the ranking bias under naive LOCF plug-in). *Rank firms at time t by the naive plug-in score $\tilde{s}_i = \beta^\top \tilde{x}_i(t)$, while the correct observed-data score (the log of the projected intensity, Proposition 12.12) is $s_i^* = \log \mathbb{E}[\exp(\beta^\top x_i(t)) \mid \mathcal{G}_{t-}]$. Under Assumption 12.1 with stationary drift and Lemma 12.8:*

- (i) *Conditional on the truth: the score error is $\tilde{s}_i - \beta^\top x_i(t) = -\beta^\top \varepsilon_i(t)$; a firm whose covariates deteriorated in the β direction since its last deal ($\beta^\top \varepsilon_i < 0$) is over-ranked, one that improved is under-ranked, with error magnitude growing in the drift, hence stochastically in a_i .*
- (ii) *Conditional on the data: by Proposition 12.6 and Lemma 12.8,*

$$s_i^* \leq \beta^\top m + e^{-\vartheta a_i} \beta^\top (\tilde{x}_i - m) + \frac{1}{2} \beta^\top V(a_i) \beta,$$

so for firms whose last-seen signal was above the sector mean ($\beta^\top \tilde{x}_i > \beta^\top m$), the naive score exceeds the correct one by at least $(1 - e^{-\vartheta a_i}) \beta^\top (\tilde{x}_i - m) - \frac{1}{2} \beta^\top V(a_i) \beta$, a quantity

that is positive and increasing in a_i once $\beta^\top(\tilde{x}_i - m)$ clears the variance term, and the survival tilt of Lemma 12.8 strictly enlarges the gap. Symmetrically, below-mean stale firms are pushed up by mean reversion and down by the tilt: their net bias is smaller and can take either sign.

- (iii) Across the risk set: *a concentrates on low-intensity firms (Proposition 12.3), so the distortion is differential — largest exactly where the naive score trusts old good news. The characteristic failure is the stale champion: a firm that looked excellent at its last round years ago and has been silent since is systematically over-ranked by LOCF plug-in, because its signal should have been shrunk toward the prior and tilted down for the silence, and was neither.*

Proof. (i) is arithmetic on Equation (12.3). (ii): the unconditional (no-tilt) projection is Equation (12.6), whose log is the displayed bound; Lemma 12.8 applied to $h = \exp$ shows the survival-conditioned expectation is strictly smaller. Subtracting from $\tilde{s}_i = \beta^\top \tilde{x}_i$ gives the stated gap. (iii) combines (ii) with Proposition 12.3(iii). $\square$

Example 12.10 (The size of the inversion). Scalar covariate, $m = 0$, $\sigma_X = 1$, $\beta = 1$, covariate half-life two years ($\vartheta = \log 2/2 \approx 0.347$ per year). Firm A last raised four years ago looking excellent, $\tilde{x}_A = +2$; firm B raised a quarter ago looking merely good, $\tilde{x}_B = +1$. Naive LOCF plug-in scores A far above B (2 vs 1). The corrected log-scores of Equation (12.6), before any survival tilt: for A , $e^{-1.386} \cdot 2 + \frac{1}{2}(1 - e^{-2.77}) \approx 0.50 + 0.47 = 0.97$; for B , $e^{-0.087} \cdot 1 + \frac{1}{2}(1 - e^{-0.173}) \approx 0.92 + 0.08 = 1.00$. The correction alone *reverses* the ranking, and the survival tilt of Lemma 12.8 — four years of silence versus one quarter — pushes A down further. At the risk-set scale this is not an edge case: with inter-deal gaps of one to four years and covariate half-lives of the same order, the attenuation factor $e^{-\vartheta a}$ ranges over $(0.25, 1)$ within a single risk set, so a model fitting one common slope is wrong for most of its members at once.

Warning 12.11 (Look-ahead through deal-time covariates is fatal). The covariates in deal row m are measured *at or after* the deal: revenue as reported in the round’s diligence, raised-to-date *including* the round, valuation — the outcome itself. Using row m ’s covariates as features for predicting deal m , or any as-of join keyed on reference date rather than availability date, imports the label into the features. This is the internal-covariate “price sticker” in its most concrete form: the covariate value exists *because* the event happened. The failure mode is silent — metrics inflate in backtest and evaporate in deployment — and no downstream remedy can undo it. Policy: features at t use deals strictly before t (Protocol 2.5); every loader ships a look-ahead audit that verifies no feature path changes value at the timestamp of the label event it feeds (Section 12.9, code hooks). The synthetic loader already enforces the strict inequality; the real-data loader must inherit it verbatim.

12.4 Remedy R1: staleness-aware observed-data modeling

The first remedy is not an attempt to undo the defect but to model what we actually observe. Condition on the pair $(\tilde{x}, a)$ and fit

$$\lambda_i(t) = \exp\left(\beta^\top \tilde{x}_i(t) + g(a_i(t)) + \tilde{x}_i(t)^\top \Gamma_g \phi(a_i(t)) + \delta^\top z_i(t)\right), \quad (12.8)$$

with g flexible (a spline or step function on an age grid with knots at the empirical staleness quantiles), ϕ a low-dimensional basis in the age whose interaction with $\tilde{x}$ lets the covariate *slope* vary with staleness, and z_i the other engineered history covariates of Definition 2.2(iii). By Proposition 12.3(i) every term is $\mathcal{G}$ -predictable, so Equation (12.8) is a well-specified

intensity model *for the observed filtration* and the likelihood Equation (2.3) applies with its full martingale estimation theory (Andersen et al., 1993; Andersen and Gill, 1982). This is not a heuristic patch:

Proposition 12.12 (Projection: the observed-data intensity is a genuine intensity). *Let N be a counting process adapted to a subfiltration $\{\mathcal{G}_t\} \subseteq \{\mathcal{F}_t\}$ and let λ be its $\mathcal{F}$ -intensity. Then the $\mathcal{G}$ -predictable projection*

$$\lambda^{\mathcal{G}}(t) = \mathbb{E}[\lambda(t) \mid \mathcal{G}_{t-}]$$

is an intensity of N with respect to $\{\mathcal{G}_t\}$: $N(t) - \int_0^t \lambda^{\mathcal{G}}(u) du$ is a $\mathcal{G}$ -local martingale. Consequently the best-fitting element of a rich class of functions of $(\tilde{x}, a, \mathcal{H}_{t-})$ estimates the projection of the true structural intensity onto the available information — fitting Equation (12.8) is estimating a well-defined object, not compensating an error (Daley and Vere-Jones, 2008, Ch. 14) (Andersen et al., 1993, II.4).

Proof. For any bounded nonnegative $\mathcal{G}$ -predictable C (which is also $\mathcal{F}$ -predictable), the $\mathcal{F}$ -intensity property gives $\mathbb{E} \int_0^\infty C dN = \mathbb{E} \int_0^\infty C(u) \lambda(u) du$. Tower on $\mathcal{G}_{u-}$: since $C(u)$ is $\mathcal{G}_{u-}$ -measurable, $\mathbb{E}[C(u) \lambda(u)] = \mathbb{E}[C(u) \mathbb{E}(\lambda(u) \mid \mathcal{G}_{u-})] = \mathbb{E}[C(u) \lambda^{\mathcal{G}}(u)]$, whence $\mathbb{E} \int C dN = \mathbb{E} \int C \lambda^{\mathcal{G}}$. This identity for all such C is the defining property of a $\mathcal{G}$ -intensity (martingale characterization via the monotone class theorem; the technical identification of $\mathbb{E}[\cdot \mid \mathcal{G}_{t-}]$ with the predictable projection is Daley and Vere-Jones, 2008, Ch. 14). $\square$

Corollary 12.13 (R1 is exactly specified in the OU world). *Under Assumption 12.1 with stationary drift and the monotone-summary approximation of Lemma 12.8, the projected intensity is $\exp\{(e^{-\Theta a} \beta)^\top \tilde{x} + g_0(a) - d(\beta^\top \tilde{x}, a)\}$ with g_0 as in Equation (12.6) and $d \geq 0$ the survival-tilt correction, a smooth function of the scalar signal and the age. The class Equation (12.8) contains the first two terms exactly (the interaction basis reproduces $e^{-\vartheta_k a} \beta_k$) and approximates the third within the same $(\tilde{x}^\top \gamma, a)$ interaction structure. R1 is therefore not merely valid but — in the working model — essentially well specified.*

Two practical points before the caveats. First, the repository already implements degenerate cases of Equation (12.8): the risk-set ranker of `hawkes_calibration/sector_ranker.py` carries a months-since-last-funding (cooldown) covariate — a parametric g inside the choice model — and the synthetic loader emits it point-in-time (`log1p_months_since_last_funding` in `CHOICE_FEATURES`). D8.1 is the commitment that this stops being a per-model accident: the pair $(\tilde{x}, a)$, with interactions where the model class allows them, becomes mandatory plumbing in every stack. Second, an identifiability warning that doubles as an interpretation warning: the fitted g absorbs *everything* that varies with the gap — drift-variance inflation (Proposition 12.4), mean-reversion offsets (Proposition 12.6), the survival tilt, *and* genuine duration dependence of the hazard itself (post-round cooldown, runway mechanics, the renewal structure of Chapter 8). These are observationally confounded in g by construction: the projection does not distinguish “covariates have drifted” from “the hazard truly depends on time since last round.” For prediction under the observed filtration this is harmless — the projection is the target (Proposition 12.12) — but it means g has no structural reading either, and separating the channels requires the state-space machinery of Section 12.6, which supplies the drift-variance component of g from fitted dynamics and leaves the remainder as an honest estimate of duration dependence.

What R1 gives up is exactly what the projection destroys. The fitted β in Equation (12.8) is a property of the *observation scheme composed with* the structural model: it mixes the true covariate effect with mean reversion, drift variance, and the informativeness tilt. It must not be read as “the effect of revenue on funding intensity,” used for counterfactuals (“if this firm’s revenue were 20% higher...”) — the projection does not move that way), or extrapolated to a vendor with a different update policy. And because $(\tilde{x}, a)$ is internal, multi-horizon functionals

require joint simulation of events and the LOCF update process, never plug-in survival formulas (Section 12.2). Within those limits, R1 is committed, cheap, and always on — decision D8.1.

12.5 Remedy R2: supervised drift modeling and regression calibration

R1 conditions on staleness; R2 tries to *reverse* it. The key observation is that the dataset labels its own drift: at every deal the firm's true drifting covariates are re-observed. Every consecutive pair of own deals $(\tau_m, \tau_{m'})$ of the same firm therefore yields a supervised example

$$\underbrace{(x_i(\tau_m), a = \tau_{m'} - \tau_m, \{x_{\text{pub}}^{\text{mac}}(u)\}_{u \in (\tau_m, \tau_{m'}]}, \text{sector activity})}_{\text{inputs}} \mapsto \underbrace{x_i(\tau_{m'})}_{\text{target}},$$

a *drift dataset* on which a conditional model $\hat{m}(\tilde{x}, a, \cdot)$ with predictive variance $\hat{V}(\tilde{x}, a, \cdot)$ can be fit — linear-Gaussian (which is exactly the OU regression of Equation (12.5), making R2 and R3 two estimators of one object) or gradient-boosted with a variance head. The input design follows Equation (12.1): the macro path enters through summaries of the anchor movement over the gap — cumulative rate change, CPI and unemployment changes between the two deal dates, as-of joined on publish dates per Protocol 2.5(a) — and sector activity through trailing deal counts, so that the linear-Gaussian fit is literally a discretized regression estimate of $(\Theta, \Gamma, m_s, \Sigma_W)$, one consecutive-pair likelihood term per row. The gradient-boosted variant keeps the same inputs and relaxes the functional form; it is admitted only if it beats the linear-Gaussian model on *temporally* held-out pairs (pairs whose second deal falls after the training era, per Protocol 2.5(e)) on both mean error and variance calibration — a drift model whose $\hat{V}$ is uncalibrated poisons Equation (12.9) through the exponent. Imputation then replaces the stale value by $\hat{x}_i(t) = \hat{m}(\tilde{x}_i(t), a_i(t), \text{macro path})$, and the intensity uses regression calibration:

Proposition 12.14 (Plug-in of the conditional mean, with variance correction). *Suppose $x(t) \mid \mathcal{G}_{t-} \sim \mathcal{N}(\hat{x}(t), V(t))$. Then the projected intensity of Proposition 12.12 is exactly*

$$\lambda^{\mathcal{G}}(t) = \exp\left(\beta^\top \hat{x}(t) + \frac{1}{2} \beta^\top V(t) \beta\right). \quad (12.9)$$

For a general conditional law with mean $\hat{x}$ and variance V , $\log \lambda^{\mathcal{G}} = \beta^\top \hat{x} + \frac{1}{2} \beta^\top V \beta + O(\|\beta\|^3)$ by the cumulant expansion: plugging in the conditional mean makes the first-order term exact, and the variance factor $\exp(\frac{1}{2} \beta^\top V \beta)$ absorbs the second order. Plugging in the raw $\tilde{x}$, or a mode or median, leaves a first-order error. This is regression calibration (Carroll et al., 2006, Ch. 4), and Equation (12.9) is the log-linear analogue of the induced-hazard formula of Prentice (1982) — including his caveat, which here is Lemma 12.8: the conditioning $\mathcal{G}_{t-}$ includes the survival information, so the honest $\hat{x}$ is the tilted mean, not the free-drift mean.

Proof. The Gaussian case is the mgf again. In general, $\log \mathbb{E}[e^{\beta^\top x}] = \sum_{r \geq 1} c_r(\beta)/r!$ with first and second cumulants $\beta^\top \hat{x}$ and $\beta^\top V \beta$, the remainder collecting third and higher cumulants of order $\|\beta\|^3$. $\square$

Proposition 12.15 (Selection: the drift dataset is drift conditional on success). *A training pair exists only if a second deal occurred. Under the monotone-summary approximation of Lemma 12.8 with $\Lambda(s) = e^s$, the drift law represented in pairs with gap a is tilted relative to the free law by $e^s e^{-ae^s}$ (event density at the endpoint), while the population the imputation serves — firms still waiting at age a — is tilted by e^{-ae^s} alone. The ratio of the two tilts is e^s ,*

increasing in $s = \beta^\top x(\tau + a)$, so the pair-conditioned law dominates the survivor-conditioned law in the likelihood-ratio order:

$$\mathbb{E}[\beta^\top x(\tau + a) \mid \text{pair with gap } a, \tilde{x}] > \mathbb{E}[\beta^\top x(\tau + a) \mid \text{no event by } \tau + a, \tilde{x}].$$

A drift model fit naively on pairs therefore learns drift conditional on funding success and systematically over-predicts the β -direction state of the silent firms it is deployed on — optimistic imputation, which re-creates the stale-champion over-ranking of Proposition 12.9 through the back door.

Proof. The next-event density at gap a given the endpoint summary is $e^s e^{-ae^s}$ (hazard times survival), the no-event probability is e^{-ae^s} ; both statements are the conditional-Poisson formulas of Proposition 12.3(iii). Their ratio e^s is increasing, giving likelihood-ratio dominance of the pair law over the survivor law, and dominance implies the strict mean inequality for the nondegenerate increasing functional s . $\square$

Three remedies for the selection, in the order they will be tried. *Censoring-aware drift estimation:* treat pair generation as a survival problem — every firm silent since τ_m contributes a right-censored drift spell $(x_i(\tau_m), a_i(T), \text{no endpoint})$, and the drift model is fit by a likelihood that includes the survival factor, i.e. a small joint model of drift and event (this is R4's machinery in miniature, and the honest default). *Inverse-probability weighting:* weight each pair by the inverse of the estimated probability that a firm with those inputs produces a second deal within the observed gap, taken from the discrete-hazard bucket model of Chapter 13; valid under the bucket model's own correctness, cheap, and diagnosable. *Exogenous restriction:* fit drift only for covariates whose evolution is plausibly exogenous to the funding process on the gap timescale — headcount and revenue drift with the business; raised-to-date, deal counts and last-round descriptors are *mechanical*, move only at events, and must never be imputed (LOCF is exact for them, Assumption 12.2). The mechanical-vs-drifting classification of every PitchBook column is maintained in Appendix B and is decision D8.3; Table 12.1 fixes the scheme on the representative columns, including the third class that Warning 12.11 makes necessary: columns that are *outcomes* of the deal they ride on.

12.6 Remedy R3: an explicit covariate state space

R2's linear-Gaussian special case deserves its own treatment, because it is simultaneously the honest uncertainty model and the supplier of the $\Sigma(a)$, $V(a)$ objects that R1 and R2 consume. Model the drifting components per firm by Equation (12.1) — OU toward a sector mean with macro loadings — observed at own deals with reporting noise: $y_{i,m} = x_i(\tau_m) + \eta_m$, $\eta_m \sim \mathcal{N}(0, R)$ (PitchBook financials are reported figures, not audited truth; $R = 0$ recovers exact observation).

Lemma 12.16 (OU prediction and deal-time update). *Under Equation (12.1) with constant mean m (time-varying $m_i(u)$ adds the deterministic integral $\int_\tau^t e^{-\Theta(t-u)} \Theta m_i(u) du$, computable from the published macro path), the exact recursions for the conditional law $x_i(t) \mid \mathcal{G}_t \sim \mathcal{N}(\hat{x}_i(t), P_i(t))$ are:*

$$\begin{aligned} \text{predict over } a = t - \tau : \quad \hat{x}_i(t) &= m + e^{-\Theta a} (\hat{x}_i(\tau) - m), \\ P_i(t) &= e^{-\Theta a} P_i(\tau) e^{-\Theta a} + V(a), \quad V(a)_{kk} = \sigma_{X,k}^2 (1 - e^{-2\vartheta_k a}), \end{aligned} \tag{12.10}$$

$$\text{update at an own deal with report } y : \quad K = P(P + R)^{-1}, \quad \hat{x}^+ = \hat{x} + K(y - \hat{x}), \quad P^+ = (I - K)P. \tag{12.11}$$

Representative columns	Class	Staleness treatment
total_raised_to_date; deal count to date; last_financing_date/_size/_deal_type; vc_round progression; firm age	mechanical	LOCF exact by construction; recomputed point-in-time from the event log; imputation forbidden (D8.3)
revenue, ebitda, gross_profit, net_income, number_of_employees, growth since last deal; valuation between rounds	drifting	carried with a always (D8.1); eligible for R2 imputation with variance; modeled by the R3 state space
deal_size, pre_money_valuation, post_valuation, up/down/flat flags of the <i>current</i> round	outcome-at-event	usable as covariates only strictly after their deal; as predictors of their own deal they are look-ahead (Warning 12.11); as targets they belong to T4
business_status	drifting, laggy	exit-proxy territory of (C3); point-in-time only, never applied retroactively

Table 12.1: The column classification of decision D8.3 on representative PitchBook columns; the full dictionary lives in Appendix B. “Mechanical” columns are deterministic functionals of the observed event history, for which LOCF is exact; “drifting” columns are the carriers of defect (C5).

With $R = 0$ the update collapses to $\hat{x}^+ = y$, $P^+ = 0$, and the filter between deals is pure prediction: $\hat{x}_i(t)$ is exactly the shrunk mean of Equation (12.5) and $P_i(t) = V(a_i(t))$ is the explicit $\Sigma(a)$ that Propositions 12.4 and 12.6 postulated.

Proof. The predict step is Equation (12.5) applied to the current posterior (linear-Gaussian closure under the OU transition); the update step is the standard Kalman gain for a linear observation with Gaussian noise (Kalman, 1960; Harvey, 1989). The $R = 0$ limit is immediate. $\square$

The state-space parameters $(\Theta, \Sigma_W, m_s, \Gamma, R)$ are estimable by exact Gaussian likelihood on the drift dataset of Section 12.5 — with the same selection caveat, treated by the same censoring-aware likelihood: Proposition 12.15 applies to $\hat{\Theta}$ too, and ignoring it biases mean-reversion estimates toward the reversion experienced by successful firms. Estimation must pool: the median tracked firm contributes a handful of deals, far too few to identify per-firm dynamics, so (Θ, Σ_W) are shared at sector level and Γ globally, fitted by summing pair likelihoods across firms; the drifting covariates enter on log scales (log revenue, log headcount), where the Gaussian working model is tenable and where the loaders already live, with missing and undisclosed reports handled by the rules of Appendix B, never ad hoc. Firm-level heterogeneity beyond the pooled parameters is precisely the shared random effect that R4 would add — another sense in which the ladder is nested rather than branching. What R3 buys beyond R2: (i) $V(a)$ in closed form, feeding R1’s offset g_0 and R2’s variance correction Equation (12.9) instead of leaving them as free spline coefficients; (ii) calibrated per-firm uncertainty that grows honestly to the stationary variance as $a \rightarrow \infty$ — the model *knows* it knows nothing about a firm silent for a decade, which is precisely the epistemic behavior the ranking layer needs; (iii) a principled bridge to the doubly-stochastic intensity tradition — default intensities driven by stochastic covariates with fitted dynamics — whose viability at scale is established in credit risk (Duffie et al., 2007). The cost is the linear-Gaussian straitjacket, which is why R3 enters the ladder as the variance model *inside* R2 rather than as a replacement for the gradient-boosted drift mean.

12.7 Remedy R4: joint modeling, gated by a measured trigger

The statistically complete treatment is a joint model of the longitudinal covariates and the event intensity with shared random effects (Rizopoulos, 2012; Tsiatis and Davidian, 2004): latent per-firm effects b_i drive both the covariate trajectory ($x_i(t) = \mu(t) + \Phi(t)b_i + \epsilon_i(t)$) and the intensity ($\lambda_i(t) = \exp(\beta^\top x_i(t) + \alpha^\top b_i + \dots)$), and the likelihood integrates b_i out over both data streams jointly. This is the construction that makes the informative observation scheme part of the likelihood rather than a correction: the two-stage shortcut — fit the longitudinal model, plug fitted trajectories into the intensity — was shown biased precisely when the event process informs the covariate observations (Tsiatis et al., 1995), which is our Proposition 12.3(iii) verbatim, and the bias direction is the one Proposition 12.15 derives: trajectories fitted to success-selected observations are optimistic for the silent. The joint likelihood repairs this by making the no-event stretches pay their survival factor inside the same integral that fits the trajectory — R2’s censoring-aware drift estimation and R3’s censoring-aware state-space likelihood are exactly the first two terms of this construction, which is why the ladder converges to R4 rather than being overturned by it. What R4 adds beyond them is the shared random effect: persistent firm-level frailty entering both streams, so that a firm’s covariate *level and trajectory shape* inform its intensity beyond the current imputed value (Tsiatis and Davidian, 2004; Rizopoulos, 2012). The price is steep: numerical integration over random effects for every firm ($\sim 7,000$ tracked firms $\times$ quadrature nodes $\times$ optimizer iterations), delicate identifiability between α , β and the drift parameters — the frailty-versus-contagion confounding of Section 11.5 returns here in longitudinal form — and a model class fragile enough that its failure modes are harder to audit than the defect it fixes. R4 is therefore *escalation, not default*, and it is gated:

Protocol 12.17 (Attenuation-gap test, the R4 trigger). On the synthetic rehearsal world with drifting firm covariates, LOCF observation, and known structural β : (i) run the committed pipeline R1 + R2 (with R3’s variance); (ii) measure the attenuation ratio $\hat{r}$ = slope of the fitted log-intensity scores regressed on the oracle scores $\beta^\top x_i(t)$ over the risk set, and the top- K ranking gap between fitted and oracle covariates; (iii) the trigger fires if the 95% interval for $\hat{r}$ excludes $[0.9, 1.1]$ and the oracle-covariate ablation attributes a material top- K loss to residual staleness (not to (C3), which is measured separately per Chapter 2), in two consecutive evaluation campaigns. Only a fired trigger authorizes the R4 build; the same harness then has to show R4 closing the measured gap on rehearsal data before any real-data use.

12.8 The bucket-model face, and the planned measurement

Chapter 13 attacks the horizon target T3 with gradient boosting on a bucketed panel whose feature rows read, in words: “it has been a since this firm’s last deal, and these are its last-known covariates.” That is $(\tilde{x}, a)$ — and the trees’ recursive partitioning of the joint feature space fits arbitrary piecewise-constant surfaces in $(\tilde{x}, a)$, i.e. nonparametric estimates of exactly the projected intensity that Corollary 12.13 derived: offset $g_0(a)$, age-attenuated slopes $e^{-\Theta a}\beta$, and the survival-tilt correction, none of which need to be prespecified (Friedman, 2001; Chen and Guestrin, 2016). Gradient boosting on $(\tilde{x}, a)$ is R1, implemented nonparametrically. This is the principled reason the discrete-time stack tolerates staleness better than a rigid log-linear form fitted without age terms: the rigid form suffers the pooled attenuation and signed omitted-variable bias of Corollary 12.7, while the trees simply learn the interaction surface. If the boosted-ranker benchmark of the evaluation plan finds a gap between trees and the linear conditional logit, part of that gap may be exactly this staleness interaction; how much is an isolation question assigned to Chapter 19.

The attenuation mechanics themselves will be measured by a planned covariate-noise campaign, reported in Chapter 19. It has two stages. Stage one applies exogenous, uninformative Gaussian noise to the covariates at fit and prediction time — the *classical-error benchmark* of Section 12.3.1 — and measures how the fitted choice weights attenuate and how the recovered β degrades while the prediction metrics move. The hypothesis, from Proposition 12.6 and Corollary 12.7, is that structure erodes before prediction does: attenuation should hit *interpretation* (structural readings of β , cross-regime transfer) long before it hits ranking or likelihood. Stage two makes the noise informative: the loader emits drifting covariates with LOCF ages, and ranking bias is reported by staleness decile. This stage measures the differential distortion of Proposition 12.9, which uninformative noise cannot see, and doubles as the standing harness for Protocol 12.17.

Because aggregate metrics are expected to dilute exactly the damage this chapter is about — fitted structure decaying while headline prediction numbers barely move — the chapter installs a standing reporting discipline:

Protocol 12.18 (Staleness accounting in every evaluation). Every model evaluation report in the repository includes: (a) the distribution of a over the evaluation risk sets (quantiles, and the share exceeding one and two estimated covariate half-lives once R3 supplies $\hat{\Theta}$); (b) the headline metrics stratified by staleness quartile of the scored firm; (c) for ranking deliverables, the share of top- K selections whose drifting covariates are older than one year, reported next to the hit rate. Rationale: Proposition 12.9 localizes the damage in the stale strata, so a model can win on average while being systematically wrong about the firms it is most confidently promoting from old information — and the consuming action (a diligence short list) is exposed to exactly those firms. The stratified report is the cheap standing detector of residual (C5) damage and the attribution input to Protocol 12.17(iii).

12.9 Defect declaration and commitments

Per decision D2.1, the models of this chapter declare their position against the ledger of Table 2.1. **(C5)** is *owned here*: R1 corrects the fitting bias exactly in the OU working model (Corollary 12.13) and validly in general (Proposition 12.12); R2/R3 additionally correct the ranking bias under a correct, selection-aware drift model; residual vulnerability is the survival-tilt approximation and drift-model error, monitored by Protocol 12.17. **(C1)**: robust — all constructions are point-in-time and censored spells enter the drift likelihood as censored, per Section 12.5; dropping them would reimport Proposition 12.15 with the sign derived there. **(C2)**: robust by construction — $(\tilde{x}, a)$ is defined only after first observation; pre-entry firms are the newcomer problem of Chapter 15. **(C3)**: *vulnerable, with a signed interaction that must be stated*. An unrecorded-dead firm accumulates maximal staleness with frozen covariates: (C3) keeps it in the risk set, (C5) keeps its last good numbers on its face. The two defects compound on the same firms — the stale champion of Proposition 12.9 is, in the worst case, dead — and both corrupt the risk set’s information state in the same direction: overstated scores for long-silent firms, hence jointly worse for ranking than either alone. The staleness terms $g(a)$ partially absorb (C3) (large age predicts death as well as drift), which improves ranking but permanently confounds “quiet” with “gone”: no structural reading of g is permitted while (C3) is open. **(C4)**: robust — staleness ages are measured in weeks-to-years and deal-date jitter is days; aggregation does not touch the LOCF map. **(C6)**: out-of-universe firms are the $a \rightarrow \infty$ limit — no covariates at all — and are handled by the outside-option construction of Chapter 15, which this chapter’s g should be checked against for continuity (a firm silent for a decade should not score very differently from an unseen firm).

Design decision (D8.1 — staleness age is a mandatory feature).

From this document forward, every firm-level model in every stack — event-time intensities, the risk-set ranker, the bucketed panel, mark regressions — includes the staleness age $a_i(t)$ (and, where the model class permits, its interactions with $\tilde{x}_i$) among its features. Rationale: a is free, predictable (Proposition 12.3), and carries the offset and attenuation structure of Proposition 12.6; omitting it produces the signed biases of Corollary 12.7 and Proposition 12.9. Reversal trigger: none — a fitted model may drive its age terms to zero, but the feature may not be withheld from it.

Design decision (D8.2 — the remedy ladder).

Committed sequence: R1 (staleness-aware observed-data modeling, Equation (12.8)) now and always on; R2 (drift dataset, supervised drift model, regression calibration per Proposition 12.14, selection handled censoring-aware first) next quarter; R3 (OU/Kalman state layer, Lemma 12.16) folded in as R2's variance model rather than a separate stack; R4 (joint model) only if Protocol 12.17 fires. Rationale: the ladder is cost-ordered, each rung consumes the previous rung's outputs, and the working hypothesis — to be tested in Chapter 19 — is that prediction is initially robust to staleness, so the expensive rungs must prove their marginal value on a measured gap, not on principle. Reversal trigger: a fired gap test promotes R4; a gap test passing twice consecutively with R1 alone demotes R2/R3 to research status.

Design decision (D8.3 — mechanical vs. drifting column classification).

Every PitchBook covariate column is classified in the data dictionary (Appendix B) as *mechanical* (deterministic functional of the observed event history: raised-to-date, deal counts, last-round descriptors, ownership accumulations — LOCF exact, imputation forbidden) or *drifting* (revenue, EBITDA, net income, employees, growth rates, latent valuation — LOCF stale, carried with a , eligible for R2/R3 treatment). The classification is maintained under vendor schema changes and consumed by the loaders and the drift-dataset builder. Rationale: Assumption 12.2 shows (C5) binds only on drifting columns, and imputing a mechanical column is not an approximation but an error. Reversal trigger: none; reclassification of individual columns requires a data-dictionary change with review.

Code hooks.

Planned module `hawkes_calibration/covstate/`: `pairs.py` — the drift-dataset builder (consecutive own-deal pairs with gap, macro path and sector-activity summaries; right-censored spells emitted with flags per Section 12.5; selection diagnostics from Proposition 12.15); `ou.py` — OU transition and $V(a)$ in closed form, Kalman predict/update of Lemma 12.16, censoring-aware Gaussian likelihood for $(\Theta, \Sigma_W, m_s, \Gamma, R)$; `impute.py` — the imputation API returning $(\hat{x}_i(t), V_i(t))$ and the calibrated intensity factor $\exp(\frac{1}{2}\beta^\top V\beta)$ of Equation (12.9). Loaders: `synthetic/loaders.py` already enforces strictly-prior

LOCF and carries a months-since-last-funding feature; it and the real-data adapter `hawkes_calibration/data/pitchbook.py` must emit the pair $(\tilde{x}_i, a_i)$ explicitly for every stack (D8.1), keyed to the mechanical/drifted classification of Appendix B (D8.3). Audits (`hawkes_calibration/data/audits.py`): the look-ahead shift test of Warning 12.11 — no feature path may change value at the timestamp of the label event it feeds — plus a staleness-age consistency check (a recomputed from the event list must match the emitted feature). The gap-test harness of Protocol 12.17 extends the planned covariate-noise campaign with LOCF-age worlds and staleness-decile ranking reports.

Part V

Discrete-Time and Ranking Alternatives

Chapter 13

Temporal Buckets and Gradient Boosting

13.1 The horizon target and the bucketed panel

Target T3 of Chapter 1 asks the question the sourcing workflow poses: will firm i have a deal within the next Δ ? The object is the fixed-horizon event probability $p_i(t, \Delta)$ of Equation (1.3), consumed by binary, capacity-constrained actions — enter diligence now, keep on the watch list, deprioritize. The natural approach is supervised learning on a bucketed panel: turn the event history into rows “firm i at time t , did it raise within Δ ,” and hand the rows to a learner that handles nonlinearity. This chapter commits to that approach, with gradient boosting as the learner, and treats it as statistics. The bucketed panel is not a machine-learning convenience outside the point-process theory of Part II; it *is* a discrete-time hazard model, with an exact link to the conditional intensity (Theorem 13.4), a censoring theory (Section 13.3), a sampling theory (Section 13.5), and a dependence structure that rules out the usual i.i.d. habits (Section 13.4). Every labeling shortcut — what to do with windows that cross the end of the data, whether a firm that died quietly is a negative, whether sliding the window weekly “adds data” — is a statistical decision with a signed bias, and this chapter makes each one explicitly.

Definition 13.1 (Bucketed panel). Fix a horizon $\Delta > 0$ and an evaluation grid of bucket start times

$$\mathcal{T}_\Delta = \{t_b = t_0 + b\Delta : b = 0, 1, \dots, B_\Delta - 1\} \subset [0, T], \quad (13.1)$$

aligned to the weekly calendar of Chapter 2. The panel for horizon Δ is the row set $\mathcal{P}_\Delta = \{(i, t_b) : i \in \mathcal{I}, Y_i(t_b) = 1\}$ — one row per tracked firm at risk at each bucket start — with binary label

$$y_{i,b} = \mathbf{1}\{N_i(t_b + \Delta) - N_i(t_b) \geq 1\}, \quad (13.2)$$

and feature vector $z_{i,b}$ assembled at t_b under Protocol 2.5:

- (i) the LOCF firm covariates $\tilde{x}_i(t_b)$ of Equation (2.1);
- (ii) the staleness age $a_i(t_b) = t_b - \tau_i(t_b)$ — “it has been a since the last deal” — carried *always*, as mandated by design decision D8.1 of Chapter 12;
- (iii) engineered history: deal counts $N_i(t_b) - N_i(t_b - w)$ over trailing windows $w \in \{26, 52, 104\}$ weeks, last round type and size (with an undisclosed flag), raised-to-date;
- (iv) sector activity summaries: trailing sector deal counts and their changes, computed from the *complete* sector event stream (including out-of-universe deals, whose counts are trustworthy even when their firms are not);

(v) as-of macro $x_{\text{pub}}^{\text{mac}}(t_b)$, joined on publication date.

Negative rows are all at-risk firm-buckets with no deal in the window; there is no other source of negatives. The committed horizon set is $\Delta \in \{26, 52, 104\}$ weeks.

One remark on the label Equation (13.2): it depends on the event history only through a bucket *count*, never through a within-bucket position. This is deliberate. Defect (C4) of Chapter 2 says interval counts are trusted while positions are not; the panel is built on exactly the trusted functional, and Corollary 13.5 below states what that choice costs.

Defect declaration (per D2.1). Against the ledger of Table 2.1: **(C1)** corrected — buckets whose window crosses T are flagged censored and either dropped or retained with the exposure-offset likelihood of Section 13.3; they are never labeled negative. **(C2)** robust by construction — no row exists before V_i ; the deeper selection (the tracked universe is conditioned on having raised at least once) restricts the estimand to tracked firms and is owned by Chapter 15. **(C3)** vulnerable with signed bias — unlabeled deaths sit in the negative class; mitigations and the residual downward bias are stated in Section 13.3. **(C4)** robust by design — labels are invariant to within-bucket date jitter (only events within jitter distance of a bucket boundary can flip a label), at the price of zero information about within-bucket timing, a price this chapter chooses deliberately (Corollary 13.5). **(C5)** vulnerable, mitigated — LOCF covariates enter only alongside their staleness age, and the tree learner can represent the interaction “informativeness of $\tilde{x}$ decays with a ” that Chapter 12 shows is the first-order correction; residual attenuation remains and is inherited from Chapter 12. **(C6)** robust for the stated estimand — labels are own-firm events of tracked firms, and out-of-universe deals enter only through complete sector counts; the panel makes no claim about untracked firms.

Design decision (D9.1 — the bucketed panel is a hazard dataset).

The T3 panel is built exactly as Definition 13.1: calendar-aligned disjoint buckets per horizon, labels from bucket counts only, features under Protocol 2.5 with the staleness age mandatory, censored buckets flagged rather than zero-labeled, and no rows before first observation. Rationale: each clause is the discrete-time image of a continuous-time censoring or truncation rule (Theorem 13.4 and Proposition 13.7); violating any one of them injects a bias with a known sign, and more data will not fix it. Reversal trigger: none for the labeling rules; the feature list may grow freely under Protocol 2.5.

Design decision (D9.2 — horizon set $\{6M, 1Y, 2Y\}$).

We commit to $\Delta \in \{26, 52, 104\}$ weeks and build one model per horizon. Rationale: the three consuming actions named in Chapter 1 (diligence now; pipeline within a year; deprioritize if nothing plausible in two years) fix the horizons — the target is chosen by the action, not by the data; and per-horizon models keep the cloglog consistency of Theorem 13.4 exact. Reversal trigger: if the decision layer of Chapter 16 starts consuming horizon *curves* $\Delta \mapsto p_i(t, \Delta)$ rather than three scalars, the per- Δ design yields to the single-model AFT alternative of Section 13.8.

13.2 The bucket model is a hazard model

We now prove the claim that organizes the chapter: the panel of Definition 13.1 is a discrete-time hazard model, and the bucket probability is an exact functional of the same conditional intensity that Part II estimates in continuous time. Nothing about this is asymptotic; the identity holds bucket by bucket.

Lemma 13.2 (Waiting-time hazard of the bucket). *Fix a bucket (i, b) with $Y_i(t_b) = 1$ and let $W_{i,b} = \inf\{u > 0 : N_i(t_b + u) > N_i(t_b)\}$ be the waiting time to firm i 's next deal. Suppose the conditional law of $W_{i,b}$ given $\mathcal{F}_{t_b}$ is absolutely continuous on $(0, \infty)$ with hazard function $r_{i,b}(u)$. Then*

$$p_{i,b} := \mathbb{P}(y_{i,b} = 1 \mid \mathcal{F}_{t_b}) = 1 - \exp(-\Lambda_{i,b}), \quad \Lambda_{i,b} := \int_0^\Delta r_{i,b}(u) du, \quad (13.3)$$

and the hazard is the predictable projection of the conditional intensity onto the no-event paths: $r_{i,b}(u) = \mathbb{E}[\lambda_i(t_b + u) \mid \mathcal{F}_{t_b}, W_{i,b} > u]$.

Proof. The first display is the hazard representation of an absolutely continuous survival law, $\mathbb{P}(W > \Delta \mid \mathcal{F}_{t_b}) = \exp(-\int_0^\Delta r)$, applied to $\{y_{i,b} = 0\} = \{W_{i,b} > \Delta\}$. For the second, the innovation theorem (Andersen et al., 1993, II.4.2) states that the intensity of N_i with respect to the coarser filtration generated by $\mathcal{F}_{t_b}$ and the post- t_b own-event history is the conditional expectation of λ_i given that filtration; on the event $\{W_{i,b} > u\}$ the post- t_b own-event history is trivial, and the hazard of the first point of a counting process equals its intensity before the first point. $\square$

The lemma has a practical reading: the bucket probability depends on the future only through an *integrated* intensity, with the unobserved evolution of covariates and competing events averaged out along the no-event paths. The panel model therefore estimates the compensator increment of Part II, coarsened. To get the classical grouped-data form we add the proportional-hazards structure.

Assumption 13.3 (Grouped proportional hazards within buckets). For every bucket (i, b) there is a baseline $\lambda_{0,b}(u) \geq 0$, common across firms, and a coefficient vector β such that $r_{i,b}(u) = \lambda_{0,b}(u) e^{\beta^\top z_{i,b}}$ for $u \in (0, \Delta]$: covariates are frozen at their bucket-start values, and their proportional effect on the no-event intensity holds throughout the bucket. Piecewise-constant intensity ($\lambda_{0,b}(u)$ constant in u) is the special case used for exposure offsets below.

Theorem 13.4 (Hazard link). *Let the panel be built as in Definition 13.1 and let $h_{i,b} = p_{i,b}$ denote the discrete-time hazard Equation (13.3).*

(i) (Exact cloglog / grouped PH.) *Under Assumption 13.3,*

$$\mathbb{P}(y_{i,b} = 1 \mid z_{i,b}) = 1 - \exp(-\Lambda_{0,b} e^{\beta^\top z_{i,b}}), \quad \text{cloglog } p_{i,b} := \log(-\log(1 - p_{i,b})) = \alpha_b + \beta^\top z_{i,b}, \quad (13.4)$$

with $\alpha_b = \log \Lambda_{0,b}$, $\Lambda_{0,b} = \int_0^\Delta \lambda_{0,b}$, and β identical to the continuous-time coefficients (Prentice and Gloeckler, 1978; Allison, 1982). The Bernoulli likelihood of the panel is the exact likelihood of the point process observed through the coarsening “at least one event per bucket”: a bucket classifier estimates a deterministic coarsening $\lambda \mapsto (\Lambda_{i,b})_b$ of the same intensity object as Part II.

(ii) (Continuum limit.) *Drop Assumption 13.3. If $u \mapsto r_{i,b}(u)$ is right-continuous at 0 then $h_{i,b}/\Delta \rightarrow r_{i,b}(0^+) = \mathbb{E}[\lambda_i(t_b) \mid \mathcal{F}_{t_b}]$ as $\Delta \downarrow 0$; and for a fixed window $[t, t + v]$ partitioned*

into buckets of mesh $\Delta \rightarrow 0$, with hazards refreshed at each bucket start and bounded above,

$$\prod_b (1 - h_{i,b}) \longrightarrow \exp \left(- \int_t^{t+v} \lambda_i(u) du \right) \quad \text{along no-event paths:}$$

the product-limit of the discrete hazards recovers the continuous-time survival factor, so the discrete model converges to the continuous intensity model, not merely to an approximation of it.

(iii) (Logit versus cloglog.) Writing $\Lambda = \Lambda_{i,b}$ for the bucket's hazard mass,

$$\text{logit } p = \log(e^\Lambda - 1) = \text{cloglog } p + d(\Lambda), \quad d(\Lambda) = \log \frac{e^\Lambda - 1}{\Lambda} = \frac{\Lambda}{2} + \frac{\Lambda^2}{24} - \frac{\Lambda^4}{2880} + \cdots, \quad (13.5)$$

so for small per-bucket probabilities the log-odds equal the log-hazard-mass up to $O(\Lambda)$: a logistic-link panel model estimates the same object as the cloglog model to first order. The divergence is material at large p : $d = 0.026$ nats at $p = 0.05$, 0.114 at $p = 0.2$, 0.367 at $p = 0.5$.

Proof. (i) Substituting Assumption 13.3 into Equation (13.3) gives $\Lambda_{i,b} = e^{\beta^\top z_{i,b}} \Lambda_{0,b}$ and hence both displays of Equation (13.4), since $\text{cloglog}(1 - e^{-\Lambda}) = \log \Lambda$. For the likelihood claim: the coarsened data for bucket (i, b) is the binary $y_{i,b}$, whose conditional law given $\mathcal{F}_{t_b}$ is Bernoulli($p_{i,b}$) by Lemma 13.2; over disjoint buckets the joint likelihood factorizes by the tower property into exactly the product of these Bernoulli terms, which is the person-period likelihood of Allison (1982). No approximation enters at any step. (ii) $h = 1 - \exp(-\int_0^\Delta r)$ with $\int_0^\Delta r = \Delta r(0^+) + o(\Delta)$ by right-continuity, and $1 - e^{-x} = x + O(x^2)$, give $h/\Delta \rightarrow r(0^+)$; the identification of $r(0^+)$ is Lemma 13.2. For the product limit, $\log \prod_b (1 - h_b) = \sum_b \log(1 - h_b) = -\sum_b h_b + O(\sum_b h_b^2)$; along a no-event path each refreshed hazard mass satisfies $h_b = \int_{t_b}^{t_b+\Delta} \lambda_i(u) du + o(\Delta)$, so $\sum_b h_b \rightarrow \int_t^{t+v} \lambda_i$, while $\sum_b h_b^2 \leq (\max_b h_b) \sum_b h_b \rightarrow 0$ because $\max_b h_b = O(\Delta)$ under the boundedness assumption. (iii) $p = 1 - e^{-\Lambda}$ gives $p/(1 - p) = e^\Lambda - 1$, hence the first equality. For the expansion, $(e^\Lambda - 1)/\Lambda = e^{\Lambda/2} \sinh(\Lambda/2)/(\Lambda/2)$, so $d(\Lambda) = \Lambda/2 + \log(\sinh y/y)$ with $y = \Lambda/2$, and $\log(\sinh y/y) = y^2/6 - y^4/180 + O(y^6)$ yields the stated series. The numerical values follow from $\Lambda = -\log(1 - p)$. $\square$

Part (iii) has a practical reading. When per-bucket probabilities are small — short horizons, ordinary firms — the logistic link is the cloglog link plus a distortion of order the probability itself, and a logistic learner (including XGBoost with its native `binary:logistic` objective) estimates a hazard-consistent object. When per-bucket probabilities are large — the two-year horizon on recently active firms — the gap $d(\Lambda)$ is covariate-dependent, so it is *not* absorbed by an intercept: logistic coefficients stop being transferable across horizons, while cloglog coefficients remain the same β at every Δ by Equation (13.4). In this precise sense cloglog, not logit, is the aggregation-coherent link. It is also why the calibration layer of Section 13.7 is mandatory: it repairs, on the validation era, exactly the link distortion bounded here.

Corollary 13.5 (Choosing Δ is choosing the aggregation defect). *Any two conditional intensities with equal bucket masses $(\Lambda_{i,b})_{i,b}$ induce the same panel law; consequently the panel Fisher information is identically zero in every parameter direction that moves λ only within buckets. In particular, timescale parameters faster than Δ (a kernel decay θ with half-life $\ll \Delta$) are unidentifiable from the panel.*

Proof. The panel likelihood in Theorem 13.4(i) depends on the underlying process only through $(\Lambda_{i,b})$; a parameter direction that leaves all bucket masses fixed leaves the likelihood fixed, so the score in that direction vanishes identically and so does its variance. $\square$

This is the κ - θ ridge of Chapter 10 returned as a design choice. There, weekly binning was an observation defect (C4) that destroys decay information while sparing the branching scale; how fast recovery degrades as the bin width grows past the kernel half-life is a question for the planned synthetic study (Chapter 19). Here the same mathematics appears with the roles reversed: T3 does not *want* within-horizon timing, so we choose Δ to integrate it out, and Corollary 13.5 states exactly what is discarded. The defect is not an accident; it is accepted at a known price because the consuming action is horizon-bound.

13.3 Censoring and eligibility

Three eligibility rules determine which (i, t_b) pairs become rows and what their labels may say. Each is the discrete image of a continuous-time censoring statement, and each has a signed bias when violated.

Right censoring at the window end (C1). Define the exposure fraction of bucket (i, b) as $e_{i,b} = \min\{1, (T - t_b)/\Delta\}$ (rows require $t_b < T$, so $e_{i,b} \in (0, 1]$). A bucket with $e_{i,b} < 1$ — any window crossing June 2024 — is *censored*: the label Equation (13.2) is not computable from data, because the window’s unobserved remainder may contain the event.

Warning 13.6. Labeling censored buckets $y = 0$ is not a conservative convention; it is a biased estimator with a known sign. It converts “unobserved” into “negative” and deflates every fitted hazard, most strongly in the most recent calendar eras — which are exactly the eras closest to deployment and the eras whose macro covariates differ most from the training bulk, so the bias looks like an era effect and contaminates the macro coefficients. This is the bucketed instance of Warning 2.4.

Proposition 13.7 (Censored-bucket bias). *Suppose bucket (i, b) has hazard mass $\Lambda_{i,b} > 0$ distributed with constant within-bucket intensity, and exposure $e = e_{i,b} < 1$. The naive label $\tilde{y}_{i,b} = \mathbf{1}\{\text{event observed in } [t_b, T]\}$ satisfies $\mathbb{E}[\tilde{y}_{i,b} \mid \mathcal{F}_{t_b}] = 1 - e^{-e\Lambda_{i,b}} < 1 - e^{-\Lambda_{i,b}} = p_{i,b}$: on the cloglog scale the fitted linear predictor is shifted by $\log e < 0$, i.e. hazards are biased down by exactly the unobserved exposure fraction.*

Proof. With constant within-bucket intensity the observed sub-window carries hazard mass $e\Lambda_{i,b}$, so the observed-event probability is $1 - e^{-e\Lambda_{i,b}}$ by Lemma 13.2 applied to the sub-window; $\text{cloglog}(1 - e^{-e\Lambda}) = \log \Lambda + \log e$. $\square$

Two legal treatments follow. *Drop*: for the pure classification pipeline, delete rows with $e_{i,b} < 1$. This restores unbiasedness — administrative censoring at fixed T is independent by construction (Chapter 2, defect C1) — at the cost of discarding the most recent eras; the cost grows with the horizon, and at $\Delta = 104$ weeks it amounts to the final two years of bucket starts. *Keep with an exposure offset*: retain censored rows with the survival-style Bernoulli likelihood implied by Proposition 13.7,

$$\ell = \sum_{(i,b)} \left[\tilde{y}_{i,b} \log \left(1 - \exp(-e_{i,b} e^{\eta_{i,b}}) \right) - (1 - \tilde{y}_{i,b}) e_{i,b} e^{\eta_{i,b}} \right], \quad (13.6)$$

where $\eta_{i,b}$ is the cloglog linear predictor (or boosted margin) and $\log e_{i,b}$ enters as a fixed offset. Under within-bucket constant intensity this is exact; it is the grouped-data analogue of keeping open spells with their survival factor. The classification stack commits to the drop rule for simplicity, and Section 13.8 names the offset likelihood and AFT as the routes that recover the discarded eras.

Entry (C2). No row is emitted for $t_b < V_i$: a firm contributes person-periods only after its first observation. This is delayed entry handled mechanically by the at-risk indicator, exactly as in Equation (2.3); emitting earlier rows would populate the panel with firms observed *because* they later raised, a truncation bias with the opposite sign to Warning 13.6.

Unlabeled exits (C3). The negative class is contaminated: a firm that quietly died before t_b has $Y_i(t_b) = 1$ in the naive risk set and contributes an endless string of structural negatives. The expected consequences follow from Chapter 2: hazards deflate by the contamination fraction, and covariates correlated with staleness can acquire *inverted* weight, because the optimal response to a pool containing dead firms is to bet against staleness. The synthetic study of Chapter 19 is designed to measure both effects. Three mitigations are committed, in order of force. First, *exposure windows*: a row (i, t_b) requires observed activity (a deal, or a point-in-time exit proxy refresh) within the trailing E years. Because the rule is a predictable function of the observed history, it is exactly independent censoring; its benefit is not repairing the mechanism but capping each dead firm’s phantom exposure at E years. Second, an explicit *active model* — a point-in-time classifier of “still alive” — used as a feature, not as a hard filter: we do not expect even a good discriminative exit model to reconstruct truthful risk sets, a hypothesis Chapter 19 will test, so it may inform but must not silently redefine the estimand. Third, the staleness age itself partially absorbs the defect: dead firms accumulate very large a , and a flexible learner routes their predicted probability toward zero through the a feature — which is also why the fitted a effect is *not* interpretable as the true renewal profile (it mixes the genuine gap dependence with the contamination). The residual bias after all three mitigations is signed: fitted hazards remain biased *downward*, most severely at large staleness ages, and every probability this chapter emits for a long-quiet firm should be read as a lower bound in that direction.

Protocol 13.8 (Panel construction and audits). The panel builder emits, per horizon: the grid Equation (13.1); rows filtered by $V_i \leq t_b < T$, the exposure-window rule with recorded E , and $Y_i(t_b) = 1$; labels Equation (13.2) for fully exposed buckets and censor flags with $e_{i,b}$ otherwise; features exactly as Definition 13.1 under Protocol 2.5, with the exp26 prohibition on end-of-sample snapshot columns (the companies-table `last_*` fields, totals, employees) enforced by schema. Audits, run on every build: (a) no row precedes its firm’s entry; (b) no label consults data past T ; (c) censor-flag retention — the count of censored rows must equal the count implied by the grid and T ; (d) as-of-join look-ahead test on macro features; (e) cross-horizon coherence — empirical event rates must be nondecreasing in Δ on shared origins; (f) base rate and row count by era, published next to every fit.

13.4 Dependence: what the rows share

The panel looks like a supervised-learning table but is not an i.i.d. sample. Rows share firms across buckets: the latent quality that Chapter 12 models as a persistent state makes $y_{i,b}$ and $y_{i,b'}$ positively dependent for all bucket pairs of the same firm. Rows share time within buckets: macro regimes and sector cycles are common shocks, and we expect this shared structure to carry much of the signal — a hypothesis the synthetic study of Chapter 19 will test. Both dependencies survive conditioning on the features whenever the features are incomplete, which defect (C5) guarantees. The effective sample size is therefore far smaller than the row count, nominal i.i.d. standard errors are wrong, and shuffled cross-validation leaks by construction (rows from the same firm and era land on both sides of the fold).

Warning 13.9. Sliding the bucket window (origins every $s < \Delta$ weeks) does not create data. Each realized deal then appears in $\lceil \Delta/s \rceil$ positive labels, and overlapping windows re-count

the same survival exposure: the resulting objective is not the likelihood of any process, and any resampling scheme that treats the rows as exchangeable understates uncertainty by a factor up to $\sqrt{\Delta/s}$. Sliding windows are pseudo-replication, not augmentation.

Design decision (D9.3 — disjoint buckets for training, sliding for evaluation only).

Training panels use disjoint (non-overlapping) buckets per horizon, as in Equation (13.1); the likelihood factorization of Theorem 13.4(i) is exact only then. Phase-shifted disjoint grids (origins offset by $s < \Delta$) may be built as robustness replicates but are never pooled into one training set. Sliding origins *are* permitted for evaluation curves — lift@ K or calibration as a function of the decision date — because evaluation is description, not estimation; their inference uses the blocked resampling below. Rationale: disjointness trades a constant factor of rows for an exact likelihood and honest uncertainty; we expect between-era variance to dominate, so the discarded overlap should buy little (Chapter 19 will check). Reversal trigger: if the $\Delta = 104$ panel’s bucket count per firm (≈ 6 over the 2011–2024 window) proves too coarse for stable fitting, revisit pooled phase-shifted grids with firm-and-era clustered inference priced in.

Uncertainty for every panel-derived quantity — metrics, calibration curves, coefficient contrasts — is computed by resampling that respects both sharing directions: firms are resampled as whole clusters (all rows of a firm move together) and eras as contiguous blocks (all rows of a calendar block move together), following the temporal split and blocked-inference protocol fixed in Chapter 11. The split itself is temporal: training era, validation era (model selection, early stopping, calibration), test era touched once. This is the same discipline the exp26 benchmark harness already implements — temporal early stopping strictly inside the training era, a single touch of the test era — and the panel stack reuses that procedure under the calendar-era boundaries fixed by Chapter 11.

13.5 Class imbalance and negative sampling

Positives are rare: most at-risk firm-buckets contain no deal, with the base rate rising in Δ . Rarity has two costs — a statistical one for the parametric benchmarks and a computational one for the full panel — and both have exact treatments; neither justifies ad hoc reweighting.

For the linear cloglog and logistic benchmarks that every boosted model must beat, rare-event finite-sample bias is real: the MLE intercept is biased in small-event-count regimes and separation becomes possible in sparse eras. The corrections are standard and committed: the King and Zeng (2001) rare-events adjustment for logistic fits, and Firth (1993) penalization whenever any era-by-feature cell is sparse enough to threaten separation.

For scale, the committed device is case-control subsampling of negatives, with its exact correction.

Proposition 13.10 (Subsampling offset). *Retain every positive row and each negative row independently with probability $r \in (0, 1]$, with retention independent of z given y ; let $S = 1$ denote retention. If $p(z) = \mathbb{P}(y = 1 \mid z)$, then*

$$\text{logit } \mathbb{P}(y = 1 \mid z, S = 1) = \text{logit } p(z) - \log r. \quad (13.7)$$

Hence if the population model is logistic with predictor $\alpha + f(z)$, the retained sample follows a logistic model with the same f and intercept $\alpha - \log r$: fitting on the subsample with the fixed

offset $-\log r$ in the predictor (equivalently, adding $\log r$ to every fitted margin at prediction time) recovers the population model exactly.

Proof. By Bayes, $\mathbb{P}(y = 1 \mid z, S = 1) = p(z) \cdot 1 / [p(z) \cdot 1 + (1 - p(z))r]$, so the retained odds are $p(z)/[(1 - p(z))r]$, i.e. the population odds divided by r ; taking logs gives Equation (13.7). The second claim follows because the shift is constant in z and the logistic family is closed under intercept shifts. $\square$

Two remarks bound the result's reach. First, the exactness is a property of the *logistic* link: the retained-sample law under a cloglog population model is not a cloglog model with shifted intercept, so subsampling lives in the logistic pipeline, where Theorem 13.4(iii) already prices the link discrepancy, and the calibration layer absorbs the residual. Second, the same logic underwrites the ranking stack: estimating a conditional logit on a *sampled subset of alternatives* is consistent when the sampling protocol satisfies the positive-conditioning property, by exactly the same Bayes correction (McFadden, 1978) — the sampled-softmax training of Chapter 14 and Equation (13.7) are the same result in two forms: a known sampling scheme induces a computable likelihood correction, so sampling is unbiased whenever the scheme is recorded.

For the boosted trees themselves the common alternative is `scale_pos_weight`: multiply positive-class gradients by a constant w . We reject it for probability work. In the unregularized, infinite-capacity limit weighting shifts every logit by $\log w$ uniformly — harmless. In an actual boosted tree, leaf values are $-G/(H + \lambda_{\text{reg}})$ (Section 13.6) and splits are accepted against γ and `min_child_weight` thresholds on H : reweighting rescales G and H heterogeneously across leaves, so the induced distortion is *not* a constant shift, and no single intercept correction undoes it. Subsampling with the offset of Proposition 13.10, followed by post-hoc calibration on the *unsubsampled* validation era, achieves the same variance-per-cost benefit with a distortion we can state and remove.

Design decision (D9.4 — subsample negatives, correct exactly, calibrate after).

Training panels may subsample negatives at a recorded rate r per era-stratum; positives are never subsampled. Corrections are mandatory and layered: the $\log r$ margin correction of Proposition 13.10 at prediction time, then the calibration map of Section 13.7 fit on the full (unsubsampled) validation era. `scale_pos_weight` and other loss reweightings are banned in the probability pipeline (permitted in pure-ranking experiments, where only order matters). Rationale: Proposition 13.10 makes subsampling exactly correctable; reweighting is not. Reversal trigger: none; this is a mathematical, not empirical, commitment.

13.6 XGBoost as the committed learner

The learner attached to the panel is gradient-boosted trees in the XGBoost formulation (Chen and Guestrin, 2016), i.e. the functional gradient view of Friedman (2001) with a second-order objective and explicit regularization. The same learners appear again in Chapter 9 with a different target: here the tree model fits a discrete-time hazard on the bucketed panel, while in Chapter 9 the same learners parameterize a continuous-time intensity through the binned likelihood — same tooling, different target. We derive the fitting rule once, because two of its internals (the Hessian denominators and the default directions) are load-bearing for our data.

The model after m rounds is $F_m(z) = \sum_{k \leq m} f_k(z)$, each f_k a regression tree with T leaves, leaf weights $w \in \mathbb{R}^T$, and leaf-assignment map q . Round m solves

$$\min_f \sum_n l(y_n, F_{m-1}(z_n) + f(z_n)) + \gamma T + \frac{1}{2} \lambda_{\text{reg}} \|w\|^2, \quad (13.8)$$

with l the per-row loss — for us `binary:logistic`, $l(y, F) = -yF + \log(1 + e^F)$, or the cloglog-exposure loss implied by Equation (13.6) as a custom objective.

Proposition 13.11 (Second-order boosting step). *Let $g_n = \partial_F l(y_n, F)|_{F_{m-1}(z_n)}$ and $h_n = \partial_F^2 l(y_n, F)|_{F_{m-1}(z_n)}$, and for a fixed tree structure q let $I_j = \{n : q(z_n) = j\}$, $G_j = \sum_{n \in I_j} g_n$, $H_j = \sum_{n \in I_j} h_n$. To second order in f , the objective Equation (13.8) is minimized over leaf weights by*

$$w_j^* = -\frac{G_j}{H_j + \lambda_{\text{reg}}}, \quad \text{with optimal value} \quad -\frac{1}{2} \sum_{j=1}^T \frac{G_j^2}{H_j + \lambda_{\text{reg}}} + \gamma T,$$

and the gain of splitting a leaf with statistics (G, H) into (G_L, H_L) and (G_R, H_R) is

$$\text{Gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda_{\text{reg}}} + \frac{G_R^2}{H_R + \lambda_{\text{reg}}} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda_{\text{reg}}} \right] - \gamma. \quad (13.9)$$

For the logistic loss, $g_n = \sigma(F_{m-1}(z_n)) - y_n$ and $h_n = \sigma(F_{m-1}(z_n))(1 - \sigma(F_{m-1}(z_n)))$.

Proof. Taylor-expanding each loss term to second order, $l(y_n, F_{m-1} + f) \approx l(y_n, F_{m-1}) + g_n f(z_n) + \frac{1}{2} h_n f(z_n)^2$, and summing within leaves (where $f \equiv w_j$) turns Equation (13.8) into $\sum_j [G_j w_j + \frac{1}{2} (H_j + \lambda_{\text{reg}}) w_j^2] + \gamma T$ up to a constant: a separable quadratic, minimized at w_j^* with the stated value. The gain formula is the difference of optimal values before and after the split, with γ charged for the extra leaf (Chen and Guestrin, 2016). $\square$

Why boosted trees. Four reasons. First, *the staleness interaction is the model*: Chapter 12 establishes that the predictive content of a LOCF covariate decays with its age, so the truth has the form $f(\tilde{x}, a)$ with effect modification in a — exactly what axis-aligned splits represent cheaply and what a linear cloglog must hand-engineer term by term. Second, PitchBook columns are mixed-type: round-type categories, flags, sizes spanning orders of magnitude, and pervasive missingness; XGBoost’s sparsity-aware default directions (Chen and Guestrin, 2016) learn, per split, which way missing values go — and since missingness here is informative (undisclosed size is a signal, not an accident), learning the direction is modeling, not imputation. Third, nonlinearity in the engineered history (trailing counts, raised-to-date) is expected and cheap to capture. Fourth, we expect the linear-utility restriction to fail. The planned B1 benchmark of Chapter 19 tests XGBoost against the linear conditional logit at strict feature parity — the same features, the same choice sets, the same temporal split — and the panel’s feature set is wider and dirtier than any setting in which linearity is plausible. The linear cloglog GLM remains in the harness as the benchmark every boosted fit must beat, but it is the challenger, not the champion.

Hyperparameters that matter. Depth (`max_depth`) bounds interaction order; we expect gains at modest depth (3–5), with deeper trees mostly overfitting firm identity. `min_child_weight` thresholds the leaf Hessian mass H_j ; under the logistic loss $h_n = p_n(1 - p_n)$, so the threshold is an effective-count guard that, under class imbalance, mechanically prevents leaves fit to a handful of positives — it is the imbalance safeguard that *is* compatible with calibration, unlike loss reweighting. Row subsampling per boosting round reduces variance but should sample firms as clusters when feasible, for the reasons of Section 13.4; column subsampling decorrelates trees in the presence of the many collinear trailing-window counts. The number of trees is chosen by early stopping on the temporal validation era — never by shuffled cross-validation — following the exp26 protocol (grid over depth and learning rate, early stop on the validation tail, refit at the chosen round count, one shot at test).

Monotonicity constraints. XGBoost accepts per-feature monotone constraints, and economics suggests candidates: probability nondecreasing in trailing deal counts, nonincreasing in the policy rate. We decline to impose the tempting constraint that intensity decreases in the staleness age a . The true gap dependence is not monotone: renewal-style maturation pushes hazard *up* over the first year-plus of a gap before frailty heterogeneity pulls the population hazard down (Vaupel et al., 1979; Aalen et al., 2008), and the apparent steep decline at large a is partly the (C3) contamination artifact of Section 13.3, not economics. Forcing monotonicity would hard-code the artifact. Constraints are therefore used as robustness probes — fit with and without, compare test loss and reliability — and imposed only where the constrained fit is not measurably worse.

13.7 Probability calibration

The decision layer of Chapter 16 consumes the panel’s outputs as probabilities: expected shortlist yield $\sum_{i \in \text{list}} \hat{p}_i$, thresholds set by expected utility, horizon-qualified pipeline pacing. A model that ranks well but emits probabilities off by a factor of two corrupts every one of those computations, so calibration is mandatory infrastructure. Three distortions are anticipated and named: the logit–cloglog gap of Theorem 13.4(iii), residual subsampling distortion after the $\log r$ correction of Proposition 13.10, and the boosted-margin distortion documented by Niculescu-Mizil and Caruana (2005), who find boosted trees among the models that benefit most from post-hoc mapping.

The committed instruments are the two standard ones. *Platt scaling* (Platt, 1999): fit $c(m) = \sigma(Am + B)$ on validation margins by logistic regression — two parameters, stable on modest validation eras, and well matched to distortions that are smooth and sigmoidal in the margin. *Isotonic regression* (Zadrozny and Elkan, 2002): the nonparametric monotone map fit by pooled adjacent violators — strictly more flexible, needs materially more validation data, and prone to piecewise-constant artifacts in the tails where our decisions live. Guidance follows Niculescu-Mizil and Caruana (2005): Platt as default; isotonic when the validation reliability curve is visibly non-sigmoidal *and* the validation era supplies enough positives per bin to estimate it — per-horizon, the 26-week panel is the most likely to qualify, the 104-week panel the least.

Protocol 13.12 (Calibration and its audit). (a) The calibration map is fit on the validation era only, after the $\log r$ subsampling correction, and is part of the model artifact: no downstream consumer ever sees raw margins. (b) Reliability diagrams (binned $\hat{p}$ versus empirical frequency, with firm-cluster, era-block resampled bands) are produced per horizon and per era, before and after calibration. (c) Scores: the Brier score with its reliability–resolution–uncertainty decomposition, and the log score; both are proper, so no calibration map can game them in expectation (Gneiting and Raftery, 2007). (d) Drift: calibration is re-audited on each new era; a reliability shift with stable discrimination triggers recalibration, not refitting.

Design decision (D9.5 — calibration is mandatory).

Every deployed horizon model carries a calibration map fit per Protocol 13.12; uncalibrated scores are internal quantities. Rationale: the consuming actions are probability arithmetic (Chapter 16), and the three distortion sources above are known and repairable at the cost of a validation-era fit. Reversal trigger: none. If a future learner is verified reliably calibrated end-to-end, the map degenerates to the identity, at zero cost.

13.8 Survival-aware boosting alternatives

The per- Δ classifier design pays two visible costs: censored buckets are dropped (Section 13.3), losing the most recent eras precisely at the long horizons; and the three horizon models are fit separately, so nothing enforces coherence of $\hat{p}_i(t, 26) \leq \hat{p}_i(t, 52) \leq \hat{p}_i(t, 104)$ beyond the audit in Protocol 13.8. Both costs are removed by boosting a survival model instead of a classifier.

XGBoost-AFT (Barnwal et al., 2022) boosts an accelerated failure time model, $\log W_i = \mu(z_i) + \sigma\varepsilon$, with ε standard normal, logistic, or extreme-value, and μ the boosted ensemble. Its likelihood consumes exactly the data the classifier discards: an observed waiting time contributes a density term, a bucket that crosses T contributes a right-censored survival term $\mathbb{P}(W > T - t_b)$, and grouped observations contribute interval-censored terms $\mathbb{P}(l < W \leq u)$ — censored rows are retained natively, no offsets required. One fitted model produces the whole curve $\Delta \mapsto \hat{\mathbb{P}}(W \leq \Delta)$, monotone in Δ by construction, serving all three horizons and any future one. *Gradient-boosted Cox* (Pölsterl, 2020) boosts the partial likelihood instead: it inherits the proportional-hazards reading of Theorem 13.4(i) and needs no error-distribution choice, but it delivers relative risks; absolute horizon probabilities require a Breslow-type baseline estimate on top, at which point its calibration story is no simpler than the classifier’s.

The costs run the other way. AFT buys its coherent curve with a global parametric assumption on ε : a misspecified error shape miscalibrates *every* horizon in a correlated way, and repairing it per horizon reintroduces exactly the per- Δ calibration maps the design was meant to avoid. AFT also consumes exact waiting times, thereby re-inheriting the date-jitter component of (C4) that the bucket design is immune to by construction. And the surrounding tooling — imbalance-aware sampling theory (Proposition 13.10 is logistic-specific), calibration instruments, evaluation conventions — is mature for binary classification and improvised for boosted AFT.

Design decision (D9.6 — per- Δ classifiers now, AFT as the benchmarked alternative).

The committed T3 stack is the per- Δ , cloglog-consistent classifier pipeline of this chapter. XGBoost-AFT is maintained in the benchmark harness on the identical feature set and splits, evaluated on the same per-horizon probabilities $\hat{\mathbb{P}}(W \leq \Delta)$ after its own calibration audit. Rationale: exact censoring and sampling theory, mature calibration, and (C4) robustness today outweigh the coherent curve and censored-row retention. Reversal trigger: if AFT beats the calibrated per- Δ classifiers on test log score at two of three horizons in two consecutive evaluation campaigns — or if D9.2’s reversal fires and the decision layer demands curves — AFT is promoted.

The repository has prior art: experiment `exp16` implements a discrete-time logistic-hazard model with a Kaplan–Meier calibration reference and a pairwise ranking-loss hybrid on synthetic firms. The panel stack of this chapter builds that sketch out fully, with the labeling, censoring, and sampling theory made explicit.

13.9 Evaluation, and the bridge to ranking

Evaluation follows the temporal split of Chapter 11, reuses the `exp26` harness conventions (selection on the validation era, one shot at test), and reports per horizon and per era. The headline metrics are chosen to proxy the consuming action, per Equation (1.1). *PR-AUC*, not *ROC-AUC*, is the discrimination summary: under heavy imbalance *ROC-AUC* is inflated by the ocean of easy negatives, while precision–recall tracks the operating region the sourcing

action lives in; because PR-AUC depends on the base rate, it is reported next to the era's base rate and never compared across eras or horizons without that context. *Lift at K* — the event rate in the top- K list divided by the base rate, at K set to realistic diligence capacities — is the closest offline proxy to sourcing value. *Reliability and proper scores* per Protocol 13.12 complete the set: discrimination without calibration is not enough here. All uncertainty is firm-clustered, era-blocked (Section 13.4).

Sorting firms by $\hat{p}_i(t, \Delta)$ is a ranking, and the T1 machinery of Chapter 14 is close enough that the relationship must be stated explicitly. The two targets differ in conditioning: the ranker estimates $\mathbb{P}(i \mid \text{deal in } s, t, \mathcal{H})$ — identity given that a deal happens, timing canceled — while the horizon model estimates a marginal that mixes identity with timing mass; under the two-layer factorization of Chapter 1, heuristically $p_i(t, \Delta) \approx 1 - \exp(-\int_t^{t+\Delta} \Lambda_s(u) \pi_i(u) du)$, so a mid-ranked firm in a hot sector can out-probability a top-ranked firm in a cold one. They differ in universe: the ranker competes firms within a sector risk set and must own the outside option (C6); the horizon model scores each tracked firm absolutely, no competition and no outside option. And they differ in unit: $\text{recall}@K$ is computed against realized deals and is calibration-free; $\text{lift}@K$ on the panel is computed on all at-risk rows and consumes calibrated levels. The two are complements, not rivals: the sector-conditional ranker is the defensible object when timing is untrusted (Chapter 1), the firm-level horizon model is the object the diligence trigger consumes, and the composed continuous-time stack furnishes a third estimate of the same $p_i(t, \Delta)$ by integrating the fitted ground intensity against the mark factor. Comparing the boosted panel's $\hat{p}_i(t, \Delta)$ with that composed structural estimate on identical origins is the cross-stack audit demanded by design decision D1.2: when the two stacks disagree systematically, one of them is wrong about something specific, and the diagnostics in this chapter exist to locate it.

Code hooks.

Panel layer (to be written): `hawkes_calibration/panel/builder.py` — grids, at-risk filtering, labels, exposure fractions $e_{i,b}$, censor flags, and feature assembly per Definition 13.1 under Protocol 2.5; `hawkes_calibration/panel/audits.py` — the audit battery of Protocol 13.8 (entry consistency, censor-flag retention, look-ahead tests, cross-horizon monotonicity, base rates by era). *Model layer* (to be written): `hawkes_calibration/models/xgb_horizon.py` — per- Δ XGBoost with the D9.4 subsampling-offset correction, an optional cloglog-exposure custom objective implementing Equation (13.6), monotone-constraint configuration, and the calibration map of Protocol 13.12 as part of the persisted artifact; `hawkes_calibration/models/xgb_aft.py` — the D9.6 AFT benchmark on identical features and splits. *Harness*: the temporal-split, validation-tail early-stopping, grid-then-one-test-shot protocol is inherited from `experiments/exp26_xgb_benchmark.py` and must not be reimplemented; `experiments/exp16_survival.py` (discrete-time logistic hazard, KM calibration, ranking-loss hybrid) is the prior art this chapter supersedes. Linear benchmarks (cloglog GLM with Firth (1993) penalization) live beside the boosted models in the same harness so that every gain is measured at feature parity.

Chapter 14

Ranking Objectives: Who Gets Funded Next

14.1 The metric is the ranking

Chapter 1 argued that the first answer to “what is valuable to know” is an identity, not a time: given finite diligence capacity, which K firms should be looked at now because they are most likely to be the next to raise. The consuming action is a short list, refreshed on a cadence; nothing in that action consumes an arrival time. The target is therefore the conditional distribution of the next deal’s identity, already fixed as Equation (1.2),

$$p(i \mid s, t) = \mathbb{P}(\text{next deal in sector } s \text{ goes to firm } i \mid \text{a deal occurs, } \mathcal{F}_{t-}), \quad i \in \mathcal{R}_s(t), \quad (14.1)$$

where $\mathcal{R}_s(t) \subseteq \mathcal{I}$ is the risk set of tracked firms at risk in sector s at t^- . This chapter sharpens that framing: when T1 is the valuable thing, the likelihood of the full marked process is not the metric — the ranking is. The full likelihood rewards timing accuracy that no action consumes and that the data defects of Chapter 2 corrupt. The object to be scored is the ordered list, and the object to be fitted is the conditional law Equation (14.1), whose own likelihood — the conditional NLL of the mark factor in Theorem 6.7 — is the proper training loss because it is a likelihood of the ranking target and of nothing else.

The bridge between intensity models and Equation (14.1) is the competing-intensities identity. Chapter 6 states it as part of the ground–mark factorization; here we give the operational proof, because every robustness claim in this chapter follows from the form of the identity: a ratio.

Assumption 14.1 (Firm-level intensity system). Fix a sector s and work conditionally on $\mathcal{F}_{t-}$. Each firm $i \in \mathcal{R}_s(t)$ has a counting process N_i with $\mathcal{F}$ -conditional intensity $\lambda_i(t) \geq 0$ (Definition 2.3), the intensities are locally integrable, no two firms jump at the same instant almost surely, and the sector ground process $N_s = \sum_i N_i$ has intensity $\lambda_g(t) = \sum_{j \in \mathcal{R}_s(t)} \lambda_j(t) \in (0, \infty)$.

Proposition 14.2 (Competing intensities). *Under Assumption 14.1:*

- (i) (Exact form, at the event.) *For almost every t , the conditional distribution of the identity of an event occurring at t is*

$$\mathbb{P}(\text{identity} = i \mid \text{event at } t, \mathcal{F}_{t-}) = \frac{\lambda_i(t)}{\sum_{j \in \mathcal{R}_s(t)} \lambda_j(t)}. \quad (14.2)$$

- (ii) (Next-event form.) *Fix a decision time τ and suppose in addition that on the no-event continuation after τ the intensities are $\mathcal{F}_\tau$ -measurable functions $u \mapsto \lambda_i^\tau(u)$ (true for*

internal-history models with exogenous covariate paths frozen at τ). Then the next event time T has conditional density $\lambda_g^\tau(u) \exp(-\int_\tau^u \lambda_g^\tau(v) dv)$ and, jointly with the identity J ,

$$\mathbb{P}(J = i, T \in du \mid \mathcal{F}_\tau) = \lambda_i^\tau(u) \exp\left(-\int_\tau^u \lambda_g^\tau(v) dv\right) du, \quad \mathbb{P}(J = i \mid T = u, \mathcal{F}_\tau) = \frac{\lambda_i^\tau(u)}{\lambda_g^\tau(u)}. \quad (14.3)$$

Proof. (i) View the system as a single marked point process on $[0, T] \times \mathcal{R}_s(t)$ with intensity kernel $\lambda(t, \{i\}) = \lambda_i(t)$; the absence of common jumps makes this kernel well defined, and superposition gives the ground intensity $\lambda_g = \sum_j \lambda_j$. The ground-mark factorization of Theorem 6.7 writes the kernel as $\lambda(t, \{i\}) = \lambda_g(t) p(i \mid t, \mathcal{F}_{t-})$, where $p(\cdot \mid t, \mathcal{F}_{t-})$ is, by construction, the conditional mark distribution given an event at t (Daley and Vere-Jones, 2003, Ch. 6). Identifying the two expressions for the kernel and dividing by $\lambda_g(t) > 0$ yields Equation (14.2).

(ii) The operational derivation. Partition $(\tau, u]$ into n intervals of mesh $\delta = (u - \tau)/n$. Given $\mathcal{F}_\tau$ and the frozen intensity functions, the probability that no firm jumps in $(v, v + \delta]$, given no jump before v , is $1 - \lambda_g^\tau(v)\delta + o(\delta)$, and these no-jump factors multiply across subintervals because, in the absence of events, the conditional jump probabilities depend only on the deterministic functions λ_j^τ — this is the independent-increment approximation, exact in the limit $\delta \rightarrow 0$ by the product-integral representation of the survivor function (Andersen et al., 1993). Hence $\mathbb{P}(T > u \mid \mathcal{F}_\tau) = \exp(-\int_\tau^u \lambda_g^\tau)$. The event $\{T \in (u, u + \delta], J = i\}$ requires no jump on $(\tau, u]$ and a jump of firm i in $(u, u + \delta]$, of probability $\lambda_i^\tau(u)\delta + o(\delta)$; multiplying and letting $\delta \rightarrow 0$ gives the joint density in Equation (14.3). Dividing by the marginal density of T — the same expression with λ_i^τ replaced by λ_g^τ — gives the ratio. When exogenous covariates evolve stochastically after τ , part (ii) holds with the intensities replaced by their conditional expectations along the no-event continuation; part (i) is unaffected, and as $u \downarrow \tau$ the two parts agree, so ranking at the decision time inherits the exact form (i). $\square$

Corollary 14.3 (Every intensity model is a ranker). *Any firm-level intensity model $\{\lambda_i\}$ induces a ranker: order $\mathcal{R}_s(t)$ by $\lambda_i(t)$, equivalently by the conditional probability Equation (14.2), of which $\lambda_i(t)$ is a monotone transform. Under the model, this ordering is optimal for every list metric considered in this chapter (Lemma 14.9). Conversely — and this is the direction the chapter exploits — the ratio Equation (14.2) can be modeled directly, without ever committing to a timing model for λ_g .*

The ratio form is the key observation. Any factor common to the whole risk set — the sector's level of activity, the calendar week's deal flow, a macro regime, an unmodeled burst — appears in every λ_j and cancels. The next section turns this cancellation into a likelihood with a proof of exactly what it buys.

14.2 The conditional-choice mark factor

The repository's mark factor models the ratio directly. Each candidate $i \in \mathcal{R}_s(t)$ receives a score built from point-in-time features $z_i(t)$ (firm covariates with staleness ages, Chapter 12; engineered history summaries, Definition 2.2) and a cooldown indicator $C_i(t) = \mathbf{1}\{\text{firm } i \text{ funded in the last } K_c \text{ weeks}\}$:

$$q_i(t) = (w_0 + u_s)^\top z_i(t) + \eta_s C_i(t), \quad \eta_s \leq 0, \quad (14.4)$$

with a global weight vector w_0 , sector deviations u_s , and sector-specific cooldown coefficients η_s . The conditional choice probability is the softmax over the *current* risk set,

$$p(i \mid s, t) = \frac{\exp q_i(t)}{\sum_{j \in \mathcal{R}_s(t)} \exp q_j(t)}, \quad (14.5)$$

and the training loss is the conditional negative log-likelihood over observed events (t_m, s_m, i_m) ,

$$-\ell_c(\vartheta) = \sum_m \left[\log \sum_{j \in \mathcal{R}_{s_m}(t_m)} q_j(t_m) - q_{i_m}(t_m) \right]. \quad (14.6)$$

The normalization over $\mathcal{R}_s(t)$ rather than over a fixed candidate list is the crucial design choice: firms enter the risk set by becoming active and leave it by the exposure rules of Chapter 13, and the model is well defined for every risk-set size. Algebraically, Equations (14.5) and (14.6) are McFadden’s conditional logit (McFadden, 1974; Train, 2009); they are simultaneously, term for term, the factors of Cox’s partial likelihood for a proportional-intensity model (Cox, 1972, 1975). The two literatures prove the same theorem twice, and we will lean on both.

Proposition 14.4 (Baseline cancellation). *Suppose the firm intensities have the multiplicative form*

$$\lambda_i(t) = \lambda_0(s, t) \exp q_i(t), \quad i \in \mathcal{R}_s(t), \quad (14.7)$$

where $\lambda_0(s, t) \geq 0$ is any sector–time baseline common to the risk set: unknown, unmodeled, nonstationary, dependent on the whole history, or misspecified in any way whatsoever. Then the conditional identity distribution given an event is Equation (14.5), free of λ_0 . Consequently the conditional likelihood Equation (14.6) involves no timing model, and any deformation of the time axis or of the baseline that acts on the risk set as a common factor — weekly aggregation, jittered or back-filled dates that preserve the assignment of events to risk-set snapshots, bursts, regime shifts, unmodeled excitation in the ground process — leaves Equation (14.6) unchanged.

Proof. Substitute Equation (14.7) into Equation (14.2):

$$\frac{\lambda_i(t)}{\sum_j \lambda_j(t)} = \frac{\lambda_0(s, t) e^{q_i(t)}}{\lambda_0(s, t) \sum_j e^{q_j(t)}} = \frac{e^{q_i(t)}}{\sum_{j \in \mathcal{R}_s(t)} e^{q_j(t)}}.$$

The baseline cancels because it is common; nothing about its form was used. For the aggregation claim: replacing t_m by its containing bin changes which snapshot $(\mathcal{R}_s, z, C)$ the event is matched to, but if features and risk sets are constructed at bin resolution from pre-bin information (Protocol 2.5), the conditional law of the winner given the bin’s snapshot is again Equation (14.5) with the bin-level baseline cancelled — interval censoring corrupts λ_0 , which does not appear. What does *not* cancel is instructive: a firm-specific multiplicative frailty $\lambda_i = \lambda_0 \xi_i e^{q_i}$ does not cancel (it is not common) and is absorbed into the score as an omitted variable; and a baseline that differs *within* the risk set — e.g. stage-specific deal clocks inside one sector — breaks the common-factor premise and must be moved into q_i as a covariate or handled by stratifying the risk set. $\square$

Proposition 14.5 (Robustness of the conditional MLE). *Let events $m = 1, \dots, M$ carry risk sets and features constructed point-in-time, with $i_m \in \mathcal{R}_{s_m}(t_m)$. Suppose (a) the conditional model is correctly specified: there exists ϑ^* such that $\mathbb{P}(i_m = i \mid \text{event } m, \mathcal{F}_{t_m}^-) = p(i \mid s_m, t_m; \vartheta^*)$ of Equation (14.5); (b) features are uniformly bounded, ϑ^* is identified and interior to a compact parameter set; (c) the averaged conditional information $\tilde{\mathcal{I}}_M = M^{-1} \sum_m \mathbb{E}[V_m]$, with V_m the conditional covariance of the score gradient under Equation (14.5), converges to a positive-definite limit. Then the maximizer $\hat{\vartheta}$ of ℓ_c is consistent and asymptotically normal, with no assumption on the ground process: the law of (t_m, s_m) may be arbitrarily misspecified, dependent, aggregated, or simply never modeled.*

Proof. Let $\mathcal{G}_m = \mathcal{F}_{t_m}$ be the event-indexed filtration. The m -th score contribution at ϑ^* is

$$U_m = \nabla q_{i_m}(t_m) - \sum_{j \in \mathcal{R}_{s_m}(t_m)} p(j \mid s_m, t_m; \vartheta^*) \nabla q_j(t_m),$$

and by (a), $\mathbb{E}[U_m \mid \mathcal{G}_{m-1}, \text{event } m \text{ at } (t_m, s_m)] = \sum_i p_i^* \nabla q_i - \sum_j p_j^* \nabla q_j = 0$: the score is a martingale difference array with respect to $\{\mathcal{G}_m\}$ *conditionally on the event stream, whatever its law*. Boundedness (b) and nondegeneracy (c) then give a law of large numbers and a martingale central limit theorem for the score, and consistency plus asymptotic normality of $\hat{\vartheta}$ follow by the standard argument — this is precisely Cox’s partial-likelihood heuristic (Cox, 1975) made rigorous by counting-process martingale theory in Andersen and Gill (1982); the conditional-logit sampling theory of McFadden (1974) gives the same conclusion when events are treated as independent choice occasions, and the martingale version shows independence is not needed. The proof fails exactly where it should: if the risk set is wrong (defect (C3): members retained who cannot win, or competitors omitted), the softmax in U_m is normalized over the wrong set and the conditional mean of the score is no longer zero at ϑ^* — the estimand silently changes to the conditional law *given the contaminated risk set*. Robustness to timing is not robustness to membership. $\square$

Propositions 14.4 and 14.5 together are the precise version of the claim previewed in Section 1.3: the mark factor needs no timing model, survives the (C4) aggregation and untrusted-date defect, and is insulated from ground-process misspecification — the three failure modes that Chapter 10 shows corrupt every timing estimate on current data. That is why T1 is the most defensible deliverable on present data quality, and why the ranking layer, not the Hawkes layer, is the repository’s headline deliverable while the timing stack matures.

14.3 Positive-only labels: manufacturing the comparison

A supervised-classification framing of T1 needs a negative class, and the data has none (Chapter 2): we observe fundings, never rejections. No row anywhere says “firm i sought capital at t and was declined,” so per-firm labels are positive-only, and any attempt to manufacture firm-level negatives by declaring non-events (“firm i did not raise in week t ”) imports the timing model through the back door — the probability of a non-event depends on λ_0 , which we just proved we do not need and cannot trust. The conditional-choice construction solves the label problem structurally: each funding event manufactures a valid comparison — the funded firm against its risk-set peers at the same sector-time — and Proposition 14.5 says these comparisons identify the score parameters without a single explicit negative. This is the same device by which Cox regression learns hazard ratios from cases and risk sets alone, and the connection to case-control methodology is exact: the risk set is the control pool, sampled at the case’s own event time.

The practical obstacle is scale: real sector risk sets contain hundreds to thousands of firms, and Equation (14.6) sums over all of them per event. Sampling the risk set fixes this without changing the estimand.

Theorem 14.6 (Consistency of the sampled softmax). *For event m , let a subset $D_m \subseteq \mathcal{R}_{s_m}(t_m)$ with $i_m \in D_m$ be drawn by a sampling protocol $\pi(D \mid i, \mathcal{R})$ — the probability of drawing D given that the winner is i — satisfying the uniform conditioning property: $\pi(D \mid i) = \pi(D \mid j) > 0$ for all $i, j \in D$. Then, under the model Equation (14.5), the conditional law of the winner given D_m is the softmax restricted to D_m ,*

$$\mathbb{P}(i_m = i \mid D_m, \mathcal{F}_{t_m^-}) = \frac{\exp q_i(t_m)}{\sum_{j \in D_m} \exp q_j(t_m)}, \quad (14.8)$$

so maximizing the subsampled conditional likelihood built from Equation (14.8) is maximum likelihood in a correctly specified model and inherits the consistency of Proposition 14.5. In particular, uniform sampling of J non-winners from $\mathcal{R} \setminus \{i_m\}$, with the winner always retained,

satisfies the property, and no correction term is needed. For a general protocol, replacing q_j by $q_j + \log \pi(D | j)$ restores Equation (14.8) (McFadden, 1978; Train, 2009).

Proof. By Bayes' rule on the joint draw (winner first, then the sampled set given the winner),

$$\mathbb{P}(i_m = i | D_m = D) = \frac{p(i | s_m, t_m) \pi(D | i)}{\sum_{j \in D} p(j | s_m, t_m) \pi(D | j)} = \frac{e^{q_i + \log \pi(D|i)}}{\sum_{j \in D} e^{q_j + \log \pi(D|j)}},$$

where the full-risk-set normalizer of $p(\cdot | s_m, t_m)$ cancels between numerator and denominator — the same ratio trick as Proposition 14.4, now applied to the sampling design. Under uniform conditioning the $\log \pi$ terms are equal across $j \in D$ and cancel as a common shift of the scores, giving Equation (14.8). Uniform sampling of J alternatives yields $\pi(D | i) = \binom{|\mathcal{R}|-1}{J}^{-1}$ for every $i \in D$, which is constant. The subsampled events (i_m, D_m) then follow a correctly specified conditional logit over D_m , and Proposition 14.5 applies verbatim with $\mathcal{R}$ replaced by D . What is lost is efficiency, not consistency: the conditional information per event shrinks with J , so J is a compute–variance dial, not a bias dial. $\square$

14.4 Cooldown: inhibition carried by the mark factor

Chapter 8 establishes that a nonnegative Hawkes kernel cannot express inhibition and enumerates the three legal routes: (a) rectified additive intensities, (b) exp-link log-intensity models, and (c) inhibition carried by the mark factor. The cooldown term of Equation (14.4) is route (c). A firm that has just raised is, for a while, less likely to raise again — not because the sector cooled, but because that firm left the market for capital. Encoding this as a negative-coefficient covariate in the score gives, by Proposition 14.4, a multiplicative suppression of the firm's *share*: while $C_i = 1$, firm i 's conditional probability is scaled by $e^{\eta_s} < 1$ relative to its own baseline share, renormalized within the risk set. The ground layer is untouched — sector excitation stays positive and the linear Hawkes/MBPP mean-field structure of Chapters 8 and 10 survives, which is exactly why route (c) was chosen as the default (the positivity link and the expectation operator do not commute for routes (a) and (b), breaking the closed-form machinery).

The economic interpretation is direct: η_s is the log of the factor by which a recent raise suppresses a firm's odds of being the next funded firm in its sector, and the constraint $\eta_s \leq 0$ (enforced as a bound in `fit_startup_ranker`) encodes the sign as a modeling assumption rather than a hope. Whether the device works under misspecification is a planned synthetic test: in regime B of the synthetic world the generator's inhibition is a continuous 30-day-half-life kernel, not a binary window, and Chapter 19 will measure how well the fitted mark-factor cooldown recovers the generator's self-inhibition *ordering* across firms. The binary window $C_i(t)$ with default $K_c = 26$ weeks is a one-column basis for a decaying curve; the natural refinements — an age-since-last-deal feature with learned shape, or a few staggered windows — belong to the gradient-boosted scorer of the next section, which can learn the shape without a parametric commitment.

14.5 Gradient-boosted scores in the same likelihood

The linear score Equation (14.4) is a restriction, and we do not expect it to hold. The B1 benchmark of Chapter 19 tests it directly: a feature-parity XGBoost ranker against the linear conditional logit, scored on held-out conditional NLL and top-5 hit rate. If the restriction fails, the design response is not to abandon the conditional-choice likelihood but to change the score family inside it: replace $q_i(t)$ by $f(z_i(t), C_i(t), a_i(t), \dots)$ with f an additive tree ensemble (Friedman, 2001; Chen and Guestrin, 2016), keeping the softmax over the current risk set and

the loss Equation (14.6) exactly as they are. Training requires only the per-candidate gradient and Hessian of the grouped-softmax loss; the “listwise” machinery of the learning-to-rank literature reduces to a short calculation.

Lemma 14.7 (Grouped-softmax gradients). *Fix one event group m with candidate scores $\{q_j\}_{j \in \mathcal{R}_m}$, softmax probabilities p_j , and one-hot label $y_j = \mathbf{1}\{j = i_m\}$. The per-event loss $\ell_m = \text{logsumexp}_j q_j - q_{i_m}$ has*

$$\frac{\partial \ell_m}{\partial q_j} = p_j - y_j, \quad \frac{\partial^2 \ell_m}{\partial q_j \partial q_k} = p_j \mathbf{1}\{j = k\} - p_j p_k, \quad (14.9)$$

so the diagonal Hessian entries are $h_j = p_j(1 - p_j)$. The full Hessian $\text{diag}(p) - pp^\top$ is positive semidefinite (it is the covariance matrix of the candidate-indicator vector under p), so ℓ_m is convex in the scores and the diagonal is a legitimate curvature surrogate.

Proof. Differentiate logsumexp: $\partial_{q_j} \text{logsumexp} = p_j$; subtracting $\partial_{q_j} q_{i_m} = y_j$ gives the gradient. Differentiating p_j once more gives $\partial_{q_k} p_j = p_j \mathbf{1}\{j = k\} - p_j p_k$. For a vector v , $v^\top (\text{diag}(p) - pp^\top) v = \mathbb{E}_p[v_J^2] - (\mathbb{E}_p[v_J])^2 = \text{Var}_p(v_J) \geq 0$. $\square$

Implementation is a custom objective for XGBoost: candidates of one event form one group; per boosting round, compute the softmax within each group from the current margin scores, emit $g_j = p_j - y_j$ and $h_j = p_j(1 - p_j)$ per candidate row, and let the tree learner take the regularized Newton step in function space (Chen and Guestrin, 2016). Dropping the off-diagonal Hessian terms is the standard diagonal approximation; by convexity of ℓ_m it costs step quality, never correctness of the stationary point. Note what this objective is and is not: it *is* the exact conditional NLL of Equation (14.6) — the listwise cross-entropy per event group — so the fitted trees produce genuine conditional probabilities through the softmax; it is *not* a pairwise surrogate in the RankNet family (Burges et al., 2005), which we treat separately in Section 14.7. Sampled groups per Theorem 14.6 keep the objective correct at scale. The same boosted learners can also parameterize a continuous-time intensity directly through a binned likelihood; Chapter 9 develops that intensity-side counterpart.

Warning 14.8 (The robustness belongs to the family, not the score). Every robustness statement of Section 14.2 — baseline cancellation, timing-free consistency, survival of aggregation — is a property of the conditional-choice *likelihood family*: any scorer, linear or boosted, normalized over the current risk set inherits them. None of them certify the *linear* score, which the B1 benchmark of Chapter 19 will test, and none of them certify a boosted scorer trained on a different loss (a pairwise surrogate, a pointwise classifier): change the likelihood family and the cancellation proofs no longer apply. When this chapter says “the ranker is robust to (C1) and (C4),” it means the grouped-softmax likelihood is; claims about any particular fitted score function are empirical and belong to Chapter 19 and Chapter 20.

14.6 Multiple deals per sector-week: Plackett–Luce

Weekly resolution creates ties: several deals land in the same sector-week cell and share a risk-set snapshot. The current implementation treats them as repeated independent softmax draws over the same risk set. Survival analysis has the same situation: tied event times in a Cox partial likelihood, where the repeated-softmax treatment is exactly Breslow’s approximation (Breslow, 1974), the exact treatment is the sequential-conditioning likelihood, and Efron’s correction sits between them (Efron, 1977; Therneau and Grambsch, 2000). The discrete-choice literature derives the exact object independently as the Plackett–Luce model (Plackett, 1975; Luce, 1959).

The derivation is a corollary of Proposition 14.2. Let d deals occur in cell (s, t) with winners $i_1, \dots, i_d$ in sequence. After the k -th deal, firm i_k ceases to compete for the remaining within-cell deals (it has just raised; within one week its continued presence in the denominator is a modeling error, not a modeling choice). Applying Equation (14.2) to each successive deal over the shrinking risk set gives the sequential softmax without replacement:

$$\mathbb{P}(i_1, \dots, i_d \mid s, t, d \text{ deals}) = \prod_{k=1}^d \frac{\exp q_{i_k}(t)}{\sum_{j \in \mathcal{R}_s(t) \setminus \{i_1, \dots, i_{k-1}\}} \exp q_j(t)}, \quad (14.10)$$

which is the Plackett–Luce likelihood of the ordered top- d under score utilities. The repeated-softmax approximation replaces every denominator by the full-risk-set sum, so its per-cell log-likelihood error is

$$\Delta \ell_{s,t} = \sum_{k=2}^d -\log(1 - P_{k-1}), \quad P_{k-1} = \sum_{l < k} p(i_l \mid s, t), \quad (14.11)$$

the cumulative softmax mass of already-funded firms. Two consequences. First, the sign is systematic: the approximation understates every later-deal probability, so it penalizes exactly the concentrated, confident score profiles the ranker is supposed to produce, biasing fitted scores toward diffuseness. Second, the size is quantifiable: to first order $\Delta \ell_{s,t} \approx \binom{d}{2} \bar{p}$, with $\bar{p}$ the typical winner probability — negligible when risk sets are large and diffuse ($\bar{p} \sim 10^{-2}$, $d \leq 2$), material when many deals hit a small or concentrated risk set in one week ($d \geq 3$ with $\bar{p} \sim 10^{-1}$ costs tenths of a nat per cell). Since same-week multiplicity is common in hot sectors — precisely the cells the business cares about — the exact likelihood is worth its implementation cost. One complication: Equation (14.10) conditions on the within-cell *order*, and (C4) says within-week order is exactly what the dates cannot support. The clean treatment is the exchangeable-order PL: average the log-likelihood over random permutations of the cell’s winners (exact summation over $d!$ orders for the typical $d \leq 3$, sampled permutations beyond), which removes the dependence on untrusted ordering at the price of a small, controlled Monte Carlo cost.

Design decision (D10.2 — Plackett–Luce in the next ranker revision).

The next revision of the mark factor replaces repeated softmax with the Plackett–Luce likelihood Equation (14.10), with within-cell order handled by exact averaging over permutations for $d \leq 3$ and sampled permutations beyond, and with cooldown state updated between within-cell draws in simulation (as `simulate_marked_paths` already does). Rationale: Equation (14.11) gives a systematic bias against concentrated score profiles that can reach tenths of a nat per hot cell; the Cox-ties literature says the exact method dominates Breslow whenever ties are heavy (Therneau and Grambsch, 2000). Reversal trigger: if on real data the measured NLL and recall@ K differences between PL and repeated softmax are inside the era-blocked bootstrap intervals of Protocol 14.10 for two consecutive evaluation campaigns, revert to repeated softmax for simplicity and record the equivalence.

14.7 Optimizing the list directly

If the consumed object is only the top- K list, one can ask whether the likelihood should be bypassed entirely in favor of a loss that targets the list metric — the learning-to-rank program:

pairwise RankNet (Burgess et al., 2005), listwise LambdaMART with its λ -gradients aimed at NDCG (Burgess, 2010; Järvelin and Kekäläinen, 2002). The question deserves a precise answer, and the answer has two halves: a lemma that says likelihood training is already list-optimal *under the model*, and an empirical clause that says surrogates can win only through score-family misspecification — whose extent Chapter 19 will measure.

Lemma 14.9 (Top- K optimality of the conditional probabilities). *Fix a sector-time (s, τ) and let $p_i, i \in \mathcal{R}_s(\tau)$, be the conditional identity distribution of the next deal. For a list $L \subseteq \mathcal{R}_s(\tau)$ with $|L| = K$, the expected recall@ K against the next deal is $\mathbb{E}[\mathbf{1}\{\text{winner} \in L\}] = \sum_{i \in L} p_i$, and it is maximized by any L consisting of K largest- p_i candidates. For an ordered list σ , the expected reciprocal rank $\sum_{r \geq 1} p_{\sigma(r)}/r$ is maximized by sorting p in descending order. Hence, under the model, producing the top- K by conditional probability is the optimal list policy for both headline metrics, and NDCG with flat gains and any decreasing discount is maximized by the same sort.*

Proof. Recall: the objective $\sum_{i \in L} p_i$ is a set function maximized by exchange: if $i \in L, i' \notin L$, and $p_{i'} > p_i$, swapping i' for i strictly increases the sum; iterating terminates at a set of K largest probabilities. MRR: the discounts $1/r$ are strictly decreasing; if an ordering places $p_{\sigma(r)} < p_{\sigma(r')}$ with $r < r'$, transposing them changes the objective by $(p_{\sigma(r')} - p_{\sigma(r)})(1/r - 1/r') > 0$; absence of improving transpositions is exactly the descending sort — the rearrangement inequality. The NDCG claim is the same argument with $1/r$ replaced by the discount sequence. Under multiple deals per window the argument applies to each successive deal with the PL-updated distribution; ranking by the current p_i remains optimal for the first hit, which is what recall@ K scores. $\square$

The lemma settles the division of labor. A correctly specified, likelihood-trained conditional model followed by a sort cannot be beaten in expectation on recall@ K , MRR, or NDCG; training on a surrogate can help only by trading bias in regions the likelihood cares about for accuracy in the top of the list when the score family cannot represent the truth. That trade can be real, but it has a cost we refuse to pay in the primary model: a LambdaMART scorer emits uncalibrated ordinal scores, while the decision layer of Chapter 16 consumes *probabilities* — conditional identity probabilities composed with ground intensities to produce statements like “firm i is on the list *and* a deal is likely within the quarter” (T1 composed with T2/T3), and the conditional NLL is the strictly proper score for that object (Gneiting and Raftery, 2007). A scorer that cannot feed the composition is a dead end in this architecture regardless of its list metrics.

Design decision (D10.3 — LambdaMART as challenger only).

LambdaMART (and pairwise RankNet as its diagnostic little sibling) is maintained as a benchmarked *challenger* in the evaluation harness, never as the primary model. Rationale: Lemma 14.9 says it can only win through score-family misspecification, which the boosted grouped-softmax primary already addresses inside the likelihood family; and its output does not feed the probability-consuming decision layer. Reversal trigger: if LambdaMART beats the likelihood-trained boosted ranker on pooled recall@5 by more than the era-blocked bootstrap interval for two consecutive campaigns, *and* a post-hoc calibration overlay (Platt, 1999; Zadrozny and Elkan, 2002; Niculescu-Mizil and Caruana, 2005) on its scores passes the calibration checks of Chapter 11 at the event-group level, it is promoted to primary and this section is rewritten around the surrogate-first architecture.

14.8 The recall at K protocol

Everything above concerns fitting; what follows is the committed evaluation contract — the repository’s headline metric, stated fully so that no future implementation “simplifies” it into something unauditable.

Protocol 14.10 (Rolling recall@ K and MRR for the mark factor).

- (a) *Origins.* Evaluation origins τ run on a weekly grid over the held-out era. At each τ and each sector s , freeze the risk set $\mathcal{R}_s(\tau)$ under the versioned risk-set definition (item (g)), compute point-in-time scores, and emit the full ordering of $\mathcal{R}_s(\tau)$; the top- K prefixes are the lists $L_{s,\tau,K}$, $K \in \{1, 5, 10\}$.
- (b) *Window and hits.* The evaluation window is the next W weeks, $(\tau, \tau + W]$; we commit to next- W -weeks rather than next- K' -deals so that empty windows are informative and the metric has a fixed temporal meaning. W is matched to the consuming action’s refresh horizon: default $W = 4$ (a monthly sourcing look-ahead), with $W = 13$ reported as a robustness column. A deal in the window scores a *hit* at level K iff its firm was on $L_{s,\tau,K}$ — membership at τ , not at deal time.
- (c) *Denominator and newcomers.* The denominator counts deals in the window whose firm was in $\mathcal{R}_s(\tau)$. Deals to firms outside every frozen risk set at τ — first appearances and out-of-universe deals — are *excluded from the denominator and reported separately* as the newcomer share, under the newcomer-vs-entrant discipline of Chapter 15. Silently scoring newcomer deals as misses (or dropping them without reporting) is forbidden: the first deflates the metric for a failure the mark factor cannot commit, the second hides the (C6) exposure.
- (d) *Metrics.* Report recall@ K for $K \in \{1, 5, 10\}$ and MRR (reciprocal rank of the funded firm in the full frozen ordering), per sector and pooled (deal-weighted), aggregated over origins rolling-origin style. Alongside, *always*, the held-out conditional NLL of Equation (14.6) on the same events: the list metrics say whether the sort is right, the NLL says whether the probabilities feeding the decision layer are right, and Lemma 14.9 is the license to demand both from one model.
- (e) *Uncertainty.* Era-blocked bootstrap: resample contiguous blocks of evaluation origins (blocks aligned with the macro-era segmentation of Chapter 11) with replacement, recompute pooled metrics, report percentile intervals. Origin-level i.i.d. resampling is forbidden — overlapping windows and regime persistence make it anti-conservative.
- (f) *Baselines.* Every report carries the same metrics for the random ranker over the same frozen risk sets (chance floor) and, on synthetic data, the oracle ranker (ceiling); synthetic reference values will be reported in Chapter 19, in the format of Table 14.1.
- (g) *Frozen, versioned risk sets.* The risk-set definition — entry rule, exposure/exit rule from Chapter 13, sector assignment — is frozen per evaluation campaign and identified by a version string recorded with every metric row; changing the rule starts a new metric series rather than overwriting one. Rationale: we expect risk-set truthfulness to be the largest sensitivity in the system — Chapter 19 will quantify the gap between observed and oracle membership — so an unversioned membership rule would make every metric movement uninterpretable.

Held-out metric	fitted	oracle (ceiling)	random (floor)
top-1 hit rate	—	—	—
top-5 hit rate	—	—	—
MRR	—	—	—

Table 14.1: Reference format for the mark-factor ranker baselines on synthetic data; the fitted, oracle, and random columns will be filled by the synthetic campaign of Chapter 19. The oracle ceiling is expected to sit well below 1: the residual is irreducible softmax stochasticity, not model error. Synthetic values calibrate expectations for real-data campaigns; they are priors and vetoes, never substitutes (Section 2.5).

Design decision (D10.1 — the T1 deliverable and its two metrics).

The primary T1 deliverable is the sector-conditional risk-set ranker: gradient-boosted scores inside the grouped-softmax conditional likelihood Equation (14.6) (per Lemma 14.7), with cooldown and staleness-age features, sampled risk sets per Theorem 14.6 where scale requires, upgraded to Plackett–Luce per D10.2. It is evaluated by Protocol 14.10 *and* held-out conditional NLL — both, always; a model improving one while degrading the other is investigated, not shipped. Rationale: the competing-intensities identity makes the conditional model the exact T1 target (Proposition 14.2); the cancellation and consistency results make it the defect-robust factor (Propositions 14.4 and 14.5); boosted scores are chosen over linear within the family because we expect the linear restriction to fail, with the B1 benchmark of Chapter 19 as the planned test; and Lemma 14.9 guarantees likelihood training plus sorting is list-optimal under the model while the NLL keeps the probabilities honest for Chapter 16. Reversal triggers: if the boosted scorer fails to beat the linear conditional logit on real data (both metrics, two campaigns), demote to the linear score for interpretability; if the challenger clause of D10.3 fires, the primary itself is reconsidered.

14.9 Defect declaration

Per design decision D2.1, the mark factor’s position against the ledger of Table 2.1, with the proofs above doing the work.

(C1) *Administrative censoring: robust.* The conditional likelihood Equation (14.6) is a sum over observed events only; no survival factor appears because timing is conditioned away (Proposition 14.4), and under independent censoring the risk set at each observed event is uncorrupted. Open spells cost the ranker nothing — there is simply no term.

(C2) *Left truncation: robust for the conditional estimand, with a scope caveat.* Comparisons are made within risk sets of firms that have entered observation; the estimand is conditional on membership, so delayed entry does not bias it. What (C2) forbids is re-reading scores as statements about the untracked population — that extension belongs to Chapter 15.

(C3) *Unlabeled exits: vulnerable — the binding defect.* Dead firms retained in $\mathcal{R}_s(t)$ occupy denominator mass and top- K slots; Proposition 14.5’s proof shows the estimand silently becomes the conditional law given the contaminated pool. We expect this to be the largest sensitivity in the whole system; Chapter 19 will quantify the gap between observed and oracle risk sets and test whether a discriminative exit classifier can close it. Mitigations are the exposure rules of Chapter 13 (trailing-activity windows closing spells point-in-time) plus staleness features that let the scorer learn to demote ghost-like profiles. One caution follows from the theory: the optimal score weighting *given a contaminated pool* differs from the oracle

weighting — the fitted model partially adapts to contamination — so cleaning the risk set changes the optimal parameters, and oracle-world or cleaned-world coefficients must never be transplanted onto contaminated risk sets or vice versa. Risk-set definition and fitted model are a matched pair; Protocol 14.10(g) versions them together.

(C4) *Untrusted dates and aggregation: robust, by proof.* Proposition 14.4: binning and jitter deform $\lambda_0(s, t)$, which cancels; the residual exposure is only through snapshot assignment (a deal jittered across a week boundary meets a one-week-shifted risk set and covariates) and through within-cell order, which D10.2’s exchangeable-order Plackett–Luce removes.

(C5) *Stale covariates: vulnerable, with signed bias, remedies inherited.* Stale z_i is classical covariate measurement error inside a choice model: score coefficients attenuate toward zero, the softmax flattens, lists diffuse, and recall@ K degrades toward the random floor — a bias against the model, never in its favor. The remedies of Chapter 12 (staleness ages as first-class features, covariate-state layer where it pays) apply verbatim to the ranker’s feature block, and the boosted scorer can learn staleness-dependent discounting that the linear score cannot.

(C6) *Out-of-universe deals: vulnerable without the outside option.* Deals from untracked firms mean the true choice set exceeds $\mathcal{R}_s(t)$; a mark factor without an outside option overstates every tracked firm’s probability by the factor $1/(1 - \pi_{\text{new}})$. The treatment — the newcomer branch with its exact separation of binary and choice likelihoods — is Chapter 15’s; this chapter’s protocol already quarantines the exposure by excluding newcomer deals from the recall denominator and reporting their share (Protocol 14.10(c)).

Code hooks.

Existing. `hawkes_calibration/sector_ranker.py`: `fit_startup_ranker` implements Equations (14.4) to (14.6) with sector deviations, the $\eta_s \leq 0$ bound of Section 14.4 (`constrain_cooldown_negative`), and dynamic risk sets via `candidate_set` and `cooldown_vector`; `ranker_predict_proba` exposes the conditional probabilities; `evaluate_ranker` computes held-out conditional NLL, MRR, and top- k hit rates and is the seed of the harness below. `hawkes_calibration/models/hawkes_ranker.py` carries the regime-B synthetic ranker used for the planned cooldown test of Section 14.4.

To be written. (i) The grouped-softmax XGBoost objective module (`hawkes_calibration/ranking/grouped_softmax_objective.py`): per-group softmax from margin scores, per-candidate (g_j, h_j) of Equation (14.9), group boundaries from event ids, uniform risk-set sampling per Theorem 14.6 with the winner retained. (ii) The Plackett–Luce upgrade of D10.2 inside `hawkes_calibration/sector_ranker.py` (sequential denominators, exchangeable-order averaging for $d \leq 3$). (iii) The evaluation harness (`hawkes_calibration/eval/ranking.py`) implementing Protocol 14.10 verbatim: weekly origins, frozen versioned risk sets (`riskset_version` recorded per metric row), W -week windows, newcomer-share reporting, era-blocked bootstrap, and the random/oracle baseline columns in the format of Table 14.1. (iv) The LambdaMART challenger of D10.3 as a harness plug-in, never as a dependency of the decision layer.

Part VI

The Open Universe

Chapter 15

Deals from Companies We Cannot See

15.1 Two populations, one deal stream

Every model developed so far has assumed that we know who could receive the next deal. This chapter treats the two ways that assumption fails. First, deals arrive from firms that are not in the tracked universe: the deal row exists, carries a date and a sector, and is flagged `company_universe = 'Out of Universe'`, but the firm has no companies-table row, no covariates, and no history — defect (C6) of Chapter 2. Second, deals arrive from firms that will be in the universe tomorrow but are invisible today: a firm enters observation *by raising*, so the tracked population is selected on the very event we model — defect (C2). Neither failure is negligible: the synthetic world is built with a material newcomer share, and Chapter 19 will measure how much of the deal stream falls outside the current risk set and how that share behaves in held-out windows. A mark factor that ignores this mass overstates every tracked firm’s probability by the factor $(1 - \pi)^{-1}$ of Equation (15.2); a ground model that ignores it cannot forecast sector totals honestly. The remedy of this chapter is structural: an explicit outside option in the mark factor, an external component in the ground process, a model-free audit of the unseen mass, and an entry model for the population side of (C2).

We first fix what is and is not observed, in the notation of Definition 2.1.

Definition 15.1 (Tracked and untracked populations; observed stream). The firm universe splits as $\mathcal{I} \cup \mathcal{I}^\dagger$: the *tracked* universe $\mathcal{I}$ (a companies-table row, covariates $\tilde{x}_i$, an event history N_i) and the *untracked* remainder $\mathcal{I}^\dagger$ (no row, no covariates, no attributable history). The observed deal stream is

$$\{(t_m, s_m, \iota_m)\}_{m \geq 1}, \quad \iota_m = \begin{cases} i_m, & i_m \in \mathcal{I}, \\ \text{'Out of Universe'}, & i_m \in \mathcal{I}^\dagger, \end{cases} \quad (15.1)$$

together with deal marks m_m : the time and sector of every deal are recorded, but firm identity is recorded only on $\mathcal{I}$. Operationally, a deal is *untracked* if its universe flag says so *or* if its firm makes its first-ever appearance in the data at t_m (no prior attributable history existed at t_m^-).

Assumption 15.2 (Complete sector stream). Every deal row carries a deal time and a sector label; no deal is dropped from the stream for being untracked. Sector labels are correct up to the small mislabel noise documented in Appendix B, whose treatment (hygiene rules; the group-level variant of Section 15.3) is orthogonal to universe membership.

Assumption 15.2 is a statement about the vendor’s delivery format, and it is the central asymmetry of this chapter: sector-level ground counts are *complete* — every deal has a sector

— while firm-level attribution is complete only on $\mathcal{I}$. We state the consequence as a result, because it decides which layers of the architecture (C6) can hurt.

Proposition 15.3 (Attribution asymmetry). *Under Assumption 15.2, with the exact factorization of Theorem 6.7:*

- (i) *The ground factor — the likelihood of times and sectors, with sector intensities $\Lambda_s(t)$ built from sector-level histories and exogenous covariates — is a functional of fully observed data. Every ground-process estimand and estimator is invariant to (C6): the ground layer is immune.*
- (ii) *The mark factor is not. Let $\pi_{s,t} = \mathbb{P}(i_{next} \notin \mathcal{R}_s(t) \mid \text{deal in } s \text{ at } t, \mathcal{F}_{t-})$ be the probability that the next deal in s goes outside the tracked risk set. A mark factor fitted on tracked-attributed deals only estimates the conditional law $\mathbb{P}(i \mid \text{deal}, i \in \mathcal{R}_s(t))$, and for every tracked firm i ,*

$$\mathbb{P}(i \mid \text{deal in } s, t) = (1 - \pi_{s,t}) \mathbb{P}(i \mid \text{deal in } s, t, i \in \mathcal{R}_s(t)). \quad (15.2)$$

Reading the restricted model's output as an absolute probability overstates every tracked firm by the factor $(1 - \pi_{s,t})^{-1}$.

- (iii) *Any firm-level intensity assembled from the two layers, $\lambda_i(t) = \Lambda_{s(i)}(t) \mathbb{P}(i \mid \text{deal})$, inherits the inflation of (ii) multiplicatively, as do all horizon probabilities (target T3) built from it.*

Proof. (i) By Theorem 6.7 the ground factor of the log-likelihood is $\sum_m \log \Lambda_{s_m}(t_m) - \sum_s \int_0^T \Lambda_s(u) du$, which depends on the data only through the pairs (t_m, s_m) and the exogenous covariate paths. By Assumption 15.2 these are observed for every deal, untracked or not; hence the ground likelihood, its maximizer, and any functional of the fitted Λ_s are unchanged by the loss of firm identities. (The caveat — that Λ_s 's history term must itself be built from *all* deals — is exactly the histories rule of Section 15.4; violate it and immunity is forfeited.) (ii) The event $\{i\} \subseteq \{i \in \mathcal{R}_s(t)\}$ for $i \in \mathcal{R}_s(t)$, so the chain rule gives $\mathbb{P}(i) = \mathbb{P}(i \in \mathcal{R}) \mathbb{P}(i \mid i \in \mathcal{R}) = (1 - \pi_{s,t}) \mathbb{P}(i \mid i \in \mathcal{R})$. A likelihood evaluated only on deals with $i_m \in \mathcal{R}$ is, term by term, the likelihood of the conditional law given $\mathcal{R}$ -membership; its MLE targets that conditional law, not the absolute one. (iii) Multiply (ii) by $\Lambda_s(t)$; the marking theorem for marked point processes (Daley and Vere-Jones, 2003, Ch. 6) identifies $\Lambda_s(t) \mathbb{P}(i \mid \text{deal})$ as firm i 's intensity, and the factor $(1 - \pi_{s,t})^{-1}$ carries through every monotone functional of λ_i , including $p_i(t, \Delta)$. $\square$

For illustration, a newcomer share of $\pi = 0.25$ makes the inflation in Equation (15.2) a 33% overstatement on every tracked firm — uniform enough to leave within-sector *rankings* intact (which is why Chapter 14 could defer the issue) but fatal to any absolute probability, any cross-sector comparison in which $\pi_{s,t}$ differs by sector, and any sector total decomposed by firm. Proposition 15.3 is also one more argument for the two-layer architecture of Theorem 6.7: the defect ledger of Table 2.1 assigns (C6) to the mark layer *alone*, so the timing deliverable (T2) survives on sector-complete counts while the identity deliverable (T1) is repaired where it is broken — in the mark factor, by the outside option of Section 15.3.

15.2 Newcomers are not entrants

Before building the repair, one distinction must be fixed, because every downstream interpretation depends on it.

Warning 15.4 (Newcomer $\neq$ entrant). The *newcomer probability* $\pi_{s,t}$ is the probability that the next deal’s firm is not in the current tracked risk set — equivalently, that the deal’s firm has no attributable prior appearance in the data. This is an *observable*: every deal either matches a tracked history or it does not, so $\pi_{s,t}$ can be estimated, calibrated, and audited. The *entrant probability* — the probability that a genuinely new economic entity enters the market — is *not identified* from this data without an entry model and auxiliary coverage information, because a first appearance conflates two channels: economic entry (a firm’s first raise ever) and first-time *tracking* of an old firm (vendor coverage catching up). The two channels are an unidentified mixture. We expect the conflation to cost little in fit and to be undetectable by fit — the synthetic world, which carries oracle channel labels, can test this (Chapter 19). What it costs, permanently, is every compositional claim: “entry accelerated in 2021,” “new company formation shifted to fintech,” “the entrant cohort of 2020 is larger than 2019’s” are all unsupported unless the coverage channel is modeled or measured. Policy: in code, documentation, and reporting, this repository says *newcomer*, never *entrant*, for the observable; *entrant* is reserved for the entry-model constructs of Section 15.6, always with their identification caveats attached.

The warning has a converse worth stating: if the two channels are indistinguishable in likelihood, a single outside option that absorbs both should lose little as a *predictive* device. The architecture below therefore models the newcomer mass as one bucket and defers the decomposition to the entry model and the coverage audit, which are the only places it can honestly live.

15.3 The mark factor with an outside option

15.3.1 The extended risk set

The repair to the mark factor is a one-row change with exact semantics. Extend each sector’s risk set with an *outside option* $\diamond_s$:

$$\mathcal{R}_s^+(t) = \mathcal{R}_s(t) \cup \{\diamond_s\}, \quad (15.3)$$

and give the outside option a score built from *context* covariates only — nothing firm-level exists on that branch by construction:

$$q_{\diamond_s}(t) = v^\top g_{s,t}, \quad g_{s,t} = [\text{trailing newcomer share, sector deal volume, macro as-ofs, vintage/era effects, } 1] \quad (15.4)$$

while tracked firms keep their ranker scores $q_i(t) = \beta^\top z_i(t)$ from Chapter 14. The mark distribution is the softmax over the extended set,

$$\mathbb{P}(m = a \mid \text{deal in } s, t, \mathcal{F}_{t-}) = \frac{e^{q_a(t)}}{e^{q_{\diamond_s}(t)} + \sum_{j \in \mathcal{R}_s(t)} e^{q_j(t)}}, \quad a \in \mathcal{R}_s^+(t), \quad (15.5)$$

so that $\pi_{s,t} = \mathbb{P}(\diamond_s \mid \cdot)$ is produced by the same softmax that produces firm probabilities. The next proposition says this is not a heuristic patch but the *correct* conditional model of the observable mark when the next deal may go outside $\mathcal{I}$.

Proposition 15.5 (The collapsed mark model is exact). *Let $P^*(\cdot \mid s, t, \mathcal{F}_{t-})$ be the true next-deal firm distribution on $\mathcal{I} \cup \mathcal{I}^\dagger$, and define the collapsed mark $\tilde{m} = i$ if $i \in \mathcal{R}_s(t)$ and $\tilde{m} = \diamond_s$ otherwise. Then:*

- (i) *$\tilde{m}$ is fully observed for every deal, including untracked ones; the extended-softmax likelihood built from Equation (15.5) is the exact mark-factor likelihood of the collapsed marked process — no approximation is introduced by the collapse, and no deal is dropped.*

- (ii) The family Equation (15.5) is well specified for the collapsed law: for any categorical distribution with full support on $\mathcal{R}_s^+(t)$ there exist scores realizing it (logits equal log-probabilities up to an additive constant), and the parameterization adds only the outside score Equation (15.4); identification is up to the usual location shift, with the outside option playing the role of the outside good's alternative-specific constant (McFadden, 1974; Train, 2009).
- (iii) Estimation is unchanged: the log-likelihood is the standard conditional-choice form of Chapter 14 with one extra row per choice set, and its MLE targets the KL projection of the true collapsed law onto the family. In particular the tracked-firm probabilities it outputs are absolute — the inflation of Equation (15.2) is removed by construction, since $\mathbb{P}(i \mid \cdot) = (1 - \pi_{s,t}) \cdot \text{softmax}_{\mathcal{R}_s(t)}(q)_i$ holds identically in the extended model.

Proof. (i) The collapse map $a \mapsto \tilde{m}$ is a deterministic, measurable coarsening of the mark space. For a tracked deal we observe i_m and hence $\tilde{m}_m$; for an untracked deal we observe precisely the event $\{i_m \notin \mathcal{R}_{s_m}(t_m)\}$, which is $\tilde{m}_m = \diamond_{s_m}$. The likelihood of a coarsened mark is the pushforward of P^* under the collapse, and Equation (15.5) is a density for exactly that pushforward; nothing unobservable enters. What is lost is only resolution *within* $\mathcal{I}^\dagger$ and among out-of-pool firms — which the data does not carry anyway. (ii) Given target probabilities $(p_a)_{a \in \mathcal{R}^+}$ with $p_a > 0$, set $q_a = \log p_a + c$: the softmax returns (p_a) for any c , and conversely the softmax determines scores up to c — the standard bijection between positive categorical laws and score differences. Restricting $q_\diamond$ to the span of $g_{s,t}$ makes the family parametric in the usual sense; whether the *true* $\pi_{s,t}$ lies in that span is an empirical question answered by the calibration protocol below, not by the algebra. (iii) Adding one alternative to a conditional-logit choice set changes neither the objective's form, nor its concavity in the linear-in-features parameterization, nor the softmax gradient; misspecified-model MLE converges to the KL projection under the usual regularity. The displayed identity is the chain rule applied inside the softmax: conditioning Equation (15.5) on $\{a \neq \diamond_s\}$ cancels the outside term from the denominator. $\square$

Two estimation remarks. First, if the newcomer share is material the outside option is not a rare event, so the intercept bias of rare-event logistic estimation (King and Zeng, 2001) is minor; if per-sector shares in thin sectors fall to a few percent, the Firth correction (Firth, 1993) costs one line and removes the concern. Second, the context features in Equation (15.4) exist to make the newcomer share *calibratable over time*; whether they do, and how much they improve on a constant share, is a planned synthetic test (Chapter 19), with the acceptance check fixed by Protocol 15.7. The trailing newcomer share must be computed on an expanding or trailing window strictly before t (Protocol 2.5), and the vintage/era terms exist to absorb the coverage drift of Section 15.7 rather than let it contaminate the macro loadings.

15.3.2 The architecture is a two-stage classifier, recognized

The committed factorization — ground process for sector timing, mark factor for firm-given-sector with an outside option — should be recognized as an instance of a familiar design pattern: *predict the group first, then the member within the group, with an outlier class*.

Proposition 15.6 (Two-stage equivalence). *Given a deal at t , let $\Lambda(t) = \sum_s \Lambda_s(t)$ be the superposed ground intensity. Then for any tracked firm i with sector $s(i)$,*

$$\mathbb{P}(i \mid \text{deal at } t, \mathcal{F}_{t-}) = \underbrace{\frac{\Lambda_{s(i)}(t)}{\Lambda(t)}}_{\text{stage 1: sector}} \times \underbrace{\frac{e^{q_i(t)}}{e^{q_{\diamond_{s(i)}}(t)} + \sum_{j \in \mathcal{R}_{s(i)}(t)} e^{q_j(t)}}}_{\text{stage 2: firm or outlier}}, \quad (15.6)$$

and the residual mass $\sum_s \frac{\Lambda_s(t)}{\Lambda(t)} \pi_{s,t}$ is the probability the deal goes outside the tracked universe altogether. Thus the two-layer model is the two-stage sector $\rightarrow$ firm classifier with a per-sector outlier bucket; conversely, any such two-stage classifier with strictly positive stage probabilities defines a valid mark distribution on $\bigcup_s \mathcal{R}_s^+(t)$.

Proof. By the marking theorem for the superposition of sector streams (Daley and Vere-Jones, 2003, Ch. 6), conditional on a deal at t the sector is distributed as $\Lambda_s(t)/\Lambda(t)$; multiplying by the conditional firm law Equation (15.5) is the chain rule. The converse direction is immediate: nonnegative stage probabilities that sum to one compose, by the same chain rule, into a categorical law on the disjoint union of extended risk sets. $\square$

The equivalence matters for one practical reason: it makes the *group-level variant* a first-class alternative rather than an architectural change. Sector labels carry mislabel noise (Assumption 15.2; Appendix B), and at the finest granularity a mislabeled deal both pollutes one sector’s ground counts and lands in the wrong choice set. Collapsing sectors into industry groups is another measurable coarsening of the mark — the same quotient construction as Proposition 15.5 — so stage 1 may equally predict the industry *group* and stage 2 rank firms within the group’s (larger) risk set, with one outlier bucket per group. Mislabels that stay within a group then cancel entirely. The trade is variance for bias: coarser groups mean larger softmax denominators and weaker sector-specific context in Equation (15.4). The decision between granularities belongs to the goodness-of-fit machinery of Chapter 11 applied per level; the mathematics is level-agnostic, provided — and this is a hard constraint — ground and mark layers use the *same* partition, or Equation (15.6) silently stops being a probability distribution.

15.3.3 Calibration of the outside option is a deliverable

With the outside option in place, the newcomer share stops being a nuisance and becomes a first-class model output: $\pi_{s,t}$ is a predicted probability, realized outcomes are observed for every deal, and the pair supports exactly the calibration discipline of Chapter 11.

Protocol 15.7 (Reliability of the outside option). On every evaluation window: (a) per quarter and per sector, compare the average predicted $\pi_{s,t}$ over deals against the realized newcomer share, reporting the mean absolute error next to a constant-share baseline; (b) pool deals into bins by predicted π and report a reliability diagram with binomial bands; (c) report the same by era, so that coverage drift (Section 15.7) is visible as an era-structured miscalibration rather than averaged away. Failures of (a) with stable context covariates are escalated to the coverage audit (Protocol 15.12) *before* any model change.

Design decision (D11.1 — outside option in the next ranker revision).

The next revision of the mark-factor ranker extends every choice set with the outside-option row Equation (15.3)–Equation (15.4): context features on the $\diamond_s$ row only, firm features on tracked rows only, one shared softmax. This extends `hawkes_calibration/sector_ranker.py`’s risk-set construction and mirrors the synthetic loader’s `build_choice_sets(newcomer_context=True)` path, with `experiments/exp28_newcomer_designs.py` (designs N1–N4) as prior art. Rationale: Proposition 15.3 (the restricted model’s probabilities are inflated by $(1 - \pi)^{-1}$), Proposition 15.5 (the extension is exact and costs nothing in estimation), and the expected calibration gain from context covariates, to be verified in Chapter 19. Reversal trigger: if on real data the realized newcomer share falls below 2% in every sector and era *and*

Protocol 15.7 shows a constant share calibrates within 1pp, the outside option may be frozen to an intercept-only score — removed it is not, since Equation (15.2) holds at any $\pi > 0$.

15.4 The ground process: the external component and the histories rule

15.4.1 Decomposing the immigrant intensity

On the ground layer, (C6) and (C2) appear not as missing labels but as a component of intensity that no tracked-firm bookkeeping can generate. Write the sector ground intensity in Hawkes form (Chapter 8) and split the baseline:

$$\Lambda_s(t) = \underbrace{\mu_s^{\text{tr}}(t) + \mu_s^{\text{ext}}(t)}_{\mu_s(t)} + \sum_{m: t_m < t} h_{ss_m}(t - t_m), \quad (15.7)$$

where μ_s^{tr} carries the part of the immigrant intensity attributable to the tracked population's state and μ_s^{ext} is the *external component*: deal flow from firms we cannot see, driven by the context covariates $g_{s,t}$ of Equation (15.4) (and, once Section 15.6 exists, by cohort-based newcomer supply). The decomposition is not identified by the ground layer alone — both components are baselines — but it is pinned by the mark layer through the marking theorem:

Corollary 15.8 (Newcomer substream intensity). *If the outside-option probability $\pi_{s,t}$ of Equation (15.5) is predictable, then the newcomer deals of sector s form a point process with conditional intensity $\pi_{s,t} \Lambda_s(t)$, and the tracked-attributed deals one with intensity $(1 - \pi_{s,t}) \Lambda_s(t)$. The sector total decomposes exactly as*

$$\Lambda_s(t) = \underbrace{(1 - \pi_{s,t}) \Lambda_s(t)}_{\text{tracked}} + \underbrace{\pi_{s,t} \Lambda_s(t)}_{\text{newcomer}}, \quad (15.8)$$

which is the position-dependent thinning of the ground process by the mark factor (Daley and Vere-Jones, 2003, Ch. 5).

Equation (15.8) is the honest version of a sector forecast: a predicted quarterly total *with its newcomer fraction attached*, rather than a tracked-only total silently presented as the market. It is also the Hawkes–Oakes immigrant/offspring split (Hawkes and Oakes, 1974) read at the mark level: the newcomer substream is the immigrant channel of the branching representation, arriving from outside the tracked system's genealogy. One caution tempers the interpretation: entry waves are where latent common factors live, so apparent “excitation” fitted on the newcomer substream must be read as clustering of unknown origin, in line with the frailty-inflation argument of Chapter 8 (fitted branching ratios are upper bounds). Fitting the two substreams separately and comparing their apparent excitation is a planned check for Chapter 19.

15.4.2 The histories rule, and the price of breaking it

The excitation term in Equation (15.7) raises the sharpest operational question in this chapter: *which deals enter the history sum?* Untracked deals are real events; they move the sector's state and excite future deal flow exactly as tracked deals do. The rule is therefore:

Design decision (D11.2 — histories include all deals).

Every history functional in every fitter — lagged sector counts in the weekly GLM, exponential-kernel sums in the event-time fitters, trailing volume covariates, cooldown clocks at sector level — is computed from the *complete* deal stream, untracked deals included. Only *firm-level attribution* (risk-set membership, firm histories N_i , firm covariates) is restricted to $\mathcal{I}$. Rationale: Proposition 15.9 below — dropping untracked deals from histories understates excitation by a factor of about the tracked share and distorts the decay estimate — and Proposition 15.3(i), whose immunity claim is conditional on exactly this rule. Reversal trigger: none; this is an accounting identity about what an event is, not a tunable choice.

The bias when the rule is broken can be derived exactly in the linear case. Model the loss of untracked deals as independent thinning: each event is retained (tracked) with probability $q = 1 - p$ independently of everything else — optimistic, since real coverage is not independent of firm type, but sufficient to sign and size the effect.

Proposition 15.9 (Thinning bias in the branching ratio). *Let N be a stationary linear Hawkes process with immigrant rate μ , kernel h with branching ratio $\kappa = \int_0^\infty h < 1$, and let N^q be its independent thinning with retention $q = 1 - p$ (Daley and Vere-Jones, 2003, Ch. 11). Then:*

- (i) (Rates) N^q has mean rate $\bar{\lambda}^q = q\mu/(1 - \kappa)$: thinning scales the mean by q exactly.
- (ii) (First generation) In the Hawkes–Oakes cluster representation (Hawkes and Oakes, 1974), the expected number of retained direct offspring of a retained event is $q\kappa = (1 - p)\kappa$. Any estimator that identifies branching by attributing observed events to observed parents — the declustering/EM estimand (Veen and Schoenberg, 2008) — therefore targets first-generation mass scaled by $(1 - p)$.
- (iii) (All generations) The expected number of retained descendants (all generations) of a retained event is $q\kappa/(1 - \kappa)$; matching the total-progeny identity $\kappa_{\text{app}}/(1 - \kappa_{\text{app}}) = q\kappa/(1 - \kappa)$ gives

$$\kappa_{\text{app}} = \frac{(1 - p)\kappa}{1 - p\kappa} \in [(1 - p)\kappa, \kappa), \quad (15.9)$$

again $(1 - p)\kappa$ to first order in κ . In both readings the apparent branching ratio is understated, and the missing excitation mass reappears as an inflated apparent baseline: $\mu_{\text{app}} = \bar{\lambda}^q(1 - \kappa_{\text{app}}) = q\mu/(1 - p\kappa) > q\mu$.

Proof. (i) The stationary mean of a subcritical linear Hawkes solves $\bar{\lambda} = \mu + \kappa\bar{\lambda}$, giving $\mu/(1 - \kappa)$ (sum the geometric cascade of generations, $\mu \sum_{k \geq 0} \kappa^k$). Independent thinning multiplies the first moment measure by q (Daley and Vere-Jones, 2003, Ch. 11). (ii) In the cluster representation each event, independently of everything else, produces direct offspring as an inhomogeneous Poisson process with mean κ . The retention marks are i.i.d. Bernoulli(q), independent of the genealogy and of each other; by Wald’s identity the expected number of retained direct offspring is $q\kappa$, and conditioning on the parent’s own retention changes nothing by independence of the marks. (iii) Generation- k descendants of a fixed event have expected count κ^k ; each is retained with probability q independently, so the expected retained progeny is $q \sum_{k \geq 1} \kappa^k = q\kappa/(1 - \kappa)$. Solving the displayed identity for κ_{app} gives Equation (15.9); monotonicity in q and the bounds follow by inspection ($q = 1$ recovers κ). The baseline claim is (i) combined with $\mu_{\text{app}} = \bar{\lambda}^q(1 - \kappa_{\text{app}})$ and $1 - \kappa_{\text{app}} = (1 - \kappa)/(1 - p\kappa)$. $\square$

Two remarks complete the picture. First, the *decay* is distorted alongside the scale: a retained grandchild whose intermediate parent was dropped presents to the fitter as a direct child at the *sum* of two kernel lags, so the apparent kernel is contaminated by convolution powers h^{*k} — heavier-tailed than h — and a fitted exponential decay θ is dragged toward slower values. Since Chapter 11 already shows θ is the fragile parameter (the κ - θ ridge), history thinning attacks the model exactly where it is weakest. Second, the repository’s actual failure mode is the *mixed* case — sector counts complete (Assumption 15.2) but the history term built from tracked deals only, as happens whenever a fitter joins deals to the companies table and drops non-matches. Then the excitation covariate is a q -scaled, noisy version of the true excitation state: an errors-in-variables regression that attenuates the excitation coefficient and reroutes the missing mass into the baseline and into any covariate correlated with activity — the same understatement, arrived at differently. If covariates dominate the likelihood, as we expect (Chapter 19 will test this on synthetic data), excitation has little likelihood mass to begin with; Proposition 15.9 says that breaking the histories rule removes a $(1 - p)$ -sized share of exactly that mass.

15.5 Unseen mass: the model-free audit

Everything so far trusts the model to measure the outside world through $\pi_{s,t}$. Good practice demands an estimate of the unseen mass that does not pass through the model at all, so that model and audit can disagree informatively. The classical tool fits our question exactly: *what is the probability that the next deal comes from a firm we have not seen before?*

Fix a sector and a window containing N deals, and regard the firm identities as draws $X_1, \dots, X_N$ from an unknown distribution (p_i) over the (countable, partly invisible) firm population. Let N_r be the number of firms appearing exactly r times in the window, and let the *missing mass* $M_0 = \sum_i p_i \mathbf{1}\{i \notin \{X_1, \dots, X_N\}\}$ be the probability mass of firms not seen at all. The Good–Turing estimator (Good, 1953) is

$$\widehat{M}_0 = \frac{N_1}{N}, \quad (15.10)$$

the fraction of deals going to firms seen exactly once.

Proposition 15.10 (Good–Turing is nearly unbiased). *If $X_1, \dots, X_N$ are i.i.d. (exchangeability suffices), then*

$$\mathbb{E}[\widehat{M}_0] = \mathbb{E}[M_0^{(N-1)}] \quad \text{and} \quad 0 \leq \mathbb{E}[\widehat{M}_0] - \mathbb{E}[M_0^{(N)}] = \sum_i p_i^2 (1-p_i)^{N-1} \leq \frac{1}{N} (1 - \frac{1}{N})^{N-1} \leq \frac{1}{2N},$$

where $M_0^{(n)}$ is the missing mass of a sample of size n : the estimator is unbiased for the missing mass of a sample one draw smaller, hence biased upward for $M_0^{(N)}$ by at most $O(1/N)$.

Proof. Leave one out. By exchangeability,

$$\mathbb{E}\left[\frac{N_1}{N}\right] = \frac{1}{N} \sum_{n=1}^N \mathbb{P}(X_n \text{ is a singleton in } X_{1:N}) = \mathbb{P}(X_N \notin \{X_1, \dots, X_{N-1}\}),$$

since X_N is a singleton in the full sample precisely when its firm is absent from the other $N - 1$ draws. Conditioning on the first $N - 1$ draws, that probability is $\mathbb{E}[\sum_i p_i \mathbf{1}\{i \notin X_{1:N-1}\}] = \mathbb{E}[M_0^{(N-1)}]$, which proves the identity; the estimator answers “will the *next* deal come from an unseen firm” almost by definition. For the bias, $\mathbb{E}[M_0^{(n)}] = \sum_i p_i (1 - p_i)^n$, so

$$\mathbb{E}[\widehat{M}_0] - \mathbb{E}[M_0^{(N)}] = \sum_i p_i [(1-p_i)^{N-1} - (1-p_i)^N] = \sum_i p_i^2 (1-p_i)^{N-1} \leq \max_{x \in [0,1]} x(1-x)^{N-1} \sum_i p_i,$$

and the maximum is attained at $x = 1/N$ with value $\frac{1}{N}(1 - \frac{1}{N})^{N-1} \leq \frac{1}{2N}$ for $N \geq 2$ (and $\rightarrow (eN)^{-1}$). When exchangeability fails the identity fails with it: this is the assumption to watch, not the algebra. $\square$

The proof makes the operative assumption explicit: exchangeability of deals *within the window*. On this data that is violated by everything this document models — excitation, macro dynamics, coverage drift — so the estimator is applied on *rolling windows per sector*, short enough that within-window dynamics are second-order, long enough that N_1 is not noise; the smoothing practice of [Gale and Sampson \(1995\)](#) applies when small counts make the raw frequencies-of-frequencies ragged. Two further caveats. The window convention must match the functional: N_1/N on a trailing window estimates the window-relative unseen mass, which is the right comparator for a model probability evaluated under the same convention — first-ever appearance versus first-in-window appearance are different events, and the audit must fix one and compute both sides accordingly. And GT estimates a *probability mass*, not a population count. For the count question — how many fundable firms are we not seeing? — the Chao lower bound ([Chao, 1984](#)) on the number of distinct firms S active in the window is

$$\hat{S}_{\text{Chao}} = S_{\text{obs}} + \frac{N_1^2}{2N_2} \leq S \quad (\text{in expectation, as a lower bound}), \quad (15.11)$$

valid as a floor precisely under the abundance heterogeneity that is certain to hold here (firms differ wildly in deal propensity), which is the honest direction: the invisible pool is *at least* this large.

Protocol 15.11 (Good–Turing versus the model). Standing diagnostic, run with every evaluation campaign and reported alongside Protocol 15.7: per sector and rolling window, (a) compute $\hat{M}_0 = N_1/N$ and $\hat{S}_{\text{Chao}}$ from the raw deal stream — no model involved; (b) compute the model’s average predicted outside-option mass over the same deals under the same window convention; (c) report the paired series and their gap. Agreement within the binomial noise band is a nontrivial consistency check (two estimators of one functional, one parametric and one not); a drifting gap with stable context covariates is escalated to the coverage audit of Protocol 15.12 before any refit.

Design decision (D11.3 — unseen-mass audit ships with the evaluation harness).

The GT/Chao audit of Protocol 15.11 is implemented once, in the shared evaluation layer (planned `hawkes_calibration/eval/unseen_mass.py`), and run by default in every campaign — not as an occasional notebook. Rationale: it is the only estimate of the outside world that does not share assumptions with the model it audits, and it is nearly free (counting singletons). Reversal trigger: none anticipated; if window choice proves contentious, the audit gains a sensitivity panel over window lengths rather than being dropped.

15.6 Modeling entry: cohorts on the Lexis diagram

Defect (C2) has appeared so far as a missing-mass problem. Its population face is different: the tracked universe is *selected on having raised*, so the data is silent about the reservoir of founded-but-not-yet-funded firms from which tomorrow’s newcomers arrive. An *entrant-profile module* models what can be modeled about that reservoir: the intensity of *first observed rounds*, organized by founding cohort.

The natural coordinate system is the Lexis diagram (Keiding, 1990): calendar time t on one axis, firm age $a = t - c$ (time since founding, cohort c) on the other; firms travel along unit-slope lines, and a first observed round at age a is a point on that line — observed only if it falls inside the data window (firms whose first raise postdates June 2024 are invisible entirely: right truncation of entry, the mirror image of (C1)). Because the number of not-yet-funded firms in a cohort is unobserved, only the *aggregate* first-round intensity is identified, and that is what the module fits:

$$\lambda_{s,c}^{\text{entry}}(t) = \exp(\alpha_s + \phi_{\text{era}(c)} + f_s(t - c) + \beta^\top x_{\text{pub}}^{\text{mac}}(t)), \quad \mu_s^{\text{ext}}(t) \supseteq \sum_{c \leq t} \lambda_{s,c}^{\text{entry}}(t), \quad (15.12)$$

with f_s an age profile (entry hazards peak in the first years after founding), ϕ era-of-founding effects (the dry-powder vintages), and macro as-ofs on the calendar axis. Founding dates come from the companies table — available, with (C4)-grade noise, for firms once they are tracked — so the cohort structure is estimable from each firm’s *first appearance*, the one moment at which entry-relevant features are observed.

What the module buys is exactly two things. First, *honest sector totals*: with Equation (15.12) feeding μ_s^{ext} , the decomposition Equation (15.8) becomes a forecast of incumbent *plus* newcomer intensity with the newcomer part driven by cohort supply rather than extrapolated share — the difference matters whenever era effects move (a 2021-vintage cohort bulge produces newcomer flow for years). Second, *cohort-based forecasts of newcomer supply*: questions like “how much first-round flow should sector s expect next year given the founding activity of 2022–2024” become answerable, with the era effects carrying the dry-powder cycle.

What it cannot buy must be stated with equal clarity: *firm-level prediction for unseen firms*. A firm with no covariates admits no firm-level score; no entry model changes that. The honest statement of the module’s output is a *sector-level newcomer intensity* together with a mark distribution over *observable-at-entry* features — stage, geography, sector, and the deal-time profile measured at first appearance — learned from tracked firms at their first rounds. That learned distribution is a *selection* distribution (features conditional on having entered), which supports predicting the profile of the next entrant; it says nothing about the latent population of firms that have not raised and may never (Warning 15.4 applies in full: the “entry” intensity Equation (15.12) is estimated from first *appearances* and inherits the coverage channel unless the audit of Section 15.7 clears it). One working hypothesis, to be tested in Chapter 19: newcomer utility is dominated by the alternative-specific constant and the context/pool-size terms rather than by entrant-profile quality. If that holds, effort belongs in $g_{s,t}$ and the cohort structure, not in rich profile models.

Design decision (D11.4 — a planned entrants/ module, scoped by identification).

The repository plans an `hawkes_calibration/entrants/` module implementing Equation (15.12): cohort–age–era first-round intensities per sector, an observable-at-entry mark distribution estimated at first appearances, and the coupling of both into μ_s^{ext} and the outside-option features $g_{s,t}$. The module’s API exposes sector-level newcomer intensity and entrant-profile distributions only — it deliberately has no per-firm scoring surface, so the identification limit is enforced by the interface rather than by discipline. Rationale: the honest-totals and cohort-supply gains above; the interface constraint is Warning 15.4 made structural. Reversal trigger: acquisition of auxiliary coverage data (a second vendor, or vendor backfill metadata) that identifies the entrant-versus-untracked split would justify promoting the module to a genuine entry model with compositional claims.

15.7 Two-sided universe drift

One final structural fact cuts across everything above: the universe boundary itself moves, from both sides. Firms cross into $\mathcal{I}$ as PitchBook’s coverage expands, and coverage has expanded materially over 2011–2024 — tracking improves, backfill happens, whole segments (regions, stages) arrive in waves. The untracked share is therefore time-varying *for reasons that are not market dynamics*: a declining `off_universe_share` is what improving coverage looks like, and it is also exactly what a genuine shift toward incumbent-dominated deal flow would look like. Every quantity in this chapter — $\pi_{s,t}$, the GT mass, the cohort intensities, the era effects of Equation (15.12) — confounds the two explanations unless coverage is audited first.

Protocol 15.12 (Coverage audit by era). The loader reports `off_universe_share` by sector and by era (fixed calendar blocks) as a first-class output, on synthetic and real data alike (Chapter 2 already requires the aggregate; this protocol requires the era breakdown). Any trend in the series is treated as a coverage artifact *candidate* until examined: check for step changes aligned with vendor events rather than market events, for divergence between the untracked share and the GT newcomer mass (Protocol 15.11) — which respond differently to backfill — and for era-localized failures of Protocol 15.7.

Design decision (D11.5 — coverage audit precedes entry claims).

No claim about entry dynamics — trends in newcomer share, cohort sizes, era effects in Equation (15.12) — is reported, internally or externally, until Protocol 15.12 has been run on the relevant eras and the trend has survived it. Rationale: the drifting-share tripwire is the one diagnostic that doubles as a data-quality monitor; its most likely cause on this vendor’s history is coverage, not the market, and Warning 15.4 already establishes that the data cannot arbitrate by itself. Reversal trigger: none; this is an ordering constraint on analysis, and it costs one audit per campaign.

15.8 Defect declarations and the ranking interface

Per Table 2.1 and decision D2.1, this chapter *owns* (C2) and (C6), and its models relate to the ledger as follows. (C6): the ground layer is robust under Assumption 15.2 and the histories rule (Proposition 15.3); the extended mark factor *corrects* the defect exactly at the collapsed-mark resolution (Proposition 15.5); fitters that break the histories rule are vulnerable with signed bias — excitation understated by a factor $\approx (1 - p)$, baseline inflated, decay dragged slow (Proposition 15.9). (C2): the outside option absorbs its deal-stream face; the entry module models its population face at sector level under the stated identification limits; compositional claims remain vulnerable permanently absent auxiliary coverage data (Warning 15.4), and the vulnerability is accepted and fenced by D11.4/D11.5 rather than assumed away. (C1): robust — all windows and spells here inherit the censoring treatment of Chapters 2 and 10; right truncation of entry is handled by construction in Equation (15.12). (C3): not corrected here, and the interaction matters: unlabeled dead firms *deflate* tracked-firm probabilities by padding the softmax denominator, while a missing outside option *inflates* them (Equation (15.2)); the two biases can partially cancel in aggregate metrics, and this document forbids reading such cancellation as health — the two defects are diagnosed separately (Chapter 14 for (C3), Protocol 15.7 here). (C4): sector mislabels perturb Assumption 15.2 marginally and motivate the group-level variant of Section 15.3.2; date jitter is immaterial to shares at weekly resolution.

(C5): the outside-option score Equation (15.4) uses only context covariates, which are never stale; the tracked rows inherit the treatment of Chapter 12 unchanged.

The interface with the ranking metrics of Chapter 14 must be fixed explicitly, because the outside option changes what recall can mean. Newcomer deals have no tracked target to recall: including them in the denominator of $\text{recall}@K$ mechanically caps the metric at $1 - \pi$ and makes scores incomparable across eras with different coverage. The coordinated protocol is therefore: $\text{recall}@K$ and MRR are computed on tracked-attributed deals only, exactly as Chapter 14 specifies; the newcomer mass is reported *alongside* as the calibrated $\pi_{s,t}$ output under Protocol 15.7; and every ranking report carries both numbers, so that a model cannot buy apparent recall by silently wishing the outside world away, nor be penalized for deals no tracked-universe model could have named.

Code hooks.

Mark factor. `hawkes_calibration/sector_ranker.py` gains the extended risk set Equation (15.3): one outside-option row per choice set carrying the context features Equation (15.4), mirroring `synthetic/loaders.py`'s `build_choice_sets(newcomer_context=True)`; prior art in `experiments/exp28_newcomer_designs.py` (N1 context score, N2 entrant moments, N3 two-step, N4 immigrant/offspring split). *Ground process.* The histories rule (D11.2) binds `hawkes_calibration/sector_stability.py`, the event-time fitters in `hawkes_calibration/eventtime/`, and the interval-censored tools in `hawkes_calibration/mbpp/`: history terms consume the complete deal stream; the external baseline component of Equation (15.7) enters as covariates until `hawkes_calibration/entrants/` exists. *Audits.* `hawkes_calibration/eval/unseen_mass.py` (to be written) implements Protocol 15.11 (Good-Turing mass, Chao floor, model comparison) and the reliability report of Protocol 15.7; `hawkes_calibration/data/audits.py` extends its universe-share check to the by-era breakdown of Protocol 15.12. *Entry.* The planned `hawkes_calibration/entrants/` module implements the cohort-age-era intensities Equation (15.12) and the observable-at-entry mark distribution, exposing sector-level outputs only (D11.4).

Part VII

Synthesis

Chapter 16

From Model Output to Investment Decisions

16.1 Closing the loop with Chapter 1

Chapter 1 defined worth by the value-of-information identity Equation (1.1): a model output matters only through the action it improves. The earlier chapters built the outputs: a sector-level ground intensity (Chapters 7, 8 and 10), a firm-level conditional ranker with an outside option (Chapters 14 and 15), and horizon event probabilities (Chapter 13), all under the evaluation rules of Chapter 11 and the covariate honesty of Chapter 12. This chapter specifies the *decision layer*: the small set of composed deliverables, the decision rules that consume them, and the feedback loop that measures whether the system earns its keep. The layer is deliberately thin. It contains no new statistical estimation — only composition, thresholds, and reporting. Every quantity it consumes arrives already calibrated and already uncertainty-qualified; anything thicker would smuggle modeling into a place with no likelihood to check it.

16.2 The three deliverables

16.2.1 The sourcing list (T1 + T3)

The headline deliverable is a per-sector ranked list, refreshed weekly. For each sector s at time τ , it lists the top- K tracked firms by conditional next-deal probability $\hat{p}(i \mid s, \tau)$ (Chapter 14). Each firm is annotated with its horizon probabilities $\hat{p}_i(\tau, \Delta)$ for $\Delta \in \{6M, 1Y, 2Y\}$ (Chapter 13), its staleness age $a_i(\tau)$ and last-known covariates (honesty about the information state, Chapter 12), and the sector's calibrated newcomer share $\hat{p}(\diamond_s)$ (Chapter 15). The two probability columns answer different questions and must not be merged: the ranker column is *relative* (who, given a deal); the horizon column is *absolute* (whether, within Δ). A firm can top the ranker while all its horizon probabilities are low, because the sector itself is quiet. That pairing tells the desk to watch rather than act.

Proposition 16.1 (Optimality of the top- K sourcing rule). *Suppose diligence capacity allows opening at most K files in sector s , each opened file on firm i yields expected utility $b \cdot \mathbf{1}\{i \text{ raises within } \Delta\}$ minus cost c , with b, c constant across firms, and outcomes are as modeled. Then the expected-utility-maximizing selection is the top- K firms by $\hat{p}_i(\tau, \Delta)$, and opening the k -th file is worthwhile iff $\hat{p}_{(k)}(\tau, \Delta) \geq c/b$.*

Proof. Expected utility of a selection A , $|A| \leq K$, is $\sum_{i \in A} (b \hat{p}_i(\tau, \Delta) - c)$, additive in members; it is maximized by taking the firms with the largest positive terms, which are the top-ranked

ones by $\hat{p}_i$, truncated at K or at the threshold $\hat{p}_i \geq c/b$, whichever binds first. The threshold form is the classical cost–benefit decision boundary for calibrated probabilities; the argument fails exactly when $\hat{p}$ is uncalibrated, which is why Chapter 13 makes calibration mandatory, and when utilities interact across firms (portfolio effects), which is out of scope by Section 16.3. $\square$

The ranking-only analogue — expected recall of the next deals is maximized by ranking on conditional probability — was proved as the top- K optimality lemma of Chapter 14. Together the two results say: *under the model, the list is the decision*. Everything else is threshold arithmetic supplied by the desk’s own c/b .

16.2.2 Sector pacing (T2)

For capital-deployment pacing, the deliverable is the predictive distribution of sector deal counts over the next 4–13 weeks: mean paths with predictive intervals. We produce them by simulating the fitted ground process forward (Chapters 8 and 10 simulators; scenario or bootstrapped macro paths per Chapter 2), with dispersion honestly propagated (negative-binomial noise where adopted, Chapter 7). Every pacing chart carries two fixed footnotes: the fitted branching mass is an upper bound on contagion (Chapter 8), and timing resolution is bounded below by the honest resolution of Chapter 10 — no intra-week timing claims.

16.2.3 Deal-terms context (T4)

Mark regressions — round size, up/down/flat probability, valuation step conditional on a deal — put a candidate in context. They use the same gradient-boosting stack as Chapter 13, applied to deal-level targets from Appendix B, with staleness features and with the undisclosed-size selection handled per the data dictionary’s imputation rules (`deal_size_status` is itself informative and enters as a feature, never silently imputed away). Terms models are context, not gates: they refine the utility b in Proposition 16.1, firm by firm, when the desk wants probability-weighted expected terms rather than a flat b .

16.3 Scope: what the decision layer refuses to do

Design decision (D12.1 — no silent portfolio optimization).

The decision layer outputs ranked lists, calibrated probabilities, and predictive count distributions. It does *not* output buy/pass verdicts, position sizes, or fund-level allocations. Rationale: portfolio-level utility involves correlations, capacity, and mandate constraints that live outside the data this document models; any such optimization must be a separate, owned system consuming our calibrated outputs. Reversal trigger: a stated fund-level objective with its own evaluation protocol.

Design decision (D12.2 — probabilities travel with their caveats).

Every number leaving the system carries: the model version, the evaluation-era metrics of Chapter 11, the risk-set definition (Chapter 14), the staleness summary of the inputs (Chapter 12), and the newcomer share (Chapter 15). The reporting discipline of Chapter 11 — no hazard without its risk-set rule, no branching ratio without its upper-bound caveat, no newcomer probability sold as an entrant probability — binds the decision layer most of all, because it is the layer other people quote. Reversal trigger: none; this is charter.

16.4 Measuring the value of information

Offline metrics are proxies. The loop closes only when we measure Equation (1.1) on the workflow itself.

Protocol 16.2 (VoI measurement). (a) *Shadow deployment*: for at least one full quarter, sourcing lists are produced and frozen weekly but not shown to the desk; realized deals score them retrospectively (recall@ K , lift over the desk’s contemporaneous shortlists). (b) *Assisted period*: lists shown; the desk logs which opened files originated from the list vs. other channels. (c) The VoI estimate is the difference in hit rates per opened file between assisted and baseline periods, era-blocked as in Chapter 11; secular market shifts are partially controlled by the shadow-period comparison. (d) Attribution is reported with its confounds stated — this is an observational estimate, not a randomized one; if the desk’s capacity permits, randomize *which sectors* get assisted lists per quarter for a cleaner contrast.

16.5 Monitoring, drift, and retraining cadence

Models decay. Coverage drifts (Chapter 15), macro regimes shift, and the 2021–2023 funding cycle sits inside our evaluation eras as a standing warning. The cadence is quarterly re-evaluation against the rolling-origin protocol of Chapter 11, with three planned tripwires, each mapped to its owning chapter: (i) calibration drift on horizon probabilities beyond the tolerance band of the reliability protocol (Chapter 13) triggers recalibration before retraining; (ii) a trend in `off_universe_share` or in the Good–Turing missing mass (Chapter 15) triggers the coverage audit before any modeling response; (iii) attenuation-gap growth in covariate effects (Chapter 12’s gap test) escalates the staleness remedy ladder. Retraining without a tripped wire is discouraged: it churns the desk’s mental model of the lists for no measured gain.

Code hooks.

The decision layer is a reporting module, not a modeling module: `hawkes_calibration/decision/` (to be written) composes existing outputs — `ranker` scores from `hawkes_calibration/sector_ranker.py` (extended with the outside option), horizon probabilities from `hawkes_calibration/models/xgb_horizon.py` (planned, Chapter 13), simulated count fans from the ground-process simulators, and terms context — into versioned weekly artifacts (per-sector JSON + rendered sheets). The VoI protocol needs a tiny `hawkes_calibration/decision/log.py` for desk-interaction logging. Tripwire metrics reuse the `hawkes_calibration/eval/` harness of Chapter 11; nothing in this layer may compute a probability itself.

Chapter 17

Implementation Roadmap

17.1 Reading the roadmap

This chapter converts the document’s decisions into ordered engineering work on `hawkes_calibration/`. Each phase lists its modules (existing, extended, or new), the chapter that owns the mathematics, and the *gate* that must pass before the phase is called done. Gates come from the evaluation constitution of Chapter 11: every new fitter passes a parameter-recovery rehearsal on synthetic data generated from its own family (and a misspecified stress world where one exists), every likelihood passes analytic-vs-numeric gradient checks at the tolerances fixed in Chapter 18, and every data transformation passes its audit. Phases are ordered by the dependency structure of the mathematics, not by expected impact. The expected impact ordering — covariates first, risk-set truthfulness above all, excitation last — is a working hypothesis; the protocols of Chapter 18 test it, and Chapters 19 and 20 record the outcomes.

17.2 Phase 0 — the real-data spine

Owens: Chapters 2, 10 and 15. Everything else blocks on this.

P0.1 `hawkes_calibration/data/pitchbook.py`: the adapter from raw PitchBook deals/companies/investors extracts (2011–June 2024) to the fitter-ready arrays of `synthetic/loaders.py` — weekly sector counts, as-of macro matrix, LOCF firm features *with staleness ages* (Chapter 12, D8.1), at-risk masks, event list with universe flags. The synthetic loader’s array contract is frozen as the interface; Appendix B is the column contract.

P0.2 `hawkes_calibration/data/audits.py`: the defect-ledger validation suite of Chapter 2 — open-spell retention, entry-date consistency, as-of join look-ahead test, duplicate-collapse idempotence, coverage-share report by era (Chapter 15), date-quality audit separating announced/closed/backfilled dates (Chapter 10).

P0.3 Honest-resolution decision executed: the date audit fixes the modeling resolution (expected: weekly) per D6.1.

Gate: audits green on the real extract; a data note recording coverage share, date-quality composition, and staleness-age distributions by sector — the empirical face of Table 2.1.

17.3 Phase 1 — baselines that must be beaten

Owens: Chapters 7, 12 and 13 (R1), Chapter 14 (linear).

- P1.1 Sector ground baseline: seasonal + macro NHPP via the binned GLM equivalence (Chapter 7), then the negative-binomial noise upgrade (D4.2); held-out NLL ladder vs. random and historical-mean baselines established as the standing leaderboard (Chapter 11).
- P1.2 `hawkes_calibration/panel/`: the bucketed panel builder of Chapter 13 — disjoint buckets per horizon $\Delta \in \{26, 52, 104\}$ weeks, censoring-aware labels, point-in-time features including staleness ages and interactions (R1), leakage audits.
- P1.3 `hawkes_calibration/models/xgb_horizon.py`: per-horizon XGBoost classifiers with temporal early stopping and mandatory post-hoc calibration (D9.5); reliability reports into `hawkes_calibration/eval/`.
- P1.4 Ranker on real data: the existing linear risk-set ranker (`hawkes_calibration/sector_ranker.py`) with staleness features, evaluated by the $\text{recall@}K$ protocol of Chapter 14 — this is the first real sourcing-list prototype.

Gate: the covariates-vs-excitation decomposition (protocol B2, Chapter 18) run on real data — a nats-ladder note comparing seasonal, macro, and history-augmented sector models, recorded in Chapter 20; horizon models calibrated within tolerance; $\text{recall@}K$ baseline numbers frozen for all later comparisons.

17.4 Phase 2 — the committed architecture

Owens: Chapters 8, 10, 14 and 15.

- P2.1 Sector-level continuous-time Hawkes at honest resolution via the MBPP likelihood (`hawkes_calibration/mbpp/` already implements the machinery; wire it to the real arrays); excitation reported with the upper-bound protocol; the κ - θ ridge diagnostics of Chapter 11 standing.
- P2.2 Grouped-softmax XGBoost objective (Chapter 14's gradient/Hessian derivation) so the ranker's score function gains interactions while keeping the conditional-likelihood family — the planned response if benchmark B1 (Chapter 18) shows a material win for the boosted variant.
- P2.3 Outside option $\diamond_s$ in the risk set (Chapter 15, D11.x): newcomer share becomes a calibrated output; $\text{recall@}K$ denominators split tracked vs. newcomer per protocol; Good-Turing and coverage audits wired into `hawkes_calibration/eval/`.
- P2.4 `hawkes_calibration/eval/`: the consolidated harness of Chapter 11 — NLL ladders, PIT/dispersion, time-rescaling, rolling-origin driver, $\text{recall@}K$ /MRR with era-blocked uncertainty — replacing per-experiment ad hoc scripts.
- P2.5 `hawkes_calibration/decision/`: the composition layer of Chapter 16 (weekly artifacts, caveat propagation, VoI logging).

Gate: end-to-end weekly run on real data producing the three deliverables of Chapter 16; a full evaluation campaign note, recorded in Chapter 20, with every metric attached to its protocol.

17.5 Phase 3 — escalations, each behind its trigger

Owens: Chapter 12 (R2–R4), Chapter 14 (PL), Chapter 15 (entrants), Chapter 8 (signed kernels, frailty).

- P3.1 `hawkes_calibration/covstate/`: the supervised drift dataset from consecutive-deal pairs and the imputation API (R2), then the OU/Kalman state layer supplying $\Sigma(a)$ (R3); joint modeling (R4) strictly behind the attenuation-gap trigger of Chapter 12.
- P3.2 Plackett–Luce likelihood for multi-deal sector-weeks (D10.x); LambdaMART challenger benchmarked, not adopted, unless it beats the likelihood family on the consumed metric two campaigns running.
- P3.3 `hawkes_calibration/entrants/`: cohort entry intensities on the Lexis diagram (Chapter 15), unlocking honest sector-total decompositions.
- P3.4 Signed continuous-time kernels (route (b) of Chapter 8) and latent-frailty models: only if a consuming question requires a structural continuous-time inhibition or contagion claim that routes (a)/(c) cannot express — each with its own recovery rehearsal first.

Gate: per item, the trigger that justified it, measured before and after.

17.6 Decision registry

The numbered decisions ($Dx.y$ boxes) of Chapters 1, 2, 7, 8 and 10 to 16 form the registry against which pull requests are reviewed: a change that contradicts a decision box must either be rejected or arrive together with a documented reversal — trigger met, evidence attached, box amended in this document. The document itself is versioned with the code; Chapters 19 and 20 append new campaign verdicts rather than overwriting old ones, so the evidentiary trail stays auditable.

Code hooks.

New top-level modules by phase: `hawkes_calibration/data/` (P0), `hawkes_calibration/panel/` and `hawkes_calibration/models/xgb_horizon.py` (P1), `hawkes_calibration/eval/` and `hawkes_calibration/decision/` and the grouped-softmax objective (P2), `hawkes_calibration/covstate/` and `hawkes_calibration/entrants/` (P3). Existing modules extended in place: `hawkes_calibration/sector_ranker.py` (staleness features, outside option, PL), `hawkes_calibration/sector_stability.py` (NB noise), `hawkes_calibration/mbpp/` (real-data wiring). The synthetic world (`synthetic/`) remains the rehearsal stage for every gate.

Chapter 18

The Code and Test Plan

Chapter 17 says *what* gets built and in which phase. This chapter says *how* each piece is tested before it is trusted. It is the engineering contract behind the roadmap’s gates. Section 18.1 fixes the four test layers and what a green result in each layer actually certifies. Section 18.2 states the test plan as frozen pass/fail protocols, each mapped to the roadmap phase that owns it. Section 18.3 summarizes the uncertainty-quantification methods and the production checklist. Section 18.4 closes with the continuous-integration gates and the definition of done, tied to the decision registry of Section 17.6. No results appear in this chapter: every protocol below is planned, not run. Synthetic-data verdicts will be recorded in Chapter 19; real-data verdicts will be recorded in Chapter 20. Evaluation-metric definitions are not restated below: the evaluation constitution of Section 11.1 is the sole authority on what “calibrated” and “better” mean.

18.1 Testing philosophy and the four-layer harness

A calibration code base fails in two distinct ways. Either the code computes the wrong number, or the code computes the right number for a procedure that cannot recover the truth. Conventional software tests catch only the first failure. We therefore run four layers of tests. Each layer certifies something the previous layer cannot, and the roadmap’s phase gates (Section 17.1) are defined in terms of them.

Layer 1: unit tests on hand-computable micro-examples. Every nontrivial data transformation has a companion example small enough to verify by hand. Three anchor cases: (i) a three-firm choice set with known funding histories, where the pools, the excluded most-recently-funded firm, the newcomer row, and the winner labels produced by `build_choice_sets` are asserted against a worked-out table; (ii) a two-bucket compensator for an exponential-kernel intensity, checked against the closed-form integral to machine precision; (iii) deduplication on a constructed deal table containing exact duplicates, near-duplicates a few days apart, and legitimate consecutive-week repeats, asserting which rows survive. These tests certify arithmetic and indexing, nothing more. But when a recovery test fails, they let us clear the plumbing in minutes.

Layer 2: equivalence tests between fast and reference fitters. The repository ships slow, transparent loop fitters and vectorized fast fitters (`fit_sector_glm_fast`, `fit_choice_fast`). The fast versions exploit structure: the sector GLM is separable per receiving sector, so twelve small concave problems replace one large one; the ranker is a padded-tensor batch over events and candidates. Speed is worthless if the optimum moves. On subsamples, both implementations must agree at the optimum — same loss to 10^{-6} ,

parameters to matching tolerance — before a fast fitter may be used in any experiment, and the check is rerun whenever either implementation changes. Every new likelihood must also pass analytic-vs-numeric gradient checks at the same fixed tolerances.

Layer 3: statistical recovery tests. These are the scientific tests. We fit on synthetic data whose ground truth we wrote down, and a run passes when the truth lies within 2 standard errors of the estimate across 5 seeds. A recovery failure with passing layers 1–2 is a finding about the *method* — identifiability, confounding, misspecification. Findings of this kind will be recorded in Chapter 19 and feed the decision registry. Recovery tests are the only layer that can certify the phrase “the pipeline works,” which is why every roadmap gate requires a recovery rehearsal before a new fitter touches real data.

Layer 4: invariant and leakage audits. Two standing assertions run over the whole feature stack. The point-in-time assertion recomputes every feature using only data truncated at time t and requires the result to be bit-identical to the production value: any divergence means some path read the future. The oracle-isolation assertion requires that no fitting path opens `ground_truth.npz` or `private_deal_map.csv`; oracle files are for evaluation only, and a fit that touches them is void regardless of its metrics. Both audits are structural: they do not check that a number is right, they check that it was legal to compute. On real data this layer becomes `hawkes_calibration/data/audits.py` of Phase 0 (Section 17.2), with the oracle role played by vendor-side fields that postdate the prediction time.

Layer	What it certifies	Cadence
Unit (micro-examples)	arithmetic and indexing of each transformation	every commit
Equivalence (fast vs. loop)	the vectorized optimum is the reference optimum	every commit
Recovery (synthetic truth)	the procedure can find truth within 2 SE, 5 seeds	nightly, small configs
Invariant / leakage	features are point-in-time; oracle files untouched	blocking, every commit

Table 18.1: The four test layers and what a green result actually means.

Three reproducibility rules bind everything above. First, every reported number must be regenerable from one seeded entry point — for the rehearsal worlds themselves,

```
PYTHONPATH=. python -m synthetic.generate --regime A --seed 0
```

and analogously one script per experiment table with a `-seed` argument. Second, temporal splits only, everywhere: train weeks [0, 416) against test weeks [416, 520) on the weekly track, an 80% day quantile on the choice track, and no random split anywhere in the stack, including model selection — this is the same temporal discipline the evaluation constitution imposes (Section 11.6). Third, runtimes are logged alongside results, because a fitter that silently becomes ten times slower is a regression even when its numbers do not move.

Code hooks.

Layer 1–2 suites live under `tests/`. Recovery campaigns are the seeded entry points under `experiments/`, whose verdicts will be transcribed into Chapter 19; the rehearsal stage itself is `synthetic/` (Section 17.6). On real data, layer 4 becomes

`hawkes_calibration/data/audits.py` (P0.2) and the consolidated metric harness is `hawkes_calibration/eval/` (P2.4); both replace per-experiment ad hoc scripts.

18.2 The forward test plan

What follows is the committed test plan. Each protocol states its purpose, its procedure in a few lines, a pass criterion frozen *before* the run so that results can fail, and the roadmap phase that owns its real-data form. Every protocol has status *planned*; this chapter reports no outcomes. Synthetic runs write their verdicts into Chapter 19; real-data reruns write theirs into Chapter 20. Unless stated otherwise, every experiment runs on both rehearsal regimes at 5 seeds with error bars, under the splits and audit rules of Section 18.1, scored only by metrics the constitution admits (Section 11.1). Table 18.2 summarizes.

Protocol 18.1 (E1: bucket width, decay identifiability, and latent confounding). *Purpose:* quantify what fine-grained timestamps buy over coarse buckets, and how badly latent confounding contaminates fitted excitation. *Procedure:* fit the weekly GLM and a multivariate event-time sector Hawkes (exponential kernel, log-linear macro baseline) on the same streams; rerun the event-time fit at 1d, 7d, and 28d bucket widths, reporting bias and SE of the branching scale κ and decay θ at each width; separately, regenerate regime A with latent-factor strength in $\{0, 0.15, 0.3\}$ at fixed true radius and plot fitted effective radius against latent strength, with and without mitigations (lagged aggregate count covariate, off-diagonal penalties). *Pass:* θ recovered within 2 SE at day resolution on regime B while the κ - θ ridge visibly reopens by 28d; the confounding curve is monotone and yields a stated bucket-width recommendation. *Status:* planned; results to Chapter 19. *Phase:* the real-data descendant is the honest-resolution decision of Phase 0 (Section 17.2, D6.1) and the ridge diagnostics standing on the Phase 2 Hawkes fit (Section 17.4).

Protocol 18.2 (E2: choice-model sub-experiments). *Purpose:* settle the four open design choices inside the conditional logit before the ranker is frozen for production. *Procedure and pass criteria, per sub-experiment:*

- E2.1 *Gap functional form.* Fit log-gap versus binned gap dummies ($<1m$, $1-3m$, $3-6m$, $6-12m$, $>12m$) versus exponential decay $\exp(-\text{days}/\tau)$ with $\tau \in \{30, 90, 180\}$ days; select by held-out NLL and plot each implied hazard-versus-gap curve against truth. *Pass:* the binned fit recovers the true 26-week step in regime A and the smooth exponential decay in regime B.
- E2.2 *Newcomer designs N1-N4.* Compare the bare ASC against the context-covariate ASC (sector heat and macro features), the representative-entrant construction (plug-in and integrated Gaussian variants), the nested two-step, and the immigrant/offspring split that aligns first financings with the Hawkes background stream. *Pass:* the recommended design calibrates quarterly newcomer share within 2 percentage points, with incumbent weights materially unchanged. The design space is owned by the outside-option construction of Chapter 15.
- E2.3 *Sampled-softmax stability.* Refit at maximum candidate counts $\{32, 64, 128\}$ and with the full denominator on a 10% subsample. *Pass:* weights stable across candidate counts, consistent with the subsampled-denominator consistency result of Appendix C.

E2.4 Exclusion-rule sensitivity. Vary the exclude-most-recent rule (sector-scoped, global, none); count dropped repeats and measure weight shifts. *Pass:* the effect is pinned with a number, and the rule is stated in the production spec either way.

Status: planned, all four; results to Chapter 19. *Phase:* E2.1, E2.3, and E2.4 belong to the Phase 1 linear ranker (Section 17.3); E2.2 rides the Phase 2 outside-option work (Section 17.4).

Protocol 18.3 (E3: two-stage factorization versus single-layer block Hawkes). *Purpose:* the central architecture question — given exact dates and truthful risk sets, what does the two-stage factorization still buy, and what does it cost in joint-intensity coherence? *Procedure:* fit the full firm-level event-time block Hawkes (log-link baseline, self-inhibition, size-normalized sector field) on day-resolution timestamps; evaluate it and the two-stage pipeline on identical held-out events for joint NLL, next-deal sector prediction, and funded-firm ranking conditional on the event. *Pass:* on regime B the single-layer fit recovers the self-inhibition ordering with rank correlation at least 0.9; the head-to-head produces a written recommendation with numbers on all three metrics, whichever way it falls. *Status:* planned; results to Chapter 19. *Phase:* behind the Phase 3 trigger for signed continuous-time kernels (Section 17.5) — it runs only if a consuming question demands a structural continuous-time claim that routes (a)/(c) cannot express, and its recovery rehearsal comes first.

Protocol 18.4 (E4: neural-operator placement). *Purpose:* establish whether and where the neural-operator surrogate for the compensator equations pays for itself, given that exact likelihoods are the committed default (Chapters 10 and 11). *Procedure:* report forward-solver accuracy versus sector count M and the wall-clock crossover against the exact solvers under repeated forward solves; benchmark differentiable-solver inverse calibration against the MLE for bias, variance, and wall-clock on identical data. *Pass:* the forward solver reaches a sub-2% relative L^2 band; the crossover point (in solves and in M) is documented; the inverse-calibration table states plainly whether the MLE wins at this scale. *Status:* planned; results to Chapter 19. *Phase:* not on the critical path; revisited under Phase 3 rules only if a repeated-forward-solve workload (scenario simulation in Chapter 16) materializes.

Protocol 18.5 (E5: walk-forward backtest with covariate ablation). *Purpose:* the end-to-end rehearsal of the production loop under strictly forward information. *Procedure:* freeze the best when/where and which-firm models; walk forward weekly over the final 20% of days, emitting sector intensities, ranked candidate lists, and newcomer probabilities; score top-1/5/10 and MRR against oracle and random, PIT count calibration with the quasi-Poisson dispersion correction, and lead time (weeks a funded firm spends in the top decile before its deal); rerun with macro covariates removed and on the covariate-scale 1.5 world. *Pass:* the pipeline beats random throughout and degrades gracefully toward oracle; the covariate ablation turns “covariates help” into a signed number in both directions. *Status:* planned; results to Chapter 19. *Phase:* its real-data form is the Phase 2 gate itself — the end-to-end weekly run producing the deliverables of Chapter 16 (Section 17.4), driven by the rolling-origin machinery of `hawkes_calibration/eval/`.

Protocol 18.6 (B1: boosted-ranker benchmark). *Purpose:* a model-free discriminative ceiling for the identity layer. *Procedure:* train XGBoost rankers (Chen and Guestrin, 2016) on the identical choice sets and splits in two variants — feature parity (exactly the six logit features, isolating functional form) and kitchen sink (all point-in-time-legal raw fields, snapshot columns explicitly forbidden); score with per-event renormalization; modest hyperparameter search with a time-ordered validation tail. *Pass (frozen decision rule):* a material win for the boosted variant means the logit is missing features or nonlinearity, and the pre-committed response is the grouped-softmax boosted ranker of Chapter 14, which keeps the conditional-likelihood family while granting the score function interactions; a narrow result keeps the interpretable

logit. *Status*: planned; results to Chapter 19. *Phase*: runs against the Phase 1 linear baseline (Section 17.3); the grouped-softmax response, if triggered, ships in Phase 2 (Section 17.4). Both variants are rerun on real data, with verdicts in Chapter 20, before any response is declared necessary there.

Protocol 18.7 (B2: non-homogeneous Poisson null and improvement decomposition). *Purpose*: show, not assume, that self-excitation earns its complexity at this data scale. *Procedure*: fit the nested ladder — covariates-only inhomogeneous Poisson (simulable by thinning (Lewis and Shedler, 1979)), covariates plus excitation, full event-time timing — plus a seasonal-naive baseline; decompose held-out improvement across the rungs with residual diagnostics per rung (Ogata, 1988), on the NLL ladder of Section 11.4.1. *Pass*: the decomposition exists with error bars over 5 seeds; if the excitation rung contributes approximately nothing, that is written into the registry, not smoothed over. *Status*: planned; results to Chapter 19. *Phase*: the real-data rerun is the Phase 1 gate — the nats-ladder note on the 2011–2024 extract (Section 17.3), recorded in Chapter 20.

Protocol 18.8 (B3: nonparametric intensity challenger). *Purpose*: test whether the non-parametric intensity models of Chapter 9 — boosted trees, forests, and small networks on the discretized likelihood — beat the parametric families they challenge. *Procedure*: fit each Chapter 9 model in the same harness as every other fitter: same temporal splits, same audits, same 5 seeds, same feature-legality rules; compare against the log-linear GLM of Chapter 7 and the parametric Hawkes of Chapter 8 on held-out likelihood (the NLL ladder of Section 11.4.1) and on calibration (the PIT and dispersion checks of Section 11.3). *Pass (frozen decision rule)*: the comparison table exists with error bars; a nonparametric model is adopted only if it wins on held-out likelihood *and* calibrates at least as well as the parametric incumbent; a likelihood win with worse calibration is a finding, not an adoption. *Status*: planned; results to Chapter 19, real-data rerun to Chapter 20. *Phase*: rides the Phase 1 baseline ladder (Section 17.3) and is rerun against the Phase 2 architecture (Section 17.4).

Protocol 18.9 (Causal-pipeline validation on known ground truth). *Purpose*: test the causal stack of Chapter 4 against a world where the estimand is known. *Procedure*: generate a treatment with known selection on observables and on the latent quality mark; estimate naive, backdoor-adjusted, and AIPW effects; attach cluster-bootstrap confidence intervals and E-values (Hernán and Robins, 2020). *Pass, asserted in code so a failure is a red test*: the naive estimate is biased in the selection direction; adjustment removes the observable component; AIPW matches backdoor with better robustness to nuisance misspecification; and the residual gap to truth is attributable to latent quality with magnitude consistent with the computed E-value. *Status*: planned; results to Chapter 19. *Phase*: a blocking rehearsal gate before any estimator of Chapter 4 runs on real data, and a precondition for the Phase 3 frailty escalations (Section 17.5).

18.3 Uncertainty quantification and the production checklist

Every interval this document reports comes from one of five techniques, and each carries a different guarantee. Table 18.3 is deliberately blunt about which.

Only the conformal row carries a finite-sample statement, and even that one is marginal (not conditional) and degrades when temporal splits break exchangeability. Every other interval is asymptotic and specification-dependent. One consequence is worth spelling out: if the held-out Pearson dispersion of a count fit is ϕ , naive Poisson standard errors are understated by a factor of about $\sqrt{\phi}$ until corrected — which is why the dispersion report is a checklist item, not a courtesy. The production checklist below is the run-time counterpart of the Phase

Test	Pass criterion (frozen)	Status	Phase
E1 bucketing	θ within 2 SE at day resolution on B; ridge visible by 28d	planned	P0/P2
E1 confounding curve	monotone fitted-radius curve; mitigations quantified	planned	P2
E2.1 gap form	binned fit recovers the true step (A), smooth decay (B)	planned	P1
E2.2 newcomer N1–N4	newcomer share within 2pp; incumbent weights stable	planned	P2
E2.3 sampled softmax	weights stable across 32/64/128/full-subsample	planned	P1
E2.4 exclusion rule	sensitivity pinned with a number	planned	P1
E3 head-to-head	rank-corr ≥ 0.9 on B; three-metric comparison written	planned	P3 (trigger)
E4 operator	sub-2% forward band; crossover and inverse tables	planned	P3 (trigger)
E5 backtest	beats random throughout; ablation deltas signed	planned	P2 gate
B1 boosted ranker	frozen decision rule applied	planned	P1→P2
B2 Poisson null	improvement decomposition with error bars	planned	P1 gate
B3 nonparametric challenger	likelihood win counts only with equal-or-better calibration	planned	P1→P2
Causal validation	all four coded assertions pass against truth	planned	pre-Chapter 4

Table 18.2: Forward test plan at a glance. All rows: both regimes where applicable, 5 seeds, temporal splits, audits of Section 18.1 in force. Synthetic verdicts go to Chapter 19; real-data verdicts go to Chapter 20; phases per Chapter 17.

Method	Technique	Guarantee	Key assumptions
Count GLM (Chapter 7)	quasi-Poisson sandwich	asymptotic SE validity for mean parameters	correct mean function; dispersion as nuisance
Two-stage / choice (Chapter 14)	cluster bootstrap by week	asymptotic CI coverage	weak dependence across clusters; clustering at the right level
Boosted prediction (Chapter 13)	split conformal / CQR (Romano et al., 2019)	finite-sample marginal coverage	exchangeability; weakened under temporal splits
Hawkes MLE (Chapter 8)	Laplace / posterior interval	local asymptotic normality	correct specification; stationarity
MBPP (Chapter 10)	parametric bootstrap	coverage under the fitted DGP	model correctness; report κ , not θ

Table 18.3: UQ summary: what each interval means and what it silently assumes.

0 audit suite (Section 17.2): the audits verify the data once, the checklist binds every fit and report thereafter.

- C1. *Point-in-time audit.* Reconstruct every feature from data knowable at the prediction date; join macro series on publish date, never reference date (Chapter 2).
- C2. *Deduplication.* Apply the validated dedup window before any counting; duplicates and date jitter alias into spurious short-lag excitation (Chapters 2 and 10).
- C3. *Risk-set truthfulness audit.* Validate the membership mask $\mathcal{R}_t$ against a labelled subsample of known entrants and exits; a contaminated mask can invert fitted weights (Chapter 14).
- C4. *Effective-radius guard.* Refuse to simulate any fitted model at effective radius ≥ 0.95 ; for log-link raw-count feedback refuse well below that, or use bounded transforms (Chapter 8).
- C5. *Dispersion correction.* Report held-out Pearson dispersion and scale standard errors accordingly; dispersion far above 1 is a specification alarm, not a nuisance (Section 11.3).
- C6. *Universe coverage and newcomer calibration.* Track the untracked-firm share and recalibrate the newcomer ASC whenever the pool definition changes; the ASC is meaningless detached from its pool (Chapter 15 and Appendix C).
- C7. *Temporal splits only.* No random splits anywhere — not for model selection, not for conformal calibration, not for the benchmarks of Section 18.2 (Section 11.6).
- C8. *Seeds and regeneration.* Every reported number carries a seed and a regeneration command; a table without an entry point under `experiments/` does not exist.

18.4 Continuous integration and the definition of done

The four layers of Section 18.1 map onto three CI gates. On every commit, the unit and equivalence suites run and must pass; they are fast because the micro-examples are tiny and the equivalence checks run on subsamples. Nightly, the recovery suite runs on reduced configurations — fewer weeks, fewer seeds — as an early-warning system; a nightly recovery regression blocks the next release, not the next commit. The leakage audit is a blocking check at both cadences: a commit that makes any feature non-point-in-time, or that lets a fitting path touch `ground_truth.npz` or `private_deal_map.csv`, does not merge, whatever its metrics say. The registry review of Section 17.6 sits on top of all three: a pull request that contradicts a decision box is rejected or arrives with a documented reversal, and CI cannot green its way around that rule.

The artifact policy is equally plain. Results tables are versioned CSVs, written by the experiment entry points and diffed like code, so a silently shifted number is a visible change in review. Figures are regenerated from those CSVs by scripts and are never hand-edited; a figure that cannot be regenerated is deleted rather than trusted. Campaign verdicts are transcribed append-only into Chapter 19 for synthetic runs and into Chapter 20 for real-data runs, so the evidentiary trail behind every decision box stays auditable.

Design decision (D-CP.1 — the harness is constitutional).

The four-layer harness of Section 18.1, the three CI gates above, and the frozen-pass-criterion discipline of Section 18.2 are binding on all work in `hawkes_calibration/`. No fast fitter runs in an experiment without a current layer-2 equivalence pass; no fitter touches real data without a layer-3 recovery rehearsal on its own family; no pass criterion is edited after its experiment has run. Rationale: the value of a test plan is precisely that results are allowed to fail, and these rules make failure visible and durable. Reversal trigger: none anticipated; amendments go through the registry (Section 17.6).

Design decision (D-CP.2 — definition of done).

The test campaign is done when three conditions hold. (i) Every row of Table 18.2 has moved from *planned* to *pass* or *fail* with a number attached — not a narrative, a number — and the verdict recorded in Chapter 19. (ii) The benchmarks Protocols 18.6 to 18.8 are run on the real extract, recorded in Chapter 20, and cited by the phase gates of Sections 17.3 and 17.4, so the architecture choices rest on measured evidence from the data they will serve, not on rehearsal priors alone. (iii) Every failed criterion is either fixed and rerun, or documented as a limitation in the registry — because a plan whose failures disappear quietly was never a test plan at all. Rationale: this is the operational meaning of the roadmap’s gates. Reversal trigger: a change in Chapter 16’s consuming actions that adds or retires a protocol, per D7.1.

Part VIII

Results

Chapter 19

Results on Synthetic Data

This chapter is intentionally empty of numbers. It is the destination for the protocols of Chapter 18: every synthetic-data run writes its verdict here, append-only, under the artifact rules of Section 18.4. Each section below mirrors one protocol of Section 18.2. A section is filled only after its protocol has run under its frozen pass criterion; until then its table shell stays empty. Real-data results never appear here; they belong to Chapter 20.

19.1 Generator sanity and stability checks

Before any fitter is judged, the rehearsal worlds themselves must pass: stable event counts across seeds, no supercritical excursions, no runaway per-firm deal concentration, and a deduplication rule that keeps legitimate repeat fundings. These checks apply the harness rules of Section 18.1 to the generator itself.

Check	Regime A	Regime B	Verdict
[TODO: fill after running the generator checks of Section 18.1]			

Table 19.1: Generator sanity and stability checks, per rehearsal regime.

19.2 Poisson baselines and the covariates-vs-excitation decomposition

Results of protocol B2 (Protocol 18.7): the nested NLL ladder from seasonal-naive through covariates-only Poisson to covariates-plus-excitation, decomposed rung by rung with error bars.

Ladder rung	Held-out NLL (A)	Held-out NLL (B)	Δ vs. previous rung
[TODO: fill after running protocol B2 (Protocol 18.7)]			

Table 19.2: Improvement decomposition across the model ladder.

19.3 Parametric Hawkes recovery and bucket-width identifiability

Results of protocol E1 (Protocol 18.1): recovery of κ and θ at 1d, 7d, and 28d bucket widths, and the fitted-radius curve against latent-factor strength.

Bucket width	κ bias (SE)	θ bias (SE)	Ridge visible	Verdict
[TODO: fill after running protocol E1 (Protocol 18.1)]				

Table 19.3: Bucket-width identifiability and latent-confounding results.

19.4 Nonparametric challenger comparison

Results of protocol B3 (Protocol 18.8): the intensity models of Chapter 9 against the log-linear GLM and the parametric Hawkes, on held-out likelihood and calibration, in the same harness.

Model	Held-out NLL	Calibration (PIT / dispersion)	Verdict
[TODO: fill after running protocol B3 (Protocol 18.8)]			

Table 19.4: Nonparametric challenger versus parametric incumbents.

19.5 Choice-model and ranking metrics

Results of protocol E2 (Protocol 18.2) and benchmark B1 (Protocol 18.6): choice-model design choices, weight recovery, and ranking quality under the recall-at- K protocol of Section 14.8.

Model	Held-out NLL	Recall@5	Recall@10	MRR
[TODO: fill after running protocols E2 (Protocol 18.2) and B1 (Protocol 18.6)]				

Table 19.5: Choice-model and ranking results under the protocol of Section 14.8.

19.6 Open-universe and newcomer calibration

Results of the newcomer-design sweep (E2.2 of Protocol 18.2) and the outside-option calibration targets of Chapter 15: quarterly newcomer-share calibration and coverage diagnostics.

19.7 Causal-pipeline validation

Results of the causal validation protocol (Protocol 18.9): naive, backdoor, and AIPW estimates against the known estimand, with the four coded assertions.

19.8 Walk-forward backtest

Results of protocol E5 (Protocol 18.5): the end-to-end weekly walk-forward, scored against oracle and random, with the covariate ablations.

Design	Newcomer-share error (pp)	Incumbent weights stable	Verdict
[TODO: fill after running E2.2 of Protocol 18.2]			

Table 19.6: Newcomer designs and open-universe calibration.

Estimator	Estimate (CI)	Gap to truth	Assertion
[TODO: fill after running the causal validation protocol (Protocol 18.9)]			

Table 19.7: Causal-pipeline validation on known ground truth.

Variant	Top-5	MRR	PIT calibration	Lead time (weeks)
[TODO: fill after running protocol E5 (Protocol 18.5)]				

Table 19.8: Walk-forward backtest with covariate ablation.

Chapter 20

Results on the Spain VC Ecosystem

This chapter is the real-data counterpart of Chapter 19. The target market is the Spanish venture-capital ecosystem. It is intentionally empty of numbers: each section below receives the real-data rerun of one protocol of Chapter 18, append-only, under the artifact rules of Section 18.4.

One structural point shapes everything here. Spain is an EU jurisdiction with statutory filing requirements: companies file annual accounts with the mercantile registry, so firm fundamentals are observable on a statutory schedule rather than only at deal events. The transparent-regime chapters (Chapters 3 and 4) therefore apply directly, alongside the deal-based machinery of the rest of the book. In practice this means registry fundamentals panels can be joined to PitchBook deals, and the fundamentals-prediction and causal questions can be asked of the same firms the deal models rank.

20.1 Data-assembly checklist

Before any fit, the Spanish extract must pass the following assembly steps. Each is an audit, not a convenience.

- *Registry fundamentals join.* Join statutory filings to PitchBook company identifiers; record match rates and the reasons for non-matches. Filing dates, not fiscal-year ends, drive feature availability.
- *Deal coverage audit.* Estimate what share of Spanish VC deals the vendor universe misses, per the model-free audit of Section 15.5; report coverage by era and by sector.
- *Macro series for the euro area.* The FactSet codes used in Chapter 2 are US series; each has a euro-area analogue (policy rate, effective rate, HICP inflation, unemployment). Joins remain as-of on publish date.
- *Point-in-time discipline.* Every joined series — deals, filings, macro — passes the point-in-time protocol of Section 2.4 before it enters any feature.

Assembly step	Metric	Status
[TODO: fill after assembling the Spanish extract]		

Table 20.1: Data-assembly checklist for the Spain extract.

20.2 Data audit

The Phase 0 audit suite (Section 17.2) on the Spanish extract: coverage share, date-quality composition, staleness-age distributions, and the honest-resolution decision.

Audit	Result	Verdict
[TODO: fill after running the Phase 0 audits (Section 17.2)]		

Table 20.2: Data audits on the Spain extract.

20.3 Baselines and the improvement decomposition

The real-data rerun of protocol B2 (Protocol 18.7): the nats ladder from seasonal-naive through covariates-only to covariates-plus-excitation, on Spanish sector counts.

Ladder rung	Held-out NLL	Δ vs. previous rung
[TODO: fill after rerunning protocol B2 (Protocol 18.7) on the Spain extract]		

Table 20.3: Improvement decomposition on Spanish deal counts.

20.4 Parametric fits

The sector-level Hawkes fit at the honest resolution fixed by the data audit, with the ridge diagnostics of Chapter 11 standing and excitation reported under the upper-bound convention.

Model	κ (interval)	Held-out NLL	Diagnostics
[TODO: fill after the Phase 2 Hawkes fit (Section 17.4)]			

Table 20.4: Parametric intensity fits on Spanish sector streams.

20.5 Nonparametric challenger

The real-data rerun of protocol B3 (Protocol 18.8): the intensity models of Chapter 9 against the log-linear GLM and the parametric Hawkes, on held-out likelihood and calibration.

20.6 Ranking and recall at K

The ranker on Spanish deals, evaluated under the recall-at- K protocol of Section 14.8, with the B1 benchmark (Protocol 18.6) rerun against the linear baseline.

20.7 Newcomer and outside-option calibration

Calibration of the outside option on the Spanish universe: quarterly newcomer share against realized first financings, with the coverage audit of Section 15.5 as denominator control.

Model	Held-out NLL	Calibration (PIT / dispersion)	Verdict
[TODO: fill after rerunning protocol B3 (Protocol 18.8) on the Spain extract]			

Table 20.5: Nonparametric challenger on Spanish data.

Model	Held-out NLL	Recall@5	Recall@10	MRR
[TODO: fill after the ranking evaluation (Section 14.8, Protocol 18.6)]				

Table 20.6: Ranking results on Spanish deals.

20.8 Fundamentals prediction and the causal question

Because Spanish filings make fundamentals observable, the transparent-regime models run directly: the success-metric and quantile-prediction stack of Chapter 3 on registry panels joined to deals, and the causal question of Chapter 4 — does financing cause growth? — on the same joined panel, gated by the causal validation protocol (Protocol 18.9).

20.9 Decision-layer output

The end-to-end weekly run of Chapter 16 on Spanish data — the Phase 2 gate (Section 17.4) and the real-data form of protocol E5 (Protocol 18.5): sourcing lists, pacing fans, and the Vol measurement of Protocol 16.2.

Quarter block	Predicted newcomer share	Realized	Error (pp)
[TODO: fill after the outside-option calibration (Chapter 15)]			

Table 20.7: Newcomer-share calibration on the Spanish universe.

Target / estimand	Estimate (interval)	Diagnostic	Verdict
[TODO: fill after the Chapter 3 fits and the Chapter 4 estimates on Spanish filings]			

Table 20.8: Fundamentals prediction and causal estimates on Spanish filings.

Deliverable	Metric	Value	Caveats attached
[TODO: fill after the end-to-end weekly run (Protocol 18.5, Chapter 16)]			

Table 20.9: Decision-layer outputs on the Spain ecosystem.

Appendix A

Notation

The table below collects the symbols used throughout the document, in the conventions fixed by Chapter 2 and `macros.tex`. Chapter references point to the definition site.

Symbol	Meaning	Defined
$(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$	filtered probability space; $\mathcal{F}_t$ holds the marked-process history and covariate paths	Chapter 2
$\mathcal{H}_{t-}$	internal history strictly before t	Chapter 2
t_m, s_m, i_m, m_m	time, sector, firm, and marks of the m -th deal	Chapter 2
$\mathcal{S}, \mathcal{I}, \mathcal{I}^\dagger$	sector set; tracked universe; untracked remainder	Chapter 2
$N_e(t)$	counting process for index e (sector, firm, transition)	Chapter 2
$\lambda_e(t)$	conditional intensity of N_e	Chapter 2
$\Lambda_e(t)$	compensator $\int_0^t Y_e \lambda_e du$	Chapter 6
$\bar{\lambda}, \xi(t)$	mean intensity $\mathbb{E}[\lambda(t)]$ (MBPP)	Chapter 10
$Y_e(t)$	at-risk indicator (observed, eligible at t^-)	Chapter 2
$M_e(t)$	compensated martingale $N_e - \Lambda_e$	Chapter 2
$\mu_s(t)$	baseline (immigrant) intensity of sector s	Chapter 8
$h(t) = \kappa \theta e^{-\theta t}$	exponential excitation kernel	Chapter 8
κ	branching ratio (kernel mass $\int h$)	Chapter 8
θ	kernel decay rate; half-life $\ln 2/\theta$	Chapter 8
$G[s, r]$	aggregated excitation matrix $r \rightarrow s$; stability $\rho(G) < 1$	Chapter 8
$\rho_{\max}$	row-sum stability budget (default 0.95)	Chapter 8
$x^{\text{mac}}(t)$	macro covariates (FactSet), published with lag	Chapter 2
$x_i(t), \tilde{x}_i(t)$	true (latent) and LOCF-observed firm covariates	Chapter 2
$\tau_i(t)$	time of firm i 's last deal before t	Chapter 2
$a_i(t) = t - \tau_i(t)$	staleness age of firm i 's covariates	Chapter 2
β, γ	covariate loadings (baseline; history features)	Chapters 7 and 8
$z_i(t)$	firm-level ranker features	Chapter 14
$q_i(t)$	ranker score of firm i	Chapter 14
$\mathcal{R}[s, t]$	sector risk set (live candidates at t)	Chapter 14
$\diamond_s$	outside option (newcomer/untracked) in sector s 's risk set	Chapter 15
V_i	entry (left-truncation) time of firm i	Chapter 2
T	end of observation window (June 2024)	Chapter 2
Δ	bucket horizon (26/52/104 weeks)	Chapter 13
$p_i(t, \Delta)$	horizon event probability $\mathbb{P}(N_i(t + \Delta) - N_i(t) \geq 1 \mid \mathcal{F}_t)$	Chapter 1
$y_{i,b}$	bucket label for firm i , bucket b	Chapter 13

Symbol	Meaning	Defined
$\ell(\vartheta), L$	log-likelihood; likelihood	Chapter 6
$\vartheta, \hat{\vartheta}$	generic parameter vector; its MLE	Chapter 6
$U(\vartheta), \mathcal{I}(\vartheta)$	score; Fisher information	Chapter 6
(C1)–(C6)	the defect ledger	Table 2.1
T1–T4	the four targets (identity, timing, horizon, marks)	Chapter 1
R1–R4	staleness remedy ladder	Chapter 12
$Dx.y$	numbered design decisions (registry: Section 17.6)	throughout

Appendix B

Data Dictionary: PitchBook and FactSet Columns

This appendix is the column-level contract between the raw data and the loader of Section 17.2. It reflects the column conventions already encoded in the repository’s observation layer (`synthetic/observe.py`, `synthetic/macro.py`, `synthetic/loaders.py`), which mirror the production PitchBook/FactSet extracts. The real adapter must treat any divergence between this dictionary and a delivered extract as an audit failure, not something to patch silently.

Deals table (one row per financing event)

Column	Notes for modeling
<code>deal_id</code>	primary key; duplicates exist upstream — the dedup rule (same company within a few days, jittered size) runs first (Protocol 2.5).
<code>company_id</code>	foreign key into companies; may reference a firm with <i>no</i> companies row when out-of-universe (C6).
<code>deal_date</code>	the untrusted timestamp of (C4): announced vs. closed conventions and backfill jitter; enters models only at the honest resolution of Chapter 10.
<code>primary_industry_sector</code> <code>/ _group / _code</code>	sector taxonomy; small mislabel rate — group-level rollups are the robustness fallback (Chapter 15).
<code>company_universe</code>	'Venture Capital' vs. 'Out of Universe'; the (C6) flag; coverage share by era is a standing audit.
<code>deal_size,</code> <code>deal_size_status</code>	size with Actual/Estimated/Undisclosed status; undisclosedness is informative — always a feature, never silently imputed.
<code>pre_money_valuation,</code> <code>post_valuation,</code> <code>post_valuation_status</code>	frequently jointly missing; missingness indicator enters T4 models.
<code>vc_round,</code> <code>vc_round_up_down_flat</code>	round sequence and direction; up/down/flat is a T4 target and a T1 feature.
<code>deal_type,</code> <code>deal_type_2,</code> <code>deal_type_3,</code> <code>deal_class</code>	deal taxonomy; defines which events count as “funding rounds” for each model’s event definition (an explicit loader config, audited).

Column	Notes for modeling
raised_to_date, total_raised_to_date, total_invested_capital, total_invested_equity, debt, total_new_debt, total_debt, total_investor_ownership, pct_acquired, type_of_stock, deal_status, sellers revenue, ebitda, gross_profit, net_income, revenue_growth_since_last_deal, number_of_employees valuation_cash_flow, valuation_ebitda, valuation_revenue, deal_size_cash_flow, deal_size_ebitda, deal_size_revenue year_founded, hq_location, company_country, hq_global_region, hq_global_sub_region	capitalization block; <i>mechanically</i> event-linked columns (Chapter 12's classification): they update by construction at deals, so their staleness is definitional, not drift. firm fundamentals as reported around the deal; the <i>drifting</i> class of Chapter 12 — prime candidates for the R2 drift model. ratio block; derived — recompute from parents rather than trusting, audit inconsistencies. static descriptors; founding year anchors cohort entry models (Chapter 15).

Companies table (one row per tracked firm)

Column	Notes for modeling
company_id, company_name, description, website financing_status, business_status, ownership_status, company_universe year_founded, hq_*, primary_industry_sector, other_industries	identity block. status block; business_status is the only exit signal and it is coarse and laggy — the (C3) discussion of Chapter 2 governs its use; it may close spells only through a point-in-time rule. static descriptors.

Column	Notes for modeling
last_financing_date,	snapshot summaries of the latest deal — the companies table is itself LOCF by construction; never join it as if current (the (C5) warning in table form).
last_financing_size,	
last_financing_valuation,	
last_financing_deal_type,	
last_known_valuation_date,	
last_known_valuation_deal_type,	
total_money_raised,	
employees	

Investors table

`company_id`, `investor_id`, `investorname`, `investorstatus`, `investorsince`: the firm–investor link with first-investment dates. Used for engineered features (investor count, investor quality proxies, syndicate recency), all as-of `investorsince`.

FactSet macro feed

Long format (`series_code`, `reference_date`, `publish_date`, `value`). Series: `FFUNDT` (Fed Funds target, daily step path), `FFUND` (effective Fed Funds, daily), `CPI_RAW` (CPI level, monthly, ~14-day publication lag), `CPI_YOY` (derived YoY inflation), `RUNEMP` (unemployment rate, monthly). Joins are as-of on `publish_date` only (Protocol 2.5); reference-date joins are look-ahead and fail the audit. Derived transforms (changes, regime dummies, real-rate proxies) are computed inside the loader so their point-in-time correctness is testable in one place.

Appendix C

Detailed Proofs

This appendix completes the book’s proof obligations under one coverage rule: *every numbered mathematical claim in this document is either proved where it is stated or proved here*. Appendix C.1 proves the deferred results of the transparent-regime chapters (Chapters 3 to 5); Appendix C.2 supplies the few machinery-chapter results stated without complete proof (general time rescaling, the cluster converse, Mechanism-B instability, the κ – θ information collapse); and Appendix C.3 adds two self-contained pool results behind the outside-option constant of Chapter 15. Restated claims keep their original numbers, which the restatements reference; helper results proved only here carry appendix numbers with labels prefixed `app:pf`.

C.1 Proofs for the transparent-regime chapters

C.1.1 Success metrics and quantile prediction (Chapter 3)

Claim (Proposition 3.3). *The arc growth rate $g = (F_{y+h} - F_y)/\frac{1}{2}(|F_y| + |F_{y+h}|)$, with the convention $g = 0$ when $F_y = F_{y+h} = 0$, is defined for all sign patterns, bounded in $[-2, 2]$, antisymmetric under swapping $F_y \leftrightarrow F_{y+h}$, and for $F_y, F_{y+h} > 0$ satisfies $g = \log(F_{y+h}/F_y) + O(\log^3(F_{y+h}/F_y))$ near zero growth.*

Proof. Write $a = F_y$, $b = F_{y+h}$, $g = (b - a)/\frac{1}{2}(|a| + |b|)$. *Definedness:* the denominator vanishes iff $a = b = 0$, the case covered by the convention. *Antisymmetry:* swapping a and b negates the numerator and fixes the denominator. *Bounds:* by the triangle inequality $|b - a| \leq |a| + |b|$, so $|g| \leq 2$; squaring, $|b - a|^2 \leq (|a| + |b|)^2$ is equivalent to $-2ab \leq 2|a||b|$, with equality iff $ab \leq 0$, so $|g| = 2$ exactly for sign-crossing (or from-zero) transitions, the reading given in Chapter 3. *Expansion:* for $a, b > 0$ set $u = \log(b/a)$, so $b = ae^u$ and

$$g = \frac{a(e^u - 1)}{\frac{1}{2}a(e^u + 1)} = 2 \frac{e^u - 1}{e^u + 1} = 2 \tanh(u/2).$$

Since $\tanh$ is analytic with bounded derivatives of every order on $\mathbb{R}$, Taylor’s theorem with Lagrange remainder gives $\tanh x = x - x^3/3 + O(x^5)$ uniformly on compacts, hence $g = u - u^3/12 + O(u^5)$: arc growth agrees with log growth to second order at zero growth. $\square$

Claim (Proposition 3.8). *Let Y have distribution function F with $\mathbb{E}|Y| < \infty$ and let $\varphi(a) = \mathbb{E}[\rho_\tau(Y - a)]$ with $\rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\})$ the pinball loss of Chapter 3. Then a^* minimizes φ if and only if $F(a^{*-}) \leq \tau \leq F(a^*)$; if F is continuous and strictly increasing, the minimizer is unique and equals $F^{-1}(\tau)$.*

Proof. Since $0 \leq \rho_\tau(u) \leq \max(\tau, 1 - \tau)|u|$, φ is finite everywhere. Fix $y \in \mathbb{R}$ and let $\psi_y(a) = \rho_\tau(y - a)$: on $\{a < y\}$ it equals $\tau(y - a)$, on $\{a > y\}$ it equals $(1 - \tau)(a - y)$, so ψ_y is Lipschitz

with constant $\max(\tau, 1 - \tau)$ and derivative $\mathbf{1}\{y < s\} - \tau$ at every $s \neq y$. By the fundamental theorem of calculus for Lipschitz functions, for $a < b$, $\psi_y(b) - \psi_y(a) = \int_a^b (\mathbf{1}\{y < s\} - \tau) ds$. Taking expectations and applying Fubini — the integrand is bounded by 1 on the product of a probability space with $[a, b]$ —

$$\varphi(b) - \varphi(a) = \int_a^b (\mathbb{P}(Y < s) - \tau) ds = \int_a^b (F(s) - \tau) ds, \quad (\text{C.1})$$

using $\mathbb{P}(Y < s) = F(s^-) = F(s)$ off the (countable, Lebesgue-null) set of discontinuities.

Let a^* satisfy $F(a^{*-}) \leq \tau \leq F(a^*)$. For $b > a^*$, monotonicity gives $F \geq \tau$ on (a^*, b) , so Equation (C.1) yields $\varphi(b) \geq \varphi(a^*)$; for $b < a^*$, $F \leq \tau$ on (b, a^*) , so $\varphi(a^*) \leq \varphi(b)$: a^* is a global minimizer. Conversely, if $F(a) < \tau$, right-continuity gives $\delta > 0$ with $F < \tau$ on $[a, a + \delta)$, so $\varphi(a + \delta) < \varphi(a)$; if $F(a^-) > \tau$, symmetrically $\varphi(a - \delta) < \varphi(a)$: in either case a is not a minimizer. When F is continuous and strictly increasing, the condition reads $F(a) = \tau$, with unique solution $a = F^{-1}(\tau)$. $\square$

Lemma C.1 (Quantile of exchangeable scores). *Let $V_1, \dots, V_{n+1}$ be exchangeable real random variables, $\alpha \in (0, 1)$, and $k = \lceil (1 - \alpha)(n + 1) \rceil \leq n$. Let Q be the k -th smallest of $V_1, \dots, V_n$. Then $\mathbb{P}(V_{n+1} \leq Q) \geq 1 - \alpha$, and if the V_j are almost surely distinct, also $\mathbb{P}(V_{n+1} \leq Q) \leq 1 - \alpha + 1/(n + 1)$.*

Proof. For $j \leq n + 1$ let $R_j = \#\{i \neq j : V_i < V_j\}$, computed in the pooled sample of size $n + 1$. First, $\{V_{n+1} > Q\} \subseteq \{R_{n+1} \geq k\}$: if V_{n+1} exceeds the k -th smallest of the first n values, then at least k of those values are $\leq Q < V_{n+1}$. Second, deterministically $\#\{j : R_j \geq k\} \leq n + 1 - k$: if $R_j \geq k$ then V_j strictly exceeds at least k pooled values, hence exceeds the k -th smallest pooled value W ; and at least k indices have $V_i \leq W$, so at most $n + 1 - k$ indices can satisfy $V_j > W$. Third, by exchangeability the events $\{R_j \geq k\}$ have a common probability (swapping coordinates j and $n + 1$ maps one to the other and preserves the joint law), so

$$\mathbb{P}(R_{n+1} \geq k) = \frac{1}{n + 1} \sum_{j=1}^{n+1} \mathbb{P}(R_j \geq k) = \frac{\mathbb{E} \#\{j : R_j \geq k\}}{n + 1} \leq \frac{n + 1 - k}{n + 1} \leq \alpha,$$

the last step because $k \geq (1 - \alpha)(n + 1)$: the lower bound. For the upper bound with almost surely distinct values: $\{V_{n+1} \leq Q\}$ equals, up to a null set, $\{V_{n+1} < Q\} \subseteq \{R_{n+1} \leq k - 1\}$ (all values below V_{n+1} lie below Q , and exactly $k - 1$ of the first n do); with distinct values $\#\{j : R_j \geq k\} = n + 1 - k$ exactly, so $\mathbb{P}(R_{n+1} \leq k - 1) = k/(n + 1) \leq 1 - \alpha + 1/(n + 1)$. $\square$

Claim (Theorem 3.9). *In split conformalized quantile regression with calibration set I_2 of size n_2 , if the calibration pairs $\{(X_j, Y_j)\}_{j \in I_2}$ and the test pair (X, Y) are exchangeable, then $\mathbb{P}\{Y \in C(X)\} \geq 1 - \alpha$; if the conformity scores are almost surely distinct, also $\mathbb{P}\{Y \in C(X)\} \leq 1 - \alpha + 1/(n_2 + 1)$ (Romano et al., 2019).*

Proof. Condition on the proper training set I_1 ; the fitted quantile functions $\hat{q}_{\alpha/2}, \hat{q}_{1-\alpha/2}$ are then fixed measurable functions. Define $s(x, y) = \max\{\hat{q}_{\alpha/2}(x) - y, y - \hat{q}_{1-\alpha/2}(x)\}$, $E_j = s(X_j, Y_j)$ for $j \in I_2$, $E_{n_2+1} = s(X, Y)$. A fixed measurable function applied coordinatewise preserves exchangeability, so these $n_2 + 1$ scores are exchangeable given I_1 . The threshold $Q_{1-\alpha}$ is the k -th smallest calibration score with $k = \lceil (n_2 + 1)(1 - \alpha) \rceil$ (if $k > n_2$, set $Q_{1-\alpha} = +\infty$ and the bound is trivial). The coverage event is exactly a score comparison:

$$Y \in C(X) \iff \hat{q}_{\alpha/2}(X) - Y \leq Q_{1-\alpha} \text{ and } Y - \hat{q}_{1-\alpha/2}(X) \leq Q_{1-\alpha} \iff E_{n_2+1} \leq Q_{1-\alpha}.$$

Lemma C.1, applied conditionally on I_1 , gives $\mathbb{P}\{Y \in C(X) \mid I_1\} \geq 1 - \alpha$ and, under almost-sure distinctness, the matching upper bound; integrating over I_1 preserves both. The guarantee is *marginal* and rests entirely on exchangeability, which temporal drift breaks — the operational consequence recorded by the exchangeability warning of Chapter 3. $\square$

C.1.2 The causal layer (Chapter 4)

We work with discrete variables and a strictly positive joint law (the continuous case is identical with densities). Let G be a DAG over $V = (V_1, \dots, V_p)$ with $P(v) = \prod_j P(v_j \mid \text{pa}_j)$. Following Pearl (1995), the interventional law is defined by the truncated factorization

$$P_a(v) = \prod_{j: V_j \neq A} P(v_j \mid \text{pa}_j) \Big|_{A=a} \quad \text{on } \{v : v_A = a\}, \quad (\text{C.2})$$

and $\mathbb{E}[Y \mid \text{do}(A = a)]$ is the mean of Y under P_a . Recall the backdoor criterion of Chapter 4: Z satisfies it relative to (A, Y) if (i) no element of Z is a descendant of A , and (ii) Z blocks every path from A to Y that begins with an edge into A .

Lemma C.2 (Interventions do not move non-descendants). *Let W be the set of non-descendants of A (excluding A). Then $P_a(w) = P(w)$; in particular $P_a(z) = P(z)$ for any $Z \subseteq W$.*

Proof. No parent of a W -variable can be A or a strict descendant of A (either would make it a descendant), and every parent of A is in W (a descendant parent would close a directed cycle). Hence (W in topological order, then A , then the strict descendants D in topological order) is a topological order of G , and

$$P(w, a, d) = \left[\prod_{j \in W} P(w_j \mid \text{pa}_j) \right] P(a \mid \text{pa}_A) \left[\prod_{j \in D} P(d_j \mid \text{pa}_j) \right], \quad P_a(w, d) = \left[\prod_{j \in W} \right] \left[\prod_{j \in D} \right] \Big|_{A=a}.$$

Summing $P_a(w, d)$ over the D -coordinates in reverse topological order telescopes (each conditional sums to one), leaving $P_a(w) = \prod_{j \in W} P(w_j \mid \text{pa}_j)$. The same telescoping in the observational law (sum over D , then over a) gives $P(w) = \prod_{j \in W} P(w_j \mid \text{pa}_j)$. Summing out $W \setminus Z$ gives the claim for Z , which contains no descendant of A by backdoor condition (i). $\square$

Lemma C.3 (Backdoor blocking survives edge deletion). *If Z satisfies the backdoor criterion for (A, Y) in G , then Z d -separates A from Y in the graph $G_{\underline{A}}$ obtained from G by deleting all edges out of A .*

Proof. Let π be any A – Y path in $G_{\underline{A}}$. Its first edge points into A (all out-edges are deleted), so π is also a backdoor path of G ; by criterion (ii) some node W on π blocks it in G : either (a) a non-collider on π with $W \in Z$, or (b) a collider on π with neither W nor any G -descendant of W in Z . Collider status depends only on the orientations of the two π -edges at W , the same in both graphs. In case (a), π is blocked at W in $G_{\underline{A}}$ as well; in case (b), the $G_{\underline{A}}$ -descendants of W are a subset of its G -descendants, so π is again blocked. Hence every A – Y path of $G_{\underline{A}}$ is blocked by Z . $\square$

Claim (Theorem 4.5). *If Z satisfies the backdoor criterion relative to (A, Y) , then $\mathbb{E}[Y \mid \text{do}(A = a)] = \mathbb{E}_Z[\mathbb{E}[Y \mid A = a, Z]]$, and consequently $\tau_{\text{ATE}} = \mathbb{E}_Z[\mathbb{E}[Y \mid A=1, Z] - \mathbb{E}[Y \mid A=0, Z]]$.*

Proof. Decompose $P_a(y) = \sum_z P_a(y \mid z) P_a(z)$. Lemma C.2 gives $P_a(z) = P(z)$. For the conditional we invoke the invariance half of Rule 2 of the do-calculus: if Z d -separates A from Y in $G_{\underline{A}}$, then $P_a(y \mid z) = P(y \mid a, z)$ — proved in Pearl (1995) from Equation (C.2) via the soundness of d -separation for DAG-factorized laws; we delegate exactly this step, and Lemma C.3 verifies its graphical hypothesis. Combining, $P_a(y) = \sum_z P(y \mid a, z) P(z)$, which is the adjustment formula; subtracting the $a = 0$ and $a = 1$ versions yields the ATE. One self-contained special case is worth recording: under the ignorability assumption of Chapter 4 with potential-outcome semantics ($\mathbb{E}[Y \mid \text{do}(A = a)] := \mathbb{E}[Y(a)]$), the formula is three lines — $\mathbb{E}[Y(a)] = \mathbb{E}_Z \mathbb{E}[Y(a) \mid Z] = \mathbb{E}_Z \mathbb{E}[Y(a) \mid A = a, Z] = \mathbb{E}_Z \mathbb{E}[Y \mid A = a, Z]$, by ignorability then consistency — and this is the form under which Chapter 4 actually operates. $\square$

Claim (Proposition 4.7). *Let $\psi = \psi_1 - \psi_0$ be the AIPW score of Chapter 4,*

$$\psi_1 = \mu_1(Z) + \frac{A(Y - \mu_1(Z))}{e(Z)}, \quad \psi_0 = \mu_0(Z) + \frac{(1 - A)(Y - \mu_0(Z))}{1 - e(Z)},$$

built from working nuisances (μ_1, μ_0, e) with $\delta \leq e(Z) \leq 1 - \delta$ for some $\delta > 0$ and $\mathbb{E}|Y| + \mathbb{E}|\mu_a(Z)| < \infty$. Under the consistency and ignorability assumptions of Chapter 4, $\mathbb{E}[\psi] = \tau_{\text{ATE}}$ if either $(\mu_1, \mu_0) = (\mu_1^, \mu_0^*)$ is the true outcome pair or $e = e^*$ is the true propensity.*

Proof. All terms are integrable because the denominators are bounded away from zero. We show $\mathbb{E}[\psi_1] = \mathbb{E}[Y(1)]$ under either hypothesis; the ψ_0 half is symmetric, and subtracting gives the claim.

True outcome model. By consistency and then ignorability, $\mu_1^*(Z) = \mathbb{E}[Y \mid A = 1, Z] = \mathbb{E}[Y(1) \mid Z]$. Conditioning on Z ($e(Z)$ is Z -measurable),

$$\mathbb{E}\left[\frac{A(Y - \mu_1^*(Z))}{e(Z)} \mid Z\right] = \frac{\mathbb{P}(A=1 \mid Z)}{e(Z)} \mathbb{E}[Y - \mu_1^*(Z) \mid A = 1, Z] = \frac{e^*(Z)}{e(Z)} \cdot 0 = 0,$$

regardless of the working e . Hence $\mathbb{E}[\psi_1] = \mathbb{E}[\mu_1^*(Z)] = \mathbb{E}[\mathbb{E}[Y(1) \mid Z]] = \mathbb{E}[Y(1)]$.

True propensity. With $e = e^*$ and arbitrary μ_1 ,

$$\mathbb{E}[\psi_1 \mid Z] = \mu_1(Z) + \frac{e^*(Z) \mathbb{E}[Y \mid A = 1, Z] - e^*(Z) \mu_1(Z)}{e^*(Z)} = \mathbb{E}[Y \mid A = 1, Z] = \mathbb{E}[Y(1) \mid Z],$$

the working μ_1 cancelling exactly; take expectations over Z . $\square$

Proposition C.4 (Influence-function variance of AIPW). *Let both nuisances be correct, $\mathbb{E}[Y^2] < \infty$, and let $O_1, \dots, O_n$ be i.i.d. Write $\tau(Z) = \mu_1^*(Z) - \mu_0^*(Z)$ and $\sigma_a^2(Z) = \text{Var}(Y \mid A = a, Z)$. Then $\hat{\tau}_{\text{AIPW}} = n^{-1} \sum_i \psi(O_i)$ satisfies $\sqrt{n}(\hat{\tau}_{\text{AIPW}} - \tau_{\text{ATE}}) \rightarrow \mathcal{N}(0, V)$ with*

$$V = \text{Var}(\psi) = \text{Var}(\tau(Z)) + \mathbb{E}\left[\frac{\sigma_1^2(Z)}{e^*(Z)}\right] + \mathbb{E}\left[\frac{\sigma_0^2(Z)}{1 - e^*(Z)}\right].$$

Proof. Asymptotic normality is the Lindeberg–Levy CLT for the i.i.d. mean of the square-integrable ψ , whose mean is τ_{ATE} by Proposition 4.7. For the variance, write $\psi = \tau(Z) + U_1 - U_0$ with $U_1 = A(Y - \mu_1^*(Z))/e^*(Z)$ and $U_0 = (1 - A)(Y - \mu_0^*(Z))/(1 - e^*(Z))$. Both U_a have conditional mean zero given Z (previous proof), so they are uncorrelated with the Z -measurable $\tau(Z)$, and $U_1 U_0 = 0$ pointwise because $A(1 - A) = 0$. Conditioning on (A, Z) ,

$$\mathbb{E}[U_1^2 \mid Z] = \frac{\mathbb{E}[A(Y - \mu_1^*(Z))^2 \mid Z]}{e^*(Z)^2} = \frac{e^*(Z) \sigma_1^2(Z)}{e^*(Z)^2} = \frac{\sigma_1^2(Z)}{e^*(Z)},$$

and symmetrically for U_0 . Adding the three orthogonal contributions gives V . This V is the semiparametric efficiency bound for the ATE, attained by the cross-fitted estimator of Chapter 4 under the product-rate condition stated there (Chernozhukov et al., 2018); the firm-clustered variance and cluster bootstrap of Chapter 4 estimate the same quantity with firms as independent blocks (Efron and Tibshirani, 1994). $\square$

C.1.3 Selection identities for the proxies chapter (Chapter 5)

Chapter 5 uses three display-level identities — the selection-bias covariance formula Equation (5.5), the worst-case bounds Equation (5.6), and the unbiasedness of the inverse-probability-of-observation weights built from Equation (5.7). We record them with proofs.

Proposition C.5 (Selection identities). *Let g be integrable and $D \in \{0, 1\}$ with $p = \mathbb{P}(D=1) \in (0, 1)$. Then: (i) $\mathbb{E}[g \mid D=1] - \mathbb{E}[g] = \text{Cov}(g, \mathbf{1}\{D=1\})/p$, which is Equation (5.5); (ii) if $g \in [g_{\text{lo}}, g_{\text{hi}}]$ almost surely, then*

$$\mathbb{E}[g] \in [p \mathbb{E}[g \mid D=1] + (1-p) g_{\text{lo}}, p \mathbb{E}[g \mid D=1] + (1-p) g_{\text{hi}}],$$

the interval Equation (5.6); (iii) under the selection-on-observed-history assumption of Chapter 5 — $g \perp\!\!\!\perp D \mid \mathcal{H}_{T_{i,k}}$ — with $\pi = \mathbb{P}(D=1 \mid \mathcal{H}_{T_{i,k}}) > 0$ almost surely as in Equation (5.7), $\mathbb{E}[Dg/\pi] = \mathbb{E}[g]$.

Proof. (i) $\text{Cov}(g, \mathbf{1}\{D=1\}) = \mathbb{E}[g\mathbf{1}\{D=1\}] - p \mathbb{E}[g] = p \mathbb{E}[g \mid D=1] - p \mathbb{E}[g]$; divide by p . (ii) By the law of total expectation, $\mathbb{E}[g] = p \mathbb{E}[g \mid D=1] + (1-p) \mathbb{E}[g \mid D=0]$, and $\mathbb{E}[g \mid D=0] \in [g_{\text{lo}}, g_{\text{hi}}]$ by the almost-sure bounds. (iii) By the tower property and the conditional independence $g \perp\!\!\!\perp D \mid \mathcal{H}_{T_{i,k}}$,

$$\mathbb{E}\left[\frac{Dg}{\pi}\right] = \mathbb{E}\left[\frac{1}{\pi} \mathbb{E}[Dg \mid \mathcal{H}_{T_{i,k}}]\right] = \mathbb{E}\left[\frac{1}{\pi} \mathbb{E}[D \mid \mathcal{H}_{T_{i,k}}] \mathbb{E}[g \mid \mathcal{H}_{T_{i,k}}]\right] = \mathbb{E}[\mathbb{E}[g \mid \mathcal{H}_{T_{i,k}}]] = \mathbb{E}[g],$$

all steps valid because π is $\mathcal{H}_{T_{i,k}}$ -measurable and positive.

Finally, the closed form Equation (5.7) for π is the survival function of the first event after $T_{i,k}$: with an internal (self-exciting) history, the intensity along the no-new-event continuation is an $\mathcal{H}_{T_{i,k}}$ -measurable function of elapsed time, and the hazard representation of the inter-event distribution (Daley and Vere-Jones, 2003, Ch. 7.2) gives $\mathbb{P}(T_{i,k+1} > \tau_{\text{end}} \mid \mathcal{H}_{T_{i,k}}) = \exp(-\int \lambda_i)$. If external covariates evolve stochastically over the gap, Equation (5.7) holds with the exponential replaced by its conditional expectation. $\square$

C.2 Proofs for the machinery chapters (results stated there without proof)

Throughout this section, $(\Omega, \mathcal{F}, \mathbb{P})$ carries a filtration $(\mathcal{F}_t)_{t \geq 0}$ satisfying the usual conditions, and N is a *simple* point process adapted to it, with nonnegative predictable intensity λ and compensator $\Lambda(t) = \int_0^t \lambda(u) du$ obeying the integrability of Assumption 6.4 ($\mathbb{E}[\Lambda(T)] < \infty$ for finite T); write $M = N - \Lambda$. Two preparatory lemmas carry the martingale bookkeeping.

Lemma C.6 (Compensator calculus). *Under the standing integrability: (i) M is a martingale on every $[0, T]$; (ii) for every bounded predictable real- or complex-valued process H , the integral $t \mapsto \int_0^t H(s) dM(s)$ is a martingale on $[0, T]$; in particular it has mean zero.*

Lemma C.6 is the defining property of the compensator (Equation (6.1)) together with the elementary theory of Stieltjes integration against point-process martingales; we use it without proof, referring to Brémaud and Massoulié (1996) and Daley and Vere-Jones (2003, Ch. 7) (complex integrands: apply the real statement to real and imaginary parts). The special case used most often we prove in full.

Lemma C.7 (Unbiasedness against deterministic integrands). *Let $f : [0, T] \rightarrow [0, \infty]$ be Borel measurable. Then $\mathbb{E} \int_0^T f(s) dN(s) = \mathbb{E} \int_0^T f(s) \lambda(s) ds$, both sides possibly infinite. For bounded signed measurable f the same identity holds with both sides finite.*

Proof. Define finite measures on the Borel subsets of $[0, T]$ by $\mu_1(B) = \mathbb{E} \int_0^T \mathbf{1}\{s \in B\} dN(s)$ and $\mu_2(B) = \mathbb{E} \int_0^T \mathbf{1}\{s \in B\} \lambda(s) ds$. For $B = (a, b]$, Lemma C.6(i) gives $\mathbb{E}[N(b) - N(a)] = \mathbb{E}[\Lambda(b) - \Lambda(a)]$, so $\mu_1 = \mu_2$ on the π -system of half-open intervals and both vanish on $\{0\}$; by Dynkin they agree everywhere. The identity extends to simple nonnegative f by linearity, to measurable $f \geq 0$ by monotone convergence (each side an integral against μ_i by Tonelli), and to bounded signed f by splitting $f = f_+ - f_-$. $\square$

C.2.1 Watanabe's characterization and the general time-rescaling theorem

The proof of Theorem 6.16 in Chapter 6 is complete for intensities whose between-event paths are determined by the history at the last event — every model fitted in this document — and delegates the general case to Watanabe's characterization, cited there without proof. We prove the direction used, then run the time-change argument in full generality.

Proposition C.8 (Watanabe's characterization, the direction used). *Let N be a simple point process adapted to $(\mathcal{F}_t)$ whose compensator Λ is continuous and deterministic. Then N is an inhomogeneous Poisson process: it has independent increments and $N(b) - N(a) \sim \text{Poisson}(\Lambda(b) - \Lambda(a))$ for all $a < b$.*

Proof. Here $N - \Lambda$ is a martingale (Lemma C.6(i); integrability is automatic since Λ is finite on compacts). Fix $u \in \mathbb{R}$ and $a \geq 0$, and define for $t \geq a$

$$Z_t = \exp \left\{ iu(N(t) - N(a)) - (e^{iu} - 1)(\Lambda(t) - \Lambda(a)) \right\} = e^{iu(N(t) - N(a))} g(t), \quad (\text{C.3})$$

with g continuous, deterministic, of bounded variation, and bounded away from 0 and ∞ on compacts. Stieltjes integration by parts (with g continuous) gives $dZ_t = e^{iu(N(t^-) - N(a))} dg(t) + g(t) d(e^{iu(N - N(a))})_t$. The first differential is $-(e^{iu} - 1) Z_{t-} d\Lambda(t)$ by the chain rule for continuous BV functions (Z_{t-} may replace Z_t because the continuous Λ charges no countable set); the second is pure-jump: at an event e^{iuN} is multiplied by e^{iu} , so it equals $(e^{iu} - 1) e^{iu(N(t^-) - N(a))} dN(t)$. Adding,

$$Z_t = Z_a + \int_{(a,t]} Z_{s-} (e^{iu} - 1) d(N - \Lambda)(s).$$

The integrand $H_s = Z_{s-}(e^{iu} - 1)$ is predictable (left limit of an adapted cadlag process) and bounded on $[a, b]$ by the deterministic constant $|e^{iu} - 1| e^{|e^{iu} - 1|\Lambda(b)}$. By Lemma C.6(ii) the integral is a martingale, so $\mathbb{E}[Z_t | \mathcal{F}_a] = Z_a = 1$, i.e.

$$\mathbb{E}[e^{iu(N(t) - N(a))} | \mathcal{F}_a] = \exp \{ (e^{iu} - 1)(\Lambda(t) - \Lambda(a)) \},$$

the characteristic function of $\text{Poisson}(\Lambda(t) - \Lambda(a))$ — and it is *deterministic*: the increment is Poisson by uniqueness of characteristic functions, and independent of $\mathcal{F}_a$ because its conditional characteristic function is constant. For disjoint ordered intervals, iterating the tower property from the last interval inward factorizes the joint characteristic function into a product of Poisson ones: joint independence. Hence N is inhomogeneous Poisson with mean function Λ ; the converse and the marked-space generality, neither used here, are in Daley and Vere-Jones (2008). $\square$

Claim (Theorem 6.16, general case). *Let N have event times $t_1 < t_2 < \dots$ and conditional intensity $\lambda(t) > 0$ with compensator Λ continuous, strictly increasing, and $\Lambda(t) \rightarrow \infty$ almost surely. Then $\tau_m = \Lambda(t_m)$ are the event times of a unit-rate Poisson process; equivalently the rescaled gaps $\Delta_m = \tau_m - \tau_{m-1}$ are i.i.d. $\text{Exp}(1)$.*

Proof. Step 1: the time change. Since Λ is continuous, strictly increasing, and $\Lambda(\infty) = \infty$ a.s., each $s \geq 0$ has a unique finite $\tau(s)$ with $\Lambda(\tau(s)) = s$, continuous and increasing in s , and each $\tau(s)$ is a stopping time: $\{\tau(s) \leq t\} = \{\Lambda(t) \geq s\} \in \mathcal{F}_t$. Define $\mathcal{G}_s = \mathcal{F}_{\tau(s)}$ (a filtration, by monotonicity) and $\tilde{N}(s) = N(\tau(s))$, which is $\mathcal{G}_s$ -measurable because the right-continuous adapted N is progressive. $\tilde{N}$ is a simple counting path with jump times exactly $\{\Lambda(t_m)\} = \{\tau_m\}$, the strictly increasing Λ mapping distinct times to distinct values.

Step 2: integrability. Fix s and let $T = \tau(s)$. For each n , $T \wedge n$ is a bounded stopping time, and Doob's optional sampling applied to $M = N - \Lambda$ on $[0, n]$ gives $\mathbb{E}[N(T \wedge n)] =$

$\mathbb{E}[\Lambda(T \wedge n)] \leq s$; monotone convergence ($T < \infty$ a.s., N and Λ nondecreasing) yields $\mathbb{E}[\tilde{N}(s)] = \mathbb{E}[\Lambda(T)] = s < \infty$.

Step 3: $\tilde{N}(s) - s$ is a $\mathcal{G}$ -martingale. Fix $s' < s$, put $T' = \tau(s')$, $T = \tau(s)$, and take $A \in \mathcal{G}_{s'} = \mathcal{F}_{T'}$. For each n , $T' \wedge n \leq T \wedge n$ are bounded stopping times, so optional sampling gives $\mathbb{E}[M(T \wedge n) \mid \mathcal{F}_{T' \wedge n}] = M(T' \wedge n)$; the event $A_n = A \cap \{T' \leq n\}$ belongs to $\mathcal{F}_{T' \wedge n}$ (check $A_n \cap \{T' \wedge n \leq t\} \in \mathcal{F}_t$ separately for $t < n$ and $t \geq n$), whence $\mathbb{E}[\mathbf{1}\{A_n\}M(T \wedge n)] = \mathbb{E}[\mathbf{1}\{A_n\}M(T' \wedge n)]$. As $n \rightarrow \infty$, $\mathbf{1}\{A_n\} \uparrow \mathbf{1}\{A\}$ ($T' < \infty$ a.s.), $M(T \wedge n) \rightarrow M(T)$ and $M(T' \wedge n) \rightarrow M(T')$ a.s., and $|M(\cdot \wedge n)| \leq N(T) + s \in L^1$ by Step 2, so dominated convergence gives $\mathbb{E}[\mathbf{1}\{A\}M(T)] = \mathbb{E}[\mathbf{1}\{A\}M(T')]$ for all $A \in \mathcal{G}_{s'}$. Since $M(T') = \tilde{N}(s') - s'$ is $\mathcal{G}_{s'}$ -measurable, this says $\mathbb{E}[\tilde{N}(s) - s \mid \mathcal{G}_{s'}] = \tilde{N}(s') - s'$.

Step 4: conclude. $\tilde{N}$ is a simple counting process adapted to $(\mathcal{G}_s)$ whose compensator is the continuous deterministic function s ; by Proposition C.8 it is a unit-rate Poisson process. Its jump times are the τ_m , and a unit-rate Poisson process is exactly one with i.i.d. standard-exponential interarrivals (Daley and Vere-Jones, 2003, Ch. 2); the residual-analysis use is Ogata (1988), and Corollary 6.17 reads the same computation backwards. $\square$

C.2.2 The cluster representation: converse direction and stationarity

Theorem 8.2 is proved in Chapter 8 in the construction direction; the converse — every Hawkes process with the given parameters is equal in law to the cluster process — is stated there with a citation. We prove it in the univariate notation of Equation (8.1); the multivariate case of Equation (8.2) repeats every step with matrix kernels. Throughout, $h \geq 0$ with $\kappa = \int h < \infty$, and offspring lags have density h/κ on $(0, \infty)$.

Lemma C.9 (Branching moments and local finiteness). *Consider a cluster rooted at a point at time v : generation 0 is the root; each point of generation n independently produces $\text{Poisson}(\kappa)$ generation- $(n+1)$ points at lags drawn i.i.d. from h/κ . Let Z_n be the size of generation n and S_n a sum of n i.i.d. lags. Then (i) $\mathbb{E}Z_n = \kappa^n$, and for $\kappa < 1$ the mean total cluster size is $\sum_n \kappa^n = 1/(1 - \kappa)$; (ii) the expected number of generation- n points within $[v, v + T]$ is $\kappa^n \mathbb{P}(S_n \leq T)$; (iii) for every T and every κ , $\sum_n \kappa^n \mathbb{P}(S_n \leq T) < \infty$, so clusters are almost surely finite on compact windows.*

Proof. (i) Wald's identity for a Poisson number of nonnegative i.i.d. terms ($\mathbb{E} \sum_{i \leq K} X_i = \mathbb{E}K \cdot \mathbb{E}X_1$ for K independent of (X_i) , by conditioning on K and Tonelli), applied generation by generation: $\mathbb{E}Z_{n+1} = \kappa \mathbb{E}Z_n$, so $\mathbb{E}Z_n = \kappa^n$; the total follows by Tonelli over n . (ii) The mean measure m_n of generation- n displacements from the root satisfies $m_0 = \delta_0$ and $m_{n+1}(B) = \iint \mathbf{1}\{u \in B\} h(u - p) \, du \, m_n(dp)$ by the same conditioning plus Tonelli, so the density of m_n is κ^n times the n -fold convolution of h/κ — the law of S_n scaled by κ^n ; evaluate on $[0, T]$. (iii) Lags are a.s. strictly positive, so $\mathbb{E}e^{-\eta S_1} \downarrow 0$ as $\eta \uparrow \infty$ (dominated convergence); choose η with $r_0 := \kappa \mathbb{E}e^{-\eta S_1} < 1$. Markov's inequality gives $\mathbb{P}(S_n \leq T) \leq e^{\eta T} (\mathbb{E}e^{-\eta S_1})^n$, hence $\sum_n \kappa^n \mathbb{P}(S_n \leq T) \leq e^{\eta T} / (1 - r_0) < \infty$; finite expectation implies almost-sure finiteness. Note (iii) holds for every branching scale, an independent route to the finite-window non-explosion of Remark 8.3. $\square$

Claim (Theorem 8.2, converse direction). *Every Hawkes process with baseline $\mu(\cdot)$ and kernel h is equal in law to the cluster process constructed in Theorem 8.2. Under constant μ it admits a stationary version iff $\kappa < 1$, with mean intensity $\mu/(1 - \kappa)$.*

Proof. Local finiteness of the construction. Let N_c be the superposition of all generations over all immigrants. By Lemma C.9(ii)–(iii) and Campbell's formula over the Poisson immigrant process, $\mathbb{E}N_c[0, T] \leq (\int_0^T \mu) \sum_n \kappa^n \mathbb{P}(S_n \leq T) < \infty$, so N_c is almost surely locally finite; it is

simple because all lags and immigrant positions have nonatomic laws and there are countably many pairs of points.

Intensity with respect to the genealogical filtration, with the L^1 detail. Let $\mathcal{G}_t$ be generated by all points born up to t together with their genealogy, and decompose N_c into the immigrant stream and the direct-offspring streams N^p of each born point p . Given p , N^p is by construction a Poisson process on (p, ∞) with intensity $h(\cdot - p)$, independent of all other building blocks, so $N^p(t) - \int_p^{t \vee p} h(u - p) du$ is a $\mathcal{G}$ -martingale, as is the compensated immigrant stream. Partial sums over the countably many streams are martingales converging in L^1 for each t (monotonically in the counting and compensating parts separately, with finite limit expectation $\mathbb{E}N_c(t) < \infty$), so the limit $N_c(t) - \int_0^t \lambda_c(u) du$, with $\lambda_c(u) = \mu(u) + \sum_{p < u} h(u - p)$, is a $\mathcal{G}$ -martingale (L^1 -limits pass through conditional expectations) — the step compressed into “additivity of compensators” in Chapter 8.

Reduction to the internal history. λ_c is exactly the Hawkes formula Equation (8.1) evaluated on the points of N_c : it depends on point positions only, not on the genealogy, so it is predictable for the internal filtration $\mathcal{H}_t = \sigma(N_c(s), s \leq t) \subseteq \mathcal{G}_t$. The tower property transfers the martingale property: for $s < t$, $\mathbb{E}[N_c(t) - \int_0^t \lambda_c \mid \mathcal{H}_s] = \mathbb{E}[\mathbb{E}[\cdot \mid \mathcal{G}_s] \mid \mathcal{H}_s] = N_c(s) - \int_0^s \lambda_c$, the last equality because the variable is $\mathcal{H}_s$ -measurable. Hence N_c has internal-history intensity Equation (8.1), with no appeal to the innovation projection theorem.

Law identification. A simple point process whose internal-history intensity coincides almost surely with a fixed predictable functional of the past configuration has its law on every $[0, T]$ uniquely determined: the intensity fixes the successive conditional hazards of inter-event times, hence all finite-dimensional joint densities (Daley and Vere-Jones, 2003, Ch. 7.2–7.3). Any Hawkes process with parameters (μ, h) and the cluster process N_c share this functional, so they are equal in law: the converse direction, originally in Hawkes and Oakes (1974).

Stationarity. Let μ be constant. For $\kappa < 1$, run the same construction with immigrants homogeneous Poisson on all of $\mathbb{R}$: the mechanism is shift invariant, so the law of N_c is stationary, with mean rate $\mu \cdot \mathbb{E}[\text{cluster size}] = \mu/(1 - \kappa) < \infty$ by Wald plus Campbell (Lemma C.9(i)) — the stationary version whose existence Corollary 8.5 assumes. Conversely, if a stationary version with finite mean intensity $\bar{\lambda} = \mathbb{E}N(0, 1]$ exists, then $\mathbb{E}\lambda(t) = \bar{\lambda}$ for a.e. t (Lemma C.7 with $f \equiv 1$), and taking expectations in Equation (8.1) via Campbell’s formula gives $\bar{\lambda} = \mu + \kappa\bar{\lambda}$: for $\mu > 0$ this has no finite nonnegative solution when $\kappa \geq 1$, and for $\kappa < 1$ it returns Equation (8.4). The multivariate statement replaces κ^n by G^n , whose Neumann series converges iff $\rho(G) < 1$ (Gelfand’s formula), with mean vector $(I - G)^{-1}\mu$; for stability beyond the linear case see Brémaud and Massoulié (1996). $\square$

C.2.3 Instability of log-link raw-count feedback (Mechanism B)

This proves the claim asserted in Warning 8.18 (previewed in Warning 7.17): no coefficient inequality stabilizes raw counts in the exponent, and the operating-point condition quoted there is genuinely local — the mean recursion escapes it. We state the univariate weekly face of Equation (7.8).

Proposition C.10 (Instability of log-link raw-count feedback). *Let $a \in \mathbb{R}$, $b > 0$, and let $Y_t \mid \mathcal{H}_t \sim \text{Poisson}(m_t)$ with $m_{t+1} = \exp(a + b Y_t)$ and any initial $m_1 > 0$. Define $G(m) = e^a \exp\{m(e^b - 1)\}$. Then: (i) $\mathbb{E}[m_{t+1} \mid \mathcal{H}_t] = G(m_t)$ exactly, with G increasing, strictly convex, and $G'(m^*) = m^*(e^b - 1)$ at any fixed point m^* ; (ii) G has a fixed point iff $e^a(e^b - 1) \leq e^{-1}$, in which case there are at most two, $m_-^* \leq m_+^*$, with $m_-^*(e^b - 1) \leq 1$ (the local stability condition of the mean recursion) and $m_+^*(e^b - 1) \geq 1$ (repelling); (iii) in all cases $\mathbb{E}[m_t] \rightarrow \infty$ as $t \rightarrow \infty$: no fixed point is globally attracting for the mean dynamics; (iv) by contrast, under linear feedback $m_{t+1} = c + \varrho Y_t$ with $c > 0$ and $0 \leq \varrho < 1$, $\mathbb{E}[m_t] \rightarrow c/(1 - \varrho)$ geometrically from every initial condition.*

Proof. (i) The Poisson moment generating function: for $Y \sim \text{Poisson}(m)$ and real s , $\mathbb{E}[e^{sY}] = \sum_{k \geq 0} e^{sk} e^{-m} m^k / k! = e^{m(e^s - 1)}$ (the exponential series at me^s). Hence $\mathbb{E}[m_{t+1} \mid \mathcal{H}_t] = e^a \mathbb{E}[e^{bY_t} \mid \mathcal{H}_t] = G(m_t)$. Differentiation gives $G' = (e^b - 1)G > 0$ and $G'' = (e^b - 1)^2 G > 0$; at a fixed point $G(m^*) = m^*$, so $G'(m^*) = m^*(e^b - 1)$.

(ii) Write $c_0 = e^a$, $d = e^b - 1 > 0$. Fixed points solve $c_0 e^{dm} = m$. The strictly convex $g(m) = G(m) - m$ has $g(0) = c_0 > 0$ and $g(m) \rightarrow \infty$; it attains its minimum where $G' = 1$, i.e. at $\tilde{m} = d^{-1} \log(1/(c_0 d))$ when positive, with $g(\tilde{m}) = 1/d - \tilde{m}$. Thus g has a zero iff $\tilde{m} \geq 1/d$, i.e. iff $c_0 d \leq e^{-1}$, the stated condition. A strictly convex function has at most two zeros; at the smaller, g crosses from positive to non-positive, so $G'(m_-^*) \leq 1$; at the larger the crossing is upward, $G'(m_+^*) \geq 1$.

(iii) Note $m_t \geq e^a$ for every $t \geq 2$. Choose an integer k with $m' := e^{a+bk}$ strictly above every fixed point (possible: fixed points are finite, or absent). Then $G(m) > m$ for all $m \geq m'$, and the iterates $G^{on}(m')$ increase to ∞ (a bounded limit would be a fixed point $\geq m'$). Poisson tails are stochastically increasing in the mean, so $p_0 := \mathbb{P}(\text{Poisson}(e^a) \geq k) > 0$ lower-bounds $\mathbb{P}(Y_2 \geq k \mid \mathcal{H}_2)$. By conditional Jensen with the convex increasing G and induction on n , $\mathbb{E}[m_{3+n} \mid \mathcal{H}_3] \geq G^{on}(m_3)$ (allowing $+\infty$). On $B = \{Y_2 \geq k\}$ we have $m_3 = e^{a+bY_2} \geq m'$, hence

$$\mathbb{E}[m_{3+n}] \geq \mathbb{E}[\mathbf{1}\{B\} \mathbb{E}[m_{3+n} \mid \mathcal{H}_3]] \geq \mathbb{P}(B) G^{on}(m') \geq p_0 G^{on}(m') \rightarrow \infty.$$

(iv) Linearity passes through expectations exactly: $\mathbb{E}[m_{t+1} \mid \mathcal{H}_t] = c + \varrho m_t$, so $\mathbb{E}[m_t] - c/(1 - \varrho) = \varrho^{t-1}(\mathbb{E}[m_1] - c/(1 - \varrho)) \rightarrow 0$. The contrast with (iii) is the Jensen gap: G is convex, so rare large counts dominate the mean. Note also $e^b - 1 > b$: the true mean-map derivative $m^*(e^b - 1)$ strictly exceeds the operating-point radius $b m^*$ (the univariate $\rho(\text{diag}(m^*)B)$ of Warning 8.18), so a configuration certified locally subcritical at the operating point can still be metastable — exactly the excursion mode that the generator stability checks of Chapter 19 are designed to catch. Ergodicity theory for log-linear count autoregressions with bounded or $\log(1 + \cdot)$ feedback is in Fokianos et al. (2009); Fokianos and Tjøstheim (2011); nonlinear Hawkes stability under Lipschitz links is in Brémaud and Massoulié (1996). $\square$

C.2.4 The κ - θ ridge, made quantitative

This proves the claims asserted in Section 10.5.4 and quoted again in Proposition 11.13: on bucketed first moments the score in θ is exponentially small in the bucket width while the score in κ is order one, so the Fisher information of the MBPP likelihood degenerates along a ridge. The setting is Proposition 10.13: kernel $h(t) = \kappa \theta e^{-\theta t}$, piecewise-constant baseline $\mu(t) = \mu_w \in [\mu_{\min}, \mu_{\max}]$ on bucket w of width Δ , $0 < \mu_{\min}$, relaxation rate $r = (1 - \kappa)\theta$, $\rho = e^{-r\Delta}$. With $h = \bar{\lambda} - \mu$, the ODE in the proof of Proposition 10.13 reads $h' = \kappa \theta \mu_w - r h$ on bucket w , with target $h_w^* = \kappa \mu_w / (1 - \kappa)$; solving on one bucket and integrating $\bar{\lambda} = \mu + h$ gives, for the state h_w at the start of bucket w ,

$$h_{w+1} = \rho h_w + (1 - \rho) h_w^*, \quad \Xi_w = \frac{\Delta \mu_w}{1 - \kappa} + (h_w - h_w^*) \frac{1 - \rho}{r}, \quad (\text{C.4})$$

where Ξ_w is the bucket compensator entering the interval-censored Poisson likelihood of Definition 10.14.

Proposition C.11 (The limiting singularity of the censored information matrix). *Fix $\kappa \in (0, 1)$ and the baseline range $0 < \mu_{\min} \leq \mu_w \leq \mu_{\max}$, take the stationary initialization $h_1 = h_1^*$, and let $C_0 = \kappa(\mu_{\max} - \mu_{\min})/(1 - \kappa)$. Then for all w : (i) $|h_w - h_w^*| \leq C_0$ and $\Xi_w = \Delta \mu_w / (1 - \kappa) + R_w$ with $|R_w| \leq C_0/r$; (ii) $|\partial \Xi_w / \partial \theta| \leq C_0(1 - \kappa)(2\Delta\rho/r + r^{-2})$; (iii) $\partial \Xi_w / \partial \kappa = \Delta \mu_w (1 - \kappa)^{-2} + r_w$ with $|r_w| \leq C_1(\Delta\rho + 1/r)$, where C_1 depends only on $(\kappa, \mu_{\min}, \mu_{\max})$. Consequently, whenever $r \geq r_1 > 0$ and $r\Delta \rightarrow \infty$,*

$$\frac{\sup_w |\partial \Xi_w / \partial \theta|}{\inf_w \partial \Xi_w / \partial \kappa} \rightarrow 0,$$

and the Fisher information of the interval-censored Poisson likelihood in (κ, θ) , $\mathcal{I} = \sum_w \Xi_w^{-1} (\nabla \Xi_w) (\nabla \Xi_w)^\top$, becomes singular: $\mathcal{I}_{\theta\theta}$ and $\mathcal{I}_{\kappa\theta}$ vanish relative to $\mathcal{I}_{\kappa\kappa}$, and the profile information for θ , $\mathcal{I}_{\theta\theta} - \mathcal{I}_{\theta\kappa}^2 / \mathcal{I}_{\kappa\kappa}$, tends to zero per bucket. In the regime $\theta \rightarrow \infty$ with Δ fixed, $\partial \Xi_w / \partial \theta \rightarrow 0$ absolutely while $\partial \Xi_w / \partial \kappa \rightarrow \Delta \mu_w (1 - \kappa)^{-2}$, bounded away from zero: θ leaves the first-moment likelihood entirely.

Proof. Fisher information formula. For independent $C_w \sim \text{Poisson}(\Xi_w(\eta))$ the score in a parameter η is $\sum_w (C_w / \Xi_w - 1) \partial_\eta \Xi_w$; the summands are independent with mean zero and variance $(\partial_\eta \Xi_w)^2 / \Xi_w$ (since $\text{Var } C_w = \Xi_w$), giving the stated $\mathcal{I}$.

(i) The levels h_w^* lie in $[\kappa \mu_{\min} / (1 - \kappa), \kappa \mu_{\max} / (1 - \kappa)]$, an interval of length C_0 . By Equation (C.4), h_{w+1} is a convex combination of h_w and h_w^* ; with $h_1 = h_1^*$ in the interval, induction keeps every h_w in it, so $|h_w - h_w^*| \leq C_0$. Then $R_w = (h_w - h_w^*)(1 - \rho)/r$ obeys $|R_w| \leq C_0/r$.

(ii) Only r and ρ carry θ ; h_w^* does not. Differentiating Equation (C.4) in θ , the derivative state $D_w = \partial h_w / \partial \theta$ satisfies $D_1 = 0$ and $D_{w+1} = \rho D_w + (h_w - h_w^*) \partial \rho / \partial \theta$ with $\partial \rho / \partial \theta = -(1 - \kappa) \Delta \rho$, so $|D_{w+1}| \leq \rho |D_w| + (1 - \kappa) \Delta \rho C_0$ and by the geometric recursion $|D_w| \leq (1 - \kappa) \Delta \rho C_0 / (1 - \rho)$ for all w . Also $\partial_\theta [(1 - \rho)/r] = (1 - \kappa) [\Delta \rho / r - (1 - \rho)/r^2]$, bounded in modulus by $(1 - \kappa)(\Delta \rho / r + r^{-2})$. Hence

$$\left| \frac{\partial \Xi_w}{\partial \theta} \right| \leq |D_w| \frac{1 - \rho}{r} + C_0 (1 - \kappa) \left(\frac{\Delta \rho}{r} + \frac{1}{r^2} \right) \leq C_0 (1 - \kappa) \left(\frac{2 \Delta \rho}{r} + \frac{1}{r^2} \right).$$

(iii) The leading term of Ξ_w differentiates to $\Delta \mu_w (1 - \kappa)^{-2}$. For the remainder, $\partial_\kappa h_w^* \leq \mu_{\max} (1 - \kappa)^{-2}$, and the derivative state $F_w = \partial h_w / \partial \kappa$ obeys $F_{w+1} = (1 - \rho) \partial_\kappa h_w^* + \rho F_w + (h_w - h_w^*) \theta \Delta \rho$, whence, using $\theta = r / (1 - \kappa)$ and the geometric recursion again, $|F_w| \leq \mu_{\max} (1 - \kappa)^{-2} + C_0 \Delta \rho r [(1 - \kappa)(1 - \rho)]^{-1}$. Combining with $\partial_\kappa [(1 - \rho)/r] = \theta [(1 - \rho)/r^2 - \Delta \rho / r]$, every term of $\partial_\kappa R_w$ is bounded by a constant multiple (depending only on $\kappa, \mu_{\min}, \mu_{\max}$) of $\Delta \rho$ or $1/r$: (iii).

The displayed ratio limit follows by dividing (ii) by (iii): under $r \geq r_1$ and $r \Delta \rightarrow \infty$ the numerator terms are $o(\Delta)$ ($\Delta \rho \rightarrow 0$ superexponentially, and $r^{-2} / \Delta = (r \Delta)^{-1} r^{-1} \leq (r \Delta)^{-1} r_1^{-1} \rightarrow 0$), while the denominator is bounded below by $\Delta \mu_{\min} (1 - \kappa)^{-2}$ minus a vanishing correction. Since Ξ_w stays in a fixed positive interval around $\Delta \mu_w / (1 - \kappa)$ by (i), the information entries inherit the same ratios: $\mathcal{I}_{\theta\theta} / \mathcal{I}_{\kappa\kappa} \rightarrow 0$, $|\mathcal{I}_{\kappa\theta}| / \mathcal{I}_{\kappa\kappa} \rightarrow 0$, and the profile information for θ is at most $\mathcal{I}_{\theta\theta} \rightarrow 0$ per bucket — in the dominant regime of order $(\Delta \rho)^2 / \Xi_w \asymp \Delta e^{-2r\Delta}$, the $\Delta^2 e^{-2(1-\kappa)\theta\Delta}$ scaling per unit time quoted in Section 10.5.4 and Proposition 11.13. By Cramer–Rao, the variance of any unbiased estimator of θ from bucket means diverges in this limit; the $\theta \rightarrow \infty$, Δ fixed statement reads off (ii) and (iii) directly. What censored data retain of θ beyond first moments lives in autocovariances, which the working likelihood of Definition 10.14 discards by construction — Warning 10.17 is the operational consequence. $\square$

C.3 Self-contained auxiliary results

The outside-option score of Section 15.3 carries an intercept; the results below supply its structural reading when the unseen pool is modeled explicitly. Neither is needed by any numbered claim in the main text: they make precise (a) when a latent Gaussian pool collapses, to first order, into a single pseudo-candidate score, and (b) why the resulting constant confounds pool size with mean pool quality — the formal content of the recalibration rule attached to Protocol 15.7 and of the newcomer-versus-entrant caution of Warning 15.4.

Lemma C.12 (Gaussian moment generating function). *If $z \sim \mathcal{N}(\mu, \Sigma)$ in $\mathbb{R}^q$ and $\beta \in \mathbb{R}^q$, then $\mathbb{E}[\exp(\beta^\top z)] = \exp(\beta^\top \mu + \frac{1}{2} \beta^\top \Sigma \beta)$.*

Proof. Factor $\Sigma = LL^\top$ and write $z \stackrel{d}{=} \mu + L\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I_q)$, so $\beta^\top z = \beta^\top \mu + a^\top \varepsilon$ with $a = L^\top \beta$; by independence of coordinates and the completed square $\mathbb{E}e^{a_i \varepsilon_i} = e^{a_i^2/2} (2\pi)^{-1/2} \int e^{-(x-a_i)^2/2} dx = e^{a_i^2/2}$, multiply and use $\|a\|^2 = \beta^\top \Sigma \beta$. $\square$

Proposition C.13 (Integrated-entrant identity, with mean-field error). *Suppose the unseen pool consists of N_{pool} latent candidates whose features z_j are i.i.d. $\mathcal{N}(\mu, \Sigma)$, independent of $\mathcal{H}_t$, entering an extended softmax through scores $\beta^\top z_j$. Replacing the pool block $\sum_j e^{\beta^\top z_j}$ of the denominator by its expectation yields the single pseudo-candidate score*

$$q_{\text{new}} = \log N_{\text{pool}} + \beta^\top \mu + \frac{1}{2} \beta^\top \Sigma \beta, \quad (\text{C.5})$$

and when the tracked block of the denominator and the expected pool block both grow like the pool size, the relative error of the replacement is $O(1/N_{\text{pool}})$.

Proof. By linearity and Lemma C.12, $\mathbb{E} \sum_{j=1}^{N_{\text{pool}}} e^{\beta^\top z_j} = N_{\text{pool}} e^{\beta^\top \mu + \beta^\top \Sigma \beta / 2}$; matching $e^{q_{\text{new}}}$ to this expectation gives Equation (C.5). For the error, condition on $\mathcal{H}_t$: any tracked candidate's choice probability is $e^{q_i} \mathbb{E}[(A + T)^{-1}]$ with $A > 0$ the fixed tracked block and $T = \sum_j e^{\beta^\top z_j}$ independent of $\mathcal{H}_t$. The map $f(x) = (A + x)^{-1}$ has $0 < f'' \leq 2A^{-3}$ on $[0, \infty)$, so a second-order Taylor expansion about $\mathbb{E}T$ (the linear term has mean zero) gives $|\mathbb{E}f(T) - f(\mathbb{E}T)| \leq \text{Var}(T)/A^3$, with $\text{Var}(T) = N_{\text{pool}} e^{2\beta^\top \mu + \beta^\top \Sigma \beta} (e^{\beta^\top \Sigma \beta} - 1)$ the lognormal variance from Lemma C.12 applied at β and 2β . When $A \asymp \mathbb{E}T \asymp N_{\text{pool}}$, $f \asymp N_{\text{pool}}^{-1}$ while the error is $O(N_{\text{pool}}^{-2})$: relative error $O(1/N_{\text{pool}})$. $\square$

Proposition C.14 (Pool-size non-identifiability, made precise). *In any parametrization in which the pool block enters the choice probabilities only through the scalar score $q_{\text{new}} = b_0 + \log N_{\text{pool}} + \beta^\top \mu + \frac{1}{2} \beta^\top \Sigma \beta$ with a constant pool, the likelihood of every dataset is invariant under $(b_0, \log N_{\text{pool}}) \mapsto (b_0 + c, \log N_{\text{pool}} - c)$ for every real c : the constant b_0 and the pool size (equivalently, the pool size and the mean pool quality) are not separately identified. If instead $N_{\text{pool},t}$ varies over events and enters as a known offset, $q_{\text{new},t} = b_0 + \log N_{\text{pool},t} + \dots$, then b_0 is identified.*

Proof. Every choice probability — tracked or outside — depends on the pool parameters only through $e^{q_{\text{new}}}$ in the softmax; the stated transformation leaves q_{new} , hence every probability, hence the likelihood of any dataset, unchanged: exact non-identification, not a flat-likelihood approximation. With known offsets, identical probabilities force $b_0 = b'_0$. This is why the calibration discipline of Section 15.3 treats the fitted outside-option constant as a statement about $b_0 + \log N_{\text{pool}}$ — never about entrant quality alone — and why any change in the pool definition forces a recalibration of that constant before the outside-option probabilities of Protocol 15.7 can be compared across eras. $\square$

Bibliography

- Odd O. Aalen, Ørnulf Borgan, and Håkon K. Gjessing. *Survival and Event History Analysis: A Process Point of View*. Springer, 2008.
- Paul D. Allison. Discrete-time methods for the analysis of event histories. *Sociological Methodology*, 13:61–98, 1982.
- Per Kragh Andersen and Richard D. Gill. Cox’s regression model for counting processes: A large sample study. *The Annals of Statistics*, 10(4):1100–1120, 1982.
- Per Kragh Andersen, Ørnulf Borgan, Richard D. Gill, and Niels Keiding. *Statistical Models Based on Counting Processes*. Springer, 1993.
- Shahriar Azizpour, Kay Giesecke, and Gustavo Schwenkler. Exploring the sources of default clustering. *Journal of Financial Economics*, 129(1):154–183, 2018.
- Emmanuel Bacry, Iacopo Mastromatteo, and Jean-François Muzy. Hawkes processes in finance. *Market Microstructure and Liquidity*, 1(1), 2015.
- Avinash Barnwal, Hyunsu Cho, and Toby Hocking. Survival regression with accelerated failure time model in XGBoost. *Journal of Computational and Graphical Statistics*, 31(4):1292–1302, 2022.
- Anna Bonnet, Miguel Martinez Herrera, and Maxime Sangnier. Maximum likelihood estimation for Hawkes processes with self-excitation or inhibition. *Statistics & Probability Letters*, 179, 2021.
- Pierre Brémaud and Laurent Massoulié. Stability of nonlinear Hawkes processes. *The Annals of Probability*, 24(3):1563–1588, 1996.
- Norman Breslow. Covariance analysis of censored survival data. *Biometrics*, 30(1):89–99, 1974.
- A. E. Brockwell. Universal residuals: A multivariate transformation. *Statistics & Probability Letters*, 77(14):1473–1478, 2007.
- Emery N. Brown, Riccardo Barbieri, Valérie Ventura, Robert E. Kass, and Loren M. Frank. The time-rescaling theorem and its application to neural spike train data analysis. *Neural Computation*, 14(2):325–346, 2002.
- Chris Burges, Tal Shaked, Erin Renshaw, Ari Lazier, Matt Deeds, Nicole Hamilton, and Greg Hullender. Learning to rank using gradient descent. In *Proceedings of the 22nd International Conference on Machine Learning (ICML)*, pages 89–96, 2005.
- Christopher J. C. Burges. From RankNet to LambdaRank to LambdaMART: An overview. Technical Report MSR-TR-2010-82, Microsoft Research, 2010.

- Raymond J. Carroll, David Ruppert, Leonard A. Stefanski, and Ciprian M. Crainiceanu. *Measurement Error in Nonlinear Models: A Modern Perspective*. Chapman & Hall/CRC, 2nd edition, 2006.
- Anne Chao. Nonparametric estimation of the number of classes in a population. *Scandinavian Journal of Statistics*, 11(4):265–270, 1984.
- Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.
- Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018.
- Richard J. Cook and Jerald F. Lawless. *The Statistical Analysis of Recurrent Events*. Springer, 2007.
- Manon Costa, Carl Graham, Laurence Marsalle, and Viet Chi Tran. Renewal in Hawkes processes with self-excitation and inhibition. *Advances in Applied Probability*, 52(3):879–915, 2020.
- David R. Cox. Regression models and life-tables. *Journal of the Royal Statistical Society, Series B*, 34(2):187–220, 1972.
- David R. Cox. Partial likelihood. *Biometrika*, 62(2):269–276, 1975.
- Claudia Czado, Tilmann Gneiting, and Leonhard Held. Predictive model assessment for count data. *Biometrics*, 65(4):1254–1261, 2009.
- Daryl J. Daley and David Vere-Jones. *An Introduction to the Theory of Point Processes, Volume I: Elementary Theory and Methods*. Springer, 2nd edition, 2003.
- Daryl J. Daley and David Vere-Jones. *An Introduction to the Theory of Point Processes, Volume II: General Theory and Structure*. Springer, 2nd edition, 2008.
- Angelos Dassios and Hongbiao Zhao. Exact simulation of Hawkes process with exponentially decaying intensity. *Electronic Communications in Probability*, 18:1–13, 2013.
- Nan Du, Hanjun Dai, Rakshit Trivedi, Utkarsh Upadhyay, Manuel Gomez-Rodriguez, and Le Song. Recurrent marked temporal point processes: Embedding event history to vector. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1555–1564, 2016.
- Darrell Duffie, Leandro Saita, and Ke Wang. Multi-period corporate default prediction with stochastic covariates. *Journal of Financial Economics*, 83(3):635–665, 2007.
- Darrell Duffie, Andreas Eckner, Guillaume Horel, and Leandro Saita. Frailty correlated default. *The Journal of Finance*, 64(5):2089–2123, 2009.
- Bradley Efron. The efficiency of Cox’s likelihood function for censored data. *Journal of the American Statistical Association*, 72(359):557–565, 1977.
- Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall/CRC, 1994.

- Paul Embrechts, Thomas Liniger, and Lu Lin. Multivariate Hawkes processes: An application to financial data. *Journal of Applied Probability*, 48A:367–378, 2011.
- Vernon T. Farewell. The use of mixture models for the analysis of survival data with long-term survivors. *Biometrics*, 38(4):1041–1046, 1982.
- David Firth. Bias reduction of maximum likelihood estimates. *Biometrika*, 80(1):27–38, 1993.
- Konstantinos Fokianos and Dag Tjøstheim. Log-linear Poisson autoregression. *Journal of Multivariate Analysis*, 102(3):563–578, 2011.
- Konstantinos Fokianos, Anders Rahbek, and Dag Tjøstheim. Poisson autoregression. *Journal of the American Statistical Association*, 104(488):1430–1439, 2009.
- Jerome H. Friedman. Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5):1189–1232, 2001.
- William A. Gale and Geoffrey Sampson. Good–Turing frequency estimation without tears. *Journal of Quantitative Linguistics*, 2(3):217–237, 1995.
- Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- Irving J. Good. The population frequencies of species and the estimation of population parameters. *Biometrika*, 40(3–4):237–264, 1953.
- Gustaf Gripenberg, Stig-Olof Londen, and Olof Staffans. *Volterra Integral and Functional Equations*. Cambridge University Press, 1990.
- Niels Richard Hansen, Patricia Reynaud-Bouret, and Vincent Rivoirard. Lasso and probabilistic inequalities for multivariate point processes. *Bernoulli*, 21(1):83–143, 2015.
- Andrew C. Harvey. *Forecasting, Structural Time Series Models and the Kalman Filter*. Cambridge University Press, 1989.
- Alan G. Hawkes. Spectra of some self-exciting and mutually exciting point processes. *Biometrika*, 58(1):83–90, 1971.
- Alan G. Hawkes and David Oakes. A cluster process representation of a self-exciting process. *Journal of Applied Probability*, 11(3):493–503, 1974.
- James J. Heckman and George J. Borjas. Does unemployment cause future unemployment? Definitions, questions and answers from a continuous time model of heterogeneity and state dependence. *Economica*, 47(187):247–283, 1980.
- Miguel A. Hernán and James M. Robins. *Causal Inference: What If*. Chapman & Hall/CRC, Boca Raton, 2020.
- Philip Hougaard. *Analysis of Multivariate Survival Data*. Springer, 2000.
- Guido W. Imbens and Donald B. Rubin. *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge University Press, 2015.
- Kalervo Järvelin and Jaana Kekäläinen. Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems*, 20(4):422–446, 2002.
- John D. Kalbfleisch and Ross L. Prentice. *The Statistical Analysis of Failure Time Data*. Wiley, 2nd edition, 2002.

- Rudolf E. Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960.
- Niels Keiding. Statistical inference in the Lexis diagram. *Philosophical Transactions of the Royal Society of London, Series A*, 332:487–509, 1990.
- Gary King and Langche Zeng. Logistic regression in rare events data. *Political Analysis*, 9(2):137–163, 2001.
- John F. C. Kingman. *Poisson Processes*. Oxford University Press, 1993.
- Erik Lewis and George Mohler. A nonparametric EM algorithm for multiscale Hawkes processes. *Journal of Nonparametric Statistics (preprint)*, 2011.
- Peter A. W. Lewis and Gerald S. Shedler. Simulation of nonhomogeneous Poisson processes by thinning. *Naval Research Logistics Quarterly*, 26(3):403–413, 1979.
- R. Duncan Luce. *Individual Choice Behavior: A Theoretical Analysis*. Wiley, 1959.
- Daniel McFadden. Conditional logit analysis of qualitative choice behavior. In Paul Zarembka, editor, *Frontiers in Econometrics*, pages 105–142. Academic Press, 1974.
- Daniel McFadden. Modeling the choice of residential location. In Anders Karlqvist, Lars Lundqvist, Folke Snickars, and Jörgen W. Weibull, editors, *Spatial Interaction Theory and Planning Models*, pages 75–96. North-Holland, 1978.
- Hongyuan Mei and Jason Eisner. The neural Hawkes process: A neurally self-modulating multivariate point process. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, 2017.
- Alexandru Niculescu-Mizil and Rich Caruana. Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning (ICML)*, pages 625–632, 2005.
- Yosihiko Ogata. The asymptotic behaviour of maximum likelihood estimators for stationary point processes. *Annals of the Institute of Statistical Mathematics*, 30:243–261, 1978.
- Yosihiko Ogata. On Lewis’ simulation method for point processes. *IEEE Transactions on Information Theory*, 27(1):23–31, 1981.
- Yosihiko Ogata. Statistical models for earthquake occurrences and residual analysis for point processes. *Journal of the American Statistical Association*, 83(401):9–27, 1988.
- Takahiro Omi, Naonori Ueda, and Kazuyuki Aihara. Fully neural network based model for general temporal point processes. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019.
- Tohru Ozaki. Maximum likelihood estimation of Hawkes’ self-exciting point processes. *Annals of the Institute of Statistical Mathematics*, 31:145–155, 1979.
- Judea Pearl. Causal diagrams for empirical research. *Biometrika*, 82(4):669–688, 1995.
- Robin L. Plackett. The analysis of permutations. *Journal of the Royal Statistical Society, Series C (Applied Statistics)*, 24(2):193–202, 1975.
- John C. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*. MIT Press, 1999.

- Sebastian Pölsterl. scikit-survival: A library for time-to-event analysis built on top of scikit-learn. *Journal of Machine Learning Research*, 21(212):1–6, 2020.
- Ross L. Prentice. Covariate measurement errors and parameter estimation in a failure time regression model. *Biometrika*, 69(2):331–342, 1982.
- Ross L. Prentice and Lynn A. Gloeckler. Regression analysis of grouped survival data with application to breast cancer data. *Biometrics*, 34(1):57–67, 1978.
- Hein Putter, Marta Fiocco, and Ronald B. Geskus. Tutorial in biostatistics: Competing risks and multi-state models. *Statistics in Medicine*, 26(11):2389–2430, 2007.
- Jakob Gulddahl Rasmussen. Bayesian inference for Hawkes processes. *Methodology and Computing in Applied Probability*, 15:623–642, 2013.
- Marian-Andrei Rizoitiu, Alexander Soen, Shidi Li, Pio Calderon, Leanne J. Dong, Aditya Krishna Menon, and Lexing Xie. Interval-censored Hawkes processes. *Journal of Machine Learning Research*, 23(338):1–84, 2022.
- Dimitris Rizopoulos. *Joint Models for Longitudinal and Time-to-Event Data: With Applications in R*. Chapman & Hall/CRC, 2012.
- Yaniv Romano, Evan Patterson, and Emmanuel J. Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems 32*, 2019.
- Steven G. Self and Kung-Yee Liang. Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association*, 82(398):605–610, 1987.
- Oleksandr Shchur, Marin Biloš, and Stephan Günnemann. Intensity-free learning of temporal point processes. In *International Conference on Learning Representations (ICLR)*, 2020.
- Aleksandr Simma and Michael I. Jordan. Modeling events with cascades of Poisson processes. In *Proceedings of the 26th Conference on Uncertainty in Artificial Intelligence (UAI)*, pages 546–555, 2010.
- J. Q. Smith. Diagnostic checks of non-standard time series models. *Journal of Forecasting*, 4(3):283–291, 1985.
- Terry M. Therneau and Patricia M. Grambsch. *Modeling Survival Data: Extending the Cox Model*. Springer, 2000.
- Kenneth E. Train. *Discrete Choice Methods with Simulation*. Cambridge University Press, 2nd edition, 2009.
- Wilson Truccolo, Uri T. Eden, Matthew R. Fellows, John P. Donoghue, and Emery N. Brown. A point process framework for relating neural spiking activity to spiking history, neural ensemble, and extrinsic covariate effects. *Journal of Neurophysiology*, 93(2):1074–1089, 2005.
- Anastasios A. Tsiatis and Marie Davidian. Joint modeling of longitudinal and time-to-event data: An overview. *Statistica Sinica*, 14(3):809–834, 2004.
- Anastasios A. Tsiatis, Victor DeGruttola, and Michael S. Wulfsohn. Modeling the relationship of survival to longitudinal data measured with error: Applications to survival and CD4 counts in patients with AIDS. *Journal of the American Statistical Association*, 90(429):27–37, 1995.

- Bruce W. Turnbull. The empirical distribution function with arbitrarily grouped, censored and truncated data. *Journal of the Royal Statistical Society, Series B*, 38(3):290–295, 1976.
- George E. Uhlenbeck and Leonard S. Ornstein. On the theory of the Brownian motion. *Physical Review*, 36(5):823–841, 1930.
- Aad W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, 1998.
- Tyler J. VanderWeele and Peng Ding. Sensitivity analysis in observational research: Introducing the E-value. *Annals of Internal Medicine*, 167(4):268–274, 2017.
- James W. Vaupel, Kenneth G. Manton, and Eric Stallard. The impact of heterogeneity in individual frailty on the dynamics of mortality. *Demography*, 16(3):439–454, 1979.
- Alejandro Veen and Frederic P. Schoenberg. Estimation of space–time branching process models in seismology using an EM-type algorithm. *Journal of the American Statistical Association*, 103(482):614–624, 2008.
- Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699, 2002.